

Release Notes: Junos[®] OS Release 20.4R3 for the ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX

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Introduction

Junos OS runs on the following Juniper Networks® products: ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion Enterprise, Junos Fusion Provider Edge, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

These release notes accompany Junos OS Release 20.4R3 for the ACX Series, cSRX Container Firewall (cSRX), EX Series, JRR Series, Juniper Secure Connect, Junos Fusion Enterprise, Junos Fusion Provider Edge, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, virtual MX Series router (vMX), Virtual Route Reflector (vRR), and vSRX Virtual Firewall (vSRX). They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

- [In Focus guide](#)—We have a document called In Focus that provides details on the most important features for the release in one place. We hope this document will quickly get you to the latest information about Junos OS features. Let us know if you find this information useful by sending an e-mail to techpubs-comments@juniper.net.
- **Important Information:**
 - [Upgrading Using ISSU on page 390](#)
 - [Licensing on page 390](#)
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These release notes accompany Junos OS Release 20.4R3 for the ACX Series. They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

You can also find these release notes on the Juniper Networks Junos OS Documentation webpage, located at https://www.juniper.net/documentation/product/en_US/junos-os.

What's New

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This section describes the new features or enhancements to existing features in Junos OS Release 20.4R3 for the ACX Series.

What's New in Release 20.4R3

There are no new features or enhancements to existing features for ACX Series routers in Junos OS Release 20.4R3.

What's New in Release 20.4R2

There are no new features or enhancements to existing features for ACX Series routers in Junos OS Release 20.4R2.

What's New in Release 20.4R1

Hardware

- We've added the following features to the ACX5448 in Junos OS Release 20.4R1.

Table 1: Features Supported by the ACX5448 Routers

Feature	Description
Authentication, Authorization and Accounting	• Support for 802.1X authentication on Layer 3 interfaces. 802.1X is an IEEE standard for port-based network access control that authenticates users connected to a LAN port. [See 802.1X Authentication.]
Automation	• Support for either WAN interfaces or management interfaces to automatically download and install the appropriate software and the configuration file on your device during the ZTP bootstrap process. [See Zero Touch Provisioning.]
Class of service (CoS)	• Support for up to three levels of hierarchical scheduling (physical interfaces, logical interfaces, and queues). Configurable buffer support is also added. By default, all interfaces on the ACX5448 use port-based scheduling (eight queues per physical port). To enable hierarchical scheduling, set the hierarchical-scheduler statement at the [edit interfaces <i>interface-name</i>] hierarchy level. [See Hierarchical Class of Service in ACX Series Routers.]
Ethernet OAM	• Support for Ethernet OAM CFM. You can now synchronize local-interface status between two connected devices with remote interface up/down trigger with OAM CFM. CFM provides end-to-end signals even if the two devices are not directly connected. [See Introduction to OAM Connectivity Fault Management (CFM).]
EVPN	• Support for EVPNs and Interfaces. In EVPN-MPLS and MC-LAG environments, the configuration of anycast gateways on ACX5448 routers that are multihomed in all-active mode is supported. [See Anycast Gateways.]
Layer 2 features	• Support for pseudowire redundancy in MC-LAG. ACX5448 routers support pseudowire redundant Layer 2 circuits in MC-LAG routers. VPLS is not supported. [See Understanding Pseudowire Redundancy Mobile Backhaul Scenarios.]
Layer 3 features	• Support for Layer 3 VPN in MC-LAG chassis. ACX5448 routers support Layer 3 VPN in VRRP over IRB interfaces in MC-LAG routers. Layer 3 routing and Layer 3 VPN are not directly supported on the MC-LAG interfaces. [See Understanding VRRP and Understanding Layer 3 VPNs.]
Network Security	• Support for control plane DDoS protection, which is enabled by default on ACX5448 routers for many Layer 2 and Layer 3 protocols. Control Plane DDoS protection uses firewall filters and policers to discard or rate-limit control plane traffic at the Routing Engine level, which prevents malicious traffic from interfering with device operations. You can disable this feature or change the default policer parameters for supported protocol groups. [See Control Plane Distributed Denial-of-Service (DDoS) Protection Overview.]

Table 1: Features Supported by the ACX5448 Routers (*continued*)

Feature	Description
Software installation and upgrade	Support for the ACX5448-M-LT, a top-of-rack router that supports only Junos Limited image. The Junos Limited image does not have data-plane encryption and is intended only for countries in the Eurasian Customs Union because these countries have import restrictions on software containing data-plane encryption. Unlike the JunosWorldwide image, the Junos Limited image supports control plane encryption through Secure Shell (SSH) and Secure Sockets Layer (SSL), thus allowing secure management of the system. [See ACX5448 System Overview.]
Timing and synchronization	Support for Precision Time Protocol (PTP) G.8275.2 enhanced profile with PTP over IPv4 and IPv6 unicast traffic. [See Understanding the PTP G.8275.2 Enhanced Profile (Telecom Profile).]

- **Support for SFP-1GE-LH-ET transceivers (ACX1100 and ACX2100)**—Starting in Junos OS Release 20.4R1, the ACX1100 and ACX2100 Universal Metro Routers support the SFP-1GE-LH-ET transceivers. [See the [Hardware Compatibility Tool \(HCT\)](#) for details.]
- **Support for SFP-GE80KT14R15 and SFP-GE80KT15R14 transceivers (ACX5448, ACX5448-D, and ACX5448-M)**—Starting in Junos OS Release 20.4R1, the ACX5448, ACX5448-D, and ACX5448-M Universal Metro Routers support the SFP-GE80KT14R15 and SFP-GE80KT15R14 transceivers. [See the [Hardware Compatibility Tool \(HCT\)](#) for details.]
- **Support for SFPP-10GE-DWDM-IT transceivers (ACX5448, ACX5448-D, and ACX5448-M)**—Starting in Junos OS Release 20.4R1, the ACX5448, ACX5448-D, and ACX5448-M Universal Metro Routers support the SFPP-10GE-DWDM-IT transceivers. [See the [Hardware Compatibility Tool \(HCT\)](#) for details.]

High Availability (HA) and Resiliency

- **NSR support for IS-IS with SR (ACX Series, MX Series)**—Starting in Junos OS Release 20.4R1, ACX Series devices support NSR for IS-IS with segment routing (SR). To use NSR, you must first enable GRES on your device. [See [Nonstop Active Routing Concepts](#)]

Junos Telemetry Interface

- **JTI support for persistent active gRPC sessions between collector and server during an SSL certificate update (ACX Series, MX Series, and PTX Series)**—Junos OS Release 20.4R1 supports persistent active remote procedure call (gRPC) sessions between the collector (client) and server during an SSL certificate update.

For secure channel authentication, the TLS protocol is used to maintain a secure channel between the collector and the server. TLS uses the server certificate and the client certificate to authenticate each other and send encrypted messages over the network. When an SSL certificate is updated, existing gRPC

sessions are abruptly terminated, forcing the collector to initiate a new gRPC connection and subscribe to sensors again.

To avoid this problem, you can enable persistent active gRPC sessions by configuring **hot-reloading** at the `[edit system services extension-service request-response grpc ssl]` hierarchy level. After you enable this feature, gRPC sessions will remain active even when authentication certificates are updated.

After the certificate is updated, any new gRPC session will use the updated certificate.

[See [gRPC Services for Junos Telemetry Interface](#) and [ssl](#).]

- **Juniper Resiliency Interface for exception reporting and null route detection (ACX Series, PTX Series, and MX Series)**—Starting in Junos OS Release 20.4R1, you can use Juniper Resiliency Interface to detect and reduce Mean Time to Repair (MTTR) first-order network issues. Juniper Resiliency Interface uses a push model for data reporting from the entities in the system which encounter packet drops. This automates the workflow for detecting, reporting, and mitigating adverse exceptions.

To collect kernel routing table and routing protocol process exceptions, configure the **set system resiliency exceptions** statement at the `[edit]` hierarchy level to specify exception reporting based on kernel exceptions, and routing exceptions.

You can display exceptions from a remote collector by means of remote procedure call (gRPC) services or gRPC network management interface (gNMI) services. Display on-box exceptions by accessing the `/var/log` file or the database at `/var/db/ResiliencyExceptions.db`. No Junos operational mode commands display these exceptions.

Routing Protocols

- **Support for multiple single-hop EBGp sessions on different links using the same IPv6 link-local address (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, you are no longer required to have unique peer addresses for Juniper devices for every EBGp session. You can now enable single-hop EBGp sessions on different links over multiple directly connected peers that use the same IPv6 link-local address.

In earlier Junos OS Releases, BGP peers could be configured with link-local addresses, but multiple BGP peers could not be configured to use the same link-local address on different interfaces.

[See [Configure Multiple Single-Hop EBGp Sessions on Different Links Using the Same Link-Local Address \(IPv6\)](#).]

Timing and Synchronization

- **Support for PTP G.8275.2 profile (ACX710)**—Starting in Junos OS Release 20.4R1, we support the Precision Time Protocol (PTP) G.8275.2 profile with node type T-BC-P (BC).

You can use the `[edit protocols ptp profile-type g.8275.2]` hierarchy level to configure the G.8275.2 profile.

[See [Understanding the Time Management Administration Guide](#) and [profile-type](#).]

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What's Changed

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Learn about what changed in these releases for ACX Series routers.

What's Changed in 20.4R3

Layer 2 Ethernet Services

- **Link selection support for DHCP**—We have introduced the **link-selection** statement at the **edit forwarding-options dhcp-relay relay-option-82** hierarchy level, which allows DHCP relay to add suboption 5 to option 82. Suboption 5 allows DHCP proxy clients and relay agents to request an IP address for a specific subnet from a specific IP address range and scope. Prior to this release, the DHCP relay dropped packets during the renewal DHCP process and the DHCP server used the leaf's address as a destination to acknowledge the DHCP renewal message.

[See [relay-option-82](#).]

Network Management and Monitoring

- **Changes in contextEngineID for SNMPv3 INFORMS (PTX Series, QFX Series, ACX Series, EX Series, MX Series, and SRX Series)**— Now the contextEngineID of SNMPv3 INFORMS is set to the local engine-id of Junos OS devices. In earlier releases, the contextEngineID of SNMPv3 INFORMS was set to remote engine-id.

[See [SNMP MIBs and Traps Supported by Junos OS](#).]

- **The configuration accepts only defined identity values for nodes of type identityref in YANG data models (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—If you configure a statement that has type identityref in the corresponding YANG data model, the device accepts only defined identity values (as defined by an identity statement) as valid input. In earlier releases, the device also accepts values that are not defined identity values.

What's Changed in 20.4R2

EVPN

- **Support for displaying SVLBNH information**—You can now view shared VXLAN load balancing next hop (SVLBNH) information when you display the VXLAN tunnel endpoint information for a specified ESI and routing instance by using **show ethernet-switching vxlan-tunnel-end-point esi esi-identifier esi-identifier instance instance svlbh** command.

Junos XML API and Scripting

- **Refreshing scripts from an HTTPS server requires a certificate (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—When you refresh a local commit, event, op, SNMP, or Juniper Extension Toolkit (JET) script from an HTTPS server, you must specify the certificate (Root CA or self-signed) that the device uses to validate the server's certificate, thus ensuring that the server is authentic. In earlier releases, when you refresh scripts from an HTTPS server, the device does not perform certificate validation.

When you refresh a script using the **request system scripts refresh-from** operational mode command, include the **cert-file** option and specify the certificate path. Before you refresh a script using the **set refresh** or **set refresh-from** configuration mode command, first configure the **cert-file** statement under the hierarchy level where you configure the script. The certificate must be in Privacy-Enhanced Mail (PEM) format.

[See [request system scripts refresh-from](#) and [cert-file](#).]

- The **jcs:invoke()** function supports suppression of root login and logout events in system log files for **SLAX commit scripts (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—The **jcs:invoke()** extension function supports the **no-login-logout** parameter in SLAX commit scripts. If you include the parameter, the function does not generate and log **UI_LOGIN_EVENT** and **UI_LOGOUT_EVENT** messages when the script logs in as root to execute the specified remote procedure call (RPC). If you omit the parameter, the function behaves as in earlier releases in which the root **UI_LOGIN_EVENT** and **UI_LOGOUT_EVENT** messages are included in system log files.

[See [invoke\(\) Function \(SLAX and XSLT\)](#).]

- The **jcs:invoke()** function supports suppression of root login and logout events in system log files for **SLAX event scripts (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—The **jcs:invoke()** extension function supports the **no-login-logout** parameter in SLAX event scripts. If you include the parameter, the function does not generate and log **UI_LOGIN_EVENT** and **UI_LOGOUT_EVENT** messages when the script logs in as root to execute the specified remote procedure call (RPC). If you omit the parameter, the function behaves as in earlier releases in which the root **UI_LOGIN_EVENT** and **UI_LOGOUT_EVENT** messages are included in system log files.

[See [invoke\(\) Function \(SLAX and XSLT\)](#).]

What's Changed in 20.4R1

General Routing

- **Support for `gether-options` statement (ACX5048, ACX5096)**—Junos OS supports the `gether-options` statement at the `edit interfaces interface-name` hierarchy on the ACX5048 and ACX5096 routers. Previously, support for the `gether-statement` was deprecated.

[See [gether-options](#) and [ether-options](#).]

High Availability (HA) and Resiliency

- **The `ntp boot-server` command is deprecated** — The `boot-server` option for the `ntp` command is no longer needed, and has been deprecated.

MPLS

- **The `show mpls lsp extensivel` and `show mpls lsp detail` commands display next-hop gateway LSPid** — When you use the `show mpls lsp extensivel` and `show mpls lsp detail` commands, you'll see next-hop gateway LSPid in the output.

Network Management and Monitoring

- **Warning changed for configuration statements that correspond to "deviate not-supported" nodes in YANG data models (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—If you configure a statement corresponding to a YANG data model node that defines the `deviate not-supported` statement, the Junos OS configuration annotates that statement with the comment **Warning: statement ignored: unsupported platform**. In earlier releases, the warning is **Warning: 'statement' is deprecated**.

Routing Protocols

- **Inet6 is disabled in VT interface (ACX5448)**—Starting in this release, the `inet6` statement at the `edit interfaces vt-interface-number unit unit-number family` hierarchy level is disabled.

User Interface and Configuration

- **Verbose format option to export JSON configuration data (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—The Junos OS CLI exposes the `verbose` statement at the `edit system export-format json` hierarchy level. The default format to export configuration data in JSON changed from `verbose` format to `ietf` format starting in Junos OS Release 16.1R1. You can explicitly specify the default export format for JSON configuration data by configuring the appropriate statement at the `edit system export-format json` hierarchy level. Although the `verbose` statement is exposed in the Junos OS CLI as of the current release, you can configure this statement starting in Junos OS Release 16.1R1.

[See [export-format](#).]

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Known Limitations

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Learn about known limitations in this release for the ACX Series.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

General Routing

- With an asymmetric network connection, a 10-Gbps MACsec port connected to a 10-Gbps channelized port, high and asymmetric T1 and T4 time errors are seen. This situation introduces a high two-way time error and also different CF updates in the forward and reverse paths. [PR1440140](#)
- With the MACsec feature enabled and introduction of traffic, the peak-to-peak value varies with the percentage of traffic introduced. Find the maximum and mean values of the Time errors with different traffic rates (two-router scenario). Can have maximum value jumps as high as 1054 nanoseconds with 95% traffic, 640 nanoseconds with 90% traffic, and 137 nanoseconds with no traffic. [PR1441388](#)
- The EVPN-VPWS, L3VPN and L2VPN FRR convergence time with aggregated Ethernet as the active core interface does not meet less than 50 microseconds and might be 100 microseconds to 150 microseconds. [PR1492730](#)
- On ACX710, the PTP lock recovery is restarted when the clksyncd process is restarted. This will result in the PTP lock state moving to freerun on the clksyncd process restart. [PR1502162](#)
- IPv6-BFD supports 256 sessions. [PR1502170](#)

- Inconsistencies in the PTP lock status behavior is observed during chassis control restart. [PR1508385](#)
- On the ACX710 router, Servo moves to the **Holdover-in/Holdover-out/Acq** state from the **Phase-aligned** state with impairment. [PR1550367](#)
- On the ACX710 router, PTP with **Vlan-id-range** does not work for specific VLANs. [PR1550482](#)
- On the ACX710 router, the holdover error **HOLDOVER OUT OF SPEC** does not reset during the Servo state change. [PR1556798](#)
- The behavior of the ACX5448 or ACX710 from the day one is to reset the login session in 10 seconds if they are configured as a TWAMP server. [PR1556829](#)
- Aggregation of all Logical interface statistics is not supported on ACX series products. It has no support and needs improvement, it is the behavior of the first day. [PR1602541](#)

Timing and Synchronization

- On the ACX5448 router, the two-way time error and CTE for 1 PPS does not meet the class A metrics. [PR1535434](#)
- On the ACX5448-M router, the 1 PPS CTE does not meet the class A performance in 1-Gigabits interface. [PR1542744](#)
- On the ACX5448 router, ping stops working even though the ARP entry is present during continuous script executions. [PR1533513](#)
- On the ACX710 router, T1 or T4 cTE should be tuned closer to two-way CTE. [PR1527347](#)
- On the ACX710 router, huge offset is observed initially with ACQ and holdover inspec and outspec conditions. [PR1534470](#)
- On the ACX710 router, the incremental PTP FPGA upgrades do not bundle along with the regular image upgrades. [PR1540799](#)
- On the ACX710 router, changing the PTP profile type from g.8275.1 to g.8275.2 requires the Packet Forwarding Engine to reboot and the clksyncd process to restart. As a workaround, you must reboot the Packet Forwarding Engine and restart the clocking process before you change the profile. [PR1546614](#)
- On the ACX710 router, the Servo transition is incorrect after chassis restart. [PR1550270](#)
- On the ACX710 router, the delay-asymmetry compensation update does not work at CLI with the G.8275.2 profile. [PR1550441](#)
- On the ACX710 router, the PTP Servo status shows holdover during transition between virtual port and PTP. [PR1510880](#)
- During FRR convergence local repair can be in seconds (>50 microseconds), if the link failure is observed at the PHP node and explicit NULL is configured on the PHP node and on the PHP node of the backup path. Global repair will resume traffic flow. [PR1515512](#)

- On the ACX710 router, the SyncE to 1PPS transient test results do not meet G.8273.2 SyncE to 1PPS transient metric. [PR1522796](#)
- On the ACX710 router, the clock parameters are incorrect in certain scenarios when the Servo is in the **FREERUN** state. [PR1548192](#)
- On the ACX710 router, the PTP Servo takes longer time to lock after the clksyncd process restarts. [PR1549952](#)
- On the ACX710 router, the **show ptp global-information** command does not display correct Clock Class or ESMC QL details when the Servo goes to the **Holdover-in** state. [PR1553213](#)
- On the ACX710 router, the Servo transition is incorrect during the T-GM switchover scenario. [PR1553439](#)
- The ISSU is not supported from Junos OS earlier releases to Junos OS Release 20.4 and later. There is a major SDK upgrade from 6.3.2 to 6.5.16, because of which the warm boot feature needed for unified ISSU is not supported. [PR1554915](#)

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Learn about open issues in this release for the ACX Series.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

General Routing

- ACX device is getting transit ARP traffic coming from Logical Interfaces that are part of a bridge-domain or Layer 2 circuit punted to the Routing Engine. Internal coded queues are supporting a maximum of 200 pps in the (ARP + ICMPv6) queue. When hitting limits in this internal queue, other protocols that depend on ARP resolution can be affected and eventually flap. [PR1263012](#)
- The **inet6-precendence** under **set class of service rewrite-rules** is not supported on ACX5448. [PR1344340](#)
- Loopback status is not shown for OT interfaces on CLI (available from vty only). [PR1358017](#)
- The SD (Signal Degrade) threshold is normally lower than the SF threshold (that is, so that as errors increase, SD condition is encountered first). For the ACX6360 optical links there is no guard code to prevent the user from setting the SD threshold above the SF threshold which would cause increasing errors to trigger the SF alarm before the SD alarm. This will not cause any issues on systems with correctly provisioned SD and SF thresholds. [PR1376869](#)
- This ping latency behavior is expected for host generated ICMP traffic due to the design of Packet Forwarding Engine queue polling the packets from ASIC. `user@host-acx5448> ping 10.0.0.4 PING 10.0.0.4 (10.0.0.4): 56 data bytes 64 bytes from 10.0.0.4: icmp_seq=0 ttl=63 time=8.994 ms 64 bytes from 10.0.0.4: icmp_seq=1 ttl=63 time=49.370 ms 64 bytes from 10.0.0.4: icmp_seq=2 ttl=63 time=47.348 ms 64 bytes from 10.0.0.4: icmp_seq=3 ttl=63 time=45.411 ms <<< 64 bytes from 10.0.0.4: icmp_seq=4 ttl=63 time=106.449 ms <<< 64 bytes from 10.0.0.4: icmp_seq=5 ttl=63 time=79.697 ms <<< 64 bytes from 10.0.0.4: icmp_seq=6 ttl=63 time=37.489 ms <<< 64 bytes from 10.0.0.4: icmp_seq=7 ttl=63 time=31.436 ms << 64 bytes from 10.0.0.4: icmp_seq=8 ttl=63 time=35.460 ms << 64 bytes from 10.0.0.4: icmp_seq=9 ttl=63 time=77.198 ms << ^C --- 10.0.0.4 ping statistics --- 10 packets transmitted, 10 packets received, 0% packet loss round-trip min/avg/max/stddev = 8.994/51.885/106.449/26.824 ms` [PR1380145](#)
- On ACX6360 routers, enhancement is needed for FRR BER threshold SNMP support. [PR1383303](#)
- On ACX6360 router, Tx power cannot be configured using + sign. [PR1383980](#)
- The ccc logs are not compressed after rotation. [PR1398511](#)
- A `jnxlfoTnOperState` trap notification is sent for all ot-interfaces. [PR1406758](#)

- Memory leaks are expected in this release. [PR1438358](#)
- Drop profile maximum threshold might not be reached when the packet size is other than 1000 bytes. [PR1448418](#)
- The CFM remote MEP is not coming up after configuration or remains in start state. [PR1460555](#)
- On the ACX6360 with NGRE platforms, the vmhost disk usage might keep increasing due to no log rotation for resild log and temperature sensor info was incorrectly written into resild log, resulting in resild log file size continuous increasing. [PR1480217](#)
- If we configure DHCP option 012 host-name in DHCP server and the actual base config file also has the host-name in it, then overwriting of the base config file's host-name with the DHCP option 012 host-name is happening. [PR1503958](#)
- On ACX710 routers, when the following steps are done for PTP, chassis does not lock: 1. Use one or two ports as source for chassis synchronization and lock both PTP and SyncE locked. 2. Disable both logical interfaces. 3. Restart clksyncd. 4. Rollback 1. As a workaround, you can avoid this issue by deleting the PTP configuration, restarting clksyncd, and then reconfiguring PTP. [PR1505405](#)
- Boot from alternate media - alarm would not be seen in ACX710, when system is booted with recovery snapshot. This feature is requires new implemented and changes in ACX710 firmware. [PR1517221](#)
- Due to BRCM KBP issue route lookup may fail. Need to upgrade KBP to address this issue, due to high risk KBP SDK upgrade planned for 21.1. [PR1533513](#)
- Due to BRCM KBP issue route lookup may fail. Need to upgrade KBP to address this issue. Due to high risk KBP SDK upgrade planned for 21.1. [PR1533557](#)
- The alarm **network-service mode mismatch between configuration and kernel setting** is observed during unified ISSU or normal code upgrade even though enhanced-ip is active. [PR1546002](#)
- Unified ISSU is not supported in earlier releases to Junos OS Release 20.4 and later. There is a major SDK upgrade from 6.3.2 to 6.5.16, because of which the warm boot feature needed for unified ISSU is not supported by our vendor. [PR1554915](#)
- The day one behavior of ACX5448 or ACX710 is to trigger the start session ack by 10 seconds when configured as TWAMP server. [PR1556829](#)
- The PTP global level ipv4-dscp configuration statement is used to configure the DSCP field for both IPv4 and IPv6 sessions. [PR1557262](#)
- On certain ACX platforms, MAC address entries might not be deleted from the MAC table at the end of **mac-table-aging-time** timer when there is active traffic destined to that MAC address. When the issue happens, it might reduce the number of new MAC addresses that can be learned. If the ethernet-switching table overflows, no new MAC addresses will be learned, which might cause traffic flooding. [PR1565642](#)
- On the ACX448 routers, the packet buffer allocation failed messages appears when you scale the CFM sessions with the SLA iterator. [PR1574754](#)

- When strict-high configured on a scheduler, it is recommended to configure queue shaper along with strict-high. If user does not want to shape the traffic on a strict-high queue, then, configure parent shaping rate as max queue shaping rate on strict-high queue. Different schedulers with strict-high and shaping rate need to be configured and attached to scheduler map based on the parent shaping rate. [PR1591851](#)
- On the ACX710 and ACX5448 platforms, if the option **no-snoop** is configured under **dhcp-relay**, DHCP packets might not be relayed after Packet Forwarding Engine restarts or when the system reboots. [PR1608125](#)
- After deactivating one of the member interface of the aggregated Ethernet, output pps is still seen on the deactivated interface. It is seen that the OSPF Hello packets initiated from the peer aggregated Ethernet interface is looped & getting transmitted out of this deactivated interface. However, there will not be any functional impact, as the packets are dropped by the peer device. [PR1608827](#)
- CFMD cores could be seen if CCM configuration is changed from aggregated Ethernet IFL to physical IFL, and if the physical IFL was previously part of aggregated Ethernet bundle. [PR1612212](#)
- On the ACX710 platforms, the CPU of the routing protocol engine gets stuck at 100%, which leads to traffic impact. [PR1612387](#)

Class of Service (CoS)

- In the class-of-service context, when a wildcard interface is configured with a classifier, upon reboot of the device, the classifier defined in the wildcard interface configuration may not be properly programmed to the interfaces. [PR1559516](#)

Interfaces and Chassis

- The mc-ae option need to have **prefer-status-control-active** set to avoid flap on the split brain case (ICCP down or peer node reboot). Configure it on the **status-control active** PE device. [PR1505841](#)

Network Management and Monitoring

- On all Junos platforms, the SNMP polling might not be working if the IS-IS protocol is disabled under the same Virtual Routing and Forwarding (VRF) through which SNMP requests are sent. [PR1527251](#)

Timing and Synchronisation

- The Precision Time Protocol (PTP) clock might fail to be locking and stuck in acquiring state at clock servo. [PR1570310](#)
- In the ACX710 routers, the size of the **jnpr-clock-recovery.log** log file is small and the archives rotate too quickly. [PR1582350](#)

User Interface and Configuration

- The crash happening in this issue is for **auditd tacplus** process. The crash is happening while reading the configuration from the database in **auditd tacplus** process. There is no impact due to this issue since **auditd tacplus** process will get restarted automatically after it gets terminated. [PR1226104](#)

Virtual Chassis

- In the ACX5000 router, the following false positive parity error messages are observed:
_soc_mem_array_sbusdma_read and might raise false alarm. [PR1276970](#)

VPNs

- Need to configure **set protocols ldp label-withdrawal-delay 0** in all the routers participating in L2ckt to avoid the issue. [PR1543408](#)

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Resolved Issues

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This section lists the issues fixed in Junos OS Release 20.4R3 for the ACX Series.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Resolved Issues: 20.4R3

General Routing

- On the ACX5448 router, the two-way time error and CTE for 1 PPS does not meet the class A metrics. [PR1535434](#)
- On the ACX5448 routers, single rate three color policer does not work. [PR1559665](#)
- When an RDI is received with a CCM packet, the sessions are not deleted. [PR1560182](#)
- Inline BFD stays down with IS-IS and static clients. [PR1561590](#)
- Loopback0 firewall might not take effect along with error logs. [PR1566417](#)
- On the ACX500 routers, service MIC does not work. [PR1569103](#)
- On the ACX5048, packets higher than MTU 1518, but inline with MTU configuration were not counted in the traffic-input-pps. [PR1569763](#)
- On the ACX710, PEM feed snmp trap support. [PR1571368](#)
- On the ACX5448, RFC2544 reflector feature could not work on a higher port. [PR1571975](#)
- The Layer 2 circuit and CFM sessions might go down with asynchronous-notification configured. [PR1572722](#)
- On ACX5048 and ACX5096 platforms, ARP traffic exceeding the policer limit is not discarded. [PR1573956](#)
- On ACX5448 and ACX710 platforms, 802.1P rewrite might not work. [PR1574601](#)
- Packets might get tagged with the default VLAN-ID and dropped at the peer under Layer 2 circuits local switching scenario. [PR1574623](#)
- ACX Series routers fail to process RSVP path message. [PR1576585](#)
- RLFA might not work because service label pops incorrectly. [PR1577460](#)
- The **LIBCOS_COS_TVP_FC_INFO_NOT_FOUND: Forwarding-class information not specified** is seen when committing scheduler-map under class-of-service. [PR1579009](#)
- On ACX710 routers, the routers might continuously reboot when auxiliary port is configured. [PR1580016](#)
- On the ACX5448 Series routers, asynchronous-notification for 1G interface fails to work. [PR1580700](#)
- There might be a traffic drop between customer edge and provider edge devices in case of ARP resolution failure. [PR1580782](#)
- The rpd process might get stuck due to race condition. [PR1582226](#)

- On ACX710, unexpected result is observed while verifying channelized interface check with snmp mib get ifHighSpeed output. [PR1583995](#)
- On ACX5448, **ACX_ASIC_PROGRAMMING_ERROR** - Detection time shows the default value (6.000) instead of the configured value for single hop BFD. [PR1585382](#)
- PTP might get stuck and not function properly on ACX710 in a certain condition. [PR1587990](#)
- DHCPv4 might not work on ACX710 and ACX5448 platforms. [PR1589135](#)
- Traffic might get forwarded through the member links in down state after new member links are added to aggregated Ethernet interface on ACX710 and ACX5400 platforms. [PR1589168](#)
- ACX5448 and ACX710 platforms running DHCP relay will not process packets arriving over MPLS with an explicit null label. [PR1590225](#)
- Traffic is not passing through the Layer 2 circuit interface when vlan-id-range configured. [PR1590969](#)
- On ACX710 router, the l2ald core files seen at l2ald_event_process_list_id, l2ald_event_proc_all_lists, and l2ald_event_periodic () at ../../../../src/junos/usr.sbin/l2ald/l2ald_event.c:757. [PR1596908](#)
- On ACX5448 and ACX710 platforms, there might be a traffic drop in the EVPN VPWS flexible cross connect. [PR1598074](#)
- MACsec traffic over the Layer 2 circuit might not work on the ACX5448 and ACX710 platforms. [PR1603534](#)
- The FPC might restart when executing the command **show firewall** on the ACX5448 platform. [PR1605288](#)
- Regression, **optics_mts_010.robot** script fails to verify SNMP and matches CLI values. [PR1605348](#)
- DHCP relay is not working in routing-instance. [PR1605854](#)
- The FEB (Forwarding Engine Board) might crash on ACX1000, ACX1100, ACX2000, ACX2100, and ACX4000 platforms. [PR1606424](#)

Platform and Infrastructure

- In Junos OS, upon receipt of specific sequences of genuine packets destined to the device the kernel will crash and restart (vmcore) (CVE-2021-0283, CVE-2021-0284). [PR1557881](#)

Routing Protocols

- BGP session carrying VPNv4 prefix with IPv6 next-hop might get dropped. [PR1580578](#)

Resolved Issues: 20.4R2

General Routing

- The IPv6 BFD sessions flap when configured below 100 ms flaps. [PR1456237](#)
- The ACX1100, ACX2000, ACX2100, ACX2200, and ACX4000 might stop forwarding transit and control traffic. [PR1508534](#)

- Transit DHCP packets drop is seen on ACX5448. [PR1517420](#)
- On the ACX500, traffic drop along with **show services stateful-firewall statistics** does not work as expected. [PR1520305](#)
- The aggregated Ethernet interface might not come up with LFM configured after reboot. [PR1526283](#)
- The l2cpd memory leak could be observed with aggregated Ethernet interface flap. [PR1527853](#)
- Packets drop might be seen after configuring PTP transparent clock. [PR1530862](#)
- Broadcast, Unknown Unicast, and Multicast (BUM) traffic could drop in the VPLS instance under certain conditions. [PR1531733](#)
- The rpd process generates core file at l2ckt_vc_adv_rcv, l2ckt_adv_rt_flash (taskptr=0x4363b80, rtt=0x4418100, rti=< optimized out>, data=< optimized out>, opcode=< optimized out>) at ../../../../src/junos/usr.sbin/rpd/l2vpn/l2ckt.c:7982. [PR1537546](#)
- On the ACX5448 router, the BGPV6LU traffic drop is observed when the node is deployed in ingress. [PR1538819](#)
- Synchronization Ethernet goes in the Holdover state and comes back to the Locked state when the PTP configuration is deleted. [PR1546681](#)
- SFP-T interface might not come up when a straight cable is used on ACX5448. [PR1547394](#)
- The ACX5448 router as transit for the BGP-labeled unicast drops traffic. [PR1547713](#)
- Traffic loss might be observed on ACX710 platforms. [PR1551063](#)
- The dcpfe process might crash and the non-channelization interfaces might not come up. [PR1552798](#)
- [dhcp] [DHCP_RELAY] acx5448 : ACX5448-X: Verifying multiple PD sync with relay configure delete and add configuration. [PR1554647](#)
- The ARP packets from the CE device are added with VLAN tag if the VLAN-ID is configured in the EVPN routing instance. [PR1555679](#)
- On the ACX710 router, the T-BC-P switch-over performance fails beyond the standard mask and servo moves to multiple Holdover-in state, Acquiring state, Holdover-in state, Holdover-out state, and Acquiring state. [PR1556087](#)
- On the ACX5448 router, you cannot downgrade to Junos OS Release 18.4 code base. [PR1556377](#)
- On the ACX5448 router, the unicast packets from the customer edge devices might be forwarded by the PE devices with additional VLAN tag if IRB is used. [PR1559084](#)
- On ACX710 and ACX5448, the three-color policer might not be working. [PR1559665](#)
- The fxpc(dc-pfe) process crash might be seen on ACX5048 and ACX5096 with analyzer configuration. [PR1559690](#)
- On ACX2100, the laser-output-power is seen after interface is disabled and reboot is performed. [PR1560501](#)

- ACX710, NPI - MC-LAG:- When Link Aggregation Control Protocol (LACP) daemon is restarted LACP local partner system id remains 0 in mc-ae output. [PR1560820](#)
- On the ACX5448 router, the following syslog message is reported every 30 seconds:
ACX_DFW_CFG_FAILED: ACX Error (dfw):dnx_dfw_dyn_entry_counter_get : Entry is invalid. [PR1562323](#)
- Analyzer (Port Mirroring) might not work on ports above 20. [PR1563774](#)
- Pushing more than 2 MPLS labels on ACX5448 and ACX710 might not work. [PR1566828](#)
- "lcklsyncd" log file is eventually empty. [PR1567687](#)
- The Designated Forwarder (DF) might not forward traffic. [PR1567752](#)
- On ACX500, service MIC does not work. [PR1569103](#)
- ACX2100 resets tunable optics to default wavelength after upgrade or reboot. [PR1570192](#)
- On ACX5448 and ACX710 platforms, untagged traffic might be incorrectly queued and marked to different Class of Service queues. [PR1570899](#)
- ACX as a LSR router, fails to process RSVP path message. [PR1576585](#)
- On ACX710 platforms, there might be continuous reboot due to configuration under auxiliary port. [PR1580016](#)
- On ACX5448, there might be IPv4 traffic loss with packet size more than 1410. [PR1584509](#)

Class of Service (CoS)

- The explicit classifier or rewrite-rule might not work as expected for a logical interface if the wildcard configuration is also applied. [PR1556103](#)
- FPC crash might be observed after the **show class-of-service** command. [PR1568661](#)

Infrastructure

- The vme/me0 management interface cannot process any incoming packets. [PR1552952](#)

Routing Protocols

- The rpd memory leak might be seen in the BGP scenario. [PR1547273](#)

Resolved Issues: 20.4R1

Forwarding and Sampling

- VLAN-ID based firewall match conditions might not work for the VPLS service. [PR1542092](#)

General Routing

- The **gigether-options** command is enabled again under the interface hierarchy. [PR1430009](#)
- Repeated powering-off or powering-on of the device, the SMBUS transactions timeout occurs. [PR1463745](#)

- On the ACX5048 router, the egress queue statistics do not work for the aggregated Ethernet interfaces. [PR1472467](#)
- On the ACX5048 router, traffic loss is observed during the unified ISSU upgrade. [PR1483959](#)
- The following syslog error message is observed: **ACX_DFW_CFG_FAILED**. [PR1490940](#)
- On the ACX5048 and ACX5096 routers, the LACP control packets might be dropped due to high CPU utilization. [PR1493518](#)
- On the ACX710 router, high convergence is observed with the EVPN-ELAN service in a scaled scenario during FRR switchover. [PR1497251](#)
- On the ACX5448 router, the EXP rewrite for the Layer 3 VPN sends all traffic with incorrect EXP. [PR1500928](#)
- The following error message is observed during MPLS route add, change, and delete operation: **mpls_extra NULL**. [PR1502385](#)
- The ACX1100, ACX2100, ACX2200, ACX2000, and ACX4000 routers might stop forwarding transit and control traffic. [PR1508534](#)
- On the ACX710 router, the Packet Forwarding Engine might crash and the fpc process might remain down. [PR1509402](#)
- The loopback filter cannot take more than 2 TCAM slices. [PR1513998](#)
- On the ACX710 router, the following error message is observed in the Packet Forwarding Engine while the EVPN core link flaps: **dnx_l2alm_add_mac_table_entry_in_hw**. [PR1515516](#)
- The VM process generates a core file while running stability test in a multidimensional scenario. [PR1515835](#)
- The l2ald process crashes during stability test with traffic on a scaled setup. [PR1517074](#)
- On the ACX710 router, whenever a copper optic interface is disabled and enabled, the speed shows 10 Gbps rather than 1 Gbps. This issue is not seen with the fiber interface. [PR1518111](#)
- Tagged traffic matching the vlan-list configuration in the vlan-circuit cross-connect logical interface gets dropped in the ingress interface. [PR1519568](#)
- The **Incompatible Media** alarm is not raised when the Synchronous Ethernet source is configured over the copper SFP. [PR1519615](#)
- On the ACX710 router, the alarm port configuration is not cleared after deleting the alarm-port. [PR1520326](#)
- PTP to 1PPS noise transfer test fails for frequency 1.985 Hz. [PR1522666](#)
- The **show class-of-service interface** command does not show the classifier information. [PR1522941](#)
- Interface does not come up with the auto-negotiation setting between the ACX1100 router and the other ACX Series routers, MX Series routers and QFX Series switches as the other end. [PR1523418](#)

- With the ACX5448 router with 1000 CFM, the CCM state does not go in the **Ok** state after loading the configuration or restarting the Packet Forwarding Engine. [PR1526626](#)
- On the ACX5448 and ACX710 routers, the **vlan-id-list** statement might not work as expected. [PR1527085](#)
- The FEC field is not displayed when the interface is down. [PR1530755](#)
- The **show class-of-service routing-instance** does not show the configured classifier. [PR1531413](#)
- Memory leak in Local OutLif in VPLS/CCC topology is observed. [PR1532995](#)
- The clksyncd process generates core file on Junos OS Release 20.3R1.3 image. [PR1537107](#)
- The rpd process generates core file at **l2ckt_vc_adv_rcv, l2ckt_adv_rt_flash (taskptr=0x4363b80, rtt=0x4418100, rtl=< optimized out>, data=< optimized out>, opcode=< optimized out>)** at **../../../../../../../../src/junos/usr.sbin/rpd/l2vpn/l2ckt.c:7982**. [PR1537546](#)
- The **Management Ethernet link down** alarm is observed while verifying the system alarms in the Virtual Chassis setup. [PR1538674](#)
- On the ACX5448 router, unexpected behavior of the **show chassis network-services** command is observed. [PR1538869](#)
- The following error message is observed while deleting the remote stream 0 0 0 0 0 along with feb core file at **0x00ae6484 in bcmdnx_queue_assert (queue=0xc599b60)** at **../../../../../../../../src/pfe/common/drivers/bcmdnx/bcmdnx_sdk_ukern_layer.c: Err] clksync_mimic_delete_clock_entry Unexpected error**. [PR1539953](#)
- The announcement or synchronization interval rate range is not as expected. [PR1542516](#)
- Synchronization Ethernet goes in the **Holdover** state and comes back to the **Locked** state when the PTP configuration is deleted. [PR1546681](#)
- The ACX5448 router as transit for the BGP labeled unicast drops traffic. [PR1547713](#)
- Multicast traffic is stopped when HQoS with multicast configurations are applied. [PR1551248](#)
- With the **no-local-switching** command, traffic between the local and remote CE devices are affected. [PR1527231](#)
- On the ACX710 router, the T-BC-P switch-over performance fails beyond the standard mask and servo moving to multiple **Holdover-in** state, **Acquiring** state, **Holdover-in** state, **Holdover-out** state, and **Acquiring** state. [PR1556087](#)
- Running SNMP MIB walk and executing the **show interfaces** command might cause the picd process to crash. [PR1533766](#)
- On the ACX5448 router, you cannot downgrade to Junos OS Release 18.4 code-base. [PR1556377](#)
- BIND does not sufficiently limit the number of fetches while processing referrals. [PR1512212](#)
- The clksyncd process generates core file during the stability test with traffic and scale. [PR1518253](#)
- The fxpc process generates core file during EEPROM read when SFP is removed. [PR1518480](#)

- On the ACX5448 routers, multicast traffic loop over ICL might be observed. [PR1521113](#)
- On the ACX710 router, PIR/CIR HQoS behavior is inconsistent. [PR1525789](#)
- Error messages are displayed while attaching tcp on physical interfaces. [PR1527541](#)
- The l2cpd memory leak might be observed with the aggregated Ethernet interface flap. [PR1527853](#)
- Upon classifying the Layer 3 packets, DSCP is not preserved and is lost at the egress due to the limitations of a chipset. [PR1535876](#)
- Other than IPv4 and IPV6, other IPs should not be forwarded. Only IP header with version 4 and 6 can pass through. [PR1550748](#)
- Profile switch between G.8275.1 and G.8275.2 works as expected. [PR1533263](#)

Interfaces and Chassis

- The fpc process might crash in the inline mode with CFM configured. [PR1500048](#)

Layer 2 Features

- On the ACX5448 routers, the VPLS traffic statistics are not displayed when the **show vpls statistics** command is executed. [PR1506981](#)
- The rpd might crash on the new primary Routing Engine after GRES in the VPLS or Layer 2 circuit scenario. [PR1507772](#)

Routing Protocols

- The rpd process might report 100 percent CPU usage with the BGP route damping enabled. [PR1514635](#)
- On the ACX5448 routers, the **family inet6** configuration under the vt- interface is disabled. [PR1514595](#)

SEE ALSO

[What's New | 14](#)

[What's Changed | 19](#)

[Known Limitations | 23](#)

[Open Issues | 25](#)

[Documentation Updates | 36](#)

[Migration, Upgrade, and Downgrade Instructions | 37](#)

Documentation Updates

There are no errata or changes in Junos OS Release 20.4R3 documentation for ACX Series routers.

SEE ALSO

[What's New | 14](#)[What's Changed | 19](#)[Open Issues | 25](#)[Known Limitations | 23](#)[Resolved Issues | 29](#)[Migration, Upgrade, and Downgrade Instructions | 37](#)

Migration, Upgrade, and Downgrade Instructions

IN THIS SECTION

- [Upgrade and Downgrade Support Policy for Junos OS Releases | 37](#)

This section contains the upgrade and downgrade support policy for Junos OS for ACX Series routers. Upgrading or downgrading Junos OS might take several minutes, depending on the size and configuration of the network.

For information about software installation and upgrade, see the [Installation and Upgrade Guide](#).

Upgrade and Downgrade Support Policy for Junos OS Releases

We have two types of releases, standard EOL and EEOL:

- Standard End of Life (EOL) releases have engineering support for twenty four months after the first general availability date and customer support for an additional six more months.
- Extended End of Life (EEOL) releases have engineering support for thirty six months after the first general availability date and customer support for an additional six more months.

For both standard EOL and EEOL releases, you can upgrade to the next three subsequent releases or downgrade to the previous three releases. For example, you can upgrade from 19.2 to the next three releases – 19.3, 19.4 and 20.1 or downgrade to the previous three releases – 19.1, 18.4 and 18.3.

For EEOL releases only, you have an additional option - you can upgrade directly from one EEOL release to the next two subsequent EEOL releases, even if the target release is beyond the next three releases. Likewise, you can downgrade directly from one EEOL release to the previous two EEOL releases, even if the target release is beyond the previous three releases. For example, 19.2 is an EEOL release. Hence, you

can upgrade from 19.2 to the next two EEOL releases – 19.3 and 19.4 or downgrade to the previous two EEOL releases – 19.1 and 18.4.

Release Type	End of Engineering (EOE)	End of Support (EOS)	Upgrade and Downgrade to subsequent 3 releases	Upgrade and Downgrade to subsequent 2 EEOL releases
Standard End of Life (EOL)	24 months	End of Engineering + 6 months	Yes	No
Extended End of Life (EEOL)	36 months	End of Engineering + 6 months	Yes	Yes

For more information about standard EOL and EEOL releases, see <https://www.juniper.net/support/eol/junos.html>.

For information about software installation and upgrade, see the [Installation and Upgrade Guide](#).

SEE ALSO

- [What's New | 14](#)
- [What's Changed | 19](#)
- [Known Limitations | 23](#)
- [Open Issues | 25](#)
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Junos OS Release Notes for cSRX

IN THIS SECTION

- [What's New | 39](#)
- [What's Changed | 40](#)
- [Known Limitations | 41](#)

- Open Issues | 41
- Resolved Issues | 41

These release notes accompany Junos OS Release 20.4R3 for the cSRX Container Firewall, a containerized version of the SRX Series Services Gateway. They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

You can also find these release notes on the Juniper Networks Junos OS Documentation webpage, located at https://www.juniper.net/documentation/product/en_US/junos-os.

What's New

IN THIS SECTION

- What's New in Release 20.4R3 | 39
- What's New in Release 20.4R2 | 39
- What's New in Release 20.4R1 | 39

What's New in Release 20.4R3

There are no new features in Junos OS Release 20.4R3 for cSRX.

What's New in Release 20.4R2

There are no new features in Junos OS Release 20.4R2 for cSRX.

What's New in Release 20.4R1

There are no new features in Junos OS Release 20.4R1 for cSRX.

What's Changed

IN THIS SECTION

- [What's Changed in Release 20.4R3 | 40](#)
- [What's Changed in Release 20.4R2 | 40](#)
- [What's Changed in Release 20.4R1 | 40](#)

Learn about what changed in the Junos OS main and maintenance releases for cSRX.

What's Changed in Release 20.4R3

There are no changes in behavior or syntax for cSRX in Junos OS Release 20.4R3.

What's Changed in Release 20.4R2

There are no changes in behavior or syntax for cSRX in Junos OS Release 20.4R2.

What's Changed in Release 20.4R1

Platform and Infrastructure

- Downloading of Signature Pack You can download the signature pack from the Signature Pack directly when the cSRX doesn't have pre-installed signature pack.
 - Configure proxy server so that IP address of proxy server is reachable from cSRX.
 - Run the following command to enter the configurational mode from CLI.
root@host> configure [edit]
root@host#
 - Configure proxy server profile on cSRX using IP address and port of proxy server.
root@host#set services proxy profile appid_sigpack_proxy protocol http host 4.0.0.1
root@host#set services proxy profile appid_sigpack_proxy protocol http port 3128
 - Attach the profile to AppID and IDP.
root@host#set services application-identification download proxy-profile appid_sigpack_proxy
root@host#set security idp security-package proxy-profile appid_sigpack_proxy
 - Commit the configuration.
root@host#commit and-quit
commit complete

- Download IDP and APPID sigpack through proxy server.
 root@host>request services application-identification download
 root@host>request security idp security-package download
- To verify if download is going through proxy server:
 Verify the logs in proxy server.

```
[root@srxdpi-lnx39 squid]# cat /var/log/squid/access.log 1593697174.470 1168 4.0.0.254
TCP_TUNNEL/200 5994 CONNECT signatures.juniper.net:443 - HIER_DIRECT/66.129.242.156 -
1593697175.704 1225 4.0.0.254 TCP_TUNNEL/200 11125 CONNECT signatures.juniper.net:443 -
HIER_DIRECT/66.129.242.156 - 1593697176.950 1232 4.0.0.254 TCP_TUNNEL/200 5978 CONNECT
signatures.juniper.net:443 - HIER_DIRECT/66.129.242.156 - 1593697178.195 1236 4.0.0.254
TCP_TUNNEL/200 11188 CONNECT signatures.juniper.net:443 - HIER_DIRECT/66.129.242.156 -
1593697198.337 1243 4.0.0.254 TCP_TUNNEL/200 6125 CONNECT signatures.juniper.net:443 -
HIER_DIRECT/66.129.242.156 -
```

 In cSRX, TLS protocol is used and traffic through proxy is encrypted.

Known Limitations

There are no known behavior for cSRX in Junos OS Release 20.4R3.

Open Issues

There are no open issues for cSRX in Junos OS Release 20.4R3.

Resolved Issues

Resolved Issues: 20.4R3

There are no resolved issues for cSRX in Junos OS Release 20.4R3.

Resolved Issues: 20.4R2

There are no resolved issues for cSRX in Junos OS Release 20.4R2.

Resolved Issues: 20.4R1

There are no resolved issues for cSRX in Junos OS Release 20.4R1.

Junos OS Release Notes for EX Series

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- What's Changed | 49
- Known Limitations | 53
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- Resolved Issues | 57
- Documentation Updates | 67
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These release notes accompany Junos OS Release 20.4R3 for the EX Series. They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

You can also find these release notes on the Juniper Networks Junos OS Documentation webpage, located at https://www.juniper.net/documentation/product/en_US/junos-os.

What's New

IN THIS SECTION

- What's New in Release 20.4R3 | 43
- What's New in Release 20.4R2 | 43
- What's New in Release 20.4R1 | 43

Learn about new features introduced in this release for EX Series Switches.

NOTE: The following EX Series switches are supported in Release 20.4R3: EX2300, EX3400, EX4300, EX4600, EX4650, EX9200, EX9204, EX9208, EX9214, EX9251, and EX9253.

What's New in Release 20.4R3

There are no new features or enhancements to existing features for EX Series switches in Junos OS Release 20.4R3.

What's New in Release 20.4R2

There are no new features or enhancements to existing features for EX Series switches in Junos OS Release 20.4R2.

What's New in Release 20.4R1

Authentication, Authorization, and Accounting

- **RADIUS attributes for dynamic VLAN assignment on colorless ports (EX2300, EX2300-MP, EX3400, EX4300, and EX4300-MP)**—We now support IETF-defined RADIUS attributes that provide VLAN assignments and also indicate whether frames on the VLAN are in tagged or untagged format. This enables the network access control server to dynamically assign VLANs on colorless ports. The VLAN assignments, which are based on device profiling, can be made on either access ports or trunk ports.

[See [Dynamic VLAN Assignment on Colorless Ports.](#)]

EVPN

- **MAC limit, MAC move limit, and persistent MAC learning with EVPN-VXLAN (EX4300-48MP)**—We support the following Layer 2 port security features in an EVPN-VXLAN overlay network:
 - **MAC limit**—You can limit the number of MAC addresses learned by network (local) interfaces.

NOTE: We don't support MAC limits on virtual tunnel endpoint (VTEP) interfaces.

- **MAC move limit**—You can limit the number of times a MAC address is moved to a different interface within 1 second. To configure this feature, you apply a limit to a VLAN. In an EVPN-VXLAN network, a VLAN's members can include network (local) and VTEP interfaces. We support the following MAC move use cases and actions:

- MAC moves between network interfaces—By default, the configured action is applied on the interface to which the MAC address is last moved. If you configured action priority on the interfaces, the action is applied on the interface with the lesser priority.
- MAC moves between network and VTEP interfaces and vice-versa—The action is applied on the network interface.

NOTE: We don't support MAC moves between the following:

- VTEP interfaces.
- A VTEP interface and a network interface on which persistent MAC learning and static MAC addresses are configured.

- Persistent MAC learning (sticky MAC)—You can enable network interfaces to retain dynamically learned MAC addresses when the switch is restarted or when an interface goes down and comes back up again.

NOTE: We don't support persistent MAC learning on VTEP interfaces.

[See [Understanding MAC Limiting and MAC Move Limiting](#) and [Understanding and Using Persistent MAC Learning](#).]

- **MC-LAG emulation in an EVPN deployment (EX Series, MX Series, and vMX)**—Starting in Junos OS Release 20.4R1, you can emulate the function of an MC-LAG in active-standby mode in an EVPN configuration without having to configure an ICCP or ICL interface. In a standard EVPN configuration, logical interfaces configured on an aggregated Ethernet interface can have different designated forwarder election roles. To emulate an MC-LAG configuration, the designated forwarder (DF) takes on the role of the aggregated Ethernet interface. The provider edge (PE) that is the non-DF will send LACP out-of-sync packets to the CE. This causes LACP to go down on the CE device, and the CE device does not use the links connected to the non-DF for sending traffic. If the connection between a CE and a DF PE fails, the PE is re-elected as a DF. If the connection between a CE and a non-DF PE fails, the current DF PE is not changed.

To enable this functionality, configure the **lACP-oos-on-ndf** statement at the **[edit interfaces interface name esi df-election-granularity per-esi]** hierarchy.

- **Support for IGMP snooping and selective multicast forwarding (EX4300-MP)**—Starting in Junos OS Release 20.4R1, the EX4300-MP switch supports IGMP snooping and selective multicast forwarding in an EVPN-VXLAN centrally-routed bridging overlay network with all-active multihoming. Selective multicast Ethernet (SMET) forwarding is part of IGMP snooping. IGMP snooping and SMET forwarding reduce the volume of multicast traffic in a broadcast domain by forwarding multicast traffic only to interfaces that have IGMP listeners. SMET forwarding sends multicast packets to the leaf devices in the

core that have expressed an interest in that multicast group. SMET forwarding is supported only in intra-VLAN replication. This feature supports EVPN Type 7 (IGMP Join Synch Route) and EVPN Type 8 (IGMP Leave Synch Routes). To configure IGMP snooping, include the **igmp-snooping proxy** configuration statement at the **[edit routing-instances routing-instance-name protocols]** hierarchy level.

[See [Overview of Multicast Forwarding with IGMP Snooping in an EVPN-VXLAN Environment](#) and [Overview of Selective Multicast Forwarding](#).]

- **Support for assisted replication (EX4300MP)**—Starting in Junos OS Release 20.4R1, the EX4300-MP switch supports assisted replication in an EVPN-VXLAN centrally-routed bridging overlay network with all-active multihoming. Assisted replication (AR) optimizes multicast traffic flow by offloading traffic replication to devices that can more efficiently handle replication and forwarding. You can configure the EX4300-MP only as an AR-leaf device. You can further optimize multicast traffic by configuring AR with IGMP snooping. To configure the EX4300-MP as an AR leaf, include the **assisted-replication leaf** statement at the **[edit routing-instances routing-instance-name protocols evpn]** or **[edit protocols evpn]** hierarchy level.

[See [Assisted Replication Multicast Optimization in EVPN Networks](#)

- **Support for sFlow in an EVPN-VXLAN network (EX4300-MP)**—Starting in Junos OS Release 20.4R1, sFlow monitoring is supported on EX4300-MP switches in an EVPN-VXLAN network. sFlow monitoring provides visibility into your EVPN VXLAN network by sampling VXLAN-encapsulated traffic at the ingress and egress interfaces. You can configure sFlow technology on a device to monitor traffic continuously at wire speed on all interfaces simultaneously. You must enable sFlow monitoring on each interface individually. Configure sFlow monitoring at the **[edit protocols sflow]** hierarchy level. Use the **show sflow collector** command to display the collector statistics and the **clear sflow collector** command to delete the collector statistics.

[See [Overview of sFlow Technology](#).]

- **Layer 3 gateway in an EVPN-MPLS environment (EX9200 with EX9200-SF3 switch fabric module and EX9200-15C line card)**—Starting in Junos OS Release 20.4R1, an EX9200 switch with an EX9200-SF3 switch fabric module and an EX9200-15C line card can act as a default Layer 3 gateway for an EVPN instance (EVI) that can span a set of devices. In this role, the EX9200 switch can perform inter-subnet forwarding. With inter-subnet forwarding, each subnet represents a distinct broadcast domain.

The Layer 3 gateway supports the following features:

- IRB interfaces through which the default gateway routes IPv4 and IPv6 traffic from one VLAN to another [See [Example: Configuring EVPN with IRB Solution](#).]
- Dynamic list next hop [See [Configuring Dynamic List Next Hop](#).]
- EVPN proxy ARP and ARP suppression, and proxy NDP and NDP suppression on IRB interfaces [See [EVPN Proxy ARP and ARP Suppression, and Proxy NDP and NDP Suppression](#).]

- Substitution of a source MAC address with a proxy MAC address in an ARP or NDP reply [See [ARP and NDP Request with a Proxy MAC Address](#).]
- Data center interconnectivity using EVPN Type 5 routes [See [EVPN Type-5 Route with MPLS encapsulation for EVPN-MPLS](#).]

Interfaces and Chassis

- **10GBASE-T SFP+ transceiver for EX4600-40F**—Starting in Junos OS Release 20.4R1, EX4600-40F switches support the 10GBASE-T SFP+ transceiver (JNP-SFPP-10GE-T), capable of working at speeds of 10 Gbps, 1Gbps, and 100Mbps, and also auto-negotiation. You can use the existing show commands such as the **show interfaces media** command to view the details of the transceivers.

[See [speed\(Ethernet\)](#).]

Junos OS XML, API, and Scripting

- **Support for Certificate Authority Chain Profile (EX2300, EX3400, EX4300, MX240, MX480, MX960, PTX-5000, VMX, vSRX and QFX5200)**—Starting in Junos OS Release 20.4R1, you can configure intermediate Certificate Authority (CA) chain profile certificate and perform https REST API request using mutual and server authentications.

To configure intermediate ca-chain certificate, configure **ca-chain ca-chain** statement at the **[edit system services rest https]** hierarchy level.

- **Start time option for interval-based internal events that trigger event policies (EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—Starting in Junos OS Release 20.4R1, when you create an interval-based internal event for triggering event policies, you can specify the start date and time for the initial event. To specify a start time, configure the **start-time** option along with the **time-interval** option at the **[edit event-options generate-event]** hierarchy level.

[See [Generating Internal Events to Trigger Event Policies](#).]

Network Management and Monitoring

- **Configuration retrieval using the configuration revision identifier (EX3400, EX4300, MX204, MX240, MX480, MX960, MX2020, PTX3000, PTX10008, QFX5100, QFX10002-60C, SRX5800, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, you can use the configuration revision identifier feature to view the configuration for a specific revision. This configuration database revision can be viewed with the CLI command **show system configuration revision**.

[See [show system configuration revision](#).]

- **Junos XML protocol operations support loading and comparing configurations using the configuration revision identifier (EX3400, EX4300, MX204, MX240, MX480, MX960, MX2020, PTX3000, PTX10008, QFX5100, QFX10002-60C, SRX5800, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, the Junos XML management protocol operations support loading and comparing configurations by referencing the configuration revision identifier of a committed configuration. You can execute the **<load-configuration>** operation with the **configuration-revision** attribute to load the configuration with the given revision identifier into the candidate configuration. Additionally, you can compare the candidate

or active configuration to a previously committed configuration by referencing the configuration revision identifier for the comparison configuration. The `<get-configuration>` operation supports the `compare="configuration-revision"` and `configuration-revision` attributes to perform the comparison.

[See [<get-configuration>](#) and [<load-configuration>](#).]

Routing Protocols

- **Support for relaxing BGP router ID format from /32 to a nonzero ID per RFC 6286 (EX9208, EX9214, MX204, VRR, QFX5120-32C)**—Starting in Junos OS Release 20.4R1, you can establish a BGP connection using a BGP identifier that is a 4-octet, unsigned, nonzero integer and it needs to be unique only within the autonomous system (AS) per RFC 6286. In earlier releases, the BGP ID of a BGP speaker was required to be a valid IPv4 host address assigned to the BGP speaker.

To enable this feature, use the `bgp-identifier identifier group bgp group name bgp-identifier identifier neighbor peer address bgp-identifier identifier` configuration statement at the `[edit protocols bgp]` hierarchy level.

[See [router-id](#)]

- **BGP Prefix-Independent Convergence (PIC) Edge for MPLS VPNs (EX9200)**—You can now install a Layer 3 VPN route in the forwarding table as an alternate path, enabling fast failover when a provider edge (PE) router fails or you lose connectivity to a PE router. This already installed path is used until global convergence through the IGP is resolved.

To enable BGP PIC Edge in an MPLS VPN, include the `protect-core` statement at the `[edit routing-instances routing-instance-name routing-options]` hierarchy level. Both IS-IS LDP and OSPF LDP are supported. When BGP PIC Edge is enabled, the `show route extensive` command now displays the weight assigned to the indirect hop.

[See [Configuring BGP PIC Edge for MPLS Layer 3 VPNs](#).]

- **Support for multiple single-hop EBGp sessions on different links using the same IPv6 link-local address (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, you are no longer required to have unique peer addresses for Juniper devices for every EBGp session. You can now enable single-hop EBGp sessions on different links over multiple directly connected peers that use the same IPv6 link-local address.

In earlier Junos OS Releases, BGP peers could be configured with link-local addresses, but multiple BGP peers could not be configured to use the same link-local address on different interfaces.

[See [Configure Multiple Single-Hop EBGp Sessions on Different Links Using the Same Link-Local Address \(IPv6\)](#).]

Software Installation and Upgrade

- **Phone-home client (EX4600, EX4650, EX9200, QFX5110, QFX5200, QFX5210, QFX5120-32C, and QFX5120-48Y)**—Starting with Junos OS Release 20.4R1, you can use either the legacy DHCP-options-based ZTP or the phone-home client (PHC) to provision software for the switch. When the switch boots up, if there are DHCP options that have been received from the DHCP server for ZTP,

ZTP resumes. If DHCP options are not present, PHC is attempted. PHC enables the switch to securely obtain bootstrapping data, such as a configuration or software image, with no user intervention other than having to physically connect the switch to the network. When the switch first boots up, PHC connects to a redirect server, which redirects to a phone home server to obtain the configuration or software image.

To initiate either DHCP-options-based ZTP or PHC, the switch must be in a factory-default state, or you can issue the **request system zeroize** command.

[See [Understanding the Phone-Home Client](#)

- **ZTP with DHCPv6 client support (EX3400, EX4300, PTX1000, PTX5000, PTX10002-60C, PTX10008, QFX5100, QFX5200, QFX10002, and QFX10002-60C)**—Starting in Junos OS Release 20.4R1, zero touch supports the DHCPv6 client. During the bootstrap process, the device first uses the DHCPv4 client to request for information regarding image and configuration file from the DHCP server. The device checks the DHCPv4 bindings sequentially. If one of the DHCPv4 bindings fails, the device continues to check for bindings until provisioning is successful. However, if there are no DHCPv4 bindings, the device checks for DHCPv6 bindings and follows the same process as for DHCPv4 until the device can be provisioned successfully. Both DHCPv4 and DHCPv6 clients are included as part of the default configuration on the device.

The DHCP server uses DHCPv6 options 59 and 17 and applicable suboptions to exchange ZTP-related information between itself and the DHCP client.

NOTE: ZTP supports only HTTP and HTTPS transport protocols.

[See [Zero Touch Provisioning](#).]

- **Phone-home client (EX4300-48MP Virtual Chassis)**—Starting in Junos OS Release 20.4R1, the phone-home client (PHC) can securely provision a Virtual Chassis consisting of all EX4300-48MP member switches without requiring user interaction. If the switches all have the factory-default configuration, you just need to:
 - Connect the switches using the Virtual Chassis ports.
 - Connect any network port or the management port to the network.
 - Power on the Virtual Chassis.

The PHC automatically starts up and connects to the phone-home server (PHS), which responds with bootstrapping information. The PHC then upgrades each member with the new image and applies the configuration, and the Virtual Chassis is ready to go.

[See [Provision a Virtual Chassis Using the Phone-Home Client](#).]

Subscriber Management and Services

- **Control plane DDoS protection against DDoS attacks (EX9200 with MPC10E)**—Starting in Junos OS Release 20.4R1, control plane distributed denial of service (DDoS) protection is enabled by default on EX9200 switches with MPC10E line cards. To prevent malicious traffic from interfering with device operations, this feature uses firewall filters and policers to discard or rate-limit control plane traffic. You can disable this feature at different levels or change the default policer parameters for many protocol groups and individual packet types in the supported protocol groups.

[See [Control Plane Distributed Denial-of-Service \(DDoS\) Protection Overview](#).]

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What's Changed

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Learn about what changed in this release for EX Series Switches.

What's Changed in Release 20.4R3

Layer 2 Ethernet Services

- **Link selection support for DHCP**—We have introduced the **link-selection** statement at the **edit forwarding-options dhcp-relay relay-option-82** hierarchy level, which allows DHCP relay to add suboption 5 to option 82. Suboption 5 allows DHCP proxy clients and relay agents to request an IP address for a specific subnet from a specific IP address range and scope. Prior to this release, the DHCP relay dropped packets during the renewal DHCP process and the DHCP server used the leaf's address as a destination to acknowledge the DHCP renewal message.

[See [relay-option-82](#).]

Network Management and Monitoring

- **Enhancement to the snmp mib walk command (PTX Series, QFX Series, EX Series, MX Series, SRX Series)**—The `ipv6IfOperStatus` field displays the current operational state of the interface. The `noIfIdentifier(3)` state indicates that no valid interface identifier is assigned to the interface. This state usually indicates that the link-local interface address failed duplicate address detection. When you specify the **Duplicate Address Detected** error flag on the interface, the new value (`noIfIdentifier(3)`) is displayed. Previously, the **snmp mib walk** command did not display the new value (`noIfIdentifier(3)`).
 - **Changes in contextEngineID for SNMPv3 INFORMS (PTX Series, QFX Series, ACX Series, EX Series, MX Series, and SRX Series)**—Now the `contextEngineID` of SNMPv3 INFORMS is set to the local engine-id of Junos devices. In earlier releases, the `contextEngineID` of SNMPv3 INFORMS was set to remote engine-id.
- [See [SNMP MIBs and Traps Supported by Junos OS](#).]
- **The configuration accepts only defined identity values for nodes of type identityref in YANG data models (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—If you configure a statement that has type `identityref` in the corresponding YANG data model, the device accepts only defined identity values (as defined by an identity statement) as valid input. In earlier releases, the device also accepts values that are not defined identity values.

What's Changed in Release 20.4R2

EVPN

- **IGMP snooping options has changed hierarchy level**—Junos OS has moved the following options from the **edit protocols igmp-snooping** hierarchy to **edit routing-instances evpn protocols igmp-snooping vlan <vlan-name/vlan-all>** hierarchy:
 - `query-interval`
 - `query-last-member-interval`
 - `query-response-interval`
 - `robust-count`

- evpn-ssm-reports-only
- immediate-leave
- **Support for displaying SVLBNH information**—You can now view shared VXLAN load balancing next hop (SVLBNH) information when you display the VXLAN tunnel endpoint information for a specified ESI and routing instance by using **show ethernet-switching vxlan-tunnel-end-point esi <varname>esi-identifier esi-identifier instance <varname>instance svlnh** command.

General Routing

- **Configure internal IPsec authentication algorithm (EX Series)**—You can configure the algorithm **hmac-sha-256-128** at the **edit security ipsec internal security-association manual direction bidirectional authentication algorithm** hierarchy level for internal IP security (IPsec) authentication. In earlier releases, you could configure the algorithm **hmac-sha-256-128** for MX Series devices only.

Junos XML API and Scripting

- **Refreshing scripts from an HTTPS server requires a certificate (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—When you refresh a local commit, event, op, SNMP, or Juniper Extension Toolkit (JET) script from an HTTPS server, you must specify the certificate (Root CA or self-signed) that the device uses to validate the server's certificate, thus ensuring that the server is authentic. In earlier releases, when you refresh scripts from an HTTPS server, the device does not perform certificate validation.

When you refresh a script using the **request system scripts refresh-from** operational mode command, include the **cert-file** option and specify the certificate path. Before you refresh a script using the **set refresh** or **set refresh-from** configuration mode command, first configure the **cert-file** statement under the hierarchy level where you configure the script. The certificate must be in Privacy-Enhanced Mail (PEM) format.

[See [request system scripts refresh-from](#) and [cert-file](#).]

- **The jcs:invoke() function supports suppression of root login and logout events in system log files for SLAX commit scripts (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—The **jcs:invoke()** extension function supports the **no-login-logout** parameter in SLAX commit scripts. If you include the parameter, the function does not generate and log UI_LOGIN_EVENT and UI_LOGOUT_EVENT messages when the script logs in as root to execute the specified remote procedure call (RPC). If you omit the parameter, the function behaves as in earlier releases in which the root UI_LOGIN_EVENT and UI_LOGOUT_EVENT messages are included in system log files.

[See [invoke\(\) Function \(SLAX and XSLT\)](#).]

- **The jcs:invoke() function supports suppression of root login and logout events in system log files for SLAX event scripts (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—The **jcs:invoke()** extension function supports the **no-login-logout** parameter in SLAX event scripts. If you include the parameter, the function does not generate and log UI_LOGIN_EVENT and UI_LOGOUT_EVENT messages when the script logs in as root to execute the specified remote procedure call (RPC). If you

omit the parameter, the function behaves as in earlier releases in which the root `UI_LOGIN_EVENT` and `UI_LOGOUT_EVENT` messages are included in system log files.

[See [invoke\(\) Function \(SLAX and XSLT\)](#).]

What's Changed in Release 20.4R1

MPLS

- The `show mpls lsp extensivel` and `show mpls lsp detail` commands display next hop gateway LSPid—When you use the `show mpls lsp extensivel` and `show mpls lsp detail` commands, you'll see next hop gateway LSPid in the output as well.

Network Management and Monitoring

- Warning changed for configuration statements that correspond to deviate not-supported nodes in YANG data models (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)—If you configure a statement corresponding to a YANG data model node that defines the `deviate not-supported` statement, the Junos OS configuration annotates that statement with the comment **Warning: statement ignored: unsupported platform**. In earlier releases, the warning is **Warning: 'statement' is deprecated**.

User Interface and Configuration

- Verbose format option for exporting JSON configuration data (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)—The Junos OS CLI exposes the `verbose` statement at the `edit system export-format json` hierarchy level. The default format for exporting configuration data in JSON changed from `verbose` format to `ietf` format starting in Junos OS Release 16.1R1. You can explicitly specify the default export format for JSON configuration data by configuring the appropriate statement at the `edit system export-format json` hierarchy level. Although the `verbose` statement is exposed in the Junos OS CLI as of the current release, you can configure this statement starting in Junos OS Release 16.1R1.

[See [export-format](#).]

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Known Limitations

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Learn about known limitations in this release for EX Series. For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

EVPN

- After a reboot during recovery process, the ESI LAGs come up before the BGP sessions and routes/ARP entries are not synced. [PR1487112](#)

Platform and Infrastructure

- Junos OS can hang trying to acquire the SMP IPI lock while rebooting when it is running as a VM on Linux and QEMU hypervisor. Device can be recovered using power-cycle of the device. [PR1385970](#)
- 10G channels show false up even when peer end is configured with different speed. The LED on the device also shows green. [PR1530061](#)
- On all Junos OS platforms, in a Q-in-Q environment, xSTP is enabled on the interface having logical interface with **vlan-id-list** configured, then it will only run on those logical interfaces whose vlan-id range includes **native-vlan-id** configured and all others will be in discarding state. This might lead to traffic drop. [PR1532992](#)
- For port-based GBP tag assignment, filter counters might not be displayed. This is a hardware limitation. [PR1547268](#)

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Learn about open issues in Junos OS Release 20.4R3 for EX Series switches. For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Forwarding and Sampling

- The configuration statement **fast-lookup-filter** with match condition is not supported in FLT hardware and might cause a traffic drop. [PR1573350](#)

General Routing

- When VLAN is added as an action for changing the VLAN in both ingress and egress filters, the filter is not installed. [PR1362609](#)
- On EX9208 switches, a few xe- interfaces go down with the error message **if_msg_ifd_cmd_tlv_decode ifd xe-0/0/0 #190 down with ASIC Error**. [PR1377840](#)
- On EX9214 switches, if the MACsec-enabled link flaps after reboot, the error **errorlib_set_error_log(): err_id(-1718026239)** is observed. [PR1448368](#)
- The following message might be seen in chassisd log after rebooting or changing the configuration.
re_tvp_builtin_fwinfo_update: Unable to get firmware version. [PR1471938](#)

- On BCM Packet Forwarding Engine-based EX Series platforms, frames between MTU+4 and MTU+8 bytes with invalid FCS, code error, or IEEE length check error are treated as Jabber frames. [PR1487709](#)
- On EX Series platforms using chipset with SFP+ implemented, interface on the platforms might be in active status when TX or RX connector is removed. As a result, traffic might drop. [PR1495564](#)
- When running the command **show pfe filter hw filter-name filter-name**, the command fails to retrieve the Packet Forwarding Engine programming details of the filter. [PR1495712](#)
- When you rename a Virtual Chassis, the SNMP POE MIB walk produce either no results or sometimes show result from the primary Virtual Chassis. [PR1503985](#)
- During flooding, MAC is learnt only on normal access port but not on the aggregated Ethernet interface trunk port. [PR1506403](#)
- On the legacy EX Series platform, when adding or removing micro BFD LAG configuration, a kernel crash might happen. The kernel crash might cause unexpected Routing Engine reboot or switchover, and even result in traffic loss until the Routing Engine is restored. [PR1524490](#)
- When the streamed telemetry data for a node is deleted during a network churn and the same node is being walked or rendered for the sensor, the rpd process might crash and generate a core file. This is a corner case where rendering and deletion of a particular node occur at the same instance. This issue might be seen only in case of a unstable network. [PR1552816](#)
- When **dot1x server-fail-voip vlan-name** is configured, ensure that both **server-fail-voip vlan-name** and **voip vlan** are configured using VLAN name and not by using vlan-id. [PR1561323](#)
- EX2300 shows high FPC CPU usage. However, the system processes and kernel CPU usage does not add up to the overall FPC usage. This is due to a cosmetic issue with calculation of FPC CPU usage that has been resolved in Junos OS Release 21.1R1 and later. [PR1567438](#)
- Observing traffic drop during unified ISSU due to LAG interface flap. [PR1569578](#)
- EX4600-40F EVPN_VXLAN get unexpected multicast traffic streams after enabling EVPN. [PR1570689](#)
- There is a remote possibility that during many reboots, the Junos VM goes into a state where NMI is needed to continue the reboot. There is no workaround for this and a subsequent reboot does not seem to hit this issue. [PR1601867](#)

Infrastructure

- On EX Series switches, if you are configuring a large number of firewall filters on some interfaces, the FPC might crash and generate core files. [PR1434927](#)
- **IFDE: Null uint32 set vector, ifd and IFFPC: 'IFD Ether uint32 set' (opcode 151)** error message is observed continuously in AD with base configurations. [PR1485038](#)
- A double free vulnerability in the software forwarding interface daemon (sfid) process of Juniper Networks Junos OS allows an adjacently-connected attacker to cause a Denial of Service (DoS) by sending a crafted ARP packet to the device. Refer <https://kb.juniper.net/JSA11162> for more information. [PR1497768](#)

- On EX Series platforms, a traffic drop might be observed after restarting the pfem process due to the stale route entry in Ternary Content Addressable Memory (TCAM). [PR1517497](#)
- User while loading the kernel would see the message **GEOM: mmc0s.enh: corrupt or invalid GPT detected**. This message has no impact on functionality. [PR1549754](#)
- When receives a unicast EAPOL (0x888e) with vlan588 tag at ae1 in this example, the packet is forwarded to ae0 without changing the vlanID to 3054. set vlans vlan588 vlan-id 588. set vlans vlan588 interface ae1.0. set vlans vlan588 interface ae0.0 mapping 3054 swap. [PR1580129](#)

Layer 2 Features

- On EX4600 platforms, if a change related to TPID is made in the Device Control Daemon, traffic might be dropped in the Packet Forwarding Engine due to failure on Layer 2 learning or interfaces flapping. [PR1477156](#)

Platform and Infrastructure

- When the dhcp relay mode is configured as no-snoop, the offer gets dropped due to incorrect ASIC programming. [PR1530160](#)
- On EX9200 line of switches, FPC gets restarted and thereby disrupting traffic when there is an out-of-order filter state. This issue might be seen only in back-to-back GRES in more than 40 to 50 iterations. [PR1579182](#)

Routing Policy and Firewall Filters

- On all Junos OS platforms with **set policy-options rtf-prefix-list** configured, if upgrade to a specific version, the device might fail to validate its configuration which eventually cause rpd to crash unexpectedly due to a software fault. [PR1538172](#)

User Interface and Configuration

- The mgd crashes and generates a core file when the image is upgraded. The issue is seen on EX Series VC. As a workaround, provide a valid package during upgrade. [PR1557628](#)

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For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Resolved Issues: 20.4R3

Class of Service (CoS)

- The buffer allocation for VCP ports might not get released in Packet Forwarding Engine after physically moving the port location. [PR1581187](#)

EVPN

- Traffic loss might be seen under EVPN-VxLAN scenario when MAC-IP moves from one CE interface to another. [PR1591264](#)
- In the EVPN/VXLAN scenario, the label field for Type-1 route is not required but it is assigned 1 instead of 0, which is in conflict with the RFC7432. [PR1594981](#)
- Traffic loss might be seen if aggregated Ethernet bundle interface with ESI is disabled on primary Routing Engine followed by a Routing Engine switchover. [PR1597300](#)

General Routing

- MPPE-Send or Recv-key attribute is not extracted correctly by dot1xd. [PR1522469](#)
- On EX Series line of switches Virtual Chassis (VC), Power over Ethernet (POE) might not be detected and hence might fail to work on VC members. [PR1539933](#)
- DHCP discover packet might be dropped if the DHCP inform packet is received first. [PR1542400](#)
- Classifier is not programmed in the hardware and error logs might be seen in syslog. [PR1548159](#)
- FPC with power related faults might get on-lined again once fabric healing has off-lined the FPC. [PR1556558](#)
- The MAC addresses learned in a Virtual Chassis might fail due to aging out in the MAC scaling environment. [PR1558128](#)
- The tunable optics SFP+-10G-T-DWDM-ZR is not working in EX4600 line of switches. [PR1561181](#)
- **LICENSE_INVALID_FEATURE_ID** syslog message is not logged if a license key has features that are not applicable on the platform (unknown features), the license key is rejected. [PR1562700](#)
- The DHCP client might not obtain IP address when **dhcp-security** is configured. [PR1564941](#)
- The new primary Routing Engine post switchover might go into DB mode (or crash) on EX Series line of switches. [PR1565213](#)
- The 40G DAC connection between EX9253 and the peers might not come up. [PR1569230](#)
- Packet loss might be observed when sample based action is used in firewall filter. [PR1571399](#)
- The fxpc process might crash and cause traffic loss in the IFBD scenario. [PR1572305](#)
- Private VLAN configuration might fail in certain scenario. [PR1574480](#)
- Protocol convergence between end nodes might fail when L2PT is enabled on transit switch. [PR1576715](#)
- The device implemented with different service image version might become VC member as unexpected. [PR1576774](#)
- On EX2300-C switches, MVR configuration cannot be configured. [PR1577905](#)
- The fxpc process might crash on EX Series line of switches. [PR1578421](#)
- The dcpfe crash is observed on Junos OS EX Series line of switches. [PR1578859](#)
- Random or silent reboot might be seen on EX2300-24MP and EX2300-48MP platforms. [PR1579576](#)
- On EX Series line of switches, some 40G ports might not be channelized successfully. .` [PR1582105](#)
- The voice VLAN might not get assigned to the access interface. [PR1582115](#)
- The l2ald crashes if a specific naming format is applied between a vlan-range and a single VLAN. [PR1583092](#)
- DSCP rewriting might fail to work on EX2300 switches. [PR1586341](#)

- The SNMP trap for MAC notifications might not be generated when an interface is added explicitly under switch-options. [PR1587610](#)
- After performing NSSU, **timeout waiting for response from fpc0** error message is seen while checking version details. [PR1584457](#)
- Process dot1xd crash might be seen and re-authentication might be needed on EX9208 platform. [PR1587837](#)
- The rpd crash might be observed on the router running a scaled setup. [PR1588439](#)
- Packet loss might be observed on dynamically assigning VoIP vlan. [PR1589678](#)
- Traffic loss might be observed for interface configured in subnet 137.63.0.0/16. [PR1590040](#)
- The LLDP packet might loss on the EX4300-MP platform if configuring LLDP on the management interface. [PR1591387](#)
- The **show pfe filter hw** statement might generate the following error message: **ERROR (dfw): Unknown group id: 21**. [PR1592096](#)
- On EX4300MP platforms, if an aggregated Ethernet interface is configured as the underlay interface for the type 5 route that is used to connect the DHCP relay and server, the DHCP relay might not work which might result in the DHCP client failing to obtain an IP address. [PR1592133](#)
- xSTP might not get configured when enabled on an interface with SP style configuration on all platforms. [PR1592264](#)
- Clients authentication failure might occur due to dot1x daemon memory leak. [PR1594224](#)
- Storm control profile might not be applied on EX2300 platforms. [PR1594353](#)
- IS-IS adjacency might fail to be formed if the MTU size of an IRB interface is configured with a value greater than 1496 bytes. [PR1595823](#)
- On EX Series line of switches such as EX2300, EX3400, EX4300, EX4600, and EX4650 with chip as Packet Forwarding Engine, if IS-IS is enabled on an integrated routing and bridging (IRB) interface and the maximum transmission unit (MTU) size of the IRB interface is configured with a value greater than 1496 bytes, the IS-IS hello (IIH) PDUs with jumbo frame size (i.e., greater than 1496 bytes) might be dropped and not sent to the IS-IS neighbors. [PR1595823](#)
- The MAC or IP withdraw route might be suppressed by rpd in the EVPN-VxLAN scenario. [PR1597391](#)
- The backup Virtual Chassis member might not learn MAC address on a primary after removing a VLAN unit from the SP style aggregated Ethernet interface which is part of multiple VLAN units. [PR1598346](#)
- On the EX4300-MP switch, the disable (interface) configuration mode command might not be available for the management port em1. The em1 interface CLI disable option is removed for all the products, but it has to be disabled only for EX4300-MP device. Devices with em1 interface enabled might be out of service. [PR1600905](#)
- On EX Series switches, the fxpc process might crash and generate a core file. [PR1607372](#)

- On EX4300 platform, the dcpfe process that handles packet forwarding might crash if the mge-* interfaces are configured with CoS and scheduler port-speed is non-zero while shaping rate becomes 0. [PR1608306](#)
- The upgrade using phone-home feature to Junos OS Release 20.3 or later might fail on EX2300 and EX4650 switches. [PR1601722](#)
- On EX4400 switches, dot1x authentication might not work on EVPN/VxLAN enabled endpoints. [PR1603015](#)
- On EX Series, QFX10008, and QFX10016 line of switches, the system reboot takes approximately 9 minutes for FPCs to come online after system reboot command is issued. [PR1605002](#)
- MAC move might be seen between the ICL and MC-LAG interface while removing and then adding VLANs on the ICL interface. [PR1605234](#)
- DHCP packets might be received and then returned back to DHCP relay through the same interface on EX2300, EX3400, and EX4300 Virtual Chassis platforms. [PR1610253](#)
- Change in commit error message while configuring the same vlan-id with different vlan-name through openconfig CLI. [PR1612566](#)
- The configured MAC address does not reflect after the device reboots. [PR1569203](#)

Infrastructure

- Some MAC addresses might not be aged out on EX4300 platforms. [PR1579293](#)
- The fxpc process might crash and generate a core file. [PR1611480](#)

Interfaces and Chassis

- MC-LAG interfaces might go down if the same VRRP group-id is configured on multiple IRB units. [PR1575779](#)
- The aggregated Ethernet interface might flap. [PR1576533](#)
- ARP resolution failure might occur during VRRP failover. [PR1578126](#)
- Incorrect advertisement threshold values are seen on VRRP groups when VRRP is configured on EX2300 switches. [PR1584499](#)

Layer 2 Ethernet Services

- Aggregated Ethernet interface flap might be seen during NSSU. [PR1551925](#)
- The DHCP client will be offline for 120 seconds after sending the **DHCPINFORM** message in the DHCP relay scenario. [PR1575740](#)
- The DHCP client might be offline for about 120 seconds after sending the **DHCPINFORM** message. [PR1587982](#)

Layer 2 Features

- MAC addresses learned from the MC-LAG client device might keep flapping between the ICL interface and MC-AE interface after one child link in the MC-AE interface is disabled. [PR1582473](#)

Junos Fusion Enterprise

- Reverting the primary Routing Engine from RE1 to RE0 might lead to l2ald daemon crash and cause an outage. [PR1601817](#)

MPLS

- Incorrect EXP bit change might be seen in certain conditions under MPLS scenario. [PR1555797](#)

Platform and Infrastructure

- On EX3400 Virtual Chassis, console access on backup Virtual Chassis member is not allowed. [PR1530106](#)
- Upon receipt of specific sequences of genuine packets destined to the device, the kernel will crash and restart (vmcore). [PR1557881](#)
- The LLDP neighbor advertisement on EX4300 switches might send an incorrect 802.3 power format with TLV length 7 instead of length 12. [PR1563105](#)
- The last flapped timestamp for interface fxp0 resets every time when you perform **monitor traffic interface fxp0**. [PR1564323](#)
- On all EX9200 platforms with EVPN-VXLAN configured, the next hop memory leak in MX Series ASIC happens whenever there is a route churn for remote MAC-IP entries learned bound to the IRB interface in EVPN-VXLAN routing instance. When the ASIC's next hop memory partition is exhausted, the FPC might reboot. [PR1571439](#)
- DHCP packets with source IP as link-local address are dropped in EX4300 switches. [PR1576022](#)
- The pfex might crash during PIC 4x 1G/10G SFP/SFP+ offline or online. [PR1582457](#)
- Firewall filter is not programmed correctly and traffic might drop unexpectedly. [PR1586433](#)
- The egress RACL firewall filter might not get programmed correctly on EX4300 platforms. [PR1595797](#)
- Broadcast traffic might be discarded when a firewall filter is applied to the loopback interface. [PR1597548](#)
- VLAN tagged traffic might be dropped with service provider style configuration. [PR1598251](#)
- When you configure **mac-move-limit** statement, forwarding the VRRP packets is not possible. [PR1601005](#)
- Adding aggregated Ethernet configuration without child member might cause MAC/ARP learning issues. [PR1602399](#)
- ZTP does not work when downgrade Junos OS Release 21.1R2.2 image to Junos OS Release 21.1R2.1 image. [PR1603227](#)

Routing Protocols

- BGP session carrying VPNv4 prefix with IPv6 next-hop might be dropped. [PR1580578](#)

- The rpd might crash in scaled routing instances scenario. [PR1590638](#)

User Interface and Configuration

- Removing the flash component from monitor interfaces and DHCP pages, removes the other flash pages. [PR1553176](#)

Virtual Chassis

- EX4300 VCP might not come up after upgrade when QSFP+-40G-SR4/QSFP+-40G-LR4/QSFP+40GE-LX4 is used. [PR1579430](#)

Resolved Issues: 20.4R2

Forwarding and Sampling

- Configuration archive **transfer-on-commit** fails on devices running Junos OS Release 18.2R3-S6.5. [PR1563641](#)

General Routing

- Traffic loss might be observed on interfaces in a VXLAN environment. [PR1524955](#)
- EX4300-48MP: Sflow: dcpfe core file is observed while using "request chassis fpc slot <slot_num> restart" command. [PR1536997](#)
- FPC might not be recognized after power cycle (hard reboot). [PR1540107](#)
- DHCP discover packet might be dropped if DHCP inform packet is received first. [PR1542400](#)
- The JNH memory leak might be observed on MX Series-based line cards. [PR1542882](#)
- The Slaac-Snoopd child process generates core file upon multiple switchovers on the Routing Engine. [PR1543181](#)
- On EX4300-48MP line of switches with Linux TVP architecture and Junos OS as VM, the Junos CLI outputs do not confirm if the Junos OS and the host kernel are compatible with each other. [PR1543901](#)
- Traffic loop when adding logical interface in the case of multihomed SP style in EVPN/VXLAN. [PR1543966](#)
- High EVENTD CPU utilization upon receiving LLMNR and MDNS traffic on EX2300. [PR1544549](#)
- The device might be out of service after configuring the em1 or em2 interface. [PR1544864](#)
- FPC(s) might not boot on EX9214 line of switches in a certain condition. [PR1545838](#)
- The static MAC on an interface might not work on EX4600 line of switches. [PR1546655](#)
- Two Routing Engine's might lose communication if they have different Junos OS versions. [PR1550594](#)
- **show pfe route summary hw** shows random high free and 'Used' column for 'IPv6 LPM(< 64)' routes. [PR1552623](#)
- The dcpfe process might crash and the non-channelization interfaces might not come up. [PR1552798](#)

- The 'action-shutdown' statement of storm control does not work for ARP broadcast packets. [PR1552815](#)
- Layer 2 multicast traffic received on the Virtual Chassis port (VCP) ports might be dropped if **igmp-snooping** and STP/VSTP are enabled. [PR1553159](#)
- OIR of CBs might result in major errors and the Packet Forwarding Engine disable action halted traffic forwarding on the FPCs. [PR1554145](#)
- The link on the Linux-based LC is not brought down immediately after the FPC process (ukern/indus.elf) crashes or the process is killed. [PR1554430](#)
- The console might hang up with the configuration statement **set system ports console log-out-on-disconnect** configured. [PR1555487](#)
- On the EX9200 device, SF3 Fabric OIR issue is observed with Junos OS Release. [PR1555727](#)
- Traffic might be dropped when a firewall filter rule uses then VLAN as the action. [PR1556198](#)
- On the EX4300 device, script fails while committing the IPsec authentication configuration because of the missing algorithm statement. [PR1557216](#)
- RPD core file is seen after Routing Engine switchover. [PR1558814](#)
- Some transmitting packets might get dropped because the **disable-pfe** action is not invoked when the fabric self-ping failure is detected. [PR1558899](#)
- Tunable optics SFP+-10G-T-DWDM-ZR is not working in EX4600 line of switches. [PR1561181](#)
- Observing error opening configuration database: could not open configuration database during USB upgrading. [PR1561741](#)
- EX3400VC smartd pollutes syslog every 5 seconds after upgrade or system reboot. [PR1562396](#)
- On EX3400VC line of switches, the DAEMON-7-PVIDB throws syslog messages for every 12 to 14 minutes after you upgrade to Junos OS Release 19.1R3-S3. [PR1563192](#)
- Client authentication fails after performing GRES. [PR1563431](#)
- The JWeb upgrade might fail on EX2300 and EX3400. [PR1563906](#)
- On EX4650 line of switches, **storm control** with IRB interface might not work approximately. [PR1564020](#)
- On the EX4600 device, the following internal comment is displayed: "Placeholder for QFX platform configuration" on performing **show config output**. [PR1567037](#)
- The designated forwarder (DF) might not forward traffic. [PR1567752](#)
- Port-mirroring might not work when the analyzer output is a trunk interface. [PR1575129](#)

Infrastructure

- On the EX4600 and EX4300 Virtual Chassis or Virtual Chassis fabric, the VSTP configurations device goes unreachable and becomes nonresponsive after commit. [PR1520351](#)
- HEAP malloc(0) is observed on EX4300 VC/VCF. [PR1546036](#)

- The DSCP rewrite is not happening correctly with wildcard interface. [PR1552372](#)
- The vme/me0 management interface cannot process any incoming packets. [PR1552952](#)
- Traffic related to IRB interface might be dropped when mac-persistence-timer expires. [PR1557229](#)
- Traffic might not be forwarded on EX3400 and EX4300-MP platforms with Layer 2 classifier rules applied. [PR1561263](#)

Interfaces and Chassis

- The pppd might crash when VRRP is configured on all Junos OS platforms. [PR1561281](#)
- MC-AE interfaces might go down if same VRRP group-id is configured on multiple IRB units. [PR1575779](#)

Layer 2 Ethernet Services

- OSPF and OSPF3 adjacency uptime is more than expected after NSSU upgrade and outage is higher than the expected. [PR1551925](#)

MPLS

- Incorrect EXP bit change might be seen in certain conditions under MPLS scenario. [PR1555797](#)

Platform and Infrastructure

- Packets transiting through multicast-based VXLAN VTEP interface might be dropped when post FPC restarts. [PR1536364](#)
- The targeted-broadcast feature might not work after a reboot. [PR1548858](#)
- The BGP session replication might fail to start after the session crashes on the backup Routing Engine. [PR1552603](#)
- The targeted-broadcast feature might send out duplicate packets. [PR1553070](#)
- ARP resolution might fail if ARP packets are received over multicast-based VXLAN access network from CE. [PR1553917](#)
- The traffic might be dropped on Layer 3 LAG after rebooting or halting any member of EX4300 Virtual Chassis. [PR1556124](#)
- PFEX might crash when soft error recovery feature is enabled on the Packet Forwarding Engine. [PR1567515](#)
- Introduce two new major CMERRORs for XM chip-based line card to stabilize the running device. [PR1574631](#)

Routing Protocols

- The rpd memory leak might be seen in the BGP scenario. [PR1547273](#)
- The OSPF neighborship get stuck in the Start state after configuring the EVPN-VXLAN. [PR1519244](#)

- Sending multicast traffic to downstream receiver on MX Series-based Virtual Chassis platforms might fail. [PR1555518](#)
- The pppd memory leak might cause traffic loss. [PR1561850](#)
- The rpd process might crash if there are more routes changed during the commit-sync processing window. [PR1565814](#)
- The untagged packets might not work on EX Series platforms. [PR1568533](#)

User Interface and Configuration

- Remove flash component from monitor, interfaces, and DHCP pages, and remove other flash pages. [PR1553176](#)
- J-Web application package cannot be auto updated for all the supported EX Series devices. [PR1563588](#)

Virtual Chassis

- On EX4600 and EX4300 mixed Virtual Chassis, an error message 'ex_bcm_pic_eth_uint8_set' is seen when changing configuration related to interface. [PR1573173](#)

Resolved Issues:20.4R1

Authentication and Access Control

- The dot1x client won't be moved to held state when the authenticated PVLAN is deleted. [PR1516341](#)

EVPN

- Unable to create a new VTEP interface. [PR1520078](#)

Infrastructure

- qmon-sw sensor is not supported in EX3400. [PR1506710](#)
- The IP communication between directly connected interfaces on EX4600 would fail. [PR1515689](#)
- The VC system might get hanged after committing the VSTP configurations. [PR1520351](#)
- OID ifOutDiscards reports zero and sometimes shows valid value. [PR1522561](#)
- Firewall policer with discard action might fail on EX4300. [PR1532670](#)
- Errors might be seen when dumping vmcore on EX2300 and EX3400 switches. [PR1537696](#)
- The LLDP neighborship with the VoIP phones can't be established. [PR1538482](#)

Layer 2 Features

- The dcpfe/FPC might crash due to the memory leak during the vlan add/delete operation. [PR1505239](#)
- On the QFX5000 line of switches, traffic imbalance might be observed if hash-params is not configured. [PR1514793](#)

- The MAC address in the hardware table might become out of synchronization between the primary and member in Virtual Chassis after the MAC flaps. [PR1521324](#)

Network Management and Monitoring

- EX4300: SNMP OID 1.3.6.1.2.1.25.3.3.1.2.0 (hrProcessorLoad) always returns 0 irrespective of the real CPU utilization. [PR1508364](#)

Platform and Infrastructure

- IPv6 neighbor solicitation packets might be dropped in a transit device. [PR1493212](#)
- DHCP Binding is not happening after Graceful switchover. [PR1515234](#)
- LLDP adjacency might fail for non-AE interfaces on EX4300 platform. [PR1538401](#)
- uRPF in the Strict mode does not work. [PR1417546](#)
- Virtual Chassis split after network topology changed. [PR1427075](#)
- IRB MAC will not be programmed in hardware when MAC persistence timer expires. [PR1484440](#)
- Authentication session might be terminated if PEAP request is retransmitted by authenticator. [PR1494712](#)
- In some cases, if we have an OSPF session on the IRB over LAG interface with 40-Gigabit Ethernet port as member, the session gets stuck in restart. [PR1498903](#)
- On the EX4300, EX3400, and EX2300 Virtual Chassis with NSB and xSTP enabled, continuous traffic loss might be observed while performing GRES. [PR1500783](#)
- The mge interface might still stay up while the far end of its link goes down. [PR1502467](#)
- LLDP is not acquired when native-VLAN-ID and tagged VLAN-ID are the same on a port. [PR1504354](#)
- The output VLAN push might not work. [PR1510629](#)
- Traffic might not flow as per configured policer parameters. [PR1512433](#)
- LACP goes down after performing Routing Engine switchover if MACsec is enabled on the LAG members on EX4300. [PR1513319](#)
- Last commit line in configuration is updated after the configuration backup has been done. [PR1513499](#)
- The 100M SFP-FX is not supported on satellite device in a Junos Fusion setup. [PR1514146](#)
- ARP learning issue might be seen on EX4300-MP platform when configuring Layer 3 gateway interfaces. [PR1514729](#)
- "dot1x" memory leak is seen. [PR1515972](#)
- The dcpfe (PFE) process might crash due to memory leak. [PR1517030](#)
- MPPE-Send/Recv-key attribute is not extracted correctly by dot1xd. [PR1522469](#)
- "Drops" and "Dropped packets" counters in the output by "show interface extensive" are double counting. [PR1525373](#)

- EX4300-48MP device might go out of service during a software upgrade operation. [PR1526493](#)
- PoE messages "poe_get_dev_class: Failed to get PD class info" seen on EX2300. [PR1536408](#)
- EX3400, EX2300 : Upgrade failure do to lack of available storage. [PR1539293](#)
- Slaac-Snoopd child process core is observed upon multiple switchovers on Routing Engine. [PR1543181](#)
- EX9200 SF3 Fabric OIR Issues with Junos 23.1R1.8. [PR1555727](#)

Routing Protocols

- The rpd process might report 100 percent CPU usage with the BGP route damping enabled. [PR1514635](#)
- Packet loss might be observed while verifying traffic from access to core network for IPv4 and IPv6 interfaces. [PR1520059](#)
- OSPFv3 adjacency should not be established when IPsec authentication is enabled. [PR1525870](#)

User Interface and Configuration

- J-Web does not display the correct Flow-control status on EX Series devices. [PR1520246](#)

Virtual Chassis

- On the EX4650 device, the following error message is observed during booting: kldload: an error occurred while loading the module. [PR1527170](#)

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Documentation Updates

There are no errata or changes in Junos OS Release 20.4R3 documentation for EX Series switches.

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Migration, Upgrade, and Downgrade Instructions

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- [Upgrade and Downgrade Support Policy for Junos OS Releases | 68](#)

This section contains the upgrade and downgrade support policy for Junos OS for EX Series switches. Upgrading or downgrading Junos OS can take several hours, depending on the size and configuration of the network. For information about software installation and upgrade, see the [Installation and Upgrade Guide](#).

Upgrade and Downgrade Support Policy for Junos OS Releases

We have two types of releases, standard EOL and EEOL:

- Standard End of Life (EOL) releases have engineering support for twenty four months after the first general availability date and customer support for an additional six more months.
- Extended End of Life (EEOL) releases have engineering support for thirty six months after the first general availability date and customer support for an additional six more months.

For both standard EOL and EEOL releases, you can upgrade to the next three subsequent releases or downgrade to the previous three releases. For example, you can upgrade from 19.2 to the next three releases – 19.3, 19.4 and 20.1 or downgrade to the previous three releases – 19.1, 18.4 and 18.3.

For EEOL releases only, you have an additional option - you can upgrade directly from one EEOL release to the next two subsequent EEOL releases, even if the target release is beyond the next three releases. Likewise, you can downgrade directly from one EEOL release to the previous two EEOL releases, even if the target release is beyond the previous three releases. For example, 19.2 is an EEOL release. Hence, you

can upgrade from 19.2 to the next two EEOL releases – 19.3 and 19.4 or downgrade to the previous two EEOL releases – 19.1 and 18.4.

Release Type	End of Engineering (EOE)	End of Support (EOS)	Upgrade and Downgrade to subsequent 3 releases	Upgrade and Downgrade to subsequent 2 EEOL releases
Standard End of Life (EOL)	24 months	End of Engineering + 6 months	Yes	No
Extended End of Life (EEOL)	36 months	End of Engineering + 6 months	Yes	Yes

For more information about standard EOL and EEOL releases, see <https://www.juniper.net/support/eol/junos.html>.

For information about software installation and upgrade, see the [Installation and Upgrade Guide](#).

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These release notes accompany Junos OS Release 20.4R3 for the JRR Series. They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

You can also find these release notes on the Juniper Networks Junos OS Documentation webpage, located at https://www.juniper.net/documentation/product/en_US/junos-os.

What's New

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- What's New in Release 20.4R2 | 70
- What's New in Release 20.4R1 | 71

Learn about new features introduced in the Junos OS main and maintenance releases for JRR Series Route Reflectors.

What's New in Release 20.4R3

There are no new features or enhancements to existing features for JRR Series in Junos OS Release 20.4R3.

What's New in Release 20.4R2

There are no new features or enhancements to existing features for JRR Series in Junos OS Release 20.4R2.

What's New in Release 20.4R1

Routing Protocols

- **Support for BGP Sharding (JRR200)**—Starting in Junos OS Release 20.4R1, we support BGP sharding. BGP sharding splits a BGP RIB into several sub RIBs and each sub RIB handles a subset of BGP routes. Each sub RIB is served by a separate RPD thread to achieve parallel processing. This results in reduced convergence time and faster performance. BGP sharding is disabled by default.

To enable BGP sharding, configure **rib-sharding** at the [edit system processes routing bgp] hierarchy level. Sharding is dependent on the update I/O thread feature. Therefore, you need to enable update I/O thread when you configure sharding. To enable update I/O, configure **update-threading** at the [edit system processes routing bgp] hierarchy level for rib-sharding configuration to pass commit check.

If you configure rib-sharding on a routing engine, RPD will create sharding threads. By default the number of sharding and update threads created is same as the number of CPU cores on the routing engine. Optionally, you can specify the number-of-shards and number-of-threads you want to create.

NOTE: BGP sharding is supported for IPv4, IPv6, L3VPN and BGP-LU. All the other RIBs are processed without sharding.

[See [rib-sharding](#) and [update-threading](#).]

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What's Changed

There are no changes in behavior and syntax in Junos OS Release 20.4R3 for JRR Series Route Reflectors.

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Known Limitations

There are no known limitations in this release for JRR200 Route Reflectors.

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Open Issues

There are no open issues in this release for JRR Series Route Reflectors.

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Resolved Issues

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This section lists the issues fixed in Junos OS Release 20.4R3 for JRR Series routers.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Resolved Issues: 20.4R3

General Routing

- JRR200: Option-60 (Vendor-Class-Identifier) is not sent during ZTP. [PR1582038](#)

Resolved Issues: 20.4R2

There are no fixed issues in the Junos OS Release 20.4R2 for JRR Series routers.

Resolved Issues: 20.4R1

General Routing

- On the JRR200 routers, the firewall filter with non-zero TTL value might cause a commit error. [PR1531034](#)
- tcp_timer_keep logs flood on JRR200. [PR1533168](#)
- Optics info of physical interfaces is not available for JRR200 on Junos OS. [PR1537261](#)
- The CLI "request system power-off" and "request system halt" commands do not work as expected on JRR200. [PR1534795](#)

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Documentation Updates

There are no errata or changes in Junos OS Release 20.4R3 documentation for JRR200 Route Reflectors.

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- [Upgrade and Downgrade Support Policy for Junos OS Releases | 75](#)

This section contains the upgrade and downgrade support policy for Junos OS for the JRR Series Route Reflector. Upgrading or downgrading Junos OS might take several minutes, depending on the size and configuration of the network.

For information about software installation and upgrade, see the [JRR200 Route Reflector Quick Start](#) and the [Installation and Upgrade Guide](#).

Upgrade and Downgrade Support Policy for Junos OS Releases

We have two types of releases, standard EOL and EEOL:

- Standard End of Life (EOL) releases have engineering support for twenty four months after the first general availability date and customer support for an additional six more months.
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For both standard EOL and EEOL releases, you can upgrade to the next three subsequent releases or downgrade to the previous three releases. For example, you can upgrade from 19.2 to the next three releases – 19.3, 19.4 and 20.1 or downgrade to the previous three releases – 19.1, 18.4 and 18.3.

For EEOL releases only, you have an additional option - you can upgrade directly from one EEOL release to the next two subsequent EEOL releases, even if the target release is beyond the next three releases. Likewise, you can downgrade directly from one EEOL release to the previous two EEOL releases, even if the target release is beyond the previous three releases. For example, 19.2 is an EEOL release. Hence, you can upgrade from 19.2 to the next two EEOL releases – 19.3 and 19.4 or downgrade to the previous two EEOL releases – 19.1 and 18.4.

Release Type	End of Engineering (EOE)	End of Support (EOS)	Upgrade and Downgrade to subsequent 3 releases	Upgrade and Downgrade to subsequent 2 EEOL releases
Standard End of Life (EOL)	24 months	End of Engineering + 6 months	Yes	No
Extended End of Life (EEOL)	36 months	End of Engineering + 6 months	Yes	Yes

For more information about standard EOL and EEOL releases, see <https://www.juniper.net/support/eol/junos.html>.

For information about software installation and upgrade, see the [Installation and Upgrade Guide](#).

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Junos OS Release Notes for Juniper Secure Connect

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These release notes accompany Junos OS Release 20.4R3 for Juniper Secure Connect. They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

You can also find these release notes on the Juniper Networks Junos OS Documentation webpage, located at https://www.juniper.net/documentation/product/en_US/junos-os.

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What's New in Release 20.4R3

There are no new features in Junos OS Release 20.4R3 for Juniper Secure Connect.

What's New in Release 20.4R2

There are no new features in Junos OS Release 20.4R2 for Juniper Secure Connect.

What's New in Release 20.4R1

There are no new features in Junos OS Release 20.4R1 for Juniper Secure Connect.

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What's Changed in Release 20.4R3

There are no changes in behavior or syntax for Juniper Secure Connect in Junos OS Release 20.4R3.

What's Changed in Release 20.4R2

There are no changes in behavior or syntax for Juniper Secure Connect in Junos OS Release 20.4R2.

What's Changed in Release 20.4R1

There are no changes in behavior or syntax for Juniper Secure Connect in Junos OS Release 20.4R1.

Known Limitations

There are no known behavior or limitation for Juniper Secure Connect in Junos OS Release 20.4R3.

Open Issues

There are no known issues for Juniper Secure Connect in Junos OS Release 20.4R3.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Resolved Issues

Resolved Issues: 20.4R3

There are no resolved issues for Juniper Secure Connect in Junos OS Release 20.4R3.

Resolved Issues: 20.4R2

There are no resolved issues for Juniper Secure Connect in Junos OS Release 20.4R2.

Resolved Issues: 20.4R1

There are no resolved issues for Juniper Secure Connect in Junos OS Release 20.4R1.

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These release notes accompany Junos OS Release 20.4R3 for the Junos fusion for enterprise. They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

You can also find these release notes on the Juniper Networks Junos OS Documentation webpage, located at https://www.juniper.net/documentation/product/en_US/junos-os.

What's New

There are no new features or enhancements to existing features in Junos OS Release 20.4R3 for Junos fusion for enterprise.

NOTE: For more information about Junos fusion for enterprise, see the [Junos Fusion for Enterprise User Guide](#).

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What's Changed

There are no changes in behavior of Junos OS features and changes in the syntax of Junos OS statements and commands in Junos OS Release 20.4R3 for Junos fusion for enterprise.

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Known Limitations

There are no known behaviors, system maximums, and limitations in hardware and software in Junos OS Release 20.4R3 for Junos fusion for enterprise.

For the most complete and latest information about known Junos OS problems, use the Juniper Networks online [Junos Problem Report Search](#) application.

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Open Issues

There are no known issues in hardware and software in Junos OS Release for 20.4R3 Junos fusion for enterprise.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

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Resolved Issues

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Learn which issues were resolved in the Junos OS main and maintenance releases for Junos fusion for enterprise.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Resolved Issues: 20.4R3

- The dot1xd process might generate a core file and make it necessary to re-authenticate on the EX9208 platform in a Fusion scenario. [PR1587837](#)
- Reverting mastership from RE1 to RE0 might cause the l2ald process to generate a core file and cause an outage. [PR1601817](#)

Resolved Issues: 20.4R2

There are no resolved issues in Junos OS Release 20.4R2 for Junos fusion for enterprise.

Resolved Issues: 20.4R1

- The 100M SFP-FX is not supported on satellite devices in a Junos fusion setup. [PR1514146](#)

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Documentation Updates

There are no errata or changes in Junos OS Release 20.4R3 for documentation for Junos fusion for enterprise.

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Migration, Upgrade, and Downgrade Instructions

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This section contains the procedure to upgrade or downgrade Junos OS and satellite software for a Junos fusion for enterprise. Upgrading or downgrading Junos OS and satellite software might take several hours, depending on the size and configuration of the Junos fusion for enterprise topology.

Basic Procedure for Upgrading Junos OS on an Aggregation Device

When upgrading or downgrading Junos OS for an aggregation device, always use the **junos-install** package. Use other packages (such as the **jbundle** package) only when so instructed by a Juniper Networks support representative. For information about the contents of the **junos-install** package and details of the installation process, see the [Installation and Upgrade Guide](#).

NOTE: Before upgrading, back up the file system and the currently active Junos OS configuration so that you can recover to a known, stable environment in case the upgrade is unsuccessful. Issue the following command:

```
user@host> request system snapshot
```

The installation process rebuilds the file system and completely reinstalls Junos OS. Configuration information from the previous software installation is retained, but the contents of log files might be erased. Stored files on the routing platform, such as configuration templates and shell scripts (the only exceptions are the **juniper.conf** and **ssh** files), might be removed. To preserve the stored files, copy them to another system before upgrading or downgrading the routing platform. See the [Junos OS Administration Library](#).

To download and install Junos OS:

1. Using a Web browser, navigate to the Download Software URL on the Juniper Networks webpage:
<https://www.juniper.net/support/downloads/>
2. Log in to the Juniper Networks authentication system using the username (generally your e-mail address) and password supplied by Juniper Networks representatives.
3. Select **By Technology > Junos Platform > Junos fusion** to find the software that you want to download.
4. Select the release number (the number of the software version that you want to download) from the Version drop-down list on the right of the page.
5. Select the **Software** tab.
6. Select the software package for the release.

7. Review and accept the End User License Agreement.
8. Download the software to a local host.
9. Copy the software to the routing platform or to your internal software distribution site.
10. Install the new **junos-install** package on the aggregation device.

NOTE: We recommend that you upgrade all software packages out of band using the console because in-band connections are lost during the upgrade process.

Customers in the United States and Canada, use the following commands, where *n* is the spin number.

```
user@host> request system software add validate reboot source/package-name.n.tgz
```

All other customers, use the following commands, where *n* is the spin number.

```
user@host> request system software add validate reboot source/package-name.n-limited.tgz
```

Replace **source** with one of the following values:

- **/pathname**—For a software package that is installed from a local directory on the router.
- For software packages that are downloaded and installed from a remote location:
 - **ftp://hostname/pathname**
 - **http://hostname/pathname**
 - **scp://hostname/pathname** (available only for Canada and U.S. version)

The **validate** option validates the software package against the current configuration as a prerequisite to adding the software package to ensure that the router reboots successfully. This is the default behavior when the software package being added is a different release.

Adding the **reboot** command reboots the router after the upgrade is validated and installed. When the reboot is complete, the router displays the login prompt. The loading process might take 5 to 10 minutes.

Rebooting occurs only if the upgrade is successful.

Upgrading an Aggregation Device with Redundant Routing Engines

If the aggregation device has two Routing Engines, perform a Junos OS installation on each Routing Engine separately to minimize disrupting network operations as follows:

1. Disable graceful Routing Engine switchover (GRES) on the master Routing Engine and save the configuration change to both Routing Engines.
2. Install the new Junos OS release on the backup Routing Engine while keeping the currently running software version on the master Routing Engine.
3. After making sure that the new software version is running correctly on the backup Routing Engine, switch over to the backup Routing Engine to activate the new software.
4. Install the new software on the original master Routing Engine that is now active as the backup Routing Engine.

For the detailed procedure, see the [Installation and Upgrade Guide](#).

Preparing the Switch for Satellite Device Conversion

There are multiple methods to upgrade or downgrade satellite software in your Junos fusion for enterprise. See [Configuring or Expanding a Junos fusion for enterprise](#).

For satellite device hardware and software requirements, see [Understanding Junos fusion for enterprise Software and Hardware Requirements](#).

Use the following command to install Junos OS on a switch before converting it into a satellite device:

```
user@host> request system software add validate reboot source/package-name
```

NOTE: The following conditions must be met before a Junos switch that is running Junos OS Release 14.1X53-D43 can be converted to a satellite device when the action is initiated from the aggregation device:

- The switch running Junos OS can be converted only to SNOS 3.1 and later.
- Either the switch must be set to factory-default configuration by using the **request system zeroize** command, or the following command must be included in the configuration: **set chassis auto-satellite-conversion**.

When the interim installation has completed and the switch is running a version of Junos OS that is compatible with satellite device conversion, perform the following steps:

1. Log in to the device using the console port.
2. Clear the device:

```
[edit]
```

```
user@satellite-device# request system zeroize
```

NOTE: The device reboots to complete the procedure for resetting the device.

If you are not logged in to the device using the console port connection, your connection to the device is lost after you enter the **request system zeroize** command.

If you lose connection to the device, log in using the console port.

3. (EX4300 switches only) After the reboot is complete, convert the built-in 40-Gbps QSFP+ interfaces from Virtual Chassis ports (VCPs) into network ports:

```
user@satellite-device> request virtual-chassis vc-port delete pic-slot 1 port port-number
```

For example, to convert all four built-in 40-Gbps QSFP+ interfaces on an EX4300-24P switch into network ports:

```
user@satellite-device> request virtual-chassis vc-port delete pic-slot 1 port 0
user@satellite-device> request virtual-chassis vc-port delete pic-slot 1 port 1
user@satellite-device> request virtual-chassis vc-port delete pic-slot 1 port 2
user@satellite-device> request virtual-chassis vc-port delete pic-slot 1 port 3
```

This step is required for the 40-Gbps QSFP+ interfaces that will be used as uplink interfaces in a Junos fusion topology. Built-in 40-Gbps QSFP+ interfaces on EX4300 switches are configured into VCPs by default, and the default settings are restored after the device is reset.

After this initial preparation, you can use one of three methods to convert your switches into satellite devices—autoconversion, manual conversion, or preconfiguration. See [Configuring or Expanding a Junos fusion for enterprise](#) for detailed configuration steps for each method.

Converting a Satellite Device to a Standalone Switch

If you need to convert a satellite device to a standalone device, you must install a new Junos OS software package on the satellite device and remove it from the Junos fusion topology. For more information, see [Converting a Satellite Device to a Standalone Device](#).

Upgrade and Downgrade Support Policy for Junos OS Releases

We have two types of releases, EOL and EEOL:

- End of Life (EOL) releases have engineering support for twenty four months after the first general availability date and customer support for an additional six more months.

- Extended End of Life (EEOL) releases have engineering support for thirty six months after the first general availability date and customer support for an additional six more months.

For both EOL and EEOL releases, you can upgrade to the next three subsequent releases or downgrade to the previous three releases. For example, you can upgrade from 19.2 to the next three releases – 19.3, 19.4 and 20.1 or downgrade to the previous three releases – 19.1, 18.4 and 18.3.

For EEOL releases only, you have an additional option - you can upgrade directly from one EEOL release to the next two subsequent EEOL releases, even if the target release is beyond the next three releases. Likewise, you can downgrade directly from one EEOL release to the previous two EEOL releases, even if the target release is beyond the previous three releases. For example, 19.2 is an EEOL release. Hence, you can upgrade from 19.2 to the next two EEOL releases – 19.3 and 19.4 or downgrade to the previous two EEOL releases – 19.1 and 18.4.

Release Type	End of Engineering (EOE)	End of Support (EOS)	Upgrade and Downgrade to subsequent 3 releases	Upgrade and Downgrade to subsequent 2 EEOL releases
End of Life (EOL)	24 months	End of Engineering + 6 months	Yes	No
Extended End of Life (EEOL)	36 months	End of Engineering + 6 months	Yes	Yes

For more information about EOL and EEOL releases, see <https://www.juniper.net/support/eol/junos.html>.

For information about software installation and upgrade, see the [Installation and Upgrade Guide](#).

Downgrading Junos OS

Junos fusion for enterprise is first supported in Junos OS Release 16.1, although you can downgrade a standalone EX9200 switch to earlier Junos OS releases.

NOTE: You cannot downgrade more than three releases.

For more information, see the [Installation and Upgrade Guide](#).

To downgrade a Junos fusion for enterprise, follow the procedure for upgrading, but replace the **junos-install** package with one that corresponds to the appropriate release.

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Junos OS Release Notes for Junos Fusion for Provider Edge

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These release notes accompany Junos OS Release 20.4R3 for Junos fusion for provider edge. They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

You can also find these release notes on the Juniper Networks Junos OS Documentation webpage, located at https://www.juniper.net/documentation/product/en_US/junos-os.

What's New

IN THIS SECTION

- [What's New in Release 20.4R3 | 89](#)
- [What's New in Release 20.4R2 | 89](#)
- [What's New in Release 20.4R1 | 89](#)

Learn about new features introduced in the Junos OS main and maintenance releases for Junos fusion for provider edge.

What's New in Release 20.4R3

There are no new features introduced in this release for Junos fusion for provider edge.

What's New in Release 20.4R2

There are no new features introduced in this release for Junos fusion for provider edge.

What's New in Release 20.4R1

Hardware

- **Support for QFX5110 as a satellite device in a Junos fusion for provider edge environment on a GNF (MX480, MX960, MX2010, and MX2020)**—With Junos node slicing, you can create guest network functions (GNFs), which are partitions where an aggregation device can be configured. The aggregation device on a GNF supports a maximum of 10 satellite devices. Starting in Junos OS Release 20.4R1, you can configure QFX5110 switches as satellite devices in a Junos fusion for provider edge environment on a GNF.

[See [Understanding Junos Fusion Provider Edge Software and Hardware Requirements](#) and [Junos Node Slicing Overview](#).]

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What's Changed

There are no changes in the behavior of Junos OS features or in the syntax of Junos OS statements and commands in this release for Junos fusion for provider edge.

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Known Limitations

There are no known behaviors, system maximums, and limitations in hardware and software in Junos OS Release 20.4R3 for Junos fusion for provider edge.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

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Open Issues

There are no open issues in the Junos OS Release 20.4R3 for Junos fusion for provider edge.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

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Resolved Issues

There are no fixed issues in the Junos OS Release 20.4R3 for Junos fusion for provider edge.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

SEE ALSO

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Documentation Updates

There are no errata or changes in Junos OS Release 20.4R3 documentation for Junos fusion for provider edge.

SEE ALSO

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Migration, Upgrade, and Downgrade Instructions

IN THIS SECTION

- [Basic Procedure for Upgrading an Aggregation Device | 93](#)
- [Upgrading an Aggregation Device with Redundant Routing Engines | 95](#)
- [Preparing the Switch for Satellite Device Conversion | 96](#)
- [Converting a Satellite Device to a Standalone Device | 97](#)
- [Upgrading an Aggregation Device | 99](#)
- [Upgrade and Downgrade Support Policy for Junos OS Releases | 100](#)
- [Downgrading from Junos OS Release 20.4 | 100](#)

This section contains the procedure to upgrade Junos OS, and the upgrade and downgrade policies for Junos OS for Junos fusion for provider edge. Upgrading or downgrading Junos OS might take several hours, depending on the size and configuration of the network.

Basic Procedure for Upgrading an Aggregation Device

When upgrading or downgrading Junos OS, always use the **jinstall** package. Use other packages (such as the **jbundle** package) only when so instructed by a Juniper Networks support representative. For information about the contents of the **jinstall** package and details of the installation process, see the [Installation and Upgrade Guide](#).

NOTE: Before upgrading, back up the file system and the currently active Junos OS configuration so that you can recover to a known, stable environment in case the upgrade is unsuccessful. Issue the following command:

```
user@host> request system snapshot
```

The installation process rebuilds the file system and completely reinstalls Junos OS. Configuration information from the previous software installation is retained, but the contents of log files might be erased. Stored files on the routing platform, such as configuration templates and shell scripts (the only exceptions are the **juniper.conf** and **ssh** files), might be removed. To preserve the stored files, copy them to another system before upgrading or downgrading the routing platform. See the [Installation and Upgrade Guide](#).

The download and installation process for Junos OS Release 20.4R3 is different from that for earlier Junos OS releases.

1. Using a Web browser, navigate to the Download Software URL on the Juniper Networks webpage:
<https://www.juniper.net/support/downloads/>
2. Log in to the Juniper Networks authentication system by using the username (generally your e-mail address) and password supplied by Juniper Networks representatives.
3. Select **By Technology > Junos Platform > Junos fusion** to find the software that you want to download.
4. Select the release number (the number of the software version that you want to download) from the Version drop-down list to the right of the page.
5. Select the **Software** tab.
6. Select the software package for the release.
7. Review and accept the End User License Agreement.

8. Download the software to a local host.
9. Copy the software to the routing platform or to your internal software distribution site.
10. Install the new **jinstall** package on the aggregation device.

NOTE: We recommend that you upgrade all software packages out-of-band using the console, because in-band connections are lost during the upgrade process.

Customers in the United States and Canada, use the following commands.

- For 64-bit software:

NOTE: We recommend that you use 64-bit Junos OS software when implementing Junos fusion for provider edge.

```
user@host> request system software add validate reboot  
source/jinstall64-20.4R3.SPIN-domestic-signed.tgz
```

- For 32-bit software:

```
user@host> request system software add validate reboot  
source/jinstall-20.4R3.SPIN-domestic-signed.tgz
```

All other customers, use the following commands.

- For 64-bit software:

NOTE: We recommend that you use 64-bit Junos OS software when implementing Junos fusion for provider edge.

```
user@host> request system software add validate reboot  
source/jinstall64-20.4R3.SPIN-export-signed.tgz
```

- For 32-bit software:

```
user@host> request system software add validate reboot  
source/jinstall-20.4R3.SPIN-export-signed.tgz
```

Replace **source** with one of the following values:

- **/pathname**—For a software package that is installed from a local directory on the router.
- For software packages that are downloaded and installed from a remote location:
 - **ftp://hostname/pathname**
 - **http://hostname/pathname**
 - **scp://hostname/pathname** (available only for the Canada and U.S. version)

The **validate** option validates the software package against the current configuration as a prerequisite for adding the software package to ensure that the router reboots successfully. This is the default behavior when the software package being added is for a different release.

Adding the **reboot** command reboots the router after the upgrade is validated and installed. When the reboot is complete, the router displays the login prompt. The loading process might take 5 to 10 minutes.

Rebooting occurs only if the upgrade is successful.

NOTE: After you install a Junos OS Release 20.4R3 **jinstall** package, you cannot return to the previously installed software by issuing the **request system software rollback** command. Instead, you must issue the **request system software add validate** command and specify the **jinstall** package that corresponds to the previously installed software.

Upgrading an Aggregation Device with Redundant Routing Engines

If the aggregation device has two Routing Engines, perform a Junos OS installation on each Routing Engine separately as follows to minimize disrupting network operations:

1. Disable graceful Routing Engine switchover (GRES) on the master Routing Engine and save the configuration change to both Routing Engines.
2. Install the new Junos OS release on the backup Routing Engine while keeping the currently running software version on the master Routing Engine.
3. After making sure that the new software version is running correctly on the backup Routing Engine, switch over to the backup Routing Engine to activate the new software.
4. Install the new software on the original master Routing Engine that is now active as the backup Routing Engine.

For the detailed procedure, see the [Installation and Upgrade Guide](#).

Preparing the Switch for Satellite Device Conversion

Satellite devices in a Junos fusion topology use a satellite software package that is different from the standard Junos OS software package. Before you can install the satellite software package on a satellite device, you first need to upgrade the target satellite device to an interim Junos OS software version that can be converted to satellite software. For satellite device hardware and software requirements, see [Understanding Junos fusion Software and Hardware Requirements](#)

NOTE: The following conditions must be met before a standalone switch that is running Junos OS Release 14.1X53-D43 can be converted to a satellite device when the action is initiated from the aggregation device:

- The switch can be converted to only SNOS 3.1 and later.
- Either the switch must be set to factory-default configuration by using the **request system zeroize** command, or the following command must be included in the configuration: **set chassis auto-satellite-conversion**.

Customers with EX4300 switches, use the following command:

```
user@host> request system software add validate reboot
source/jinstall-ex-4300-14.1X53-D43.3-domestic-signed.tgz
```

Customers with QFX5100 switches, use the following command:

```
user@host> request system software add reboot
source/jinstall-qfx-5-14.1X53-D43.3-domestic-signed.tgz
```

When the interim installation has completed and the switch is running a version of Junos and OS on one line that is compatible with satellite device conversion, perform the following steps:

1. Log in to the device by using the console port.
2. Clear the device:

```
[edit]
user@satellite-device# request system zeroize
```

NOTE: The device reboots to complete the procedure for resetting the device.

If you are not logged in to the device by using the console port connection, your connection to the device is lost after you enter the **request system zeroize** command.

If you lose your connection to the device, log in using the console port.

3. (EX4300 switches only) After the reboot is complete, convert the built-in 40-Gbps QSFP+ interfaces from Virtual Chassis ports (VCPs) into network ports:

```
user@satellite-device> request virtual-chassis vc-port delete pic-slot 1 port port-number
```

For example, to convert all four built-in 40-Gbps QSFP+ interfaces on an EX4300-24P switch into network ports:

```
user@satellite-device> request virtual-chassis vc-port delete pic-slot 1 port 0
user@satellite-device> request virtual-chassis vc-port delete pic-slot 1 port 1
user@satellite-device> request virtual-chassis vc-port delete pic-slot 1 port 2
user@satellite-device> request virtual-chassis vc-port delete pic-slot 1 port 3
```

This step is required for the 40-Gbps QSFP+ interfaces that will be used as uplink interfaces in a Junos fusion topology. Built-in 40-Gbps QSFP+ interfaces on EX4300 switches are configured into VCPs by default, and the default settings are restored after the device is reset.

After this initial preparation, you can use one of three methods to convert your switches into satellite devices—autoconversion, manual conversion, and preconfiguration. See [Configuring Junos fusion for provider edge](#) for detailed configuration steps for each method.

Converting a Satellite Device to a Standalone Device

If you need to convert a satellite device to a standalone device, you must install a new Junos OS software package on the satellite device and remove the satellite device from the Junos fusion topology.

NOTE: If the satellite device is a QFX5100 switch, you need to install a PXE version of Junos OS. The PXE version of Junos OS is software that includes *pxe* in the Junos OS package name when it is downloaded from the Software Center—for example, the PXE image for Junos OS Release 14.1X53-D43 is named `install-media-pxe-qfx-5-14.1X53-D43.3-signed.tgz`. If the satellite device is an EX4300 switch, you install a standard `jinstall-ex-4300` version of Junos OS.

The following steps explain how to download software, remove the satellite device from Junos fusion, and install the Junos OS software image on the satellite device so that the device can operate as a standalone device.

1. Using a Web browser, navigate to the Junos OS software download URL on the Juniper Networks webpage:
<https://www.juniper.net/support/downloads>
2. Log in to the Juniper Networks authentication system by using the username (generally your e-mail address) and password supplied by Juniper Networks representatives.
3. Select **By Technology > Junos Platform > Junos fusion** from the drop-down list and select the switch platform series and model for your satellite device.
4. Select the Junos OS Release 14.1X53-D30 software image for your platform.
5. Review and accept the End User License Agreement.
6. Download the software to a local host.
7. Copy the software to the routing platform or to your internal software distribution site.
8. Remove the satellite device from the automatic satellite conversion configuration.

If automatic satellite conversion is enabled for the satellite device's member number, remove the member number from the automatic satellite conversion configuration. The satellite device's member number is the same as the FPC slot ID.

[edit]

```
user@aggregation-device# delete chassis satellite-management auto-satellite-conversion
satellite member-number
```

For example, to remove member number 101 from Junos fusion:

[edit]

```
user@aggregation-device# delete chassis satellite-management auto-satellite-conversion
satellite 101
```

You can check the automatic satellite conversion configuration by entering the **show** command at the **[edit chassis satellite-management auto-satellite-conversion]** hierarchy level.

9. Commit the configuration.

To commit the configuration to both Routing Engines:

[edit]

```
user@aggregation-device# commit synchronize
```

Otherwise, commit the configuration to a single Routing Engine:

```
[edit]
user@aggregation-device# commit
```

10. Install the Junos OS software on the satellite device to convert the device to a standalone device.

```
[edit]
user@aggregation-device> request chassis satellite install URL-to-software-package fpc-slot member-number
```

For example, to install a PXE software package stored in the `/var/tmp` directory on the aggregation device onto a QFX5100 switch acting as the satellite device using FPC slot 101:

```
[edit]
user@aggregation-device> request chassis satellite install
/var/tmp/install-media-pxe-qfx-5-14.1X53-D43.3-signed.tgz fpc-slot 101
```

For example, to install a software package stored in the `var/tmp` directory on the aggregation device onto an EX4300 switch acting as the satellite device using FPC slot 101:

```
[edit]
user@aggregation-device> request chassis satellite install
/var/tmp/jinstall-ex-4300-14.1X53-D30.3-domestic-signed.tgz fpc-slot 101
```

The satellite device stops participating in the Junos fusion topology after the software installation starts. The software upgrade starts after this command is entered.

11. Wait for the reboot that accompanies the software installation to complete.
12. When you are prompted to log back into your device, uncable the device from the Junos fusion topology. See *Removing a Transceiver from a QFX Series Device* or *Remove a Transceiver*, as needed. Your device has been removed from Junos fusion.

NOTE: The device uses a factory-default configuration after the Junos OS installation is complete.

Upgrading an Aggregation Device

When you upgrade an aggregation device to Junos OS Release 20.4R3, you must also upgrade your satellite device to Satellite Device Software version 3.1R1.

Upgrade and Downgrade Support Policy for Junos OS Releases

We have two types of releases, standard EOL and EEOL:

- Standard End of Life (EOL) releases have engineering support for twenty four months after the first general availability date and customer support for an additional six more months.
- Extended End of Life (EEOL) releases have engineering support for thirty six months after the first general availability date and customer support for an additional six more months.

For both standard EOL and EEOL releases, you can upgrade to the next three subsequent releases or downgrade to the previous three releases. For example, you can upgrade from 19.2 to the next three releases – 19.3, 19.4 and 20.1 or downgrade to the previous three releases – 19.1, 18.4 and 18.3.

For EEOL releases only, you have an additional option - you can upgrade directly from one EEOL release to the next two subsequent EEOL releases, even if the target release is beyond the next three releases. Likewise, you can downgrade directly from one EEOL release to the previous two EEOL releases, even if the target release is beyond the previous three releases. For example, 19.2 is an EEOL release. Hence, you can upgrade from 19.2 to the next two EEOL releases – 19.3 and 19.4 or downgrade to the previous two EEOL releases – 19.1 and 18.4.

Release Type	End of Engineering (EOE)	End of Support (EOS)	Upgrade and Downgrade to subsequent 3 releases	Upgrade and Downgrade to subsequent 2 EEOL releases
Standard End of Life (EOL)	24 months	End of Engineering + 6 months	Yes	No
Extended End of Life (EEOL)	36 months	End of Engineering + 6 months	Yes	Yes

For more information about standard EOL and EEOL releases, see <https://www.juniper.net/support/eol/junos.html>.

For information about software installation and upgrade, see the [Installation and Upgrade Guide](#).

Downgrading from Junos OS Release 20.4

To downgrade from Release 20.4 to another supported release, follow the procedure for upgrading, but replace the 20.4 **jinstall** package with one that corresponds to the appropriate release.

NOTE: You cannot downgrade more than three releases.

For more information, see the [Installation and Upgrade Guide](#).

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Junos OS Release Notes for MX Series

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These release notes accompany Junos OS Release 20.4R3 for the MX Series 5G Universal Routing Platforms. They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

You can also find these release notes on the Juniper Networks Junos OS Documentation webpage, located at https://www.juniper.net/documentation/product/en_US/junos-os.

What's New

IN THIS SECTION

- [What's New in Release 20.4R3 | 102](#)
- [What's New in Release 20.4R2 | 102](#)
- [What's New in Release 20.4R1 | 102](#)

This section describes the new features and enhancements to existing features in Junos OS Release 20.4R3 for the MX Series routers.

What's New in Release 20.4R3

There are no new features or enhancements to existing features for MX Series routers in Junos OS Release 20.4R3.

What's New in Release 20.4R2

There are no new features or enhancements to existing features for MX Series routers in Junos OS Release 20.4R2.

What's New in Release 20.4R1

Hardware

- We've added the following features to the MX Series routers in Junos OS Release 20.4R1.

Table 2: Features Supported by MPC10E and MPC11E Line Cards on MX Series Routers

Feature	Description
EVPN	• Support for configuring an Ethernet VPN Ethernet Tree (E-Tree) service on MX240, MX480, and MX960 routers using MPC10E-15C-MRATE line cards. [See EVPN-ETREE Overview.] • Support for configuring an EVPN point-to multipoint (P2MP) label switch path (LSP) as a provider tunnel on a bud router. The bud router functions both as an egress router and a transit router. [See Configuring Bud Node Support.] • Support for configuring and signalling a P2MP LSP for the EVPN Inclusive Provider Tunnel for BUM traffic. [See Understanding P2MPs LSP for the EVPN Inclusive Provider Tunnel.]
Interfaces and chassis	• Support for configuring VLAN rewrite operations on CCC interfaces. [See Stacking and Rewriting Gigabit Ethernet VLAN Tags Overview and Stacking and Rewriting Gigabit Ethernet VLAN Tags.] • Support for 100GE AOC optics on MPC10E-15C-MRATE and MPC10E-10C-MRATE (with SCBE3-MX) in the MX240, MX480, and MX960 routers. [See Hardware Compatibility Tool.] • Support for 4X100G FR transceivers and the channelization option on the 400G-DR4 transceiver on MPC10E-15C-MRATE and MPC10E-10C-MRATE (with SCBE3) in the MX240, MX480, and MX960 routers. [See Hardware Compatibility Tool.] • Support for configuring dynamic learning of the source and destination MAC addresses on aggregated Ethernet interfaces on the MPC10E-15C-MRATE, MPC10E-10C-MRATE, and MX2K-MPC11E line cards. [See MAC Address Accounting for Dynamically Learned Addresses.] • Support for monitoring link degradation of the 25GbE interfaces and 400GbE interfaces on the MPC10E (MPC10E-15C-MRATE and MPC10E-10C-MRATE) line cards. [See Link Degrade Monitoring Overview.] • Support for Layer 2 address learning process (ALD). [See Understanding Layer 2 Learning and Forwarding.] • Support for a bandwidth of 500 Gbps per Packet Forwarding Engine with four fabric planes on MPC10E-10C-MRATE and MPC10E-15C-MRATE (with the Packet Forwarding Engine 2 powered off) line cards. [See MPC10E-10C-MRATE and MPC10E-15C-MRATE.]

Table 2: Features Supported by MPC10E and MPC11E Line Cards on MX Series Routers (*continued*)

Feature	Description
General routing	• Support for configuring the TCP maximum segment size (MSS). [See Configure TCP Options.] • Support for configuring the GRE key to identify the traffic flows in a GRE tunnel on the MPC10E-10C-MRATE, MPC10E-15C-MRATE, and MX2K-MPC11E line cards. [See dynamic-tunnel-gre-key.]
Layer 2 features	• Support for packet mirroring with Layer 2 headers for Layer 3 forwarded traffic. [See Firewall Filter Nonterminating Actions.] • Support for Layer2 Ethernet services over GRE tunnel interfaces. [See Configuring Layer 2 Ethernet Services over GRE Tunnel Interfaces.] • Support for Spanning-Tree Protocol (STP), Rapid Spanning-Tree Protocol (RSTP), Multiple Spanning-Tree Protocol (MSTP), and VLAN Spanning-Tree Protocol (VSTP). [See Configuring STP Protocol.] • Support for the base bridging feature commands. NOTE: You can configure propagate option under the mac-flush command. [See clear bridge mac-table, global-mac-move, global-no-mac-learning, mac-flush, global-no-control-mac-aging, and global-no-hw-mac-learning .]
Multicast	• Support for redundant virtual tunnels (RVTs) and fast re-route (FRR) for both active/backup and active/active redundancy models (MX240, MX480, MX960, MX2010, and MX2020). RVT interfaces are used in Multicast Layer 3 VPNs (MVPN) to facilitate virtual routing and forwarding (VRF) table lookup based on MPLS labels and to provide resiliency. [See Resiliency in Multicast L3 VPNs with Redundant Virtual Tunnels.] • Support for verifying the global table multicast (GTM) with IPv6 and Type-7 on MPC10 and MPC11 line cards. [See Multicast Overview.]
Network management and monitoring	• Support for configuring ITU-T Y.1731 standard-compliant Ethernet synthetic loss measurement (ETH-SLM) and Ethernet delay measurement (ETH- DM) capabilities on MPC10E-10C-MRATE, MPC10E-15C-MRATE, and MX2K-MPC11E line cards. [See ITU-T Y.1731 Ethernet Service OAM Overview.]

Table 2: Features Supported by MPC10E and MPC11E Line Cards on MX Series Routers (*continued*)

Feature	Description
Services Applications	- Support for inline monitoring services to provide the flexibility to monitor different streams of traffic at different sampling rates on the same interface. [See Inline Monitoring Services Configuration.] - Support for Aggregated Multiservices Interfaces (AMS) on the MPC10E-10C-MRATE, MPC10E-15C-MRATE, and MX2K-MPC11E line cards to provide load balancing (LB) and high availability (HA) features for stateful firewall and NAT services. You can configure AMS with next-hop style service-sets and with MS-MPC or MS-MIC only. [See Understanding Aggregated Multiservices Interfaces.]

- **Support for QSFP-100G-FR, QSFP-100G-DR, and QSFP-100G-LR transceivers (MX2010 and MX2020 with MX2K-MPC11E)**—Starting in Junos OS Release 20.4R1, the MX2K-MPC11E MPCs in the MX2010 and MX2020 routers support the QSFP-100G-FR, QSFP-100G-DR, and QSFP-100G-LR transceivers.

[See the [Hardware Compatibility Tool \(HCT\)](#) for details.]

EVPN

- **MAC-VRF with EVPN-VXLAN (MX Series and vMX routers; QFX5100, QFX5110, QFX5120, QFX5200, QFX5210, QFX10002-60C, QFX10008, and QFX10016 switches)**—Data center service providers must support multiple customers with their own routing and bridging policies in the same physical network. To accommodate this requirement, you can now configure multiple customer-specific EVPN instances (EVIs) of type **mac-vrf**, each of which can support a different EVPN service type. This configuration results in customer-specific virtual routing and forwarding (VRF) tables with MAC addresses on each Juniper Networks device that serves as a virtual tunnel endpoint (VTEP) in the EVPN-VXLAN network.

NOTE: We support MAC-VRF routing instances for EVPN unicast routes only.

To support this feature, we introduce a uniform routing instance configuration, which complies with RFC 7432, *BGP MPLS-Based Ethernet VPN*. The uniform configuration eliminates hardware restrictions that limit the number of EVIs and combinations of EVIs with their respective policies that can simultaneously exist. The common configuration includes the following new CLI elements:

- The **mac-vrf** keyword at the **[edit routing-instances name instance-type]** hierarchy level.
- The **service-type** configuration statement at the **[edit routing-instances name]** hierarchy level. We support VLAN-based, VLAN-aware, and VLAN-bundle service types.
- (QFX10000 line of switches only) The **forwarding-instance** configuration statement at the **[edit routing-instances name]** hierarchy level. With this optional configuration statement, you can map multiple routing instances to a single forwarding instance. If you don't include this configuration statement, the default forwarding instance is used.

We continue to support the existing method of routing instance configuration along with the new uniform routing instance configuration.

[See [EVPN User Guide](#).]

- **MC-LAG emulation in an EVPN deployment (EX-Series, MX-Series, and vMX)**—Starting in Junos OS Release 20.4R1, you can emulate the function of an MC-LAG in active-standby mode in an EVPN configuration without having to configure an ICCP or ICL interface. In a standard EVPN configuration, logical interfaces configured on an aggregated Ethernet interface can have different designated forwarder election roles. To emulate an MC-LAG configuration, the designated forwarder (DF) takes on the role of the aggregated Ethernet interface. The provider edge (PE) that is the non-DF will send LACP out-of-sync packets to the CE. This will cause LACP to go down on the CE device, and the CE device will not use the links connected to the non-DF for sending traffic. If the connection between a CE and a DF PE fails, the PE is re-elected as a DF. If the connection between a CE and a non-DF PE fails, the current DF PE is not changed.

To achieve this functionality, configure the **lACP-oos-on-ndf** statement at the **[edit interfaces interface name esi df-election-granularity per-esi]** hierarchy.

- **Support for EVPN E-Tree service (MX240, MX480, and MX960)**—Starting in Junos OS 20.4R1, on MX240, MX480, and MX960 routers using MPC10E-15C-MRATE line cards you can configure an Ethernet VPN Ethernet-Tree (E-Tree) service.

[See [EVPN-ETREE Overview](#).]

High Availability (HA) and Resiliency

- **Support for pause and resume options with unified ISSU (MX Series)**—Starting in Junos OS Release 20.4R1, MX Series routers support pausing and resuming unified ISSU operations. Use the **pause** and **resume** options with the **request system software in-service-upgrade** command to control when to pause and resume unified ISSU.

[See [request system software in-service-upgrade](#)]

- **NSR support for IS-IS with SR (ACX Series, MX Series)**—Starting in Junos OS Release 20.4R1, MX Series routers support NSR for IS-IS with segment routing (SR). To use NSR, you must first enable GRES on your device.

[See [Nonstop Active Routing Concepts](#)]

Interfaces and Chassis

- **464XLAT support for mobility on MS-MPC (MX Series)**—Starting in Junos OS Release 20.4R1, you can specify the IPv6 prefix length for the CLAT source address using the new command **clat-ipv6-prefix-length**. When you configure this command, NAT rules apply 464XLAT based on destination-address of the traffic, and source-address and source-prefix are no longer required. The **clat-ipv6-prefix-length** command is available at the **[edit services nat rule rule-name term term-name then translated]** hierarchy level.

[See [translated](#) and [clat-ipv6-prefix-length](#).]

Juniper Extension Toolkit (JET)

- **Juniper Extension Toolkit (JET) support for 64-bit applications (MX5, MX10, MX40, MX80, MX104, MX150, MX204, MX240, MX480, MX960, MX2008, MX2010, MX2020, MX10003, MX10008, MX ELM, JunosV Firefly, cSRX, SRX100, SRX110, SRX210, SRX220, SRX240, SRX300, SRX320, SRX340, SRX345, SRX550, SRX550HM, SRX650, SRX720E, SRX750E, SRX1400, SRX1500, SRX3400, SRX3600, SRX4100, SRX4200, SRX4400, SRX4600, SRX4800, SRX5400, SRX5600, SRX5800, SRX7X0E, SRX-ES7, SRX-ES8, VMX, and VSRX)**—Starting in Junos OS Release 20.4R1, JET supports 64-bit applications. Use the following commands to compile 64-bit applications for use with the AMD64 or ARM64 64-bit processor architecture.

- **mk-amd64**: Compiles the application for use with AMD64 and Junos OS with FreeBSD.
- **mk-amd64,bsd**: Compiles the application for use with AMD64 and Junos OS with upgraded FreeBSD.
- **mk-arm64,bsd**: Compiles the application for use with ARM64 and Junos OS with upgraded FreeBSD.

[See [Develop On-Device JET Applications](#).]

- **Configure inner source MAC address for flexible VXLAN tunnels (MX Series and vMX with MPC1-MPC9E or LC2101)**—Starting in Junos OS Release 20.4R1, you can use the Juniper Extension Toolkit (JET) RIB Service API to configure the source MAC address used in IPv4 and IPv6 flexible VXLAN tunnel encapsulation profiles. The source MAC addresses is stored in the inner Ethernet header of VXLAN encapsulation. If you don't specify a source MAC address, the default source MAC address 00:00:5e:00:52:01 is used to encapsulate IPv4 and IPv6 flexible VXLAN tunnels.

Use the **show route detail**, **show route extensive**, and **show flexible-tunnels profiles** CLI commands or the **get-route-information** and **get-flexible-tunnels-profiles** RPC/NETCONF commands to view the source MAC address that is specified in the flexible tunnel profile.

[See [Understanding Programmable Flexible VXLAN Tunnels](#) and [JET APIs on Juniper EngNet](#).]

Junos OS, XML, API, and Scripting

- **Support for Certificate Authority Chain Profile (EX2300, EX3400, EX4300, MX240, MX480, MX960, PTX-5000, VMX, vSRX and QFX5200)**—Starting in Junos OS Release 20.4R1, you can configure intermediate Certificate Authority (CA) chain profile certificate and perform https REST API request using mutual and server authentications.

To configure intermediate ca-chain certificate, configure **ca-chain ca-chain** statement at the **[edit system services rest https]** hierarchy level.

- **Start time option for interval-based internal events that trigger event policies (EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—Starting in Junos OS Release 20.4R1, when you create an interval-based internal event for triggering event policies, you can specify the start date and time for the initial event. To specify a start time, configure the **start-time** option along with the **time-interval** option at the **[edit event-options generate-event]** hierarchy level.

[See [Generating Internal Events to Trigger Event Policies](#).]

Junos Telemetry Interface

- **JTI support for inline Junos Traffic Vision sensors with gRPC services (MX Series and PTX Series)**—Junos OS Release 20.4R1 supports inline Jflow sensors for FPC3 and MPC 1 through 9. This feature enables you to monitor inline Junos Traffic Vision (previously known as Jflow) service statistics on a router and to export statistics to an outside collector at configurable intervals using remote procedure call (gRPC) services.

Use the resource path **/junos/system/linecard/services/inline-jflow/** in a subscription to export statistics.

You can view statistics in the collector output under **/components/**. The collector component ID in the statistics output will include the FPC slot number for which inline Junos Traffic Vision statistics are exported. For example, inline Jflow statistics for FPC 0 will be under **component id 0**, and inline Jflow statistics for FPC 1 will be under **component id 1**.

Inline Junos Traffic Vision statistics are slightly different, depending on the routing platform.

[See [Guidelines for gRPC and gNMI Sensors \(Junos Telemetry Interface\)](#).]

- **JTI support for persistent active gRPC sessions between collector and server during an SSL certificate update (ACX Series, MX Series, and PTX Series)**—Junos OS Release 20.4R1 supports persistent active remote procedure call (gRPC) sessions between the collector (client) and server during an SSL certificate update.

For secure channel authentication, the TLS protocol is used to maintain a secure channel between the collector and the server. TLS uses the server certificate and the client certificate to authenticate each other and send encrypted messages over the network. When an SSL certificate is updated, existing gRPC sessions are abruptly terminated, forcing the collector to initiate a new gRPC connection and subscribe to sensors again.

To avoid this problem, you can enable persistent active gRPC sessions by configuring **hot-reloading** at the **[edit system services extension-service request-response grpc ssl]** hierarchy level. After you enable this feature, gRPC sessions will remain active even when authentication certificates are updated.

After the certificate is updated, any new gRPC session will use the updated certificate.

[See [gRPC Services for Junos Telemetry Interface](#) and [ssl](#).]

- **BGP neighbor telemetry with sharding (MX Series, PTX Series, and QFX Series)**—Starting in Junos OS Release 20.4R1, BGP neighbor telemetry with sharding (multi-threading) is supported.

[See [Guidelines for gRPC and gNMI Sensors \(Junos Telemetry Interface\)](#).]

- **LACP sensors for actor partner states on JTI (MX Series and PTX Series)**—Starting in Junos OS Release 20.4R1, you can use Junos telemetry interface (JTI) and gRPC Network Management Interface (gNMI) services to export LACP actor partner states (also known as LACP port states). When a subscription is configured, ON_CHANGE or periodic streaming statistics are sent from devices to an outside collector.

You can subscribe to `/lacpd/` to collect all statistics or include the following resource paths individually in a subscription:

- `/lacpd/ae/member/partner_collecting`
- `/lacpd/ae/member/partner_synchronization`
- `/lacpd/ae/member/partner_timeout`
- `/lacpd/ae/member/partner_aggregatable`
- `/lacpd/ae/member/partner_distributing`
- `/junos/system/linecard/interface/traffic/`

[See [Guidelines for gRPC and gNMI Sensors \(Junos Telemetry Interface\)](#).]

- **Juniper Resiliency Interface for exception reporting and null route detection (ACX Series, PTX Series and MX Series)**—Starting in Junos OS Release 20.4R1, you can use Juniper Resiliency Interface to detect and reduce Mean Time to Repair (MTTR) first-order network issues. Juniper Resiliency Interface uses a push model for data reporting from the entities in the system which encounter packet drops. This automates the workflow for detecting, reporting, and mitigating adverse exceptions.

To collect kernel routing table and routing protocol process exceptions, configure the **set system resiliency exceptions** statement at the `[edit]` hierarchy level to specify exception reporting based on kernel exceptions, and routing exceptions.

You can display exceptions from a remote collector by means of remote procedure call (gRPC) services or gRPC network management interface (gNMI) services. Display on-box exceptions by accessing the `/var/log` file or the database at `/var/db/ResiliencyExceptions.db`. No Junos operational mode commands display these exceptions.

MPLS

- **Re-engineering of SR-TE (MX Series, PTX Series)**—Starting with Junos OS Release 20.4R1, you can incorporate the following features to enhance the debugging capability of segment routing traffic-engineering (SR-TE):
 - rib-group import functionality.
 - Display of SR-TE routes installed from various tunnel sources using the `show spring-traffic-engineering` command.

- Template map for BGP SR-TE tunnels.
- Compute profile in template with distributed Constrained Shortest Path First (CSPF) for dynamic SR-TE tunnels.
- 6PE (IPv6 over IPv4 SR-TE tunnel)
- no-chained-composite-next-hop option

[See [source-packet-routing](#) and [show spring-traffic-engineering](#).]

- **Support for optimizing auto-bandwidth adjustments for MPLS LSPs (MX Series and PTX Series)**—Starting in Junos OS Release 20.4R1, you can configure faster auto-bandwidth adjustment for MPLS LSPs under overflow or underflow conditions. This feature decreases the minimum allowed **adjust-threshold-overflow-limit** and **adjust-interval** to 150 seconds when **adjust-threshold-overflow-limit** and **adjust-threshold-underflow-limit** cross the configured threshold values. In releases earlier than Junos OS Evolved Release 20.4R1, the **adjust-interval** is 300 seconds under overflow or underflow conditions.

You can configure faster in-place LSP bandwidth update that avoids signaling of a new LSP instance as part of make-before-break. To configure faster in-place LSP bandwidth update, include the **in-place-lsp-bandwidth-update** configuration statement at the **[edit protocols mpls label-switched-path *lsp-name*]** hierarchy level.

You can also configure RSVP interfaces to support subscription percentage per priority. To configure subscription percentage per priority, include the **subscription priority *priority* percent *value*** configuration statement at the **[edit protocols rsvp interface *interface-name*]** hierarchy level.

[See [Configuring Optimized Auto-bandwidth Adjustments for MPLS LSPs](#).]

- **Support for express segments to establish end-to-end segment routing path (MX Series and PTX Series)**—Starting in Junos OS Release 20.4R1, express segments can be used to establish end-to-end TE paths between interconnected TE networks. Express segments (also known as virtual TE links) are generated dynamically through policies matching the underlay LSPs. Express segments and the corresponding abstracted topology (required by RFC7926) is generated with policies.

To apply a policy, include the **policy *policy-name*** statement at the **[edit protocols express-segment traffic-engineering]** hierarchy level.

To configure express segment, include the **express-segment** statement under the **[edit protocols]** hierarchy level.

[See [How to Establish End-to-End Segment Routing Paths Using Express Segments](#).]

Network Management and Monitoring

- **Configuration support to prevent drifting of accounting records (MX Series routers, vMX)** —You can configure accounting records to record data in accounting files and archive the accounting files to analyze the information collected. Drifting of the accounting records happens if the time at which the records are written to the accounting file spills beyond the transfer window of the file. Starting in Junos OS Release 20.4R1, to prevent drifting of accounting records:

- Use the **start-time** statement with the accounting profiles (class-usage-profile, filter-profile, flat-file-profile, interface-profile, mib-profile, and routing-engine-profile) to have a predictable start time of the profiles.
- Use the **timestamp** statement with the **request accounting add records** command to record the timestamp externally instead of epoch timestamp when the command is executed.

[See [routing-engine-profile](#), [class-usage-profile](#), [interface-profile](#), [filter-profile](#), [mib-profile](#), [flat-file-profile](#).]

- **Configuration retrieval using the configuration revision identifier (EX3400, EX4300, MX204, MX240, MX480, MX960, MX2020, PTX3000, PTX10008, QFX5100, QFX10002-60C, SRX5800, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, you can use the configuration revision identifier feature to view the configuration for a specific revision. This configuration database revision can be viewed with the CLI command **show system configuration revision**.

[See [show system configuration revision](#).]

- **Junos XML protocol operations support loading and comparing configurations using the configuration revision identifier (EX3400, EX4300, MX204, MX240, MX480, MX960, MX2020, PTX3000, PTX10008, QFX5100, QFX10002-60C, SRX5800, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, the Junos XML management protocol operations support loading and comparing configurations by referencing the configuration revision identifier of a committed configuration. You can execute the **<load-configuration>** operation with the **configuration-revision** attribute to load the configuration with the given revision identifier into the candidate configuration. Additionally, you can compare the candidate or active configuration to a previously committed configuration by referencing the configuration revision identifier for the comparison configuration. The **<get-configuration>** operation supports the **compare="configuration-revision"** and **configuration-revision** attributes to perform the comparison.

[See [<get-configuration>](#) and [<load-configuration>](#).]

- **Support for an extension to the rpm-tracked static routes (MX Series, PTX Series, and vMX)**—Starting in Junos OS Release 20.4R1, you can configure route preference and tag values for each destination-prefix. This feature supports both IPv4 and IPv6 rpm-tracked static routes.

[See [show route rpm-tracking](#).]

Limitations

Qualified next hop is not supported with rpm-tracked static routes. Hence, the setting of preference, metric, and tags applies only to the **rpm-tracking** static route and not to the related next hops.

Routing Policy and Firewall Filters

- **Support for route's next-hop weight in policy match condition (MX Series, PTX Series, and QFX Series)**—Starting in Junos OS Release 20.4R1, a route with multiple next-hop paths can use the weight associated with a path to identify primary and backup paths. The path with the lowest weight is used as the primary path, and any paths with higher weights are treated as backup paths. You can use the next-hop weight as a match condition in export policies to redistribute IGP and BGP routes based on whether the primary or backup paths are active.

Configure this match condition using the `[edit policy-options policy-statement policy-name term term-name from]` statement.

[See [policy-statement](#) and [show policy](#).]

Routing Protocols

- **Support for relaxing BGP router ID format from /32 to a nonzero ID per RFC 6286 (EX9208, EX9214, MX204, VRR, QFX5120-32C)**—Starting in Junos OS Release 20.4R1, you can establish a BGP connection using a BGP identifier that is a 4-octet, unsigned, nonzero integer and it needs to be unique only within the autonomous system (AS) per RFC 6286. In earlier releases, the BGP ID of a BGP speaker was required to be a valid IPv4 host address assigned to the BGP speaker.

To enable this feature, use the `bgp-identifier identifier group bgp group name bgp-identifier identifier neighbor peer address bgp-identifier identifier` configuration statement at the `[edit protocols bgp]` hierarchy level.

[See [router-id](#)]

- **Support for multiple single-hop EBGp sessions on different links using the same IPv6 link-local address (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, you are no longer required to have unique peer addresses for Juniper devices for every EBGp session. You can now enable single-hop EBGp sessions on different links over multiple directly-connected peers that use the same IPv6 link-local address.

In earlier Junos OS Releases, BGP peers could be configured with link-local addresses, but multiple BGP peers could not be configured to use the same link-local address on different interfaces.

[See [Configure Multiple Single-Hop EBGp Sessions on Different Links Using the Same Link-Local Address \(IPv6\)](#).]

- **Support for IPv6 L3VPN over IPv6 SR-TE and IPv6 Underlay (MX Series)**—Starting in Junos OS Release 20.4R1, You can configure an IPv6 Layer3 VPN connection with an IPv6 local address and an IPv6 neighbor address. We have extended BGP support for IPv6 Layer 3 VPN over BGP IPv6 SR-TE in IS-IS networks. You can connect an IPv6 provider edge device with a colored or non-colored IPv6 penultimate nexthop (PNH) address mapped to IPv6 SR-TE tunnels.

To configure an IPv6 address for Layer 3 VPN connection, include the `family inet6-vpn` configuration statement at the `[edit protocols bgp group name]` hierarchy level.

[See [Understanding Static Segment Routing LSP in MPLS Networks](#).]

- **Support for BGP Labeled Unicast prefix SID (MX Series and PTX Series)**—Starting in Junos OS 20.4R1, BGP labeled unicast can carry segment routing global block label range and index information through the prefix segment attribute. With this feature we support segment routing using the BGP labeled unicast prefix segments and the MPLS data plane in medium to large scaled data centers. The controller directs the server to assign a stack- of labels to an incoming packet based on the available network state information. The assigned label stack avoids congested paths and steers the packet through a best available path.

To configure and advertise the SRGB label range specifically for BGP include the **source-packet-routing srgb start-label *start-label* index-range *index-range* advertise-srgb** configuration statements at the **[edit protocols bgp]** hierarchy level.

To advertise prefix SIDs to external BGP peers, include the **advertise-prefix-sid** configuration statement at the **[edit protocols bgp]** hierarchy level. You can configure this statement globally or for specific BGP groups or BGP neighbors.

[See [srgb](#).]

- **Support for SRv6 network programming and Layer 3 Services over SRv6 in BGP (MX Series)**—Starting in Junos OS Release 20.4R1, you can configure BGP based Layer 3 service over SRv6 core. You can enable Layer 3 overlay services with BGP as control plane and SRv6 as dataplane. SRv6 network programming provides flexibility to leverage segment routing without deploying MPLS. Such networks depend only on the IPv6 headers and header extensions for transmitting data.

To configure IPv4 and IPv6 transport over SRv6 core, include the **end-dt4-sid *sid*** and the **end-dt6-sid *sid*** statements at the **[edit protocols bgp source-packet-routing srv6 locator *name*]** hierarchy level.

To configure IPv4 VPN and IPv6 VPN service over SRv6 core, include the **end-dt4-sid *sid*** and the **end-dt6-sid *sid*** statements at the **[edit routing-instances *routing-instance name* protocols bgp source-packet-routing srv6 locator *name*]** hierarchy level.

[See [Understanding SRv6 Network Programming and Layer 3 Services over SRv6 in BGP](#).]

- **Support for unicast ARP request on table entry expiration (MX Series)**—Starting in Junos OS Release 20.4R1, you can configure the device to send a unicast ARP request instead of the default broadcast request when an ARP table entry is about to expire. The retry requests are unicast at intervals of 5 seconds. Without this option, the retry requests are broadcast at intervals of 800 milliseconds. This behavior reduces overall ARP broadcast traffic. It also supports the use case where access nodes are configured not to forward broadcast ARP requests toward customer CPEs for security reasons and to instead translate ARP broadcasts to unicast requests. You can verify whether this feature is configured by using the following command: **show configuration system arp | grep unicast-mode-on-expire**.

[See [arp](#).]

- **IPv6 support in TED (MX Series, PTX Series)**—Starting in Junos OS Release 20.4R1, you can configure IS-IS traffic engineering to store IPv6 information in the traffic engineering database (TED) in addition to IPv4 addresses. BGP-LS distributes this information as routes from the TED to the Isdist.0 routing table using the TED import policies. These routes are advertised to BGP-TE peers as network layer reachability information (NLRI) with IPv6 router ID type, length, and value (TLV).

With this enhancement, you can benefit from obtaining the complete network topology in the TED.

[See [Link-State Distribution Using BGP Overview](#).]

Services Applications

- **Support for passive flow monitoring (MX Series)**—Starting in Junos OS Release 20.4R1, you can configure passive flow monitoring on these routers:

- MX240/MX480/MX960/MX2008/MX2010/MX2020 routers with either the MPC7E-MRATE or MPC7E-10G line card
- MX10008 router with the JNP10K-LC2101 line card

[See [Understanding Passive Flow Monitoring on T Series, M Series and MX Series Routers](#) and [Enabling Passive Flow Monitoring on M Series, MX Series or T Series Routers](#).]

Software Defined Networking

- **PCEP support for color (MX480, QFX5200)**—Starting in Junos OS Release 20.4R1, the Path Computation Element Protocol (PCEP) supports color for colored segment routing LSPs. This includes Path Computation Element (PCE)-initiated, Path Computation Client (PCC)-controlled, and PCC-delegated segment routing LSPs. With this PCEP extension, you can configure candidate paths based on color and endpoints, where the active candidate path is the path with the highest segment routing preference, or based on source priority.

[See [Understanding Static Segment Routing LSP in MPLS Networks](#).]

- **Support for ECMP on multiple flexible routes (MX80, MX104, MX204, MX10003, and vMX routers)**—Starting in Junos OS Release 20.4R1, we support load balancing of traffic over multiple flexible routes with 64-way ECMP. A flexible route is a static route with a tunnel encapsulation profile, which has the flexible tunnel interface (FTI) attribute. Flexible routes are installed on Juniper gateway devices using the Juniper Extension Toolkit (JET) APIs. Multiple flexible routes can go over the same logical interface. When a packet is received with the flexible route as the destination address, the packet is processed using the profile associated with a flexible route. Traffic across multiple flexible routes is load-balanced based on the traffic priority.

Use the **show route** and **show route extensive** CLI commands or the **get-route-information** RPC/NETCONF command to view details about a flexible route for a destination address.

[See [Understanding Programmable Flexible VXLAN Tunnels](#).]

- **Static VXLAN at VLAN or bridge domain level (MX5, MX10, MX40, MX80, MX150, MX240, MX480, MX960, MX2008, MX2010, MX2020, MX10003, MX10008, MX10016 routers and QFX5120-32C, QFX5120-48T, and QFX5120-48Y switches)**—In Junos OS Release 20.3R1 and earlier, we supported the configuration of static VXLAN at the global level only. By including the **remote-vtep-list** configuration statement at the **[edit switch-options]** or **[edit routing-instances name]** hierarchy level, you can map all local VLANs or bridge domains to the remote virtual tunnel endpoints (VTEPs) in the list.

Starting in Junos OS Release 20.4R1, you can also configure static VXLAN at the VLAN or bridge domain level using the **static-remote-vtep-list** configuration statement at the **[edit vlans name vxlan]**, **[edit**

bridge-domains *name vxlan*], or **[edit routing-instances *name* bridge-domains *name vxlan*]** hierarchy level.

When specifying remote VTEPs at the VLAN level in the default switching instance, you must also specify the same VTEPs at the global level in the default switching instance. Or when specifying remote VTEPs at the bridge domain level in a routing instance, you must also specify the same VTEPs at the global level in the same routing instance. For example, if you specify a VTEP in the **static-remote-vtep-list** at the **[edit routing-instances *name* bridge-domains *name vxlan*]** hierarchy level, you must also specify the VTEP in the **remote-vtep-list** at the **[edit routing-instances *name*]** hierarchy level.

To replicate and flood BUM traffic, you must specify the **ingress-node-replication** configuration statement at the **[edit vlans *name vxlan*]**, **[edit bridge-domains *name vxlan*]**, or **[edit routing-instances *name* bridge-domains *name vxlan*]** hierarchy level. This configuration restricts the BUM traffic flood domain to only those VTEPs mapped to a particular bridge domain or VLAN.

[See [Static VXLAN](#) and [static-remote-vtep-list](#).]

Software Installation and Upgrade

- **Zero touch provisioning (ZTP) with IPv6 support (MX Series)**—Starting in Junos OS Release 20.4R1, ZTP supports the DHCPv6 client. During the bootstrap process, the device first uses the DHCPv4 client to request for information regarding image and configuration file from the DHCP server. The device checks the DHCPv4 bindings sequentially. If there is a failure with one of the DHCPv4 bindings, the device will continue to check for bindings until provisioning is successful. If there are no DHCPv4 bindings, however, the device will check for DHCPv6 bindings and follow the same process as for DHCPv4 until the device can be provisioned successfully. Both DHCPv4 and DHCPv6 clients are included as part of the default configuration on the device.

The DHCP server uses DHCPv6 options 59 and 17 and applicable suboptions to exchange ZTP-related information between itself and the DHCP client.

[See [Zero Touch Provisioning](#).]

Software Licensing

- **MX Series devices using SPC3 services card for IPsec VPN services requires a feature license (MX Series)**—Starting in Junos OS Release 20.4R1, you must have a valid license to use the IPsec VPN feature running on MX Series devices with SPC3 services card. For high availability, you must install IPsec base license on both the nodes.

This is a binary license, without an installed license your license count is 0, and your license count is 1 when a valid license is installed in the system.

IPsec VPNs tunnels doesn't establish without a valid license in the device, current active tunnels will stay up if the license expires. IPsec VPN tunnels that are brought down after the license expiry doesn't re-establish until you install a valid license.

[See [Software Features That Require Licenses on MX Series Routers Only](#), [MX FLEX Software License Model](#), and [Managing Licenses](#).]

Subscriber Management and Services

- **Support for mobility on Junos Multi-Access User Plane (MX204, MX240, MX480, MX960, MX10003)**—For Junos OS Release 19.4R1, we introduced Junos Multi-Access User Plane supporting a combined SGW-U/PGW-U (SAEGW-U) on MX Series routers in accordance with 3GPP Release 14 CUPS architecture. This provided high-throughput 4G and 5G fixed-wireless access service with support for 5G non-stand-alone (NSA) mode.

For Junos OS Release 20.4R1, we introduce support for running an MX router as either a standalone SGW-U or a standalone PGW-U or a combined SAEGW-U to provide high-throughput 4G and 5G mobility service (relocation of a UE to a new eNodeB, new SGW-U, or new SAEGW-U). This includes support for GTP-U based S5-U and S8-U interfaces, to provide links between SGW-U and PGW-U devices, and tunnel relay functionality to forward user plane traffic between S1-U and S5-U/S8-U interfaces or between S5-U/S8-U and SGi interfaces respectively. We support the following mobility scenarios:

- Handover with eNodeB and no SGW change

- Handover with SGW change (direct forwarding)
- Handover with SGW change (indirect forwarding)

[See [Junos Multi-Access User Plane Overview](#).]

- **Support for 5G Junos Multi-Access User Plane (MX204, MX240, MX480, MX960, MX10003)**—Starting with Junos OS Release 20.4R1, Junos Multi-Access User Plane supports 3GPP TS 29.244 Release 15, which includes support for the 5G user plane function (UPF). Specifically, these enhancements are provided:
 - PDI optimization for Sx messages
 - GTP path management via heartbeats
 - Support for User ID in PFCP Session Establishment Request
 - Support for QoS control/enforcement at the bearer level
 - Support for DDOS over Sx interface.

[See [Junos Multi-Access User Plane Overview](#).]

- **Support for increased number of pseudowire logical interface devices (MX2010 and MX2020)**—Starting in Junos OS Release 20.4R1, you can configure up to 18,000 pseudowire logical interface devices on the MX2010 and MX2020 routers with the MX2K-MPC9E or MX2K-MPC11E line card. Use the **device-count** statement at the **[edit chassis pseudowire-service]** hierarchy level.

[See [Configuring the Maximum Number of Pseudowire Logical Interface Devices Supported on the Router](#) and [device-count](#).]

- **IPv4 reassembly for fragmented soft GRE packets on the WAG (MX Series)**—Starting in Junos OS Release 20.4R1, you can enable a Wi-Fi Access Gateway (WAG) to reassemble fragmented GRE packets that the WAG receives from a Wi-Fi access point over a soft GRE tunnel.

[See [dynamic-profiles](#) and [Wi-Fi Access Gateways](#).]

System Management

- **1-Gbps support on all ports of MPC7E-10G line cards (MX240, MX480, and MX960)**—Starting in Junos OS Release 20.4R1, you can configure 1-Gbps speed on all 40 10-Gbps Ethernet ports of the MPC7E-10G line cards. The 1-Gbps interface supports the following features:
 - Synchronous Ethernet
 - Link aggregation group (LAG)
 - G.8275.1 Precision Time Protocol (PTP) profile
 - Hybrid mode

To configure an interface to operate at the 1-Gbps speed, use the **set interfaces interface-name gigether-options speed 1g/10g** command at the **[edit]** hierarchy level.

[See [Precision Time Protocol Overview](#), [Synchronous Ethernet Overview](#), and [Hardware Compatibility Tool](#).]

System Logging

- **Support for time averaged watermark (MX Series, PTX Series, and QFX Series)**—Starting in Junos OS Release 20.4R1, you can capture steady state data of routing and forwarding (RIB/FIB) table routes using the **time-averaged-watermark-interval** configuration statement at the **[edit routing-options]** hierarchy level. Time averaged watermark is calculated whenever the time averaged interval is changed from CLI. Time averaged watermark is logged in syslog if the logs are enabled in the system at **LOG_NOTICE** level. The default time averaged watermark interval is 1 day. You can see the timed averaged watermark using the existing **show route summary** command.

[See [routing-options](#) and [show route summary](#).]

SEE ALSO

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What's Changed

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This section lists the changes in behavior of Junos OS features and changes in the syntax of Junos OS statements and commands in Junos OS Release 20.4R3 for MX Series routers.

What's Changed in Release 20.4R3-S1

General Routing

- **Log messages are removed (MX Series)**—When PTP aggregate Ethernet primary is configured, and PTP aggregate Ethernet secondary is not configured, the log message **Profiles are being modified** is removed.

What's Changed in Release 20.4R3

General Routing

- **Support for multiple proxy-id list (MX5, MX10, MX40, MX80, MX104, MX240, MX480, MX960, MX2008, MX2010, and MX2020)**—MX Series routers do not support ID list except for the following two cases:
 - MX Series routers accept any-any traffic selector in proxy-id list from the remote device that supports ID lists.
 - MX Series routers accept the ID list if list can be reduced by removing duplicates to specific ID. For example, reduce ID list having 80.0.0.1 and 80.0.0.0/24 to super set ID 80.0.0.0/24.
`list(any:0,ipv4(any:0-65535,[0..3]=80.0.0.1), ipv4_subnet(any:0-65535,[0..7]=80.0.0.0/24))`
- **ISSU is not supported**—Unified in-service software upgrade (ISSU) is not supported when clock synchronization is configured for Precision Time Protocol (PTP) and Synchronous Ethernet.
- **CLI commit error resolved**—CLI commit error for configuration statement **no-filter-check** for **family any** port mirroring is now resolved.
 [See [no-filter-check](#).]
- **Use primary-asymmetry and secondary-asymmetry CLI commands to compensate asymmetry (MX Series)**—On PTP over aggregated Ethernet, when there is an asymmetry on the primary link, secondary link, or both the links of an aggregated Ethernet bundle, use the **primary-asymmetry**, **secondary-asymmetry**, or both the CLI commands respectively to compensate the asymmetry. This reduces the phase-offset and improves the performance of PTP clock synchronization.

Layer 2 Ethernet Services

- **Link selection support for DHCP**—We have introduced the **link-selection** statement at the **[edit forwarding-options dhcp-relay relay-option-82]** hierarchy level, which allows DHCP relay to add suboption 5 to option 82. Suboption 5 allows DHCP proxy clients and relay agents to request an IP address for a specific subnet from a specific IP address range and scope. Prior to this release, the DHCP relay dropped packets during the renewal DHCP process and the DHCP server used the leaf's address as a destination to acknowledge the DHCP renewal message.

[See [relay-option-82](#).]

Network Management and Monitoring

- **Enhancement to the snmp mib walk command (PTX Series, QFX Series, EX Series, MX Series, SRX Series)**—The `ipv6IfOperStatus` field displays the current operational state of the interface. The `noIfIdentifier(3)` state indicates that no valid Interface Identifier is assigned to the interface. This state usually indicates that the link-local interface address failed Duplicate Address Detection. When you specify the 'Duplicate Address Detected' error flag on the interface, the new value (`noIfIdentifier(3)`) is displayed. Previously, the `snmp mib walk` command did not display the new value (`noIfIdentifier(3)`).
- **Changes in contextEngineID for SNMPv3 INFORMS (PTX Series, QFX Series, ACX Series, EX Series, MX Series, and SRX Series)**—Now the `contextEngineID` of SNMPv3 INFORMS is set to the local engine-id of Junos devices. In earlier releases, the `contextEngineID` of SNMPv3 INFORMS was set to remote engine-id.

[See [SNMP MIBs and Traps Supported by Junos OS](#).]

- **The configuration accepts only defined identity values for nodes of type identityref in YANG data models (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—If you configure a statement that has type `identityref` in the corresponding YANG data model, the device accepts only defined identity values (as defined by an identity statement) as valid input. In earlier releases, the device also accepts values that are not defined identity values.

What's Changed in Release 20.4R2-S1

General Routing

- **Use primary-asymmetry and secondary-asymmetry CLI commands to compensate asymmetry (MX Series)**—On PTP over aggregated Ethernet, when there is an asymmetry on the primary link, secondary link, or both the links of an aggregated Ethernet bundle, use the **primary-asymmetry**, **secondary-asymmetry**, or both the CLI commands respectively to compensate the asymmetry. This reduces the phase-offset and improves the performance of PTP clock synchronization.

What's Changed in Release 20.4R2

EVPN

- **Support for displaying SVLBNH information**—You can now view shared VXLAN load balancing next hop (SVLBNH) information when you display the VXLAN tunnel endpoint information for a specified ESI and routing instance by using **show ethernet-switching vxlan-tunnel-end-point esi esi-identifier esi-identifier instance instance svlbnh** command.

General Routing

- **Update to the show chassis errors active output (MX2010 and MX2020 routers with MPC11E)**—We have updated the **show chassis errors active** output for the MPC11E line card (MX2K-MPC11E) to display the correct error information. Previously, this CLI command displayed duplicate or incorrect output when the MPC11E line card is not installed in slot 0 of the MX2010 or MX2020 routers.

[See [show chassis errors active..](#)]

- **ISSU is not supported**—Unified in-service software upgrade (ISSU) is not supported when clock synchronization is configured for Precision Time Protocol (PTP) and Synchronous Ethernet.

Interfaces and Chassis

- **Change in <range> XML tag (MX480)**—We've changed the XML tag <range> string </range> is changed to <transport-range> <transport-range-info> string </transport-range-info> <transport-range-suspect-flag> string </transport-range-suspect-flag> <transport-range-reason> string </transport-range-reason> </transport-range> under **[show interfaces transport pm optics current <interface> | display]** hierarchy in the XML output. Hence, the new XML tags that associate the values to the range-info, range-suspect-flag, and range-reason tags map the information to the given **[show interfaces transport pm optics current interface | display]** entry.

[See [Supported OTN Options on MX Series Routers.](#)]

- **Blocking duplicate IP detection in the same routing instance (ACX Series, EX Series, MX Series, NFX Series, PTX Series, QFX Series, and SRX Series)**—Junos will no longer accept duplicate IPs between different logical interfaces in the same routing instance. Refer to the table mentioned in the topic inet (interfaces). When you try to configure same IP on two logical interfaces inside same routing instance, the commit will be blocked with the error displayed as shown below: **[edit] user@host# set interfaces**

```
ge-0/0/1 unit 0 family inet address 2.2.2.2/24 [edit] user@host# commit commit complete [edit]
user@host# set interfaces ge-0/0/2 unit 0 family inet address 2.2.2.2/24 [edit] user@host# commit
[edit interfaces ge-0/0/2 unit 0 family inet] 'address 2.2.2.2/24' identical local address found on rt_inst
[default], intfs [ge-0/0/2.0 and ge-0/0/1.0], family [inet]. error: configuration check-out failed.
```

[See [inet\(interfaces\)](#).]

Junos XML API and Scripting

- Refreshing scripts from an HTTPS server requires a certificate (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)—When you refresh a local commit, event, op, SNMP, or Juniper Extension Toolkit (JET) script from an HTTPS server, you must specify the certificate (Root CA or self-signed) that the device uses to validate the server's certificate, thus ensuring that the server is authentic. In earlier releases, when you refresh scripts from an HTTPS server, the device does not perform certificate validation.

When you refresh a script using the **request system scripts refresh-from** operational mode command, include the **cert-file** option and specify the certificate path. Before you refresh a script using the **set refresh** or **set refresh-from** configuration mode command, first configure the **cert-file** statement under the hierarchy level where you configure the script. The certificate must be in Privacy-Enhanced Mail (PEM) format.

[See [request system scripts refresh-from](#) and [cert-file](#).]

- The **jcs:invoke()** function supports suppression of root login and logout events in system log files for SLAX commit scripts (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)—The **jcs:invoke()** extension function supports the **no-login-logout** parameter in SLAX commit scripts. If you include the parameter, the function does not generate and log UI_LOGIN_EVENT and UI_LOGOUT_EVENT messages when the script logs in as root to execute the specified remote procedure call (RPC). If you omit the parameter, the function behaves as in earlier releases in which the root UI_LOGIN_EVENT and UI_LOGOUT_EVENT messages are included in system log files.

[See [invoke\(\) Function \(SLAX and XSLT\)](#).]

- The **jcs:invoke()** function supports suppression of root login and logout events in system log files for SLAX event scripts (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)—The **jcs:invoke()** extension function supports the **no-login-logout** parameter in SLAX event scripts. If you include the parameter, the function does not generate and log UI_LOGIN_EVENT and UI_LOGOUT_EVENT messages when the script logs in as root to execute the specified remote procedure call (RPC). If you omit the parameter, the function behaves as in earlier releases in which the root UI_LOGIN_EVENT and UI_LOGOUT_EVENT messages are included in system log files.

[See [invoke\(\) Function \(SLAX and XSLT\)](#).]

Layer 2 Ethernet Services

- **Active leasequery-based bulk leasequery (MX Series)**—The **overrides always-write-option-82** and **relay-option-82 circuit-id** configurations at the **[edit forwarding-options dhcp-relay]** hierarchy level are not mandatory for active leasequery based bulk leasequery. For earlier releases, the **overrides always-write-option-82** and **circuit-id** configurations are mandatory for active leasequery based bulk leasequery. For regular bulk leasequery between relay and server without any active leasequery, the **overrides always-write-option-82** and **relay-option-82 circuit-id** configurations are mandatory.

[See [bulk-leasequery \(DHCP Relay Agent\)](#).]

Network Management and Monitoring

- **Support for specifying the YANG modules to advertise in the NETCONF capabilities and supported schema list (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—You can configure devices to emit third-party, standard, and Junos OS native YANG modules in the capabilities exchange of a NETCONF session by configuring the appropriate statements at the **[edit system services netconf hello-message yang-module-capabilities]** hierarchy level. In addition, you can specify the YANG schemas that the NETCONF server should include in its list of supported schemas by configuring the appropriate statements at the **[edit system services netconf netconf-monitoring netconf-state-schemas]** hierarchy level.

[See [hello-message](#) and [netconf-monitoring](#).]

- **Change in OID ifHighSpeed**—Now, the object identifier (OID) ifHighSpeed displays the negotiated speed once negotiation is completed. If the speed is not negotiated, ifHighSpeed displays the actual maximum speed of the interface. In earlier releases, ifHighSpeed always displayed the actual speed of the interface.

[See [SNMP MIBs and Traps Supported by Junos OS](#).]

VPNs

- **View the traffic selector type for an IPsec tunnel (SRX Series and MX Series)**—You can run the **show security ipsec security-associations detail** command to display the traffic selector type for a VPN. The command displays **proxy-id** or **traffic-selector** as a value for the **TS Type** output field based on your configuration.

[See [show security ipsec security-associations](#).]

What's Changed in Release 20.4R1

Class of Service (CoS)

- We've corrected the output of the **show class-of-service interface | display xml** command. Output of the following sort: `<container><leaf 1> data <\leaf 1><leaf 2> data </leaf 2><leaf 3> data </leaf 3> <leaf 1> data <\leaf 1><leaf 2> data </leaf 2><leaf 3> data </leaf 3> </container>` will now appear correctly as: `<container><leaf 1> data <\leaf 1><leaf 2> data </leaf 2><leaf 3> data </leaf 3></container><container><leaf 1> data <\leaf 1><leaf 2> data </leaf 2><leaf 3> data </leaf 3></container>`.

EVPN

- **Updated XML output for show evpn p2mp**—Starting with this release, when you pipe the output of the **show evpn p2mp** command to the **display xml** option, Junos OS now returns an XML output with a subtree structure for each neighbor. Prior to this release, the display XML returns an XML output with all the neighbors under one tree structure.
- **New output flag for the show bridge mac-ip table command**—The Layer 2 address learning daemon does not send updated MAC and IP Address advertisements to the Routing Protocol daemon when an IRB interface is disabled in an EVPN-VXLAN network. Junos has added the NAD flag in the output of the **show bridge mac-ip-table** command to identify the disabled IRB entries where the MAC and IP address advertisement will not be sent.

[See [show bridge mac-ip-table](#).]

General Routing

- **Change in show oam ethernet connectivity-fault-management mep-statistics command (MX Series)**— You can now view the real time statistics for continuity check messages (CCM) inline sessions for MPC10E (MPC10E-10C-MRATE and MPC10E-15C-MRATE) and MPC11E (MX2K-MPC11E) line cards only when you execute the **show oam connectivity-fault-management mep-statistics local-mep local-mep-id maintenance-domain name maintenance-association name** twice in immediate succession. If you execute the command once, the values are incorrectly displayed.

[See [show oam ethernet connectivity-fault-management mep-statistics](#).]

Change in show oam ethernet connectivity-fault-management interface command (MX Series)— You can now view the counter values for continuity check messages (CCM) inline sessions sent messages for MPC10E (MPC10E-10C-MRATE and MPC10E-15C-MRATE) and MPC11E (MX2K-MPC11E) line cards only when you execute the **show oam connectivity-fault-management interfaces** command three times. If you execute the command twice, the values are incorrectly displayed.

[See [show oam ethernet connectivity-fault-management interfaces](#).]

- **MS-MPC and MS-MIC service package (MX240, MX480, MX960, MX2020, MX2010, and MX2008)**—PICs of the MS-MPC and MS-MIC do not support any service package other than extension-provider. If you try to configure any other service package for these PICs by using the **set chassis fpc slot-number pic pic-number adaptive-services service-package** command, an error is logged. Use the **show chassis pic fpc-slot slot pic-slot slot** command to view the service package details of the PICs.

[See [extension-provider](#).]

- **Round-trip time load throttling for pseudowire interfaces (MX Series)**—The Routing Engine supports round-trip time load throttling for pseudowire (ps) interfaces. In earlier releases, only Ethernet and aggregated Ethernet interfaces are supported.

[See [Resource Monitoring for Subscriber Management and Services](#).]

- **Updates to ON-CHANGE and periodic dynamic subscriber interface metadata sensors (MX Series routers and EX9200 line of switches)**—We've made the following updates to the `/junos/system/subscriber-management/dynamic-interfaces/interfaces/meta-data/interfacesid='sid-value'/` sensor:

- Notifications are sent when subscribers log in on either IP demux or VLAN demux interfaces. In earlier releases, login notifications are sent only for IP demux logins.
- The **interface-set** end path has been added to the logical interface metadata. The interface-set field appears in both ON-CHANGE and periodic notifications. In earlier releases, this field is not included in the sensor metadata or notifications.

[See [gRPC Sensors for Subscriber Statistics and Queue Statistics for Dynamic Interfaces and Interface-Sets \(Junos Telemetry Interface\)](#). gRPC Sensors for Subscriber Statistics and Queue Statistics for Dynamic Interfaces and Interface-Sets (Junos Telemetry Interface).]

- **New commit check for MC-LAG (MX Series)**— We've introduced a new commit check to check the values assigned to the redundancy group identification number on the MC-AE interface (

redundancy-group-id) and ICCP peer (**redundancy-group-id-list**) when you configure multichassis aggregation groups (MC-LAGs). If the values are different, the system reports a commit check error. In previous releases, if the configured values were different, the l2ald process would crash.

[See [iccp](#) .]

- **Support for unicast ARP request on table entry expiration**—You can configure the device to send a unicast ARP request instead of the default broadcast request when an ARP table entry is about to expire. The retry requests are unicast at intervals of 5 seconds. Without this option, the retry requests are broadcast at intervals of 800 milliseconds. This behavior reduces ARP overall broadcast traffic. It also supports the use case where access nodes are configured not to forward broadcast ARP requests toward customer CPEs for security reasons and instead translate ARP broadcasts to unicast requests. To confirm whether this is configured, you can issue the following command: **show configuration system arp | grep unicast-mode-on-expire**.

[See [arp](#) .]

- **Change in show oam ethernet connectivity-fault-management mep-statistics command (MX Series)**—You can now view the real-time statistics for continuity check messages (CCM) inline sessions for MPC10E (MPC10E-10C-MRATE and MPC10E-15C-MRATE) and MPC11E (MX2K-MPC11E) line cards only when you execute the **show oam connectivity-fault-management mep-statistics local-mep local-mep-id maintenance-association name** twice in immediate succession. If you execute the command once, the values are incorrectly displayed.

[See [show oam ethernet connectivity-fault-management mep-statistics](#) .]

- **New TLV types and TLV type values in output field (MX960 and vMX)**—We've introduced TLV SR policy identifier, TLV SR candidate path identifier, and TLV SR preference fields in the output for the **show path-computation-client tlv-types** command. These new output fields help you in easily fetching the TLV type values used by PCCD irrespective of whether the type values are experimental or standardized.

High Availability (HA) and Resiliency

- **IPv6 address in the prefix TIEs displayed correctly**—The IPv6 address in the prefix TIEs are displayed correctly in the **show rift tie** output.
- **The ntp boot-server command is deprecated** — The **boot-server** option for the **ntp** command is no longer needed, and has been deprecated.

Interfaces and Chassis

- **Change in <range> XML tag (MX480)**—Starting in Junos OS, we've changed the <range> string </range> XML tag to <transport-range> <transport-range-info> string </transport-range-info> <transport-range-suspect-flag> string </transport-range-suspect-flag> <transport-range-reason> string </transport-range-reason> </transport-range> under the output of the **show interfaces transport pm optics current interface | display hierarchy** command. Hence, the new XML tags that associate the values to the range-info, range-suspect-flag, range-reason tags map the information to the given **show interfaces transport pm optics current | display entry** command.

[See [Supported OTN Options on MX Series Routers](#).]

- **Hardware assisted timestamping**—By default, hardware assistance is used for timestamping Ethernet frame delay frames on AFT based MX Series line cards, even if the hardware-assisted-timestamping is not configured.

J-Web

- **Adobe Flash Player support (MX Series)**—Adobe Flash Player support ends on December 31, 2020. As a result, starting in Junos OS Release 20.4R1, the following J-Web pages will not be supported:
 - Monitor > System View > Process Details
 - Monitor > Routing > OSPF Information

The Monitor > Interfaces page is supported. However, the Flash components are removed. In addition, these monitor pages will not load correctly for Junos OS Release 20.3R1 and earlier releases.

MPLS

- **The show mpls lsp extensivel and show mpls lsp detail commands display next-hop gateway LSPid**—When you use the **show mpls lsp extensivel** and **show mpls lsp detail** commands, you'll see next-hop gateway LSPid in the output.
- **Disable back-off behavior on PSB2 (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**— We've introduced the **cspf-backoff-time** statement globally for MPLS and LSP to delay the CSPF by configured number of seconds, on receiving bandwidth unavailable PathErr on PSB2. If the configured value is zero, then the CSPF starts immediately for PSB2, when bandwidth-unavailable PathErr is received. If the statement is not configured, the default exponential back-off occurs.

[See [cspf-backoff-time](#).]

Network Management and Monitoring

- **Warning changed for configuration statements that correspond to "deviate not-supported" nodes in YANG data models (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—If you configure a statement corresponding to a YANG data model node that defines the **deviate not-supported** statement, the Junos OS configuration annotates that statement with the comment **Warning: statement ignored: unsupported platform**. In earlier releases, the warning is **Warning: 'statement' is deprecated**.

Platform and Infrastructure

- **Firewall Filters Application**—You can configure the proto family as part of the filter name.

[See [Understanding Multiple Firewall Filters Applied as a List](#).]

Routing Protocols

- **Loading of the default configurations in a RIFT package causes the following changes**—
 1. Output of the **show rift node status** command displays the node ID in hexadecimal number even though the node ID is configured in decimal, hexadecimal, or octal number.
 2. Some of the DDoS default configurations change because of the DDoS protection interferes with the RIFT BFD operation.

User Interface and Configuration

- **Verbose format option to export JSON configuration data (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—The Junos OS CLI exposes the **verbose** statement at the **edit system export-format json** hierarchy level. The default format to export configuration data in JSON changed from **verbose** format to **ietf** format starting in Junos OS Release 16.1R1. You can explicitly specify the default export format for JSON configuration data by configuring the appropriate statement at the **edit system export-format json** hierarchy level. Although the **verbose** statement is exposed in the Junos OS CLI as of the current release, you can configure this statement starting in Junos OS Release 16.1R1.

[See [export-format](#).]

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Known Limitations

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Learn about known limitations in this release for MX Series routers. For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Class of Service (CoS)

- If you have to take an interface out of aggregated Ethernet bundle and configure it to operate in stand alone mode, then doing this in single commit might render the operation ineffective and could lead to connectivity issues. This is seen due to a race condition between Routing Engine daemons (COSD, DCD/Chassisd), Packet Forwarding Engine, and kernel. This issue is seen when there is explicit CoS configuration to be made on the interface. However, the problem can be seen without explicit CoS too as there is default CoS that is always present. In some cases, it is possible that a single shot commit will send out multiple operational messages down to kernel and might confuse the kernel to do unintended optimization that could lead to a message being consumed at kernel and not being sent to the Packet Forwarding Engine. The result is the same even in this case. [PR1504287](#)

General Routing

- In some scenarios with MPC, major alarm and following messages are generated. This major error is triggered due to parity error, and the impacted queue might drop packets. This might impact the forwarding. To recover, MPC card needs to be rebooted. **messages log: fpcx XQCHIP(46):XQ-chip[0]: DROP protect_regs error (status=0x8) alarmd[3158]: Alarm set: FPC color=RED, class=CHASSIS, reason=FPC x Major Errors Major alarm set, FPC x Major Errors fpcx XQCHIP(46):XQ-chip[0]: DROP protect_regs error (status=0x8) cli> show chassis alarms 1 alarms currently active Alarm time Class Description 2019-01-25 15:18:03 UTC Major FPC x Major Errors.** [PR1303489](#)
- RPD slips are noticed on MX104 routers that have large configuration load on the box with multiple services enabled (for example, L2circuits, VPLS, L3VPN, firewall-filters configuration, SNMP-polling, and so on). The following should be considered to avoid RPD slips for longer time duration:

Configure delta-export, persist-groups-inheritance, fast-synchronize.

Reduce the possible configuration size.

Remove any traceoptions and reduce the logging pressure on the NAND-flash storage.

Analyze the load from processes such as snmpd, mib2d, pfed processes if you are running SNMP.

[PR1361250](#)
- On MX10000 Series of routers, FPCs report /var/run/sensord.rrd messages. An illegal attempt to update using time message can be observed during the normal operation of a router. The cause is time difference between Routing Engine and FPCs. There should be no service impact. [PR1420927](#)
- Subscriber access facing CPU utilization of FPC remains 100 percent for 56 minutes after making changes to the service firewall filter configuration. [PR1447003](#)
- On the MPC11E line card, the following error message is observed when the line card is online: **i2c transaction error (0x00000002).** [PR1457655](#)
- Traffic stops after reaching the volume limit but the traffic resumes after the aggregated Packet Forwarding Engine fails. Threshold and quota values are not updated to the secondary aggregated Packet

Forwarding Engine and if quota is hit on primary aggregated Packet Forwarding Engine and traffic starts dropping due to quota and switchover happens, traffic will continue to flow until quota is hit. [PR1463723](#)

- EVPN-VPWS, L3VPN, and L2VPN FRR convergence time with aggregated Ethernet as the active core interface is not meeting <50 ms and it might be 100 ms to 150 ms. [PR1492730](#)
- On a fully scaled system where all the slices are utilized by different families of CLI filters, if you try to delete one family and add/change for another family with a higher number of filter terms that requires either expansion of the filter or creation of a new filter, the Packet Forwarding Engine fails to add the new filter as we are getting messages out of sequence, that is, add/change of filter is called earlier than the delete of another filter that will free up the slices. [PR1512242](#)
- Some memory leaks have been observed in the JET Service Daemon (JSD) process when one or more collectors are connecting and disconnecting to and from the router. These are observed in the gRPC stack code which is third party. The amount of memory leaked is relatively small. However, these leaks could increase with more frequent collector connects and disconnects. As a result of the memory leaks, the JSD process memory size can increase to a value that is higher than normal (for example, when the gRPC connections are established and stable) but is unlikely to cause any adverse effects to the system with streaming telemetry. [PR1512296](#)
- The link fault management (LFM) might flap during MX Virtual Chassis unified ISSU to and from this release. [PR1516744](#)
- On MX2020 routers with the MPC11E line card, link flap for 100G-DR/FR/LR will take longer time: more than 20 seconds. [PR1527441](#)
- When an image with the third party SDK upgrade (6.5.x) is installed, the CPU utilization might go up by around 5 percent. [PR1534234](#)
- Currently Layer 2.5 traffic support is only for P2P and not for TS. So Routing Engine traffic does not work with traffic selector case. Only transit traffic will work with traffic selector based tunnels. Routing Engine traffic needs to use P2P tunnels. [PR1534248](#)
- The NPC process continuously generates core file at **Trinity_Ktree::Trinity_FourWayBlock, Trinity_Ktree::walkSubTree** due to the NH memory exhaustion with the NH explosion. The rpd and srrd processes start hogging and the system becomes unstable. [PR1538029](#)
- 1 pps CTE performance might not meet class B metrics for MPC7E-1G and MPC7E-10G line cards. [PR1546219](#)
- The ISSU upgrade fails due to the Packet Forwarding Engine restart issue. [PR1554915](#)
- The SyncE to PTP transient response is a stringent mask to be met with two way time error. The SyncE to PTP transient response mask might not be met for MPC7E-1G and MPC7E-10G line cards. [PR1557999](#)
- On the MX960 routers, **spring-traffic-engineering lsp count** is not displayed as expected while validating 32000 inter-domain DCPSF LSPs. [PR1561947](#)
- The PTP FPGA is kept in reset during BIOS boot. During boot, the PTP FPGA is taken out of reset and pcie-tree is reenumerated. Hence you would be seeing the link-up or link-down during this sequence. [PR1572061](#)

- The known multicast traffic received over a VLAN from the core on VTEP does not get forwarded to the downstream CE interfaces. [PR1575841](#)
- On all Junos platforms, the l2ald crash might be observed on changing the routing instance from VPLS to non-Layer 2 routing instance, with the same routing instance name being used for both VPLS and non-Layer 2 routing instance. [PR1586516](#)
- SyncE source across multiple line cards cannot be used in PTP-Hybrid source as fallback clock due to PTP lanes limitations used for SyncE clock as well in SCBE2 and MPC1-9 types. [PR1536013](#)

Interfaces and Chassis

- For MC-LAG to work properly, the mc-ae interface should be configured on both the PE devices. A scenario where the mc-ae interface is deleted, deactivated, or not configured on one of the devices is a case of misconfiguration. Juniper Networks does not support such a scenario because it can lead to traffic loss and other unexpected behavior. [PR1536831](#)
- On the MPC10 line cards, DMRs or SLRs are not received with EVPN up mep on the aggregated Ethernet interface with normalization. [PR1543641](#)
- UP MEP CFM sessions over bridge-domains or VPLSs, which have ports hosted on FPC(s) does not support ISSU. The sessions needs to be explicitly deactivate and activate to recover post ISSU. [PR1543656](#)
- The issue is seen in the scaled setup with 296 LM sessions with iterator cycle time interval (100 ms). It seems there is degradation in scale number (OAM packet rate at ~5500). At this qualified PPS, now LMR packet loss is observed, but the functionality seems to be fine. To avoid LMR packet loss, reduce the scale number, and keep the OAM packet rate to less than 5500 pps. [PR1561397](#)

J-Web

- The Firefox browser displays an unsaved changes error message in the J-Web basic settings page if the autofill login and password options are selected under the browser privacy and security settings. [PR1560549](#)

MPLS

- The rpd process might crash. [PR1461468](#)
- With local reversion ON, there is a possibility of transit router not informing headend of RSVP disabled link when link flaps more than once. As a workaround, remove the **local-reversion** configuration. [PR1576979](#)

Network Management and Monitoring

- On the MPC11E line card, the following trap message is not observed after a line card reboot when the scaled interfaces are present: **SNMP Link up**. [PR1507780](#)

Platform and Infrastructure

- MPC equipped with QX-chip can completely stop forwarding traffic after QX-chip internal memory error and MQChip DDRIF WO Checksum Error. [PR1197475](#)
- Interoperability failure between Junos OS Evolved as RPM client, and Junos OS TVP platforms as RPM server (and vice versa). [PR1508127](#)

VPNs

- In some scenario (for example, configuring firewall filter), routers might show obsolete IPsec SA and NHTB entry even when the peer tear down the tunnel. [PR1432925](#)

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Open Issues

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Class of Service (CoS)

- On all Junos platforms with LT or PS interface configured, default classifier in wildcard will get attached to the LT and PS interface, even if no classifier is configured. It could be observed when there is a wildcard interface that matches the LT and PS interface. [PR1542559](#)
- When a wildcard interface is configured with a classifier, upon reboot of the device, the classifier defined in the wildcard interface configuration might not be properly programmed to the interfaces. [PR1559516](#)

EVPN

- A few duplicate packets might be seen in an AA EVPN scenario when the remote PE device sends a packet with an IM label due to MAC not learned on the remote PE device, but learned on the AA local PE device. The nondesignated forwarder sends the IM-labeled encapsulated packet to the PE-CE interface after MAC lookup instead of dropping the packet, which causes the duplicate packets to be seen on the CE side. [PR1245316](#)
- VXLAN OAM host bound packets are not throttled with DDoS policers. [PR1435228](#)

- On all Junos platforms in the EVPN-VXLAN to EVPN-MPLS stitching scenario, traffic loss might be seen with data forwarder (DF) changes when traffic flows from VXLAN to MPLS. The traffic loss will occur till MAC IP ages out. [PR1515096](#)
- On all Junos platforms that support EVPN-VXLAN, if aggregated Ethernet interface or redundant logical tunnel (RLT) interface is configured in the underlay network for EVPN-VXLAN, when there is an ARP request generated and flooded to the core side, a kernel crash might occur. [PR1524485](#)
- In a PBB-EVPN environment, the ARP suppression feature, which is not supported by the PBB might be enabled unexpectedly. This might cause MAC addresses of remote CEs not to be learned and hence traffic loss might be seen. [PR1529940](#)
- When there is no **router-id** configured on the router and if there is any configuration change done on the router, then EVPN T-3 IM routes are advertised with ingress replicator IP set to 0.0.0.0. As this is an invalid IP address, the receiving PE ignored this T-3 IM route and does not create dummy egress-IM route in mpls.0 table for that PE device. This results in I2ald not creating VE mesh-grp flood route for the remote PE device and thus all BUM packets get dropped locally. [PR1576943](#)
- In an EVPN-VXLAN scenario, the label field for Type-1 route is not required, but it is assigned with 1 instead of 0, which is in conflict with the RFC7432. [PR1594981](#)
- On Junos OS platforms with EVPN-VXLAN to EVPN-VXLAN DCI interconnect deployment, the Data Center Interconnect (DCI) InterVNI and IntraVNI traffic might get discarded or dropped silently in a gateway node due to the tagged underlay interfaces. [PR1596462](#)
- In a scenario with EVPN-VXLAN in the a datacenter and EVPN-MPLS is in the WAN, and the stitching is done with an LT interface, then the bridge mac-table learning entries are not as expected for EVPN-VXLAN routing instance. This might occur after the **restart interface-control** command is issued on gateways. [PR1600310](#)
- On all Junos OS platforms. with the **proxy-macip-advertisement** statement configured, at times during longevity tests, there are missing ARP and ND entries in the kernel while the I2ald and rpd have the entry. [PR1609322](#)
- This problem happens only with the translation VNI when MAC is moved from DC1 to DC2. VM moves across DC where there is no translate VNI configuration in the interconnect works as designed [PR1610432](#)

Flow-based and Packet-based Processing

- Use an antireplay window size of 512 for IPv4 or IPv6 in fat-tunnel. The ESP sequence check might otherwise report out-of-order packets if the fat-tunnel parallel encryption is within 384 packets (12 cores * 32 packets in one batch). Hence, there are no out-of-order packets with 5PR122397912 antireplay window size. [PR1470637](#)

Forwarding and Sampling

- Packet length for ICMPv6 is shown as 0 in the output of the **show firewall log detail** CLI command. [PR1184624](#)
- The **fast-lookup-filter** with match not supported in FLT hardware might cause the traffic drop. [PR1573350](#)

General Routing

- Some non-fatal interrupts (for example, CM cache or AQD interrupts) are logged as fatal interrupts. The following log messages will be shown on CM parity interrupt:

fpc0 TQCHIP 0: CM parity Fatal interrupt,Interrupt status:0x10

fpc0 CMSNG: Fatal ASIC error, chip TQ

fpc0 TQCHIP 0: CM cache parity Fatal interrupt has occurred 181 time(s) in 180010 msecs

TQCHIP 0: CM cache parity Fatal interrupt has occurred 181 time(s) in 180005 msecs. [PR1089955](#)

- On the MX104 platform, when using **snmpbulkget** or **snmpbulkwalk** (for example, used by the SNMP server) on a chassisd-related component (for example, jnxOperatingEntry), high CPU usage for chassis process and slow response might be seen because of a hardware limitation, which might also lead to a query time out on the SNMP client. In addition, the issue might not be seen while using an SNMP query for interface statistics. As a workaround, to avoid the issue, use either of the following approaches:

Use **snmpget** or **snmpwalk** instead of **snmpbulkget** or **snmpbulkwalk** and include the **-t 30** option when doing the SNMP query. For example, **snmpget -v2c -c XX -t 30**.

Use the **-t 30** option with **snmpbulkget** or **snmpbulkwalk**. For example, **snmpbulkget -v2c -c XX -t 30**. [PR1103870](#)

- Egress sFlow sampling is supported on only 8 ports out of every 12 ports on 40G line cards and on only 8 ports out of every 48 ports on 10G line cards. There is no such limitation on ingress sFlow sampling. [PR1202870](#)
- On the MX104 platform, when Routing Engine CPU usage increases, sporadic I2C error message might be seen. Since the situation is temporary, the I2C access might succeed in the next polling and there would be no impact. [PR1223979](#)

- You might encounter a single event upset (SEU) event that might cause a linked-list corruption of the TQCHIP. The following syslog message gets reported: **Jan 9 08:16:47.295 router fpc0 TQCHIP1: Fatal error pqt_min_free_cnt is zero Jan 9 08:16:47.295 router fpc0 CMSNG: Fatal ASIC error, chip TQ Jan 9 08:16:47.295 router fpc0 TQ Chip::FATAL ERROR!! from PQT free count is zero jan 9 08:16:47.380 router alarmd[2427]: Alarm set: FPC color=RED, class=CHASSIS, reason=FPC 0 Fatal Errors - TQ Chip Error code: 0x50002 Jan 9 08:16:47.380 router craftd[2051]: Fatal alarm set, FPC 0 Fatal Errors - TQ Chip Error code: 0x50002.**

The Junos OS chassis management error handling detects such a condition, raises an alarm, and disables the affected Packet Forwarding Engine entity. To recover this Packet Forwarding Engine entity, restart the FPC. Contact your Juniper Networks support representative if the issue persists even after an FPC restart. [PR1254415](#)

- If a vmhost snapshot is taken on an alternate disk and there is no further vmhost software image upgrade, the expectation is that if the current vmhost image gets corrupted, the system boots with the alternate disk so the user can recover the primary disk to restore the state. However, the host root file system and the node boots with the previous vmhost software instead of the alternate disk. [PR1281554](#)
- When you issue a **show interface** command to check the interface details, the system will not check whether the interface name provided is valid or invalid. The system will not generate an error message if the interface name is invalid. [PR1306191](#)
- On 30 port MACsec-enabled line card (such as, LC1101-M-30C, LC1101-M-30Q, and LC1101-M-96X), when the **exclude-protocol lacp** statement configured at the **[edit security macsec connectivity-association connectivity-association-name]** hierarchy level is deleted or deactivated, the LACP protocol's **Mux State** shown under the output of CLI command **show lacp interface**, might remain as attached or detached and might not change to distributing state. [PR1331412](#)
- Source MAC and TTL values are not updated for routed multicast packets in EVPN-VXLAN. [PR1346894](#)
- Backup Routing Engine might crash after GRES occurs continuously for more than 10 times. [PR1348806](#)
- In some cases, online insertion and removal (OIR) of a MIC installed in an MPC might lead to the silent dropping of traffic destined to the MPC. The only way to recover from this is to restart the MPC. The issue is not seen if using the corresponding CLI commands to take the MIC offline and then bring it back online. [PR1350103](#)
- The log message **SMART ATA Error Log Structure error: invalid SMART checksum** might be seen on FPC with third party mSata SSD. [PR1354070](#)
- When VLAN is added as an action for changing the VLAN in both ingress and egress filters, the filter is not installed. [PR1362609](#)
- A few xe- interfaces go down with the following error message: **if_msg_ifd_cmd_tlv_decode ifd xe-0/0/0 #190 down with ASIC Error.** [PR1377840](#)
- Ping latency behavior is expected for host generated ICMP traffic due to the design of Packet Forwarding Engine queue polling the packets from ASIC. **user@router-acx5448> ping 10.0.0.4 PING 10.0.0.4 (10.0.0.4): 56 data bytes 64 bytes from 10.0.0.4: icmp_seq=0 ttl=63 time=8.994 ms 64 bytes from**

10.0.0.4: icmp_seq=1 ttl=63 time=49.370 ms 64 bytes from 10.0.0.4: icmp_seq=2 ttl=63 time=47.348 ms 64 bytes from 10.0.0.4: icmp_seq=3 ttl=63 time=45.411 ms <<< 64 bytes from 10.0.0.4: icmp_seq=4 ttl=63 time=106.449 ms <<< 64 bytes from 10.0.0.4: icmp_seq=5 ttl=63 time=79.697 ms <<< 64 bytes from 10.0.0.4: icmp_seq=6 ttl=63 time=37.489 ms <<< 64 bytes from 10.0.0.4: icmp_seq=7 ttl=63 time=31.436 ms << 64 bytes from 10.0.0.4: icmp_seq=8 ttl=63 time=35.460 ms << 64 bytes from 10.0.0.4: icmp_seq=9 ttl=63 time=77.198 ms << ^C --- 10.0.0.4 ping statistics --- 10 packets transmitted, 10 packets received, 0% packet loss round-trip min/avg/max/stddev = 8.994/51.885/106.449/26.824 ms. [PR1380145](#)

- Due to transient hardware condition, single-bit error (SBE) events are corrected and have no operational impact. Reporting of those events had been disabled to prevent alarms and possibly unnecessary hardware replacements. This change applies to all platforms using hybrid memory controller (HMC). [PR1384435](#)
- In scaled configurations, an FPC might be shown as online and permits revert of a RLT when it is actually not yet ready to pass the traffic. This might lead to traffic loss for up to 8 minutes. [PR1394026](#)
- FPC core files are generated on multiple additions or deletions of hierarchical CoS from pseudowire devices. As a workaround, remove the pseudowire device without changing the hierarchical CoS configuration. [PR1414969](#)
- If HTTPS header enrichment function is used, the traffic throughput decreases when traffic passes through header enrichment. [PR1420894](#)
- If a firewall filter with multiple match conditions is configured on interfaces which are up and the firewall filter is modified (either a new action is added or a condition is added or removed etc.), the FPC might crash and restart. It might affect the service or traffic. [PR1432116](#)
- A timing issue during the sxe interface bring up (with respect to i40e driver) is seen. To recover, reboot the complete board. [PR1442249](#)
- The error messages **Err] mqss_ea_host_misc_wanio_intr_handler: Process ing the PTC interrupts failed - status 1, Err] mqss_pio_read_u32: Reading 32-bit register fa iled - status 1, pio_handle 0xe1b445c0, addr 0x400115** is observed after BGP flap. [PR1450358](#)
- The CFM remote MEP is not coming up after configuration or remains in **Start** state. [PR1460555](#)
- On VXLAN VNI (multicast learning) scaling scenario, traffic issue is seen from VXLAN tunnel to Layer 2 interface. [PR1462548](#)
- Introduction of backport jemalloc profiling CLI support to all releases where jemalloc is present. [PR1463368](#)
- Either static routes or implicit filters should be configured for forwarding DNS traffic to service PIC. It solves DNS packet looping issue. [PR1468398](#)
- The following message might be seen in chassisd log after rebooting or changing the configuration, etc.: **re_tvp_builtin_fwinfo_update: Unable to get firmware version.** [PR1471938](#)
- For the MPC10E card line, the IS-IS and micro BFD sessions do not come up during baseline. [PR1474146](#)

- The SNMP index for bundle interface might become zero in Packet Forwarding Engine after restarting the FPC. This could cause the sFlow records to have either input interface value (IIF) or output interface value (OIF) as 0 value. [PR1484322](#)
- The following critical syslog error messages is observed: **[Critical] Em: Possible out of order deletion of AftNode #012#012#012 AftNode details - AftIndirect token:230791 group:0 nodeMask:0xffffffffffffff indirect:333988 hwInstall:1#012.**[PR1486158](#)
- High scale login and logout (around 1M bearers) might prevent some sessions from logging in again. [PR1489665](#)
- On MX204 and MX10003 routers with MPCs MPC7E, MPC8E, MPC9E, MPC10E, and JNP10K-LC2101, an error syslog **unable to set line-side lane config (err 30)** appears occasionally. This does not impact any service and can be ignored. [PR1492162](#)
- Scheduler ingress Packet Forwarding Engine VOQ drop counters do not match egress queue drop counters. [PR1494785](#)
- When running the **show pfe filter hw filter-name filter name** command, the command fails to retrieve the Packet Forwarding Engine programming details of the filter. [PR1495712](#)
- After backup Routing Engine halt, CB1 goes offline and comes back online; this leads to the backup Routing Engine booting up, and it shows the reboot reason as **0x1:power cycle/failure**. This issue is only for the Routing Engine reboot reason, and there is no other functional impact of this. [PR1497592](#)
- When a VLAN member is specified as a string, the **IF_MSG_IFL_VADDR** TLV is not generated with the VLAN information, and the TRIO afttristream is not updated with the **nativevlanid** and **nativevlanenable** flags. Thus, the packets are still treated as untagged, and when it reaches the trunk egress interface, it is dropped because the trunk interface does not allow untagged traffic to pass through. The issue is specific to platforms with ZT line cards. As a workaround, functionally will work if the **interface-vlan-members** statement contains only numeral value for VLANs. The VLAN members with input as a string is not supported in this release. [PR1506403](#)
- The WAN-PHY interface continuously flaps with the default hold-time down of value 0. This is not applicable to an interface with the default framing LAN-PHY. [PR1508794](#)
- BGP-SRTE binding-sid with more than one label stack needs enhancement. For routers that do not support more than one **chain-composite-next-hop**, it can configure this CLI statement to disable **chain-composite-next-hop** as a workaround. With this statement configured, all labels will be pushed from egress. [PR1512213](#)
- The log file to log the activities associated with the **request rift package activate** command is created with the permissions of the CLI user. If multiple users run the command, it might fail due to problems with permissions writing to the log file. [PR1514046](#)
- A delay of 35 seconds is added in reboot time in Junos OS Release 20.4R3 compared to Junos OS Release 19.4R2. [PR1514364](#)

- Traffic drop is observed when multicast traffic on a group with 4000 egress aggregated Ethernet ports is sent. The drop is always on the egress port that are on same Packet Forwarding Engine as ingress. The PPE times out before the multicast packet is processed and that causes the packet drop. [PR1514646](#)
- The **show configuration** command does not display the actual version information. [PR1517231](#)
- The SNMP trap of power failure might not be sent out when power cable is removed from PSU. The output of the **show chassis environment** command might not display the information of the power failure. [PR1520144](#)
- Number of TCAM entries on lo0 interface was limited to 128, which caused commit error. [PR1521171](#)
- When an AMS physical interface is configured for the first time or any member of the AMS bundle is removed or added, the PICs on which the members of AMS bundle are present go for a reboot. There is a timer running in the AMS kernel which is used as a delay for the PIC reboot to complete and once that timer expires, AMS assumes that the PICs might have been rebooted, and it moves into next step of AMS finite state machine (FSM). In scaled scenarios, this rebooting of the PIC is delayed due to DCD. This is because when a PIC goes down, DCD is supposed to delete the physical interfaces on that PIC and the PIC reboot happens. But DCD is busy processing the scaled configuration and the physical interface deletion is delayed. This delay is much greater than the timer running in AMS kernel. When the timer expires, the FSM in AMS kernel incorrectly assumes the PIC reboot would be completed by then, but the reboot is still pending. By the time DCD deletes this physical interface, the AMS bundles are already up. Because of this, there is a momentary flap of the bundles. [PR1521929](#)
- Increase the number of queues in scale configuration using the **set chassis fpc <> max-queues 768k**. [PR1522136](#)
- Rpd sensors generate core file during defer-continue case on network churn. This is a timing issue and will happen only when a particular node sensor information is being rendered and the same node went through some modification. [PR1526503](#)
- SSD core files are seen at **rtslib_iflm_update_internal**. Since SSD is not needed, disable this process using the **set system processes sdk-service disable**. [PR1527741](#)
- FIPS mode is not supported. [PR1530951](#)
- In draft-ietf-bess-srv6-services-04, it is stated that if SRv6 SID received is not in the range of locator, ingress PE should perform a reachability check for the SRv6 service SID in addition to the BGP next-hop reachability procedures. But even if rpd process does this check and then honor the SID and its next hop, egress PE might loop the packet back. So for current implementation both end SID and DT4 SID needs to be extended from the same locator. [PR1532227](#)
- Due to BRCM KBP issue route lookup might fail. Need to upgrade KBP to address this issue. [PR1533513](#)
- After performing unified ISSU on the Junos node slicing, the unified ISSU unsupported field replaceable unit (FRU) will stay offline until brought up online manually once unified ISSU finishes. This issue causes a service or traffic impact for the offline FRUs. [PR1534225](#)
- When an image with the third party SDK upgrade (6.5.x) is installed, the CPU utilization might go up by around 5 percent. [PR1534234](#)

- As we do not have an option to configure router ID for IPv6, IS-IS will pick up the router ID as minimum interface address and this is a day one behavior for IS-IS. With the current implementation, the interface with minimum interface address is not used to send the packet out, but if this interface flaps even though its not used to send the packet out, still there might be a traffic glitch since router ID is going to be changed. [PR1534430](#)
- Flap might be observed on channelized ports during ZTP when one of the ports is disabled on the supporting device. [PR1534614](#)
- On MX2010 and MX2020 platforms running in Junos OS node slicing scenario, when the base system (BSYS) has Routing Engine switchover which is followed by the addition or deletion of MPC11 line card to/from the guest network function (GNF), if the slot number in use for MPC11 line card is 8 or above, the MPC11 line card might be stuck in ready state due to this issue. [PR1535588](#)
- The error **Socket to sflowd closed** error comes up when the ukern socket to sflowd daemon (server) is closed. The error is rectified by itself as the client successfully reestablishes the connection in the subsequent attempts. When these errors are consistent, it indicates a communication issue between sflowd and the sFlow running on the FPC. [PR1538863](#)
- On an EVPN-VXLAN scenario, vmcore files are generated on master and backup Routing Engine with Layer 2 or Layer 3 multicast configuration. [PR1539259](#)
- On a scaled MX2020 router with **vrf localisation** enabled, 4 million next hop scale, and 800k route scale, FPCs might go offline on GRES. Post GRES, router continues to report many fabric related CM_ALARMS. FPC might continue to reboot and might not come online. Rebooting master and backup Routing Engine will help recovering and get the router back into a stable state. [PR1539305](#)
- When running continuous sync (show interfaces aex extensive) and async (SNMP polling) queries on an aggregated Ethernet interface in parallel, spikes in aggregated Ethernet interface framing errors counter might be observed between correct values. [PR1539537](#)
- BCMX calls are deprecated and needs to be replaced with BCM calls. [PR1541159](#)
- PTP to PTP noise transfer is passing for impairments profile **400nsp-p_1Hz**", but failing for profile **400nsp-p_0.1Hz** and lower bandwidth profiles as well. This issue is common to 10 G also. [PR1543982](#)
- A new alarm **network-service mode mismatch between configuration and kernel setting** was introduced. When unified ISSU or normal code upgrade is performed from images without new alarm commit to images with new alarm commit, then the transient false alarm will be seen. [PR1546002](#)
- 1 PPS CTE performance cannot meet class B metrics for MPC7E-1G and 10G line cards.. [PR1546219](#)
- 100 G AOC from third party does not come up after multiple reboots. It recovers after interface enable or disable. [PR1548525](#)
- The following error message is observed: **Feb 27 20:26:40 xolo fpc3 Cannot scan phys_mem_size.out. Please collect /var/log/*.out (0;0xdd3f6ea0;-1) (posix_interface_get_ram_size_info): Unknown error: -1.** This log is harmless. [PR1548677](#)
- In synce configuration, ESMC transmit is configured or if the **chassis synchronization source** configuration is deactivated or there are no active **chassis synchronization source** configurations present, it might lead

to a commit error **esmc-transmit**. To avoid the error, include the **chassis synchronization source**.
[PR1549051](#)

- The PKI CMPv2 (RFC 4210) client certificate enrolment does not properly work when using root-CA.
[PR1549954](#)
- When unified ISSU on MX Virtual Chassis is performed, error log is seen mostly because of timing issue during unified ISSU. This can cause ksyncd restart or jsr unreplications on the MX Virtual Chassis, but this will not abort unified ISSU. [PR1550492](#)
- On MX platforms, the Packet Forwarding Engine might get disabled when the major CMERROR occurs due to the parity error in the DRD memory block's SRAM. [PR1551353](#)
- When NDP entries are scaled to 32,000 over IRB in one shot, the NDP process might reach to 100 percent CPU utilization and unicast next hops for all 32,000 entries might not be present. This will result in traffic drops for entries for which unicast next hop is not present. [PR1551644](#)
- When the telemetry data for a node which is streamed is deleted during a network churn and the same node is being walked or rendered for the sensor, rpd process might generate core files. This is a corner case where the rendering and deletion of a particular node has to happen at the same instance. This issue can occur only in case of a unstable network. [PR1552816](#)
- Unified ISSU is not supported due to a major SDK upgrade from 6.3.2 to 6.5.16. Due to the upgrade, the warm reboot feature needed for unified ISSU is not supported by our vendor. [PR1554915](#)
- Phone home supports captive portal with factory default configuration. Captive portal was used to enter activation code and to monitor bootstrap status of device using phone home feature. Starting Junos OS Release 20.4, support for captive portal for phone home bootstrap process is removed. [PR1555112](#)
- On high availability systems, when FPC0 (when node0 is primary) or FPC7 (when node1 is primary) is restarted (for example, with the **request chassis fpc slot <> restart node local** CLI command or due to dcpfe core files on the primary), that might cause FPC1 or FPC8 to restart, which might cause the preexisting TCP sessions to break and might not get reestablished by itself. The TCP sessions might need to be manually reestablished. [PR1557607](#)
- With T-BC across multiple line cards, the cTE cases are failing as there are extra delays introduced during boot-up causing phase variation across the line cards. Also, boot to boot variations are different.
[PR1557636](#)
- The SyncE to PTP transient response is a stringent mask to be met with 2way-time error. The SyncE to PTP Transient response mask shall not be met for MPC7E 1G and MPC7E 10G line cards. [PR1557999](#)
- VE and CE mesh groups are default mesh groups created for a given routing instance. On adding VLAN or bridge domain, flood tokens and routes are created for both VE and CE mesh-group and flood-group. Ideally, VE mesh-group does not require a CE router where IGMP is enabled on CE interfaces. MX Series based CE boxes have unlimited capacity of tokens, so this would not be a major issue. [PR1560588](#)
- On all Junos platforms, all traffic coming from the remote end of a dynamic tunnel will not be processed and dropped when a GRE-based dynamic tunnel is configured and the tunnel preference is configured

as 1. This issue is seen because of the missing programming of de-encapsulation mode in the internal system. [PR1561721](#)

- There is traffic loss on some SRv6 flows post FPC restart. [PR1562066](#)
- This is a day one issue. When a new p2mp template (for example, template1) is added, it overwrites the default_p2mp template for the LSPs. When we update from template1 to template2, the change is not taking effect. The issue has a workaround by removing p2mp template for template1 and then readding the template2. [PR1564795](#)
- The chassisd process logs are flooded with the **pic_create_ifname: 0/0/0 pic type F050 not supported** messages for every port that is connected. This happens repeatedly in a few seconds. [PR1566440](#)
- The problem is with L1 node not reflecting correct bandwidth configured for tunnel services. When baseline has 1G configuration on some FPC or PIC in groups global chassis and if we override with local chassis tunnel service in 10 G bandwidth scaled scenario. Out of 10 Gbps bandwidth configured only 1 Gbps is allowed per 1 G speed configured in baseline configuration. [PR1568414](#)
- Traffic might be dropped on MX platforms when the default route is changed in the inet.0 table. It might take 2 to 3 seconds to be updated in Packet Forwarding Engine . This issue can be recovered automatically. [PR1568944](#)
- The Precision Time Protocol (PTP) clock might fail to be locking and be stuck in the acquiring state at clock servo. [PR1570310](#)
- BUM traffic replication over VTEP is sending out more packets than expected and there seems to be a loop. [PR1570689](#)
- PIM rib-group failures to add in VRF. PIM: ribgroup VRF is not usable in this context; all RIBs are not in instance. [PR1574497](#)
- On MX204 platforms, to use port profile configuration at port level, both PICs need to be in port mode profile or the PIC should have number of port configured as 0. [PR1575441](#)
- When the scheduler configuration is not applied to all 8 egress queues of an interface and one or more egress queues has the **buffer size remainder** configuration, the distribution of buffer to egress queues with buffer size remainder is not distributed correctly, which might lead to unexpected tail drops. [PR1575798](#)
- The known multicast traffic received over a VLAN from the core on VTEP does not get forwarded to the downstream CE interfaces. [PR1575841](#)
- An alarm raised due to a transient hardware problem with MIC does not get cleared automatically after MIC restart. [PR1576370](#)
- Max ports used is not getting displayed properly in the **show services nat pool pool name detail**. [PR1576398](#)
- When firewall is configured with both discard and port-mirror as actions in the same term, mirrored packet will be corrupted because it has two Layer 2 headers. [PR1576914](#)

- If NSR is configured, in a large-scale configuration with most of the next hops (the scale numbers are 23,000 unicast next hops, 24,000 unicast nexthops and 18,000 aggregate next hops, 24,000 transit MPLS routes, 1M IPv4 routes and 215,000 IPv6 routes), egressing out of one (physical/aggregate) port and unilists are created with ports of two FPCs, transient traffic loss is observed when remote FPC is rebooted. [PR1578635](#)
- FPC status LEDs are not turning red with power fault. [PR1579466](#)
- In a fully loaded device, the firewall programming fails at times due to a scaled prefix configuration with more than 64800 entries. [PR1581767](#)
- During reboot in certain instances, the device might get into a state where Junos virtual machine hangs until the NMI is triggered and reboots fully. The system recovers after ~30 mins. [PR1584902](#)
- Platforms using SkyATP with **security-intelligence** configured, might disconnect from the cloud after several days with the **Connection status: Request client certificate failed** error. The issue can be recovered by the **restart pki-service** CLI command. [PR1585362](#)
- Sensor statistics might not be displayed accurately for sensors with producers in multiple nodes in the **show network-agent statistics** command. It is only the statistics that are not accurate. All other information displayed in the command output is fine. There is no impact on streaming functionality. Data is streamed correctly from all nodes to the collector. [PR1590249](#)
- Currently, SyncE configurations are allowed during unified ISSU, but trigger a warning since SyncE state might not be maintained during unified ISSU. PTP configurations, however, need to be deactivated, else the unified ISSU will be aborted. [PR1592234](#)
- Port related component sensors under path **/components/component/state/** are not exported when subscribed to **/components/component/state/**. As a workaround, user can subscribe to **/components/path** to get these data exported. [PR1593031](#)
- Post unified ISSU, issue will be seen on deleting and adding back sample configuration below: **interfaces { ge-0/2/5 { flexible-vlan-tagging; encapsulation flexible-ethernet-services; gigether-options { ethernet-switch-profile { tag-protocol-id 0x0800; } } unit 1 { vlan-id 1; family inet { filter { input inet4filter; output inet4filter; } address 16.0.6.65/30; } family inet6 { filter { input inet6filter; output inet6filter; } address 2002:0000:0000:0000:0000:0000:1000:0641/126; } } }** [PR1596483](#)
- With the **shared-tunnels** statement enabled in a DCI-GW along with MAC-VRF configurations, traffic loss is observed. As a workaround, disable the **shared-tunnels** statement which act as DCI-GW role. [PR1597181](#)
- On all MX platforms, changing configuration AMS 1:1 warm-standby to load-balance or deterministic NAT might result in generating vmcore and cause traffic loss. [PR1597386](#)
- Read write lock is not acquired during the sysctl invocation. The assert triggered in the interface state function call leads to Routing Engine 1 goes to debug (db>) prompt. [PR1598814](#)
- On MX platforms using MPC10 and MPC11E line cards with IPv6 configured, duplicate address detection flags are seen for IRB interfaces. This happens when a device is configured with multiple member Layer

2 interfaces and IRB interfaces, with one or two Layer 2 interfaces going into STP blocked state. This issue can cause potential service impact on the device. [PR1601065](#)

- When PTP is on default profile & PTPoE is configured in stateful with ordinary clock-mode configuration is not supported. Below unsupported configuration does not throw commit error. There are no error logs reported with below unsupported configuration. Un-supported PTP configuration: root@accio# show protocols ptp clock-mode ordinary; stateful { interface xe-0/0/0.0 { multicast-mode { transport { ieee-802.3; } } } } Stateful port configuration for PTP over Ethernet and default profile is supported only on boundary clock mode and not on ordinary clock mode. Work around is to change the clock-mode or to remove stateful configuration. [PR1601843](#)
- With the network address translation (NAT) and endpoint-independent mapping (EIM) enabled, traffic unsupported by EIM might not be translated due to packets injected back to NAT gateway. When this issue happens, EIM unsupported traffic might be dropped. Also, the issue might cause looping at NAT gateway. In the end, looping occurred at NAT gateway affects device performance. [PR1601890](#)
- Under scaling scenarios, delayed responses from kernel (for SVTEP create) might come after the new configuration deletes the logical interface object reference from RTT. Processing of this stale response might cause a core file generation. This is a very rare scenario, and l2ald recovers automatically by a restart. [PR1602244](#)
- Convergence time degradation is seen in IS-ISv6, OSPFv2, and OSPFv3. [PR1602334](#)
- J-Flow-syslog for CGNAT is using 0x0000 in the IPv4 identification field. This might have issues for some jflow-syslog-collectors especially when jflow-syslog packets get fragmented along the path to collector. [PR1602528](#)
- On MX960 platform with MPC10 and MPC11 line cards, the field numbers of the firewall sensors from MPC10E line card might not align with other MPCs and Junos Telemetry Interface data model files. This might cause the server to be unable to parse the firewall sensors from MPC10E line card. [PR1604313](#)
- When performing downgrade on VMHost platform, the following harmless error messages might be seen when issuing the **request vmhost software add** command: **mkdir: cannot create directory '/tmp/partdisk-V6pHko/jrootfs/junos': File exists mkdir: cannot create directory '/tmp/partdisk-V6pHko/jrootfs/vm': File exists mkdir: cannot create directory '/tmp/partdisk-V6pHko/jrootfs/spare': File exists.** [PR1605915](#)
- On MX104 routers, if the SFP-T optic connected interface negotiates a speed other than 1G and is part of an aggregate interface, the interface's negotiated speed will not be shown after the interface-control (dcd) daemon restart or a Routing Engine switchover. [PR1607734](#)
- When the LLDP is used and if interfaces flap, memory leaks might be observed and it causes the l2cpd process to crash. [PR1608699](#)
- In a Layer 2 bit stream access or Layer 2 wholesale services using the **auto-sensed** VLAN subscriber interfaces over a dynamic underlying VLAN interface scenario, if the **auto-config** is used on pseudowire physical interface and the logical interface of it, the Junos OS next hop might not be properly installed for the new requested tagged VLAN after deleting the pseudowire physical interface or clearing the subscriber's sessions. Then it might not be recreated single-vlan tag for the subsequent subscribers again,

the access services on the pseudowire physical interface and the logical interface might be unavailable due to this. [PR1609844](#)

- In a subscriber management scenario, if JSU package for broadband edge subscriber management daemon (bbe-smgd) is installed on backup Routing Engine when it is syncing subscriber information from master, then the authd process and RADIUS might have stale Layer 2 Bit Stream Access (L2BSA) subscriber entries which might cause subscribers to logout and login again. [PR1610476](#)
- In a subscriber management environment, statically configured shaping rate for the parent interface set might be incorrectly adjusted by PPPoE IA tags adjustment application if the CoS adjustment based on DHCP tags and PPPoE IA tags is enabled and the first subscriber that triggered creation of the interface set is DHCP. [PR1613126](#)
- When a large number of BGP routing updates (for example, 2M BGP IPv4 routes and 500,000 BGP IPv6 routes) triggered by interface flapping are pushed to the Packet Forwarding Engine at the same time, the IGP routing updates might be delayed to program in the Packet Forwarding Engine, which might cause the sessions (for example, LDP and RSVP) that rely on IGP to flap. [PR1613160](#)
- On MX platforms in a high scaled subscriber management scenario, the system might not go GRES ready after backup Routing Engine reboot. This impacts the GRES functionality. [PR1616611](#)
- With 1 pps measurement port output and PTP or hybrid mode configured by default during bootup, clk syncd might crash and generates core files. [PR1618929](#)

High Availability (HA) and Resiliency

- If performing GRES with the interface em0 (or fxp0) disabled on the master Routing Engine, then enabling the interface on the new backup Routing Engine might result in losing network access. [PR1372087](#)
- When MTU is configured on an interface, a rare ifstate timing issue might occur at a later point resulting in ksyncd process crash on backup Routing Engine. When ksyncd process crashes on the backup Routing Engine, a live kernel core file is also generated on both the Routing Engines. There is no service impact due to this issue. [PR1606779](#)

Infrastructure

- An interface is configured for single VLAN or multiple VLANs, if all these VLANs of this interface have the **igmp-snooping** is enabled, then this interface will drop hot standby router protocol for IPv6 (HSRPv2) packets. But if some VLANs do not have the **igmp-snooping** enabled, then this interface is working fine. [PR1232403](#)
- The following messages are seen during FTP: `ftpd[14105]: bl_init: connect failed for '/var/run/blacklistd.sock' (No such file or directory)`. [PR1315605](#)

- The following error message is observed continuously in AD with base configurations: **IFDE: Null uint32 set vector, ifd and IFFPC: 'IFD Ether uint32 set' (opcode 151).** [PR1485038](#)
- While loading the kernel, user sees the **EOM: mmc0s.enh: corrupt or invalid GPT detected** error message. This message has no impact on functionality. [PR1549754](#)

Interfaces and Chassis

- The mc-ae needs to have the **prefer-status-control-active** to avoid flap on the split brain case (ICCP down or peer node reboot). Configure it on the **status-control active** PE device. [PR1505841](#)
- The CLI output for **show interfaces transport pm otn current interface** command has a formatting issue with the interval range. The correct range information is returned in the commands XML message. The information can be displayed by redirecting the command output to **display xml**. [PR1560533](#)
- The issue is seen in the scaled setup with 296 LM sessions with iterator cycle time interval (100ms). It seems there is degradation in scale number (OAM packet rate at ~5500). At this qualified PPS, now LMR packet loss is observed but the functionality seems to be fine. To avoid LMR packet loss, reduce the scale number and keep the oam packet value to less than 5500 pps. [PR1561397](#)
- On all Junos platforms, if XE/GE interface that have the **set interfaces xe/ge disable** configuration and XE/GE interface is added as a member of aggregated Ethernet bundle and the **delete interfaces xe/ge disable** command is committed, then in some rare scenario, it might generate vmcore file and reboots the system. This impact the traffic. After the vmcore file is generated, system boots up and comes to normal state. [PR1569399](#)
- With aggregated multiservices interface (AMS) configured, the memory leak on dcd daemon occurs when making configuration changes on any interface. The leak rate is slow and depends on the scale of the logical interfaces on AMS interfaces (for example, if there are 8 AMS physical interfaces with 8000 logical interfaces, the leak is about 5 MB on each commit), which might lead to a dcd crash. [PR1608281](#)
- On MX960 routers, the following syslog messages are found: **dcd[40867]: %DAEMON-5: lo0 family maximum labels is non-adjustable.** [PR1611098](#)

Juniper Extension Toolkit (JET)

- The jsd process might take some time to detect abrupt termination of the socket at the collector or client side in certain cases. This can occur when flapping the interface on which the collector is connected to the router or when a firewall terminates the client port. In such cases, the client must wait for the connection termination to be detected, which could take around 1 hour, or restart the jsd process before being able to reconnect with the same client ID. [PR1549044](#)

Layer 2 Ethernet Services

- On Junos OS devices configured as a DHCP forwarder, the jdhcp process might crash when receiving a malformed DHCP packet. [PR1430874](#)
- It is observed rarely that issuing the **request system zeroize** command does not trigger zero-touch provisioning (ZTP). A workaround is to reinitiate the ZTP. [PR1529246](#)
- The subscriber login might fail on backup Broadband Network Gateway (BNG) running Active Leasequery (ALQ) if the authentication is dependent on relay-agent-remote-id. In the issue state, the redundancy services will not be available for these DHCP clients. [PR1583445](#)

MPLS

- When we configure the **minimum-bandwidth** statement, the LSP is still resigalled with the previous **configured minimum-bandwidth** and not with the current **configured minimum-bandwidth**.. [PR1526004](#)
- The single hop BFD sessions might flap sometimes after GRES in a highly scaled setup which have RSVP link or **link-node-protection** bypass enabled. This happens because the RSVP neighbor goes down sometimes after GRES if RSVP hellos are not received before neighbor time out happens. As a result of the RSVP neighbor goes down, RSVP installs a /32 route pointing to bypass tunnel which is required to signal backup LSPs. This route is removed when all LSPs stop using bypass after the link comes back. The presence of this /32 route causes BFD to flap. [PR1541814](#)
- If IS-IS-TE or OSPF-TE is enabled, but extended admin groups (which is configured under routing-options) are configured after the peer router advertises the extended admin groups, the LSP with extended admin groups constraints might fail to be established. [PR1575060](#)
- With local reversion ON, there is a possibility of transit router not informing headend of RSVP disabled link when link flaps more than once. As a workaround, remove the **local-reversion** configuration. [PR1576979](#)
- When a protected link goes down, MPLS gets tunnel local repair message from RSVP and trigger CSPF computation. Next, MPLS gets link protection information through RRO notification. If MPLS receives TED notification before RRO notification, then CSPF computation fails. Since the link protection flag is not set, MPLS considers it as an unprotected link and brings down the LSP. [PR1598207](#)

Network Management and Monitoring

- On Junos platform with SNMP enabled, snmpd process might stop or become very slow once in a very long period of time. For example, it occurs once in 6 months or so. If snmpd process restarts, then it responds faster again. [PR1328455](#)
- On all Junos platforms, the SNMP polling might not work if the IS-IS protocol is disabled under the same Virtual Routing and Forwarding (VRF) through which the SNMP requests are sent. [PR1527251](#)
- The SNMP polling failures timeout might be observed when the number of outstanding requests to any subagent (mib2d, snmpd-subagent) reaches 500. This will impact the SNMP polling functionality. [PR1585409](#)

Platform and Infrastructure

- OSPF flapping occurs sometime during unified ISSU. [PR1371879](#)
- On MX Series platforms with MPC7, MPC8, MPC9 line card or on MX204 and MX10003, when the packets exceeds the MTU and whose DF-bit is set to go into a tunnel (such as GRE and LT), they might be dropped in the tunnel egress queue. [PR1386350](#)
- The error messages are the result of PPE commands injected from the host. One possible reason might be Layer 2 BD code, which is trying to decrement BD MAC count in the data plane. It is unlikely that there is a packet loss during this condition. This might happen during unified ISSU and this might be due to a problem with unified ISSU counter morphing used for LU-based cards, where certain counters are not disabled or disabled too late during unified ISSU. [PR1426438](#)
- Due to a software implementation, firewall filter is reapplied duration GRES. This might result in a short duration of time when the filter is not applied, provoking side effects such as drop or loss of traffic. [PR1487937](#)
- On all Junos OS platforms that support EVPN-MPLS or EVPN-VXLAN, when an existing ESI interface flaps or is newly added to the configuration, sometimes designated forwarder (DF) election happens before the local bias feature is enabled. During this time, existing broadcast, unknown unicast, and multicast (BUM) traffic might be looped for a short time duration (less than several seconds). [PR1493650](#)
- With GRES and NSR functionality with VXLAN feature, the convergence time might be slightly higher than expected for Layer 2 domain to Layer 3 VXLAN. [PR1520626](#)
- A rmopd reports false TCP errors in the log messages: **RMOPD_TWAMP_SOCKET_FAILURE setsockopt(TCP_KEEP.*) failed, error: Invalid argument**. This is a cosmetic issue. [PR1522488](#)
- When the DHCP relay mode is configured as **no-snoop**, we are observing the offer gets dropped due to incorrect ASIC programing. [PR1530160](#)
- Load balancing hash logic is changed in latest, which means for the same source IP and destination IP combination, the hash result will be of one link for releases prior to Junos OS Release 20.4 and will be a different one for Junos OS Release 20.4 and later, it will point to a different link/PIC. The overall load

will be still shared among those links, only those sessions will get moved from one PIC to another PIC. [PR1549259](#)

- On the MX platforms with XM chipset based line card is installed, when the line card experiences the `CMERRORXMCHIP_CMERROR_DDRIF_PROTECT_WR_RD_SRAM_RUNN_CHKSUM`, the disable-pfe action will be involved. This issue will cause the Packet Forwarding Engine to be disabled and traffic lost. [PR1568072](#)
- On MX platforms, FPC gets restarted and thereby disrupting traffic when there is an out-of-order filter state and its terms, this issue might be seen only in back to back GRES in more than 40 to 50 iterations. [PR1579182](#)
- When the **deactivate services rpm** and the **deactivate routing-options rpm-tracking** commands are applied together and committed, some of the rpm tracked added routes are not deleted from the routing table. The issue cannot be seen when using the following steps:
 1. Issue the **deactivate routing-options rpm-tracking** command.
 2. Commit the configuration, then all the rpm tracked routes will be deleted.
 3. If the RPM service needs to be deactivated, issue the **deactivate services rpm** and commit. [PR1597190](#)
- MS-PIC RPM probes with large data-size is failing at random. The issue on MS-PIC is not consistent and fails at random. Continuing further debugging for root cause. [PR1602508](#)

Routing Policy and Firewall Filters

- On all Junos OS platforms with the **set policy-options rtf-prefix-list** configured, if you upgrade to a specific version, the device might fail to validate its configuration, eventually causing rpd to crash unexpectedly due to a software fault. [PR1538172](#)

Routing Protocols

- While interoperating with other vendors in a draft rosen multicast VPN, by default Junos OS attaches a route target to multicast distribution tree (MDT) subsequent address family identifier (SAFI) network layer reachability information (NLRI) route advertisements. But some vendors do not support attaching route targets to the MDT-SAFI route advertisements. In this case, the MDT-SAFI route advertisement without route target extended communities are prevented from propagating if the BGP route target filtering is enabled on the device running Junos OS. [PR993870](#)
- If delegated BFD sessions flap continuously, packet buffer memory might be leaked. The automatic memory leak detection process reports this within the syslog once a certain threshold is reached, like **fpc7 SHEAF: possible leak, ID 8 (packet(clones)) (10242/128/1024)** on MX-MPC or **fpc4 SHEAF: possible leak, ID 9 (packet(clones)) (255/1/5)** on other platforms. Note that BFD sessions operating in centralized mode are not exposed. [PR1003991](#)

- Certain BGP traceoption flags (for example, open, update, and keepalive) might result in (trace) logging of debugging messages that do not fall within the specified traceoption category. which results in some unwanted BGP debug messages being logged to the BGP traceoption file. [PR1252294](#)
- LDP OSPF are in synchronization state because the IGP interface is down with **ldp-synchronization** enabled for OSPF. As per the current analysis, the IGP interface goes down because although LDP notified OSPF that LDP synchronization was achieved. OSPF is not able to take note of the LDP synchronization notification because the OSPF neighbor is not up yet. [PR1256434](#)
- In rare cases, RIP replication might fail as a result of performing NSR Routing Engine switchovers when the system is not NSR ready. [PR1310149](#)
- On MX platforms, unexpected log message will appear if the **show version detail** or **request support information** CLI command is executed: `user@router> show version detail *** messages *** Oct 12 12:11:48.406 re0 mcsnoopd: INFO: krt mode is 1 Oct 12 12:11:48.406 re0 mcsnoopd: JUNOS SYNC private vectors set.` [PR1315429](#)
- SCP command with routing instance -JU is not supported. [PR1364825](#)
- BFD session flaps during unified ISSU only in MPC7E line card. The issue is not seen frequently. [PR1453705](#)
- On all Junos platforms with scaling MVPN scenario, some PIM join or prune messages might not be processed for the first attempt. For instance, a dedicated PIM router receives more than 2500 PIM hello packets from the new neighbors, followed by PIM join packets for the same multicast group in a very short period of time. [PR1500125](#)
- Due to behavior change, if there is no IFA present in the interface, we do not encode the router ID in the hello packet by default. In current scenario between R1 and R2, we do not have any inet or inet6 address set for interfaces forming the adjacency in question. Then, in the **show isis adjacency detail** command output, we do not see IPv4 or IPv6 address and it is shown that the adjacency is missing an IP address. [PR1559079](#)
- In a Virtual Chassis or Virtual Chassis fabric scenario, inconsistent MCSNOOPD core file is seen when **igmp-snooping** configuration is removed. [PR1569436](#)
- On all Junos platforms with the **static defaults** statement configured under the **routing-options** hierarchy, if the IPv4 static route configuration is added, and then deleted, the IPv4 static route might not be removed from routing table and still forward traffic unexpectedly. [PR1599084](#)
- When performing Routing Engine switchover with NSR enabled, MSDP peering flaps for ~10 seconds and issue is seen intermittently. For instance, when first switchover is performed, issue is not seen. But on second switchover it is observed. But, this reset of MSDP peering sessions during NSR has no impact on multicast traffic flow. Multicast states are maintained properly across the switchover. [PR1616991](#)

Services Applications

- In a subscriber management environment, the **show services l2tp tunnel extensive** and the **show services l2tp session extensive** CLI commands provide incorrect outputs on LTS (L2TP tunnel switch). [PR1601886](#)

Subscriber Access Management

- There is configuration option to set slow interval to 1 min. By this approach, Packet Forwarding Engine uses the slow interim which is set to 1 min. [PR1515899](#)
- On DHCPv6 over PPPoE scenario, where the PPPoE sessions have delegated IPv6 prefix assigned from a local pool, when a DHCP session comes up over the PPPoE leading to a change to the PPPoE session's address, an address-change notification might be triggered. The processing of this notification by general-authentication-service might result in incorrect marking of the delegated IPv6 prefix as available. Once this happens, DHCP service might reassign the same prefixes which might be rejected because of IP duplication. [PR1609403](#)

User Interface and Configuration

- The auditd process might crash due to a rare timing issue. The crash is happening while reading the configuration from the database in the auditd tacplus process. There is no impact due to this issue as the auditd tacplus process gets restarted automatically after it gets terminated. [PR1226104](#)
- When a user tries to deactivate the MPLS related configuration, the commit fails on backup Routing Engine. [PR1519367](#)
- The mgd process generates core file upon simultaneous rollback command in two different terminals of same router. It is a rare and corner case and is a timing issue. If this happens, the CLI session ends abruptly. [PR1554696](#)
- The issue is seen on Virtual Chassis only which can be avoided with a simple workaround as to providing a valid package during upgrade command [PR1557628](#)
- SCP command does not work from CLI. Users can use **file copy** CLI command instead of SCP CLI command as a workaround. [PR1582916](#)
- When there is groups configured under **apply-path**, the configuration will not be expanded, and the expected feature programming of apply-path does not occur. [PR1592032](#)
- When performing commit check for the firewall and interface related configurations, if an operator uses the Ctrl+C to abort it, the dfwc and dcd might crash after performing another commit check. This issue will happen only with those daemons that follow the message-based commit check model (such as dfwc, dcd, rdmd and fwa), and has no impact on other daemons. [PR1600435](#)

VPNs

- During unified ISSU, the IPsec tunnels flap, causing a disruption of traffic. The IPsec tunnels recover automatically after the unified ISSU process is completed. [PR1416334](#)
- In some scenario (for example, configuring firewall filter), sometimes routers might show obsolete IPsec SA and NHTB entry even when the peer tear down the tunnel. [PR1432925](#)
- The MC-AE Layer 2 circuit states are not updated instantly and for some time after disabling the core interface on the MC-LAG active node, double hit in traffic is observed. To avoid the issue, configure the **set protocols ldp label-withdrawal-delay 0** in all the routers participating in l2ckt. [PR1543408](#)

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This section lists the issues fixed in Junos OS Release 20.4R3 for MX Series routers.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Resolved Issues: 20.4R3

Authentication and Access Control

- The ssh crashes on all Junos platforms. [PR1601150](#)

Class of Service (CoS)

- On MPC7E, MPC8E, and MPC9E line cards, the BPS counter of the egress queue displays the incorrect BPS value when the cell mode is configured on the static interface. [PR1568192](#)
- Child mgd processes might get stuck when multiple sessions are continuously asking for interface information. [PR1599024](#)
- Traffic loss might be observed if **per-unit-scheduler** is configured on aggregated Ethernet interface. [PR1599857](#)

EVPN

- The rpd process might crash under EVPN-VPWS environment. [PR1562160](#)
- ESI preference is not preferred when configured on lo0 interface for multicast VXLAN. [PR1570618](#)
- The multicast traffic loss might be seen in an EVPN-VXLAN scenario with CRB multicast snooping. [PR1570883](#)
- Configuring **static-mac** and **no-mac-learning** simultaneously on the VXLAN interface causes stale MAC and IP entry in the EVPN database. [PR1576147](#)
- The mustd process generates core file during upgrade or while committing a configuration. [PR1577548](#)
- The rpd process might crash if EVPN routing instances or BGP connections are flapping. [PR1581674](#)
- Multicast traffic loss might be seen in EVPN setup with IGMP snooping used. [PR1582134](#)
- After device reboot in an EVPN-VXLAN setup with graceful restart, EVPN routes are not advertised to EVPN peers until rpd is up for 180 seconds. [PR1586246](#)
- The BUM traffic might lose after triggering NSR in an EVPN-MPLS or EVPN-ETREE scenario. [PR1586402](#)
- The traffic might be dropped when an EVPN and L3VPN routes are resolved using the same MPLS-over-UDP tunnel. [PR1587204](#)
- The traffic might be dropped in an EVPN-VXLAN multihomed scenario. [PR1590128](#)
- Traffic loss might be seen under an EVPN-VxLAN scenario when MAC-IP moves from one CE interface to another. [PR1591264](#)
- Transit traffic gets dropped post disabling one of the PE-CE links on a remote multihome PE in an EVPN-MPLS AA setup with dynamic-list next hop configured. [PR1594326](#)
- EVPN might not work properly in a multi-homing setup. [PR1596723](#)
- The device announces router-mac, target, and EVPN-VXLAN community to BGP IPv4 NLRI. [PR1600653](#)

Forwarding and Sampling

- Logical interface statistics for aggregated sonet displays double value than expected. [PR1521223](#)
- User defined ARP policer is not applied on aggregated Ethernet interface until firewall process is restarted. [PR1528403](#)
- In a VXLAN scenario, the locally originated packets have UDP source port 0. [PR1571970](#)
- The pfed memory leak might be observed. [PR1573285](#)
- The l2ald process might crash on changing the routing instance. [PR1584737](#)
- The snmpwalk might not get polling the MIB for some logical interfaces. [PR1601761](#)

General Routing

- Junos OS does not provide any logging for non-SNI sessions. [PR1442391](#)
- FPC might crash after performing unified ISSU on the device which equips the type of 3D 20x 1GE MIC. [PR1480212](#)
- On MX204 routers, incorrect log message might be displayed when changing the configuration from PIC mode to port mode. [PR1500429](#)
- Transit IPv4 traffic forwarding over BGP SR-TE might not work. [PR1505592](#)
- Sometimes external 1 pps cTE is slightly above class B requirement of the ITU-T G.8273.2 specification. [PR1514066](#)
- Kernel crash might occur after NSSU while performing GRES. [PR1533874](#)
- The dcpfe process might crash and cause FPC to restart due to the traffic burst. [PR1534340](#)
- The spcd process might crash during early initialization. [PR1535536](#)
- Certain Linux based FPCs might reboot if TNP neighbor towards backup Routing Engine continuously flaps on dual Routing Engine platforms. [PR1537869](#)
- The BFD neighborship fails with the EVPN-VXLAN configuration after starting the Layer 2 learning. [PR1538600](#)
- DHCP discover packet might be dropped if the DHCP inform packet is received first. [PR1542400](#)
- Sessions creation rate is set to minimal rate after IDS and CPU throttling in place during DDoS attack. [PR1544489](#)
- The kmd process might crash when the interface flaps. [PR1544800](#)
- The kernel crash might happen if NSR is enabled. [PR1545143](#)
- The performance of Packet Forwarding Engine process on MX204 platforms might be degraded. [PR1545989](#)
- The 40 G or 100 G interfaces might flap during unified ISSU if PTP is deactivated on the interfaces on MX platforms. [PR1546704](#)

- The PTP protocol might get stuck at initializing state on MX platforms. [PR1547423](#)
- FPC crash might occur after flapping the multicast traffic. [PR1548972](#)
- Deleting or deactivating the PS interface should not be allowed when used by BBE subscriber. [PR1550915](#)
- Silent compact flash (/dev/ada1) failure might occur during reboot or startup of router. [PR1551171](#)
- The interface might not come up with 1 Gigabit optics. [PR1554098](#)
- Error logs might be observed on the platforms using next generation Routing Engine. [PR1555872](#)
- FPC with power related faults might get online again once fabric healing has turned the FPC offline. [PR1556558](#)
- The dcpfe process might crash and restart with a dcpfe core file created while running the Type5 EVPN-VXLAN with 2000 VLANs. [PR1556561](#)
- Multiple FPC crash might be seen when performing GRES or FPC reboot repeatedly in a subscriber scenario. [PR1557294](#)
- The l3static license is required though it is included in the base license. [PR1557631](#)
- The MAC addresses learned in a Virtual Chassis might fail due to aging out in the MAC scaling environment. [PR1558128](#)
- The device might run out of service post GRES or unified ISSU. [PR1558958](#)
- The subscriber management infrastructure daemon (smid) process might be stuck at 100 percent. [PR1559402](#)
- Single rate three color policer does not work. [PR1559665](#)
- Zero suppression disabling for Junos OS Release 20.4. [PR1559882](#)
- The untagged traffic routed over native VLAN might be dropped. [PR1560038](#)
- The VXLAN queue DDoS violation and RARP packets flood might happen if receiving the RARP packets more than the supported DDoS bandwidth. [PR1560243](#)
- The PIC in MX-SPC3 card might get stuck in offline status after flowd crash occurs on it. [PR1560305](#)
- Filters are not allowed on family any port-mirroring destination interface. [PR1560624](#)
- Interface cannot send or receive packets after repeated 100 G link flaps on MPC10E line cards. [PR1560772](#)
- The tunable optics SFP+-10G-T-DWDM-ZR does not work. [PR1561181](#)
- SPC3 is not supported on MX for deployment. [PR1561188](#)
- The dcpfe process might crash after deleting VXLAN configuration. [PR1562692](#)
- **LICENSE_INVALID_FEATURE_ID** syslog message is not being logged. [PR1562700](#)
- The rpd process might crash when the routing instances are deleted and recreated quickly. [PR1562905](#)
- The following error message might be seen after unified ISSU: **Turbotx process not running.** [PR1564418](#)

- Commit error observed when the **tunnel-service** is configured on a PIC without explicit bandwidth. [PR1565034](#)
- The MX150 device might reboot after performing the **request system snapshot recovery** command. [PR1565138](#)
- On the MX2010 or MX2020 routers, the following error message might be observed after switchover with GRES and NSR: **CHASSISD_IPC_FLUSH_ERROR**. [PR1565223](#)
- The KRT log file might continue to grow after removing the KRT log configuration. [PR1565425](#)
- TLB composite next hop is installed incorrectly in other routing instances. [PR1567568](#)
- MAC addresses might not be relearned successfully after MAC address age time out. [PR1567723](#)
- MAC addresses might not be installed in the EVPN MAC table due to route churn. [PR1568130](#)
- LLDP out-of-bounds read vulnerability in l2cpd. [PR1569312](#)
- Wi-fi mPIM is reaching out to NTP and DNS servers. [PR1569680](#)
- The MPLS traffic passed through the back to back PE router topology might match the incorrect CoS queue. [PR1569715](#)
- The mspmand process might crash if the packet flow control issue occurs on MS-MPC and MS-MIC. [PR1569894](#)
- The log message **/tmp//mpci_info: No such file or directory :error[1]** might be seen on VM host platform. [PR1570135](#)
- The jinsightd process might be stuck with high CPU process utilization. [PR1570526](#)
- The bbe-smgd process might crash after committing several thousand addresses in a filter term. [PR1570536](#)
- Improve deletion handling of static demux interface with active subscribers. [PR1570739](#)
- PDB pull or synchronization does not occur in new primary during unified ISSU. [PR1570841](#)
- Upgrading with unified SSDs (2x32G SSD) might result in boot loop in certain scenario. [PR1571275](#)
- Packet loss might be observed when sample based action is used in the firewall filter. [PR1571399](#)
- Packets with the MAC address of eth0 and macvlan0@eth0 interface might be sent out to the management interface on VM host platform with Next Generation Routing Engine. [PR1571753](#)
- With NSR configurations and switchover-on-routing crash statement is configured, Routing Engine switchover will be performed on an rpd crash if rpd was NSR ready before crashing. [PR1571914](#)
- The gRPC session is hanging in CLOSED state. [PR1571999](#)
- High CPU usage might occur on rpd for routes that use static subscriber. [PR1572130](#)
- DCI traffic loss of 100 percent is observed in transit spine devices. [PR1572238](#)
- The fxpc process might crash and cause traffic loss in the IFBD scenario. [PR1572305](#)

- On the MX960 routers, the **Require a Fan Tray upgrade** alarm is raised when the top fan tray 0 is removed, even though the enhanced fan tray is already used. [PR1572778](#)
- DCPFE and FPC crash might be observed if ARP MAC move happens [PR1572876](#)
- A traffic loop might be observed after the VCP interface flap. [PR1573047](#)
- The CFP unplugged message is not logged. [PR1573209](#)
- Fabric errors are observed and FPC processes might get offline when MPC3-NG or MPC3E line cards are installed along with MPC7 or MPC10 and SCBE3 or SCB4 operating in increased-bandwidth fabric mode. [PR1573360](#)
- On devices with MPC4E-3D line card shows **si5374 clock PLL lock timed out** error at boot up. [PR1573729](#)
- ARP traffic exceeding the policer limit is not discarded. [PR1573956](#)
- Only root user is allowed to execute commands on host using vhclicent. [PR1574240](#)
- QSFP 4x10G interface might not come up after FPC reboot. [PR1574279](#)
- DS-Lite throughput degradation might be seen on MS-MPC. [PR1574321](#)
- Slow FPC heap memory leak might be triggered by flapping the subscribers terminated over multiple pseudowires. [PR1574383](#)
- On MX204 and MX10003 routers, the chassisd process memory leak issue might be seen and causes traffic loss. [PR1575029](#)
- On the EA-based cards IGMP group membership is displayed incorrectly. [PR1575031](#)
- PTP might be stuck in phase acquiring state after unified ISSU. [PR1575055](#)
- The rpd process might continuously crash if you delete the **forwarding-class** policy with discard action. [PR1575177](#)
- The MPC10E line cards generates the following error message: **user.err aftd-trio: [Error] Em: root: Insert entry failed, entry:parentToken:747441 entryMask:ffffffffffffff index:52**. [PR1575310](#)
- On MX150 routers, the interface might take a long time to power down while rebooting, powering-off, halting, or upgrading. [PR1575328](#)
- The **show services service-sets statistics syslog** command returns the following error message as the service-set does not have the syslog configuration: **error: usp_ipc_client_rcv_ 1237: ipc_pipe_read fails! error:No error: 0(0), tries:1**. [PR1576044](#)
- IPsec tunnel is not established when receiving the proxy-id list. [PR1576071](#)
- The **CHASSISD_FRU_IPC_WRITE_ERROR: fru_send_msg: FRU GNF 2, errno 40, Message too long** logs might appear periodically in the chassisd logs. [PR1576173](#)
- On MX10016 routers, when the fan X failed alarm is cleared in the fan tray 1, the fan/blower OK SNMP traps are generated for the fan tray 0. [PR1576521](#)

- The LLDP neighbor information displays hex string instead of chassis ID when subtype 1 is used. [PR1576721](#)
- The MS-MPC and SPC3 might reset on receiving the subscriber traffic. [PR1576946](#)
- The following commit failure-error is observed: **Modified IFD "ae0" is in use by targeted BBE subscriber, commit denied - mtu config changed (1522), (1514).** [PR1577007](#)
- Repd core file might be seen in subscribers setup with scaled around 32,000 connections. [PR1577085](#)
- The OSPF session over IRB might not come up in the EVPN-VXLAN scenario. [PR1577183](#)
- Traffic loss might be seen when the subscriber service over aggregated Ethernet bundle interface. [PR1577289](#)
- When line card is booted on Routing Engine 1 being master, next generation statistics fails to fetch the value of backup MAC address correctly. [PR1577611](#)
- Native sensors do not work for LDP LSP and LDP p2mp sensor. [PR1577931](#)
- The bbe-smgd process might crash when the RADIUS server sends multiple CoA. [PR1578162](#)
- The DHCP ALQ is not working as expected. [PR1578543](#)
- TACACS traffic might be dropped. [PR1578579](#)
- High FPC CPU usage might be seen when signal on the link is unstable. [PR1579173](#)
- Random or silent reboot might be seen. [PR1579576](#)
- The dcpfe process might crash when any interface flaps. [PR1579736](#)
- On the MPC11E line cards, system resource monitor does not list some of the available Packet Forwarding Engines. [PR1579975](#)
- Authentication might fail if the password contains special characters. [PR1580003](#)
- On MX Virtual Chassis, gRPC-based, components, or sensor output is missing a lot of data. [PR1580120](#)
- While mapping analyzers to the channelized port, mirror might not work properly. [PR1580473](#)
- More than one subscriber on the same VLAN fails to apply same FWF template. [PR1580826](#)
- The following error is observed: **kern.ipc.maxpipekva exceeded; see tuning error.** [PR1581192](#)
- Memory leak might happen due to stale NAT64 entries. [PR1581231](#)
- Vmcore.0 file is observed at 0xffffffff80443eef in kern reboot. [PR1581260](#)
- The rpd process might crash on the new primary Routing Engine after performing graceful switchover. [PR1581878](#)
- Changing the bandwidth statement does not take effect for SNMP ifHigSpeed oid until a PSX interface is disabled and enabled. [PR1582060](#)
- The rpd process generates core file after Routing Engine switchover. [PR1582095](#)
- The voice VLAN might not get assigned to the access interface. [PR1582115](#)

- Communication between two CEs might be failed when BGP **rib-sharding** is enabled. [PR1582210](#)
- The rpd process might get stuck in 100 percent due to race condition. [PR1582226](#)
- The bbe-smgd crash might be seen after subscriber logout due to a rare timing issue on MX platforms. [PR1582356](#)
- Traffic drop might be observed on MX platforms with SPC3 in the DS-Lite scenario. [PR1582447](#)
- Destination port might be incorrectly set on MS-MPC and MS-MIC in DS-Lite scenario. [PR1582595](#)
- Node locked license addition fails. [PR1582704](#)
- Configuring or removing **hierarchical-scheduler** or **per-unit-scheduler** might cause traffic to stop forwarding. [PR1582724](#)
- Load balancing is not working correctly on AMS interfaces for CGNAT traffic on MX USF mode with SPC3. [PR1582764](#)
- The firewall filter logs are incorrectly populated with entries for protocol 8847. [PR1582780](#)
- On MX platforms, gRPC core file might be seen on a large number of telemetry subscriptions. [PR1583161](#)
- The bcmd process might crash on the MX150 platform. [PR1583281](#)
- Reset JBS, JAS, and JPS definition to align with the new license model. [PR1583438](#)
- Reset PFL and AFL definition to align with the new license model. [PR1583439](#)
- The firewall filter might not be programmed after deleting a large filter and adding a new one in a single commit. [PR1583440](#)
- SNMP SysObjectID.0 is empty with the **unified-services** enabled. [PR1583534](#)
- TCP connection to syslog server might fail to be established after adding the **tcp-log** configuration for an existing service-set. [PR1583979](#)
- Layer 2 multicast VXLAN instance is down since local VTEP logical interface is not associated to EVPN instance. [PR1584109](#)
- The jsd process leads to a high CPU usage. [PR1584357](#)
- Traffic might not get filtered properly when security-intelligence profile is configured on the MX platforms. [PR1584377](#)
- After performing NSSU, the **timeout waiting for response from fpc0** error message is seen while checking version detail. [PR1584457](#)
- The rpd process might crash due to a rare timing issue if both BGP Local RIB and adjacency-RIB-in route monitoring are enabled in BMP. [PR1584560](#)
- Bridge domain names information is not displayed properly in the **show bridge statistics instance** command output. [PR1584874](#)
- After changing configuration, the **show bridge statistics** command shows extremely larger value. [PR1584876](#)

- Traffic impact might be seen when **tunnel-services** bandwidth is configured. [PR1584969](#)
- Secure web proxy continue sending DNS query for unresolved DNS entry even after the entry is removed. [PR1585542](#)
- Traffic might be impacted due to a memory leak caused by the constantly flapping of EVPN route. [PR1586120](#)
- GRE OAM packets are sent through queue 0 with the **force-control-packets-on-transit-path** statement enabled. [PR1586169](#)
- Traffic drop is observed after enabling the **flexible-queuing-mode** on the MPC2E line cards. [PR1586403](#)
- The l2ald process might crash on changing the **routing-instance**. [PR1586516](#)
- Inter and intra VNI traffic drop might occur in spine with EVPN-VXLAN CRB configuration. [PR1586537](#)
- The rpd process generates core file if executing the **show igmp continuous stats** command after GRES. [PR1587023](#)
- The SNMP trap for MAC notifications might not be generated when an interface is added explicitly under **switch-options**. [PR1587610](#)
- The bbe-smgd process might crash if the staled ACI based subscribers are not cleaned up properly. [PR1587792](#)
- The na-grpc process crash might be seen and existing telemetry connections will be disconnected. [PR1587956](#)
- The rpd process crash might be observed on the router running a scaled setup. [PR1588439](#)
- The bbe-statsd memory leak might be observed on backup Routing Engine during subscribers login or logout. [PR1589081](#)
- The jsd process crash might be seen in a rare condition in a telemetry scenario. [PR1589103](#)
- The l2cpd process might crash. [PR1589216](#)
- The MPLS traffic might not be forwarded after the aggregate interface flap. [PR1589840](#)
- Allow default license for FBF, CFM, VRRP, QINQ, MC-LAG, TIMING, IGMP, PIM, GRE-TUNNEL, RIP, OSPF, Virtual Chassis, sFlow. [PR1589920](#)
- Traffic loss might be observed for interfaces configured in subnet /16. [PR1590040](#)
- VXLAN DDoS violation might occur when disabling the port mirror analyzer output interface. [PR1590150](#)
- Traffic loss might be observed due to FPC crash in a scaled subscriber scenario. [PR1590374](#)
- NAT service might not happen after performing AMS switchover or deactivating and activating NAT service. [PR1590890](#)
- If the CoS CR features used by VBF service is configured, MPC might crash with subscriber. [PR1591533](#)
- Frequent phydriver sync_state toggling resulting high two way time errors. [PR1591667](#)

- The **clear-ipsec-sas-for-duplicate-ts** command is not clearing secure access (SA) for duplicate traffic selectors (TS). [PR1591735](#)
- On Junos OS, xSTP might not get configured when enabled on an interface with SP style configuration on all platforms. [PR1592264](#)
- The mobile daemon might crash after switchover is performed for an AMS interface or crash occurs on service PIC where the AMS member interfaces are present. [PR1592345](#)
- The AMS warm standby with deterministic NAT functionality might not work properly. [PR1592437](#)
- Routing Engine kernel might crash due to logical interface of aggregated interface adding failure in Junos OS kernel. [PR1592456](#)
- The l2cpd-agent might go unresponsive after starting telemetry service. [PR1592473](#)
- Using the BITS interface from backup Routing Engine for clock recovery might not work. [PR1592657](#)
- The packet coming from the PS interface and forwarding to the SPC3 might be dropped. [PR1592706](#)
- Any mmcq based services might crash due to shared memory queue issue happens in a rare condition. [PR1592889](#)
- The TCP connections to the telemetry server might be stuck in CLOSE_WAIT status. [PR1593113](#)
- The TCP keepalive might not be processed by the private network host. [PR1593226](#)
- IPv6 neighbor might remain unreachable in VRRP for IPv6 scenario. [PR1593539](#)
- J-web deny log nested-application is UNKNOWN instead of specific application. [PR1593560](#)
- Fabric errors will be generated after swapping MPC10E with MPC7E line cards in the same slot. [PR1593821](#)
- The dcpfe process might crash in an EVPN-VxLAN scenario. [PR1593950](#)
- Node name should not be attached to the system hostname under LLDP. [PR1593991](#)
- Packet drop might be seen when traffic is moving from one FPC to another FPC. [PR1594244](#)
- On MX5, MX40, and MX80 routers, TEB gets stuck in present state. [PR1595107](#)
- With EVPN-VXLAN with shared-tunnel configuration, when there is BGP flap or restart of l2ald, info logs appear. [PR1595203](#)
- The interface down might be delayed after performing the **set interface *interface name* disable** command. [PR1595682](#)
- Firmware might fail to download to MIC on MX Virtual Chassis setup. [PR1595693](#)
- Packet Forwarding Engine wedge might be seen if received many IPv4 packets that need to be fragmented. [PR1596100](#)
- The l2ald process might crash on all leaves and spines after a new leaf is added to the EVPN fabric. [PR1596229](#)

- Traffic loss might happen periodically in MACsec used setup if Routing Engine is working under a pressure situation. [PR1596755](#)
- SR-TE tunnel initiated from a non-Juniper PCE might fail. [PR1596821](#)
- The bbesmgd core files are generated after Routing Engine goes down. [PR1596848](#)
- Redundancy problem with CLI CPU statistics lost data after PIC failover. [PR1596976](#)
- Major alarms on all FPCs in chassis might be seen after some time from bootup. [PR1597066](#)
- The MAC and IP withdraw route might be suppressed by rpd in the EVPN-VXLAN scenario. [PR1597391](#)
- ALG traffic might be dropped. [PR1598017](#)
- The AFEB crash might be observed with MIC-3D-8DS3-E3. [PR1598411](#)
- The packet loop might be seen after receiving the PCP request packets which are destined to software-concentrator address. [PR1598720](#)
- MX SPC3 applications for ICMP is not detected and does not allow user to modify inactivity-timeout values. [PR1599603](#)
- The configuration check would fail if more than 8 FCs are configured and CBF is enabled. [PR1600544](#)
- The multiservices card does not drop the TCP ACK packet received as a reply to the self-generated TCP keepalive. [PR1600619](#)
- A few line cards might not come up online with the **increased-bandwidth** mode. [PR1602080](#)
- Traffic drop might be observed in a Virtual Chassis scenario when the firewall filter is configured. [PR1602914](#)
- The Packet Forwarding Engine might be disabled by a detected major CMERROR event while ungracefully removing the MIC from MPC2E-3D-NG or MPC3E--3D-NG. [PR1602939](#)
- Interface hold-time up is not working on MX150 platforms. [PR1604554](#)
- The interface on MCP3-NG HQoS/MPC7E flaps continuously after enabling LACP on aggregated Ethernet interface. [PR1605446](#)
- The MPLS transit router might push an extra entropy label to the LSP. [PR1605865](#)
- The multicast streams might stop flooding in a VXLAN setup. [PR1606256](#)
- After an FPC oversubscription, new subscribers might not be able to connect. [PR1607056](#)
- The IPv6 BGP flowspec filter matching icmp-type 2 only drops the TCP traffic on source port range 512 to 767. [PR1607185](#)
- In a subscriber management scenario, under a rare condition, the Routing Engine might reboot and generates a vmcore file. [PR1607282](#)

- The following commit error is observed: **foreign file propagation (ffp) failed, reported for any type of change to dynamic-profiles**. [PR1607494](#)
- Memory might be exhausted when both the BGP RIB sharding and the BGP ORR features are enabled. [PR1613104](#)

Infrastructure

- Some MAC addresses might not be aged out. [PR1579293](#)
- The fxpc process might crash and generate core files. [PR1611480](#)

Interfaces and Chassis

- The input errors counter command on the **monitor interface** command does not work. [PR1561065](#)
- Unable to set **member-id** as Routing Engine is in synching mode forever when it has having invalid Virtual Chassis data. [PR1569556](#)
- The if-media-type is missing from interface XML output on MX Series routers. [PR1574035](#)
- There might be increase in memory for the fabspoked process. [PR1574391](#)
- On MX Virtual Chassis, unified ISSU incompatible FRU offline can result in unexpected FPC restarts after unified ISSU completes. [PR1575687](#)
- The errors are seen during GRES. [PR1575689](#)
- MC-LAG interfaces might go down if the same VRRP **group-id** is configured on multiple IRB units. [PR1575779](#)
- ARP resolution failure might occur during VRRP failover. [PR1578126](#)
- The facing configuration check-out failed with the following error message: **identical local address found on rt_inst [default], intfs**. [PR1581877](#)
- Newly added MC-LAGs do not come up after Routing Engine switchover. [PR1583547](#)
- Add configuration for PPP NCP max-failure number of retry count. [PR1584168](#)
- Incorrect advertisement threshold values are seen on VRRP groups when the **global-advertisements-threshold** is set. [PR1584499](#)
- Unable to configure pseudowire interface on an MX10003 platform in Virtual Chassis mode. [PR1587499](#)
- The dcd process crash might be seen after performing Routing Engine switchover, reboot, or management interface configuration change. [PR1587552](#)
- The dcd process might crash after removing aggregated Ethernet logical interface from the targeted distribution database. [PR1591032](#)
- Removing the configuration from interface stanza might cause the dcpfe process to crash. [PR1594356](#)
- On MX240 platforms, difference between the statistics of DMM sent and DMR received is not as expected. [PR1595780](#)

- The VRRP host cannot be reached if the **native-vlan-id** is configured. [PR1595896](#)
- The dcd process might crash and FPC might be stuck in ready state on MX platforms. [PR1601566](#)

J-Web

- J-Web allows a locally authenticated attacker to escalate their privileges to root. [PR1511853](#)
- To improve performance in Monitoring > Network > Interfaces page, admin status is removed, services and protocols data are merged into one host inbound traffic. [PR1574895](#)

Layer 2 Features

- LACP does not come up in the non-oversubscribed mode for a set of ports. [PR1563171](#)
- Traffic forwarding for VLAN 2 might not be correct when a VLAN member is removed from the ESI interface. [PR1570446](#)

Layer 2 Ethernet Services

- Aggregated Ethernet interface flap might be seen during NSSU. [PR1551925](#)
- The DHCP client will be offline for 120 seconds after sending the DHCPINFORM message in the DHCP relay scenario. [PR1575740](#)
- DHCP relay drops packets during the DHCP renewal process. [PR1576417](#)
- The jdhcpd process might crash if **relay-source lo0** is enabled in DHCP relay. [PR1580724](#)
- There is ALQ synchronization issue on master BNG and backup BNG with loss of subscriber session redundancy via PS interface. [PR1583310](#)
- The DHCP ALQ queue might get stuck causing subscriber flap. [PR1590421](#)
- The jdhcpd process might not respond to any discover message when it is in the **clients waiting to be restored** state. [PR1592552](#)
- The jdhcpd process generates core file post Junos OS upgrade. [PR1594371](#)

MPLS

- The rpd process might crash in corouted bidirectional RSVP LSP scenario. [PR1544890](#)
- Incorrect EXP bit change might be seen in certain conditions under MPLS scenario. [PR1555797](#)
- Suboptimal routing issues might be seen in case LDP route with multiple next hops. [PR1582037](#)
- Add **lsp-ping-multiplier** option support for LDP-OAM similar to RSVP-OAM. [PR1582254](#)
- MBB is not triggered when LSP is reverting back to primary path. [PR1587704](#)
- Rpd core file in backup Routing Engine at **mirror_process_recvd_data_queue** with MLDP NSR configuration. [PR1594405](#)
- The LDP replication session might not get synchronized when **dual-transport** is enabled. [PR1598174](#)
- The rpd process might crash with LSP external controller configuration. [PR1601763](#)

- VPLS connection might get down if the **dual-transport** statement is configured. [PR1601854](#)
- RSVP detour LSP might fail to come up when an LSR in the detour path goes down. [PR1603613](#)
- The rpd process might crash on standby_re LDP module when VPLS mac-flush is enabled on peer by default or by configuration. [PR1610638](#)

Multicast

- Multicast traffic in an MVPN setup might be silently dropped and discarded on some platforms acting as transit LSR. [PR1555274](#)
- FPC might crash in a multicast scenario. [PR1569957](#)

Network Address Translation (NAT)

- Services NAT mappings and sessions are incorrect while checking the SIP sessions from public to private and RTP from private to public. [PR1577922](#)

Network Management and Monitoring

- Slow memory leak could be observed for snmpd process. [PR1575790](#)
- SNMP reflects outdated ARP entries. [PR1606600](#)

Platform and Infrastructure

- Interwork failure between RPM client and TVP platforms as RPM server (and vice versa). [PR1508127](#)
- Console access on backup Virtual Chassis member is not allowed. [PR1530106](#)
- COoS queue egress interface forwarding-class might not work as expected. [PR1538286](#)
- The following major error message might cause the Packet Forwarding Engine to disable:
XQ_CMERROR_SCHED_L3_PERR_ERR. [PR1538960](#)
- The following error might be seen when adaptive **load-balancing** is configured on a LAG: **HEAP malloc(0) detected!.** [PR1547240](#)
- FPCs might go to **ISSU error** state post performing enhanced ISSU. [PR1553961](#)
- Upon receipt of specific sequences of genuine packets destined to the device, the kernel will crash and restart. [PR1557881](#)
- The BUM frame might be duplicated on an aggregate device if the extended port on the satellite device is an aggregated Ethernet interface. [PR1560788](#)
- Multicast traffic with incorrect source MAC address might be observed from IRB interface. [PR1561313](#)
- Traffic loss might be observed due to FPC crash on MX platforms. [PR1563144](#)
- Last flapped timestamp for interface fxp0 gets reset every time when the **monitor traffic interface fxp0** is executed. [PR1564323](#)
- The L2TP tunnel might not work with filter-based encapsulation. [PR1568324](#)

- On MX series routers, hosting subscribers error logs might be seen. [PR1570631](#)
- FPCs might crash randomly while deleting the interface-set in the system. [PR1571192](#)
- Platforms with EVPN-VXLAN configured, the next hop memory leak in MX Series ASIC happens whenever there is a route churn for remote MAC-IP entries learned bound to the IRB interface in EVPN-VXLAN routing instance. When the ASIC's next hop memory partition is exhausted, the FPC might reboot. [PR1571439](#)
- Scale subscriber license might be not updated properly on the backup Routing Engine which leads to **License grace period for feature scale-subscriber(44) is about to expire** alarm after GRES. [PR1573289](#)
- FPC reports following error log message: **cassxr_err_addr(8593): Uninitialized Read Error @ EDMEM[0x7cb601b0]**. [PR1573920](#)
- Memory partitioning issue might happen on Packet Forwarding Engine after applying sampling and **flex-flow-sizing** to the MX Series with MPCs/MICs line cards. [PR1575994](#)
- On all Junos platforms with private edit mode enabled, if you commit the **source-address address routing-instance** and then delete **source-address routing-instance**, the commit fails with a warning message: **warning: patch removes statement that is not empty**. [PR1582529](#)
- VRRP device originally taking backup role might cause destination IP unreachable after VRRP mastership switchover. [PR1584115](#)
- FPC might crash in a scaled-firewall configuration. [PR1586817](#)
- The traffic might not failover with the **shared-bandwidth-policer** enabled on aggregated Ethernet. [PR1588708](#)
- The audit core file is generated while changing TACACs and login user passwords. [PR1589953](#)
- Junos OS: Upon receipt of specific sequences of genuine packets destined to the device, the kernel will crash and restart. [PR1595649](#)
- VLAN tagged traffic might be dropped with service provider style configuration. [PR1598251](#)
- The service filter might get incorrectly programmed in the Packet Forwarding Engine due to a rare timing issue in enhanced subscriber management environment. [PR1598830](#)
- When a local PE device has to terminate two tunnels from same remote PE device but on different VRFs, FPC0 core files might be seen during deletion of decap prefix. [PR1600030](#)
- The kernel core file might be seen if restarting BGP connections after deleting BGP authentication. [PR1601492](#)
- The ZTP does not work when downgrade the image. [PR1603227](#)
- Multicast traffic is dropped when forwarded over VPLS via IRB. [PR1607311](#)

Routing Policy and Firewall Filters

- The DNS name cannot be resolved if customer-defined routing instance is configured under name-server. [PR1539980](#)

- The rpd process might crash when the deletion of routing table occurs. [PR1565629](#)
- The bbe-smgd fails when reading the configuration for address mask prefix-length configured in a policy statement. [PR1583535](#)

Routing Protocols

- The route validation states might flip between VALID, INVALID, and UNKNOWN in some corner case. [PR1556656](#)
- Incorrect active, received, accepted counters might be seen in the output of the **show bgp summary**. [PR1558678](#)
- There might be traffic loss when GRE interface flaps. [PR1566428](#)
- The rpd memory leak might be observed during CLI or ephemeral commits in OSPFv2 scenario. [PR1568157](#)
- After the first parallel ISSU, subsequent ISSU aborts with **Aborting Daemon Prepare** error message. [PR1572265](#)
- The BFD session of DHCP subscriber does not come up on the MPC2E line card and gets stuck in the Down state. [PR1572577](#)
- The DHCP packets might be dropped in the static VXLAN scenario. [PR1576168](#)
- The ppmmd might crash when enabling MD5 authentication on OSPF with BFD flapping. [PR1576893](#)
- BGP session flap might be observed after the Routing Engine switchovers when the VRRP virtual address is used as the local address for the BGP session. [PR1576959](#)
- Multicast traffic loss might be observed as logical PIM de-encapsulation interface is not created as expected. [PR1577461](#)
- The rpd process might crash when two or more routing instances are deleted in one shot. [PR1578740](#)
- The BGP signaled dynamic tunnels might be still up after deactivating the BGP export policy from the remote peer. [PR1579225](#)
- Short multicast packets drop using PIM when multicast traffic received at a non-RPT or non-SPT interface. [PR1579452](#)
- BGP session carrying VPNv4 prefix with IPv6 next hop might be dropped. [PR1580578](#)
- BGP replication might be stuck in rare and timing conditions. [PR1581578](#)
- The rpd process might crash in BGP and MPLS scenario. [PR1581794](#)
- With IGMP snooping implemented, there is unexpected jitter issue that could cause traffic loss. [PR1583207](#)
- The rpd process crash might be seen in certain IS-IS scenario. [PR1583484](#)
- On rare occasion, rpd core file might be observed on backup Routing Engine after loading a new image. [PR1583630](#)

- Origin validation replication status shows up in the **show task replication** command output even when not configured. [PR1583692](#)
- The rpd process might crash in BGP multipath scenario if the single hop EBGP peer goes down. [PR1585265](#)
- The rpd process crash might be seen when BGP RPKI session **record-lifetime** is configured to be less than the hold-time. [PR1585321](#)
- Traffic drop might be occurred on link flap when IS-IS is configured. [PR1585471](#)
- The rpd process might crash after committing with the configured static group. [PR1586631](#)
- Incorrect BGP next hop advertisement in the Layer 3 VPN scenario. [PR1587879](#)
- The multicast traffic loss might be observed after unified ISSU is being performed. [PR1588555](#)
- The rpd process might crash in a scaled routing instances scenario. [PR1590638](#)
- The rpd process might crash post GRES. [PR1590912](#)
- PIM joins might not be synchronized between master and backup Routing Engines because of pcmd restart. [PR1591685](#)
- If BGP disable or enable in a short time interval on a scaled NSR router might result in backup rpd process restart. [PR1591717](#)
- The rpd process crash might be seen if BGP peer flaps. [PR1592123](#)
- The remote loop-free-alternate (LFA) backup path might not be formed. [PR1592424](#)
- The traffic might get silently dropped and discarded or forwarded through not-best path in BGP setup. [PR1592550](#)
- The routing process might crash due to memory corruption while processing BGP multipath route. [PR1594626](#)
- Some routes might get incorrectly programmed in the forwarding table in the kernel with the next- hop installed as dead. [PR1601163](#)
- The rpd process might be stuck at 100 percent in OSPFv3 scenario. [PR1601187](#)
- After changing MTU on interface BGP routes that resolve over IS-IS will be installed in kernel as dead and traffic will drop. [PR1605376](#)
- On MPC10E line cards, the rpd process generates core files after deactivate and activate interfaces: **rt_table_flash_job_cancel,rt_instance_set_lsi_ifl_data_shard,rt_flash_all_internal**. [PR1605620](#)
- The multicast traffic might be duplicated on a subscriber interface. [PR1607493](#)
- With **rib-sharding** enabled, any commit will flap all BGP sessions with 4 byte peer-as (AS number 65536 or greater). [PR1607777](#)
- The rpd process might generate core files if TI-LFA or MLA feature ends up having more than 5 SIDs in the SRVv6 SIDs stack. [PR1613384](#)

Services Applications

- The CoA with LI-on or LI-off message might be dropped during CoA process. [PR1554618](#)
- IWF AVP value might not be reflected properly on LTS. [PR1581096](#)
- On MX480 routers, vmcore might be seen due to doadump (textdump=1) at `/volume/build/junos/occam/llvm-5.0/sandbox-204ab-20210401/freebsd/stable_11-204ab/20210401.22`. [PR1595088](#)
- The **show services l2tp tunnel extensive**, **show services l2tp session extensive**, and **show subscribers accounting-statistics** commands do not work on LTS. [PR1596972](#)

Subscriber Access Management

- CoA request might not be processed correctly from time to time. [PR1571501](#)
- Subscribers might be stuck in terminated state when the RADIUS server is unreachable. [PR1600655](#)
- The **Service session entry creation failed** errors are seen during ephemeral commit. [PR1603030](#)
- The DHCP session fails with the **session-limit-per-username** statement configuration. [PR1612196](#)

User Interface and Configuration

- NTP might be stuck in AUTH state if authentication-key in a hexadecimal format configured by **load** command. [PR1486498](#)
- Removing the flash component from Monitor > Interfaces and DHCP pages, removes the other flash pages. [PR1553176](#)
- On MX10008 platforms, smaller database size when compared to other platforms such as MX10003 and MX960. [PR1572768](#)
- The mustd process might crash with multiple core files due to memory issue. [PR1599641](#)

Virtual Chassis

- Virtual Chassis port might not come up after upgrade when QSFP+-40G-SR4, QSFP+-40G-LR4, or QSFP+40GE-LX4 is used. [PR1579430](#)

VPNs

- The rpd process might crash when link-protection is added or deleted from LSP for MVPN ingress replication selective provider tunnel. [PR1469028](#)
- The iked process might crash when IKEv2 negotiation fails on MX devices. [PR1577484](#)
- The rpd process might crash in the NG-MVPN scenario on all Junos platforms. [PR1579963](#)
- The traffic of the draft rosen multicast VPN might lose after switching over the Routing Engines. [PR1584720](#)
- The rpd core is found at `tag_dpvc_create_p2p_pvc,ir_ucast_tnl_p2p_init_new,ir_p2p_ucast_tnl_init,ir_ucast_tnl_add,mvpn_p_tnl_ir_`

leaf_add,mvpn_p_tnl_ir_sel_leaf_add,mvpn_spmsi_leaf_add_del,mvpn_spmsi_replay_leaf_ad_upds etc. [PR1588486](#)

- The DDoS protection reason **packets failed the multicast RPF check** might be seen in NG-MVPN scenario with GRE transport. [PR1591228](#)
- The rpd process might crash if the interface goes down in the BGP-MVPN scenario. [PR1597387](#)

Resolved Issues: 20.4R2

General Routing

- Need to be able to show which shard a given route is hashed to. [PR1430460](#)
- The MPC2E-NG or MPC3E-NG line card with specific MIC might crash after a high rate of interface flaps. [PR1463859](#)
- The following line card errors are seen: **HALP-trinity_nh_dynamic_mcast_add_irb_topo:3520 snooping-error: invlaid IRB topo/ IRB ifl zero in l2 nh 40495 add IRB**. [PR1472222](#)
- Subscribing to **/linecard/packet/usage** and triggering the UDP decoder: the hardware statistics are exported with improper hierarchy. [PR1485739](#)
- Aggregate Ethernet interfaces do not display member links' statistics. [PR1505596](#)
- MX150 router might go into debug prompt mode after software upgrade or downgrade. [PR1510892](#)
- On MX960 routers, the **show interfaces redundancy rlt0** statement shows current status as primary down as FPC is still in the Ready state after RLT failover (restart FPC). [PR1518543](#)
- No response from the other Routing Engine for the last 2 seconds triggers **SNMP trap generated: Fru Offline** messages. [PR1524390](#)
- Problem with static VLAN deletion with active subscribers and the FPC might be stuck at Ready state during restart. [PR1525036](#)
- The aggregated Ethernet interface might not come up with LFM configured after reboot. [PR1526283](#)
- L2TP subscribers might fail to establish a session on MX if the CPE is a virtual host. [PR1527343](#)
- Removing superflous XML tags within syslog strings. [PR1528116](#)
- Error message **FAILED(-1) read of SFP eeprom for port** might be seen. [PR1529939](#)
- After performing unified ISSU with a high-scale bridge-domain configuration, less than 0.0254 percent of traffic loss is observed for a single bridge-domain interface. [PR1531051](#)
- Configuring **no-flow-control** statement under gigether options does not work on MX150 router. [PR1531983](#)
- Need support for **show chassis fabric summary** output. [PR1532163](#)
- Wavelength unlocked alarm is set as On while using the SFP+-10G-T-DWDM-ZR optics. [PR1532593](#)

- The dcpfe process might crash and cause FPC to restart due to the traffic burst. [PR1534340](#)
- Some routes might get incorrectly programmed in the forwarding table in the kernel which is no longer present in rpd. [PR1534455](#)
- CFM sessions go down during FRU upgrade stage of unified ISSU in MX Virtual Chassis. [PR1534628](#)
- Multiple vmxt processes might generate core files. [PR1534641](#)
- All SFBs might go offline due to fabric failure and fabric self-ping probes performing the disable-pfe action. [PR1535787](#)
- Mixed primary and backup Routing Engine types alarm reported on MX240 with NG-Routing Engine. [PR1536184](#)
- Enhancements are needed for debugging l2ald. [PR1536530](#)
- The chassisd memory leak might cause traffic loss. [PR1537194](#)
- The error message **vmhost-platform-grub-install.sh: line 140: [: ==: unary operator expected** might be seen during upgrade on VM host platform. [PR1537980](#)
- NGMPC2 core files are seen at **bv_entry_active_here::bv_vector_op::gmph_reevaluate_group::gmph_destroy_client_group**. [PR1537846](#)
- On AFT-based MX Series with FPCs (MPC10 and MPC11) Packet Forwarding Engine, the **show jnh exceptions inst inst-number** command might cause FPC to crash. [PR1538138](#)
- The BFD sessions might not come up in a VXLAN scenario. [PR1538600](#)
- The rpd memory leak might be observed on the backup Routing Engine due to link flaps. [PR1539601](#)
- FPC might not be recognized after power cycle (hard reboot). [PR1540107](#)
- The dcpfe process might crash in the specific MAC move cases and traffic loss might be observed in the EVPN-VXLAN scenario. [PR1542709](#)
- The JNH memory leak could be observed on MX Series with MPC/MIC based line cards. [PR1542882](#)
- Continuous rpd errors might be seen and new routes will fail to be programmed by rpd. [PR1545463](#)
- **jnxDomAlarmSet** and **jnxDomAlarmClear** traps are getting generated at 15 minutes intervals after a link on the transceivers support DOM becomes up or down. [PR1545514](#)
- FPC might not boot-up on MX960 routers in certain condition. [PR1545838](#)
- OSPFv3 session might keep flapping and OSPFv3 hellos might be dropped in the host-path. [PR1547032](#)
- Backup Routing Engine vmcore might be seen due to absence of NH ACK Infra. [PR1547164](#)
- PTP backup device might discard the PTP packets from primary when MPLS **explicit-null** is configured. [PR1547901](#)
- SR-TE might stay up when the routes are deleted through policy. [PR1547933](#)
- Validation of OCSP certificate might not go through in case of certain CA servers. [PR1548268](#)

- The adapted sample rate might be reset to the configured sample rate without changing the sampling rate information in sFlow datagrams after enabling sFlow technology on a new interface. [PR1550603](#)
- The rpd process crash might be seen when BGP service route is resolved over color-only SR-TE policy. [PR1550736](#)
- The PPPoE subscribers might fail to login. [PR1551207](#)
- The IRB interface might not work after chassisd and l2ald reboot in an EVPN scenario. [PR1551631](#)
- The LCM peer absent alarm might be seen on TVP-based platforms. [PR1551760](#)
- The **action-shutdown** statement of storm control does not work for ARP broadcast packets. [PR1552815](#)
- The fabric errors are observed and the FPC processes might go offline with SCBE3, MPC3E-NG, MPC3E, MPC7, and MPC10 in the increased-bandwidth fabric mode. [PR1553641](#)
- The following error messages are observed: **Disable-pfe with intermittent ipc_pipe_get_packet(): packet_get() failed error message and CM_CMERROR_FABRIC_SELFPING failure.** [PR1554209](#)
- The following errors are seen when interpolate is configured under drop profile: **CoS WRED Curve: Create Expr Curve: No curve data points!!.** [PR1554220](#)
- Medium-high and medium-low loss-priority actions are not supported under firewall policer. [PR1554362](#)
- During ISSU, BNG loses subscriber sessions without sending session stop but stays in authd. [PR1554539](#)
- The chassisd might crash with repeated configuration commits on MX204 and MX10003 platforms. [PR1555271](#)
- MACsec session might remain down after CA applied or reapplied to different interfaces. [PR1555736](#)
- Configuring HFRR, that is, link protection on an interface might cause rpd to crash. [PR1555866](#)
- Chassisd SNMP trap FRU Offline not getting generated on MPC11E line card due to no power. [PR1556090](#)
- The dcpfe might crash and restart with a dcpfe core file created while running Type 5 EVPN VXLAN with 2000 VLANs. [PR1556561](#)
- The framed route installed for a demux interface has no MAC address. [PR1556980](#)
- Script fails while committing the IPsec authentication configuration as the algorithm statement is missing. [PR1557216](#)
- ISSU will be aborted and chassisd core might be seen on backup Routing Engine during Junos OS upgrade in MX Series devices. [PR1557413](#)
- Packet corruption on 100G/40G interface is configured with PTP. [PR1557758](#)
- Application identity unknown packet capture utility does not function when **enhanced-services** mode is enabled. [PR1558812](#)
- Rpd core files are seen after Routing Engine switchover. [PR1558814](#)

- On MX150 routers, the following license error is observed continuously:
`licinfra_set_usage_nextgen_async:1733] Invalid input parameters.` [PR1559361](#)
- Three-color policer might not work. [PR1559665](#)
- On MX960 routers, mismatch is found between YANG schema and RPC output. [PR1559810](#)
- When the system has only one plane (in the process of plane offline/online), the MPC10-10C line card shows destination errors. [PR1560053](#)
- The **request system software validate** command might corrupt installation of Junos OS openconfig package. [PR1560234](#)
- The PIC in SRX5K-SPC3/MX-SPC3 card might get stuck in offline status after flowd process crash occurs on it. [PR1560305](#)
- On the MX240 router, the following error message is observed: **On R0 Overlay Ping FAILS tunnel-src 10.255.0.53 tunnel-dst 10.255.0.139 vni 1, invalid VNI: '1'.** [PR1560408](#)
- The l2cpd core file might be seen on reboot. [PR1561235](#)
- On MX240 routers, VIA headers are not getting changed properly when SIP ALG is enabled. [PR1561312](#)
- The following error is Observed opening configuration database: **Could not open configuration database during usb upgrading.** [PR1561741](#)
- After recovering from restart routing immediately, object-info anomalies are observed on rpdagent. [PR1561812](#)
- The rpd process crash might be observed during processing huge amount of PIM prune messages. [PR1561984](#)
- Commit issue is seen after loading limited-signed image through USB. [PR1562723](#)
- The rpd process might crash when routing instances are deleted and recreated quickly. [PR1562905](#)
- Client authentication is failing after performing GRES. [PR1563431](#)
- MX Series platforms with MX-SCBE3 might reboot continuously. [PR1564539](#)
- The PPPoE service-name-tables does not correctly count active sessions matching agent-specifier ACI/ARI used for delay. [PR1565258](#)
- Pushing more than two MPLS labels might not work. [PR1566828](#)
- Need to allow tunnel interface as the peer-address for ALQ. [PR1567735](#)
- The MX204 router FPC might show high CPU utilization because the JGCI background thread runs for a long period. [PR1567797](#)
- On MX150 routers, the **request system software add** CLI command is disabled. [PR1568273](#)
- Core files are generated at `export_svc_set_nat_idl@nsd_calloc` while verifying no-translation with destination-nat. [PR1568997](#)
- The rpd process might crash while using BFD API to bring up BFD sessions. [PR1569040](#)

- Traffic loss might be observed when SCU accounting is configured and **logical-systems** is enabled. [PR1569047](#)
- The agent sensor **__default_fabric_sensor__** seems to be partly applied to some FPCs, which causes zero payload issue **AGENTD received empty payload for pfe sensor __default_fabric_sensor__**. [PR1569167](#)
- The MPLS traffic passed through the back-to-back PE topology might match the incorrect CoS queue. [PR1569715](#)
- OAM might not work as expected after FPC reboot or flapping. [PR1569790](#)
- The mspmand process might crash if the packet flow-control issue happens on MS-MPC/MS-MIC. [PR1569894](#)
- The log message **/tmp//mpci_info: No such file or directory :error[1]** might be seen on VM host platform. [PR1570135](#)
- The following na-grpcd.re core file is observed: **net::juniper::appSupport::BufferStream::Byteliterator,net::juniper::sysmanmsg::sm_node_data::deserializeInternal,net::juniper::sysmanmsg::sm_node_data::deserialize (itr=...) at src/sysman_msg.cpp:576**. [PR1572107](#)
- The TFEB/FPC might fail to be online after rebooting the system or the FPC if **interface-set** is configured for CoS. [PR1572348](#)
- Fabric errors are observed and FPC processes might go offline when MPC3-NG, MPC3E, and SRX5K-IOC2 line cards are installed along with MPC7, MPC10, SRX5K-IOC04, and SCBE3/SCB4 line cards operating in increased-bandwidth fabric mode. [PR1573360](#)
- The rpd process on the transit node might crash when performing MPLS traceroute on the ingress node. [PR1573517](#)
- Slow FPC heap memory leak might be triggered by flapping subscribers terminated over multiple pseudowires. [PR1574383](#)
- Interfaces might take a long time to powerdown while performing reboot/power-off/halt/upgrade on MX150 routers. [PR1575328](#)
- The **show services service-sets statistics syslog** command returns an error **usp_ipc_client_rcv_ 1237: ipc_pipe_read fails! error:No error: 0(0), tries:1** when service-set does not have syslog configuration. [PR1576044](#)
- When **Fan Tray 1 Fan <x> Failed** alarm is cleared, **Fan/Blower OK** SNMP traps are generated for Fan Tray 0 [Fan 31 - 41] and Fan Tray 1 [Fan 11 - 41]. [PR1576521](#)
- The following commit failure-error is observed: **Modified IFD "ae0" is in use by targeted BBE subscriber, commit denied - mtu config changed (1522), (1514)**. [PR1577007](#)
- Issue is seen in telemetry when the **set services analytics streaming-server** configuration is present, and the server is not reachable. [PR1581192](#)
- IPv4 traffic loss with packet size more than 1410. [PR1584509](#)

Class of Service (CoS)

- The explicit classifier or rewrite-rule might not work as expected for a logical interface if the wildcard configuration is also applied. [PR1556103](#)
- On the MPC7E line card, bps counter of egress queue shows wrong bps value when cell mode is configured on static interface. [PR1568192](#)

EVPN

- The rpd process memory leak might occur when changing EVPN configuration. [PR1540788](#)
- The rpd process crash might be seen after adding route-target on a dual Routing Engine system under EVPN multihoming scenario. [PR1546992](#)
- The l2ald process might crash under VLAN-based EVPN-VXLAN scenario. [PR1550109](#)
- ARP replies from EVPN CE might get dropped incorrectly on MX Series platforms if the EVPN routes are resolved via MPLS-over-UDP tunnels. [PR1563802](#)
- The mustd process crash can be seen during upgrade or while committing a configuration. [PR1577548](#)

Forwarding and Sampling

- After routing restarts, the remote mask that the routing daemon sends might be different from the existing remote mask that the Layer 2 learning daemon had before restart. [PR1452990](#)
- MAC learning issue might happen when EVPN-VXLAN is enabled. [PR1546631](#)
- All traffic might be dropped on an aggregated Ethernet bundle without VLAN configuration if bandwidth-percent policer is configured. [PR1547184](#)
- The l2ald process might crash due to next-hop issue in EVPN-MPLS. [PR1548124](#)
- Configuration archive transfer-on-commit fails. [PR1563641](#)

General Routing

- The DHCP relay-reply packets are dropped in a DHCPv6 relay scenario. [PR1352613](#)
- DHCP discover packet might be dropped if DHCP inform packet is received first. [PR1542400](#)
- The **show dynamic-profile session client-id** command displays only one IPv6 framed-route information. [PR1555476](#)

Infrastructure

- On Virtual Chassis and Virtual Chassis fabric, the following error message is observed: **HEAP malloc(0) detected**. [PR1546036](#)
- Ethtraceroute local privilege escalation vulnerabilities in SUID binaries. [PR1529209](#)
- In a multi-tenant environment, a tenant host administrator might configure logical firewall isolation affecting other tenant networks. [PR1537491](#)

Interfaces and Chassis

- The **startup-silent-period** command might not work. [PR1548464](#)
- The VCP port is marked as administratively down on the incorrect MX Series Virtual Chassis member. [PR1552588](#)
- The dcd process might leak memory on pushing the configuration to the ephemeral database. [PR1553148](#)
- An lacpd core is observed after router reboot. [PR1553196](#)
- Block duplicate IP across different logical interfaces inside same routing instance. [PR1555861](#)
- Sessions might flap after applying the Action profile on the router. [PR1561044](#)
- The ppmmd process might crash when VRRP is configured on all Junos OS platforms. [PR1561281](#)
- MAC address entry issue might be seen after MC-LAG interface failover. [PR1562535](#)
- Traffic loss issue might be seen while verifying VRRP state machine functionality. [PR1564551](#)
- Unused memory allocation removed from CCL toolkit. [PR1574391](#)
- MX Series Virtual Chassis ISSU incompatible FRU offline can result in unexpected FPC restarts after ISSU completion. [PR1575687](#)
- The following multiple errors **VRRPMAN_PATRICIA_GROUP_ADD_FAIL: vrrp_ifcm_send_bulk: Failed to add group to patricia tree key,VRRPMAN_ENTRY_KEY_PRESENT: vrrp_ifcm_send_bulk: Already an entry present with the key** are found during GRES. [PR1575689](#)
- MC-AE interfaces might go down if same VRRP group-id is configured on multiple IRB units. [PR1575779](#)

Juniper Extension Toolkit (JET)

- TCP connection might not be established while creating default gRPC channel with name **fw_channel**. [PR1559064](#)

Layer 2 Ethernet Services

- The copying of files to the RCB over WAN ports is slow. [PR1496895](#)
- Receipt of a crafted DHCP packet will cause the jdhcpd DHCP service to generate core file. [PR1534814](#)
- OSPF and OSPFv3 adjacency uptime is more than expected after NSSU upgrade and outage is higher than expected. [PR1551925](#)
- DHCP packet drop might be seen when the DHCP relay is configured on leaf device. [PR1554992](#)
- Receipt of malformed DHCPv6 packets causes jdhcpd to crash and restart. [PR1564434](#)
- `jnxJdhcpLocalServerMacAddress (.1.3.6.1.4.1.2636.3.61.61.1.4.3)` returns incorrect format of MAC address. [PR1565540](#)
- The option 82 information might be incorrectly cleared by DHCP relay agent. [PR1568344](#)
- The jdhcpd process might crash in a subscriber management scenario with DHCPv6 subscribers. [PR1568940](#)

MPLS

- Traffic loss might be observed due to rpd process crash in MPLS scenario. [PR1528460](#)
- The RSVP graceful-restart might fail. [PR1533161](#)
- A new LSP might not be up even if bypass LSP is up and **setup-protection** is configured. [PR1555774](#)
- Unexpected LSP packet count is observed for the ingress MPLS LSP statistics. [PR1570382](#)

Network Management and Monitoring

- Commit error while deleting the routing instance when SNMP trap-group also has the same routing instance referred. [PR1555563](#)
- The mib2d process crashes and generates a core file on backup Routing Engine. [PR1557384](#)
- After the l2cpd service is restarted, the context of registration from l2cpd to snmpd was failing due to incorrect reinitialization. [PR1561736](#)
- SSH connection might become unresponsive and logs show **kern.maxfiles limit exceeded by uid** messages. [PR1567634](#)

Platform and Infrastructure

- The state of the flow detection configuration might not be displayed properly if DDOS-SCFD is configured globally. [PR1519887](#)
- The following error message is observed when alarms after interface reset: **7836 ifl 567 chan_index 8 NOENT & jnh_ifl_topo_handler_pfe(13015): ifl=567 err=1 updating channel table nexthop**. [PR1525824](#)
- PPE errors or traps might be observed in Layer 2 flooding scenarios. [PR1533767](#)
- MPC might crash in the subscriber management environment. [PR1534542](#)
- CoS queue egress interface forwarding-class might not work as expected. [PR1538286](#)
- Core files are found at **I2_metro_bd_host_inject_del bd_platform_delete bd_handle_msg**. [PR1538516](#)
- Major error **XQ_CMERROR_SCHED_L3_PERR_ERR** might cause Packet Forwarding Engine to disable. [PR1538960](#)
- In rare occurrence Routing Engine kernel might crash while handling TCP sessions if GRES and NSR are enabled. [PR1546615](#)
- An internal timer on backup Routing Engine might cause an ARP storm upon GRES switchover on new primary (old backup) Routing Engine. [PR1547583](#)
- The kernel might crash if performing GRES in either new iteration or after swapping Routing Engine and restoring HA configuration. [PR1549656](#)
- The BGP session replication might fail to start after the session crashes on the backup Routing Engine. [PR1552603](#)
- Traffic is not forwarded over IRB to Layer 2 circuit on -lt interfaces. [PR1554908](#)

- IPv4 EXP rewrite might not work properly when inet6-vpn is enabled. [PR1559018](#)
- The BUM frame might be duplicated on the aggregate device if extended-port on satellite device is aggregated Ethernet. [PR1560788](#)
- DHCPv4 request packets might be wrongly dropped when DDoS attack happens. [PR1562474](#)
- The **enforce-strict-scale-limit-license** configuration enforces subscriber license incorrectly in the ESSM subscriber scenario. [PR1563975](#)
- The pfe might crash when soft error recovery feature is enabled on Packet Forwarding Engine. [PR1567515](#)
- The following error message is observed: **pfe err-jnh_physmem_add_resvd_to_cntr(18014): PFE 0 jnh_app 0x08020860, add 0x00080000 from 0x00b00000-0x00b80000 to baMask 0x1.** [PR1570631](#)

Routing Policy and Firewall Filters

- The policy configuration might be mismatched between the rpd and mgd processes when deactivating policy-options prefix-list is involved in the configuration sequence. [PR1523891](#)
- Global variable policy_db_type not set to correct value on failure. [PR1561931](#)
- Generate route goes to hidden state when protect core statement is enabled. [PR1562867](#)

Routing Protocols

- The BFD session might get stuck in the Init or Down state after the BFD session flaps. [PR1474521](#)
- Traffic might be silently discarded when the BGP route gets deleted which is part of multipath. [PR1514966](#)
- The rpd process crashes when a fresh router is configured with IS-IS and RIB-group to leak inet.3 routes from no-forwarding to master instance in single commit. [PR1534486](#)
- Traffic loss might be seen in next-hop-based dynamic tunnels of L3VPN scenario after changing the dynamic-tunnel preference. [PR1542123](#)
- With BGP rib-sharding enabled, rpd memory exhaustion might be seen. [PR1546347](#)
- The BGP session neighbor shutdown configuration does not affect the non-established peer. [PR1554569](#)
- The changes do not get effective when the values are set under static default hierarchy. [PR1555187](#)
- The BGP session might not come up if **extended-nexthop** is enabled by default on the other vendor remote peer. [PR1555288](#)
- Sending multicast traffic to downstream receiver on MX Series with MPCs/MICs based Virtual Chassis platforms might fail. [PR1555518](#)
- 6PE prefixes might not be removed from RIB upon reception of withdrawal from a BGP neighbor when RIB sharding is enabled. [PR1556271](#)
- Multipath info is still shown for BGP route even after disabling interface for one path. [PR1557604](#)
- BGP LU session flap might be seen with AIGP used scenario. [PR1558102](#)

- When admin-color-based policy evaluation happens with the policy LFA configuration, the backup next hop chosen (among the different backup next hops possible) might not be correct. [PR1558581](#)
- VPNv4 routes learned from the core might not get advertised to CE router when BGP sharding is configured. [PR1560661](#)
- Unexpected packet loss might be happen when configuring a new VRF or after performing GRES. [PR1560827](#)
- Duplicate LSP next hop is shown on inet.0, inet.3, and mpls.0 route table when OSPF traffic-engineering shortcuts and MPLS bgp-igp-both-ribs are enabled. [PR1561207](#)
- Incorrect SPF calculation might be observed for OSPF with **ldp-synchronization hold-time** configured after interface flap. [PR1561414](#)
- BGP routes might be stuck in routing table with **Accepted DeletePending** state when the BGP peering session goes down. [PR1562090](#)
- The rpd process might crash on backup Routing Engine after the rpd restart is triggered on primary Routing Engine. [PR1563350](#)
- The rpd process might crash when there is BGP session re-establishing or flapping. [PR1567182](#)
- VRF table does not get refreshed after a change to **maximum-prefixes** in the VRF. [PR1564964](#)
- Traffic might be lost during mirror data transmit from the primary ppmdd/bfdd. [PR1570228](#)
- There might be 10 seconds delay to upload the LSP on the point-to-point interface if rpd is restarted on its direct neighbor. [PR1571395](#)
- SNMP MIB ospfv3NbrState is returning drifted value. [PR1571473](#)
- The ppmdd process might crash when enabling MD5 authentication on OSPF with BFD flapping. [PR1576893](#)
- BGP replication might be stuck in rare and timing conditions. [PR1581578](#)

Services Applications

- Executing CLI command repetitively might cause the system to run out of disk space. [PR1537772](#)

User Interface and Configuration

- Remove Adobe Flash-dependent elements from J-Web for MX Series devices. [PR1553176](#)

VPNs

- The PIM (S,G) join state might stay forever when there are no MC receivers and source is inactive. [PR1536903](#)
- Type 7 messages might not be sent from egress PE resulting in Type 3/5 messages are not created for some S, Gs in source PE devices. [PR1567584](#)

Resolved Issues: 20.4R1

EVPN

- EVPN-VXLAN core isolation does not work when the system is rebooted or the routing is restarted. [PR1461795](#)
- Configuring the **proxy-macip-advertisement** command for EVPN-MPLS leads to functionality breakage. [PR1506343](#)
- With the EVPN-VXLAN configurations, the IRB MAC does not get removed from the route table after disabling IRB. [PR1510954](#)
- With dynamic list next hop configured, a forwarding problem occurs after performing graceful switchover. [PR1513759](#)
- ARP might break when multicast snooping is enabled in EVPN for the VLAN-based and VLAN-bundle service scenarios. [PR1515927](#)
- no-arp-suppression is required for MAC learning across the EVPN domain on the static VTEP. [PR1517591](#)
- Unable to create a new VTEP interface. [PR1520078](#)
- The ARP resolution to the gateway IRB address fails if **decapsulate-accept-inner-vlan** or **encapsulate-inner-vlan** is configured. [PR1526618](#)
- The rpd process might crash when **auto-service-id** is configured in the EVPN-VPWS scenario. [PR1530991](#)
- All the ARP reply packets toward some address are flooded across the entire fabric. [PR1535515](#)
- The **GE LOS** alarm logs on the change in IFF_CCCDOWN are not logged in the syslog message file. [PR1539146](#)
- VLAN ID information is missed while installing the EVPN route from the BGP type 2 route after modifying a routing-instance from the instance-type EVPN to instance-type virtual-switch. [PR1547275](#)
- The BUM traffic might get dropped in the EVPN-VXLAN setup. [PR1525888](#)
- The route table shows additional paths for the same EVPN or VXLAN type 5 destination after upgrading from Junos OS Release 18.4R2 S3 to Junos OS Release 19.4R1 S2. [PR1534021](#)
- The ARP table might not be updated after VMotion or network loop is performed. [PR1521526](#)
- The l2ald process might generate core file when changing the EVPN-VXLAN configuration. [PR1541904](#)

Forwarding and Sampling

- The DHCP subscribers might get stuck in the **Terminated** state for around 5 minutes after disabling the cascade ports. [PR1505409](#)
- Traffic might get dropped for not exceeding the configured bandwidth under policer. [PR1511041](#)
- The srrd process might crash in a high route churns scenario or if the process flaps. [PR1517646](#)
- The commit might fail if a filter enabled with **enhanced-mode to et- interface** is configured. [PR1524836](#)

- The l2ald process might crash when a device configuration flaps frequently. [PR1529706](#)
- VLAN-id based firewall match conditions might not work for the VPLS service. [PR1542092](#)
- MAC learning issue might happen when EVPN-VXLAN is enabled. [PR1546631](#)
- All traffic would be dropped on the aggregated Ethernet interface bundle without VLAN configuration if the bandwidth-percent policer is configured. [PR1547184](#)
- The l2ald process might crash due to next-hop issue in the EVPN-MPLS. [PR1548124](#)

General Routing

- New subscribers might fail to connect due to the following error message: **Filter index space exhausted.** [PR1531580](#)
- In some MX Series deployments running Junos OS, the following random syslog messages are observed for FPCs: **fpcx ppe_img_ucode_redistribute Failed to evict needed instr to GUMEM - xxx left.** These messages might not have a service impact. These messages are addressed as INFO level messages. On a Packet Forwarding Engine, there are dedicated UMEM and shared GUMEM memory blocks. This informational message indicates some evicting events between UMEM and GUMEM and can be safely ignored. [PR1298161](#)
- The **max-drop-flows** statement is not available. [PR1375466](#)
- On the MX2000 router, the following error message might be observed if the MPC7 line card is offline when Routing Engine switchover occurs: **Failed to get xfchip.** [PR1388076](#)
- The RPD scheduler slips might be observed upon executing the **show route resolution extensive 0.0.0.0/0 | no-more** command if the number of routes in the system is large (several millions). [PR1425515](#)
- Application and removal of 1-Gbps speed results in the channel being down. [PR1456105](#)
- Random packet drop with flow cache is disabled, when NIC is mapped to NUMA node 1. [PR1458742](#)
- In the MVPN instance, the traffic drops on multicast receivers within the range of 0.1 to 0.9 percent. [PR1460471](#)
- The following error message is observed after GRES: **[user.err aftd-trio: [Error] IF:Unable to add member to aggregate member list, member already exists, aggIfName:ps1.0 memberIfName:lt-3/0/0.32767].** [PR1466531](#)
- Dynamic SR-TE tunnels do not get automatically recreated at the new primary Routing Engine after the Routing Engine switchover. [PR1474397](#)
- Expected number of 512,000 MAC entries are not re-learned in the bridge table after clearing 512,000 MAC entries from the table. [PR1475205](#)
- The syslog reports simultaneous zone change reporting for all green, yellow, orange, and red zones for one or more service PICs. [PR1475948](#)
- Fabric healing logic incorrectly makes all MPC line cards go offline in the MX2000 router while the hardware fault is located on one specific MPC line card slot. [PR1482124](#)

- The vmcore process crashes sometimes along with the mspmand process on MS-MPC or MS-MIC if large-scale traffic flows are processed. [PR1482400](#)
- Traffic decreases during throughput testing. [PR1483100](#)
- Prolonged flow control might occur with MS-MPC or MS-MIC. [PR1489942](#)
- The following error message is observed on the MPC card in the manual mode:
clksync_as_evaluate_synce_ref: 362 - Failed to configure clk. [PR1490138](#)
- The MX10003 router might shut itself down automatically after the system upgrades or downgrades. [PR1492121](#)
- VPLS flood next hop might not get programmed correctly. [PR1495925](#)
- Some of the virtual services might not come up after GRES or rpd restart. [PR1499655](#)
- As prefix is not emitted, the path emitted for **te-lsp-timers/state/cleanup** contains incorrect value. [PR1500690](#)
- The following error message is observed after deactivating the demux logical interface: **configuration check-out failed.** [PR1501002](#)
- The packets from a nonexisting source on the GRE or UDP designated tunnel might be accepted. [PR1503421](#)
- On the vMX instances, configuring the statement ranges for auto-sensed VLANs (either stacked VLANs or single-tag VLANs) might not work. This is because the VLANs are not programmed on the NIC drivers. [PR1503538](#)
- The gNMI stream does not follow the frequency on the subscription from the collector. [PR1504733](#)
- After sending the Layer 4 or Layer 7 traffic, the HTTP redirect messages are not captured as expected. [PR1505438](#)
- The heap memory utilization might increase after extensive subscriber login or logout. [PR1508291](#)
- Outbound SSH connection flap or memory leak issues might be observed during push configuration to the ephemeral database with a high rate. [PR1508324](#)
- The disabled QSFP transceiver might fail to get turned on. [PR1510994](#)
- PFCP message acknowledgment or non-acknowledgment responses are not tracked without the fix. If the CPF peer drops an acknowledged UPF response message and CPF retries the request, the reattempts do not get an acknowledgment by the response cache at UPF and get silently dropped. This causes the CPF state machine to constantly retry requests with that message being dropped at UPF, which leads to the **Established** state at both CPF and UPF. [PR1511708](#)
- Static subscribers are logged out after creating a unit under the demux0 interface. [PR1511745](#)
- Memory leak on l2ald might be observed when adding or deleting the routing-instances or bridge-domains configuration. [PR1512802](#)

- The wavelength configured through the CLI might not be set on the SFP+-10G-T-DWDM-ZR optics when the optics is used on the MPC7E line card. [PR1513321](#)
- Subscribers might not be able to bind again after performing back-to-back GRES followed by an FPC restart. [PR1514154](#)
- Not able to forward traffic to VCP FPC after the MX Virtual Chassis reboots, FPC reboots, or adding VCP link. [PR1514583](#)
- The MACsec session might fail to establish if 256-bit cipher suite is configured for MACsec connectivity association assigned to a logical interface. [PR1514680](#)
- Duplicate prefix in secondary tunnel table is observed. [PR1514947](#)
- On the MX2010 and MX2020 routers, the SPMB CPU is elevated when an SFB3 is installed. [PR1516287](#)
- Active sensor check fails while checking the **show agent sensors |display xml** command. [PR1516290](#)
- Used-Service-Unit of the CCR-U has output-bytes counter zero. [PR1516728](#)
- yin2tlv sets unsupported command as hidden deprecated. [PR1516910](#)
- The MPC7E line card with QSFP installed might get rebooted when the **show mtip-chmac <1|2> registers vty** command is executed. [PR1517202](#)
- There might be memory leak in cfmd if both the CFM and inet or IPv4 interfaces are configured. [PR1518744](#)
- The vgd process might generate a core file when the OVSDB server restarts. [PR1518807](#)
- During an upgrade, vSRX3.0 would display the following incorrect license warnings when utilizing licensable features even if the license was present on the device: **such as warning: requires 'idp-sig' license**. [PR1519672](#)
- The PADI packets might be dropped when the interface encapsulation VPLS is set along with accepted protocol configured as PPPoE. [PR1523902](#)
- The PSM firmware upgrade must not allow multiple PSM upgrades in parallel to avoid the firmware corruption and support multiple firmware for different hardware. [PR1524338](#)
- Commit is successful while deactivating CB0 or CB1 interfaces with GNF. [PR1524766](#)
- According to the OC data model, the **openconfig-alarms.yang** subscription path must be used as system, alarms, or an alarm. [PR1525180](#)
- Addition and removal of an aggregated Ethernet interface member link might cause the PPPoE subscriber session and traffic to drop. [PR1525585](#)
- The following error message is observed during GRES if an IRB interface is configured without a profile: **RPD_DYN_CFG_GET_PROF_NAME_FAILED**. [PR1526481](#)
- The MPC10E line card might crash with the sensord process generating a core file due to a timing issue. [PR1526568](#)
- WAG control route prefix length is observed. [PR1526666](#)

- On the MX150 router, IFDs stay up during vmhost halt or power-off. [PR1526855](#)
- Commit error messages come twice while validating the **physical-cores** command. [PR1527322](#)
- The cpdd process might generate the core file after upgrading to Junos OS Release 19.4 and later. [PR1527602](#)
- The transit PTP packet might be unexpectedly modified when passing through MPC2E-NG, MPC3E-NG, and MPC5E line cards. [PR1527612](#)
- Commit confirmed rollback does not work. [PR1527848](#)
- The l2cpd process might crash when removing LLDP on an aggregated Ethernet interface. [PR1528856](#)
- The **speed** command cannot be configured under the interface hierarchy on an extended port when the MX204 or MX10003 router works as an aggregation device. [PR1529028](#)
- Non-impacting error message is observed in the message logs: **IFP error>/..../..../..../src/pfe/usp/control/applications/interface/ifp.c@3270:(errno=1000) tunnel session add failed.** [PR1529224](#)
- The multicast traffic might get dropped due to hash mismatch when there are aggregated Ethernet and ECMP links involved in the multicast tree. [PR1529475](#)
- In the subscriber management environment, the RADIUS interim accounting records do not get populated with the subscriber statistics. [PR1529602](#)
- The SFP-LX or SFP-SX optics on MIC-3D-20GE-SFP-E/EH might show as unsupported after ISSU. [PR1529844](#)
- After performing ISSU with a high-scale bridge-domain configuration, less than 0.0254 percent of traffic loss is observed for a single bridge-domain interface. [PR1531051](#)
- On the MX10003 router, PEM 0 always shows as **Absent** or **Empty** even if PEM 0 is present. [PR1531190](#)
- Deleting the address of the jmgmt0 interface might fail if the shortened version of the CLI command is used. [PR1532642](#)
- VRRP synchronization does not occur in the backup Routing Engine with NSR in the **Steady** state. [PR1533357](#)
- The **clear ike statistics** command with remote gateway does not work. [PR1535321](#)
- Certain BGP SR-TE segment lists cause the rpd process to generate the core file during tunnel attribute parsing. [PR1535632](#)
- Multicast traffic might be observed even through unexpected interfaces with distributed IGMP is enabled. [PR1536149](#)
- Enhancements are needed for debugging l2ald. [PR1536530](#)
- The following error message might be observed when the JAM packages for the MX204, MX10003, and MX10008 are installed: **JAM: Plugin installed for summit_xxx PIC.** [PR1537389](#)
- Version-alias gets missed for subscribers configured with dynamic profiles after ISSU. [PR1537512](#)

- On the MPC10 and MPC11 line cards, the `aftd` process might crash in case of composite chain next hop creation failures. [PR1538559](#)
- The accounting interim-updates for subscriber does not work after GRES and subsequent reboot of FPCs in the node-slicing setup. [PR1539474](#)
- With hold time configuration, the `ge` Interfaces remain down on reboot. [PR1541382](#)
- Subscriber might not come up on some dynamic VLAN ranges in a subscriber management environment. [PR1541796](#)
- After changing addresses in the source pool, if the carrier-grade NAT traffic does not stop, the source pool cannot perform the NAT translation from the new pool. [PR1542202](#)
- Port mirroring with **maximum-packet-length** configuration does not work over the GRE interface. [PR1542500](#)
- The `nsd` daemon crashes after configuring the inline NAT44 in the USF mode. [PR1547647](#)
- The **verbose** command unexpectedly becomes hidden after Junos OS Release 16.1 for **set system export-format json**. [PR1547693](#)
- **SENSOR APP DWORD** leak is observed during the period of churn for routes bound to the sensor group. [PR1547698](#)
- Family IPv6 does not come up for the L2TP subscriber when the additional attributes are not passed in the **Framed-IPv6-Route** VSA. [PR1526934](#)
- The **show dynamic-profile session client-id** command displays only one IPv6 framed-route information. [PR1555476](#)
- The ERO update by the controller for branch LSP might cause issues. [PR1508412](#)
- The `mspmnd` process leaks memory in relation to the MX telemetry reporting the following error message: **RLIMIT_DATA exceed**. [PR1540538](#)
- The `mspmnd` process might generate core file on activating or deactivating the interface. [PR1544794](#)
- In the syslog output, the **sylog-local-tag** name is truncated (as **SYSLOG_SF**) when the **sylog-local-tag** name is configured as **SYSLOG_SFW**. [PR1547505](#)
- **SENSOR APP DWORD** leak is observed during the period of churn for routes bound to the sensor group. [PR1547698](#)
- Multicast traffic drop might be seen after ISSU. [PR1548196](#)
- The PPPoE subscribers might fail to login. [PR1551207](#)
- The fabric errors are observed and the FPC processes might get offlined with SCBE3, MPC3E-NG, or MPC3E and MPC7 or MPC10 in the increased-bandwidth fabric mode. [PR1553641](#)
- The `l2ald` process might crash with traffic on the scaled set-up. [PR1517074](#)

- Difference between the port count and terminated count might be observed upon login or logout of the subscribers indicated by the output of the **show subscribers summary port extensive** command. [PR1523813](#)
- False positive TSensor errors are observed on vjunos0. [PR1508580](#)
- Snmp mib walk for jnxSubscriber OIDs returns the general error. [PR1535754](#)
- Delay in disabling the Packet Forwarding Engine might be seen on MPC7, MPC8, and MPC9 line cards. [PR1481879](#)
- The **next hop learning** statement is enabled by default in MPC10 and MPC11 line cards irrespective of the configuration statement. [PR1489121](#)
- The AMS bundle might remain inactive while adding the member interface to the AMS bundle with the scaled service sets. [PR1489607](#)
- Slow response might be observed when the **show | compare** or **commit check** action is executed in a large-scale configuration environment. [PR1500988](#)
- Sensord crashes on MPC10E line cards even when telemetry is not enabled. [PR1502260](#)
- Transit IPv4 traffic forwarding over BGP SR-TE might not work. [PR1505592](#)
- The l2cpd might crash if the ERP is deleted after the switchover. [PR1517458](#)
- The fxpc process might generate core file during EEPROM read when SFP is removed. [PR1518480](#)
- Traffic loss might occur when an uncorrected (Fatal) AER error is detected. [PR1519530](#)
- The VMXs might go to the amnesiac mode if they are deployed on the OpenStack based platforms. [PR1519668](#)
- The phc daemon might crash while committing the phone-home client configuration. [PR1522862](#)
- The BFD session status remains down at the non-anchor FPC even though the BFD session is up after the anchor FPC reboots or panics. [PR1523537](#)
- The rpd process might crash while restarting the routing gracefully with MPLS LSPs configured. [PR1527172](#)
- CFM does not consider the 8021AD configuration for the rewrite and classification tables. [PR1527303](#)
- BiDi 1G SFP optics displays wrong value in JVision for **optics/laser_rx_power_*_thresholds**. [PR1530120](#)
- The unilists are incorrectly formed and the list of the forwarded next-hops are not resolved properly if the value of the ECMP is set to 128. [PR1530803](#)
- The interface with the pic-mode 10GE configuration might not come up if upgraded to Junos OS Release 18.4R3-S4 or later. [PR1534281](#)
- Deactivating or activating PTP or SyncE in the upstream router causes the 100G links on the LC2103 to flap. [PR1538122](#)
- Traffic drop might be seen when executing the **request system reboot** command. [PR1538252](#)

- Upon receipt of a specific BGP FlowSpec message, network traffic might be disrupted. [PR1539109](#)
- The KRT queue might get stuck after the Routing Engine switchovers. [PR1542280](#)
- On the MX2010 and MX2020 devices, traffic loss might be observed when the Switch Fabric Board 3 and MPC8E 3D combination is used. [PR1544953](#)
- The Broadcom chip FPC might crash during the system booting. [PR1545455](#)
- Unexpected log messages appearing related to Neighbor Solicitation (NS) messages with multicast as source address is observed. [PR1546501](#)
- SR-TE might stay UP when the routes are deleted through policy. [PR1547933](#)
- The **LCM Peer Absent** error message might be seen on all TVP platforms. [PR1551760](#)
- ISSU might be aborted on MX Series devices. [PR1557413](#)

Infrastructure

- If the serial number of the PEM starts with 1F1, the following alarm might be generated: **Minor FPC PEM Temp Sensor Failed**. [PR1398128](#)
- Unknown MIB OIDs 1.3.6.1.2.1.47.2.0.30 are referenced in the SNMP trap after upgrading to Junos OS Release 18.4R3-S3. [PR1508281](#)
- SNMP polling might return an unexpectedly high value for the ifHCOutOctets counter for a physical interface when any jnxDom OID is processed at the same time. [PR1508442](#)
- The kernel might crash if a file or directory is accessed for the first time and is not created locally. [PR1518898](#)
- The telnetd.real local privilege escalation vulnerabilities in SUID binaries is observed. [PR1525318](#)
- The output drops in the **show interfaces extensive** command ' might display 0 temporarily during a race condition when the SNMP query for JnxCos is also issued. [PR1533314](#)

Interfaces and Chassis

- The **sonet-options configuration** statement is disabled for the xe interface that works in the wan-phy mode. [PR1472439](#)
- Fail to configure proactive ARP detection. [PR1476199](#)
- The fpc process might crash in an inline mode with CFM configured. [PR1500048](#)
- The following error message is observed: **Request failed: OID not increasing: ieee8021CfmStackServiceSelectorType**. [PR1517046](#)
- Buffer overflow vulnerability in a device control daemon is observed. [PR1519334](#)
- The configuration might not be applied after deleting all existing logical interfaces and adding a new logical interface for an IFD in a single commit. [PR1534787](#)

- Inline Y.1731 SLM or DM does not work in an enhanced-cfm-mode for the EVPN UP MEP scenario. [PR1537381](#)
- The following error message might occur after commit for configuration under interface hierarchy: **should have at least one member link on a different fpc**. [PR1539719](#)
- The following commit error is observed while trying to delete unit 1 logical system interfaces: **ae2.1: Only unit 0 is valid for this encapsulation**. [PR1547853](#)
- The **startup-silent-period** command might not work in Junos OS Release 20.3R1 or later. [PR1548464](#)
- The dcd process might leak memory on pushing the configuration to the ephemeral database. [PR1553148](#)
- Distribution fails for few sessions when VRRP is configured in a large-scale with active-inherit scenario. [PR1505998](#)
- Backup router generates **VRRP_NEW_BACKUP** syslog during bring up. [PR1539277](#)
- The rpd memory leak might be observed on the backup Routing Engine due to flapping of the link. [PR1539601](#)

Intrusion Detection and Prevention (IDP)

- The CLI provides helpful remarks about the tunable detector parameters of IDP. [PR1490436](#)

Juniper Extension Toolkit (JET)

- The JET application configuration must be disabled before upgrading Junos OS vmhost images. [PR1488769](#)

J-Web

- Privilege escalation in J-Web is observed due to arbitrary command and code execution through information disclosure from another users active session. [PR1518212](#)

Layer 2 Ethernet Services

- The DHCPv6 lease query is not as expected while verifying the DHCPv6 server statistics. [PR1506418](#)
- The **show dhcp relay statistics** command displays **DHCPLEASEUNASSIGNED** instead of **DHCPLEASEUNASSINGED**, which is a spelling error. [PR1512239](#)
- The **show dhcpv6 relay statistics** command must display **DHCPV6_LEASEQUERY_REPLY** instead of **DHCPV6_LEASEQUERY_REPL** for the messages sent. [PR1512246](#)
- The DHCP6 lease query is not as expected while verifying the DHCPv6 relay statistics. [PR1521227](#)
- Memory leak in the jdhcpd process might be seen if access-profile is configured under the **dhcp-relay** or **dhcp-local-server** statement. [PR1525052](#)

Layer 2 Features

- The rpd process might crash on the new primary Routing Engine after GRES in the VPLS or Layer 2 circuit scenario. [PR1507772](#)

- The host generated traffic might get lost as the current forwarding member nexthop is down while another member nexthop is up. [PR1516514](#)

MPLS

- Committing might trigger externally provisioned LSP MBB mechanism. [PR1546824](#)
- A same device responds twice for traceroute in case of the device going through an MPLS network under specific conditions. [PR1494665](#)
- Traffic loss might occur if ISSU is performed when P2MP is configured for an LSP. [PR1500615](#)
- The rpd scheduler might slip after the link flaps. [PR1516657](#)
- The rpd process might crash after upgrading Junos OS Release 18.1 to a later release. [PR1517018](#)
- The SNMP trap is sent with the incorrect OID jnxSpSvcSetZoneEntered. [PR1517667](#)
- The LDP session-group might throw a commit error and flap. [PR1521698](#)
- The inter-domain LSP with loose next hops path might get stuck in the **Down** state. [PR1524736](#)
- The **ping mpls rsvp** command does not take into account lower MTU in the path. [PR1530382](#)
- The rpd process might crash when the LDP route with indirect next hop is deleted on the aggregated Ethernet interface. [PR1538124](#)
- The LDP routes might be deleted from the MPLS routing table after the Routing Engine switchover. [PR1527197](#)
- The rpd process might crash during the restart routing when the MPLS LSPs are present. [PR1530213](#)

Network Address Translation (NAT)

- Need to improve the maximum eNode connections for one persistent NAT binding from 8 to 32. [PR1532249](#)

Network Management and Monitoring

- Unable to poll dot1dStp objects with l2cpd registered context after l2cpd restarts. [PR1561736](#)

Platform and Infrastructure

- With multiple different fixed-sized traffic streams configured at 10,000,00 fps (40-Gbps combined rate) on aggregated Ethernet0 along with another independent aggregated Ethernet interface (aggregated Ethernet1, 50 percent line rate 4 streams bidirectional => 118-Gbps combined traffic rate), both hosted on a single Packet Forwarding Engine instruction of the MPC11E line card, small varying packet drops occur for every iteration on aggregated Ethernet1 on disabling aggregated Ethernet0. [PR1464549](#)
- Traffic to VRRP virtual IP or MAC addresses might be dropped when ingress queuing is enabled. [PR1501014](#)
- Traffic originated from another subnet is sent out with 0x8100 instead of 0x88a8. [PR1502867](#)

- The kernel might crash causing the router or the Routing Engine to reboot when performing virtual IP related change. [PR1511833](#)
- The output of the **show jnh qmon queues-sensor stats 0** command has no content. [PR1514881](#)
- The VPLS connection might be stuck in the **Primary Fail** status when a dynamic profile is used on the VPLS pseudowire logical interface. [PR1516418](#)
- Configured scheduler-map is not applied on the ms- interface if the service PIC is in the **Offline** state during commit. [PR1523881](#)
- Flow programming issue for lt- interface in the Packet Forwarding Engine level is observed. [PR1525188](#)
- The following error message is observed when alarms after interface reset: **7836 ifl 567 chan_index 8 NOENT & jnh_ifl_topo_handler_pfe(13015): ifl=567 err=1 updating channel table nexthop**. [PR1525824](#)
- There is a TWAMP interoperability issue between Junos OS releases. [PR1533025](#)
- The fpc process might crash when the next hop memory of ASIC is exhausted in the EVPN-MPLS scenario. [PR1533857](#)
- Packet loss might be observed when the RFC2544 egress reflector session is configured on the non-zero Packet Forwarding Ethernet interface. [PR1538417](#)
- Trio-based FPC might crash when the underlying Layer 2 interface for ARP over IRB interface is changed from the physical interface to the LSI interface. [PR1542211](#)
- Subscribers does are not come up on VPLS PS interface. [PR1536043](#)
- The rmopd process memory leak might be seen if TWAMP client is configured. [PR1541808](#)
- The PE and CE devices OAM CFM might have issues in the aggregated Ethernet interface. [PR1501656](#)
- The VXLAN encapsulation over IPv6 underlay might not work. [PR1532144](#)
- The ISSU might fail on Junos platforms with the LUCHIP based line cards. [PR1535745](#)
- Dynamic filter fails to match IPv6 prefix. [PR1536100](#)
- TWAMP interoperability issue are observed. [PR1536939](#)
- The ARP expired timer on the backup Routing Engine is not the same as the primary Routing Engine if the aging-timer is configured. [PR1544398](#)

Routing Policy and Firewall Filters

- For setting the IPv6 router ID, the **routing-options** statement is added. [PR1523283](#)
- The policy configuration might be mismatched between the rpd and mgd process when **deactivate policy-options prefix-list** is involved in the configuration sequence. [PR1523891](#)

Routing Protocols

- The output of the **show isis interface detail** command might be incorrect if **wide-metrics-only** is enabled for IS-IS and the ASCII representation of the metric in decimal is more than 6 characters long. [PR1482983](#)
- The BGP RPKI ROA withdrawal might lead to an unexpected BGP route flap. [PR1483097](#)
- The rpd process might report 100 percent CPU usage with the BGP route damping enabled. [PR1514635](#)
- The IS-IS SR routes might not be updated to reflect the change in the SRMS advertisements. [PR1514867](#)
- The BGP link-bw of the non-multipath routes are included in an aggregation. [PR1515264](#)
- The rpd process might crash after deleting and then adding a BGP neighbor. [PR1517498](#)
- The rpd process might crash if there is a huge number of SA messages in an MSDP scenario. [PR1517910](#)
- Tag matching in the VRF policy does not work properly when the independent-domain option is configured. [PR1518056](#)
- The BFD sessions might flap continuously after disruptive switchover followed by GRES. [PR1518106](#)
- NLRI handling improvements for BGP-LS ID TLV is needed. [PR1521258](#)
- BFD with authentication for BGP flaps after GRES or NSR switchover on the NG-RE and SCBE2 setup. [PR1522261](#)
- The IS-IS LSP database synchronization issue might be observed while using the flood-group feature. [PR1526447](#)
- The rpd process generates core file at **is_srv6_delete_locator_end_sid_data isis_srv6_end_sid_local_data_delete isis_srv6_locator_config_check**. [PR1531830](#)
- Transit labels for Layer 3 VPN routes are pushed momentarily to the MPLS.0 table. [PR1532414](#)
- Configuring **then next hop** and **then reject** on a route policy for the same route might cause the rpd process to crash. [PR1538491](#)
- After moving peer out of the protection group, the path protection does not get removed from the PE router. Multipath routes are still present. [PR1538956](#)
- The rpd process generates the core file at **gp_rtarget_tsi_update,bgp_rtarget_flash_rt,bgp_rtarget_flash**. [PR1541768](#)
- Continuous rpd crash might be observed if a static group is added to protocol pim. [PR1542573](#)
- The metric of prefixes in intra-area-prefix LSA might be changed to 65535 when the metric of one of the OSPFv3 p2p interfaces is set to 65535. [PR1543147](#)
- IS-IS does not call **ted_add_halfink** for P2P IPv6-only links for traffic engineering topology. [PR1548506](#)
- Telemetry key value for transport or remote-address field for link-local IPv6 peer is incorrect and logical interface is absent. [PR1548754](#)
- The BGP session with VRRP virtual address might not come up after a flap. [PR1523075](#)
- The VRF label is not assigned at ASBR when the inter AS is implemented. [PR1523896](#)

- The BGP session neighbor shutdown configuration does not effect the non-established peer. [PR1554569](#)
- The BGP session might not come up if **extended-nexthop** is enabled by default on the other vendor remote peer. [PR1555288](#)
- The rpd process might crash with BGP RPKI enabled in a race condition. [PR1487486](#)
- The ppmmd process might generate core file after FPC restarts. [PR1490918](#)
- The virtual-router option is not supported under a routing-instance in a lean rpd image. [PR1494029](#)
- Traffic loss might be seen in the next-hop-based dynamic tunnels of Layer 3 VPN scenario after changing the dynamic-tunnel preference. [PR1542123](#)
- Six PE device prefixes might not be removed from RIB upon reception of withdrawal from a BGP neighbor when RIB sharding is enabled. [PR1556271](#)
- BGP routes might be stuck in routing table in the **Accepted DeletePending** state when the BGP peering session goes down. [PR1562090](#)

Services Applications

- The following error message is observed: **SPD_CONN_OPEN_FAILURE: spd_pre_fetch_query: unable to open connection to si-1/0/0**. [PR1550035](#)

Subscriber Access Management

- Subscriber accounting message retransmissions exist even after configuring the accounting retry 0. [PR1405855](#)
- CCR-T does not contain the usage-monitoring information. [PR1517507](#)
- The **show network-access aaa subscribers statistics username "<>"** command fails to fetch the subscriber-specific AAA statistics information if a subscriber username contains a space. [PR1518016](#)

User Interface and Configuration

- NETCONF service over SSH might not work on the device that runs Junos OS if in-band management is used. [PR1517160](#)
- The command injection vulnerability in the **request system software** command is observed. [PR1519337](#)
- The dexp local privilege escalation vulnerabilities in SUID binaries is observed. [PR1529210](#)

VPNs

- The MPLS label manager might allow configuration of a duplicated VPLS static label. [PR1503282](#)
- The rpd process might crash after removing the last configured interface under the Layer 2 circuit neighbor. [PR1511783](#)
- MVPN multicast route entry might not be properly updated with the actual downstream interfaces list. [PR1546739](#)
- The Junos image upgrade or installation with **validate** fails with XML errors. [PR1525862](#)

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Documentation Updates

There are no errata or changes in Junos OS Release 20.4R3 documentation for MX Series routers.

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Migration, Upgrade, and Downgrade Instructions

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This section contains the procedure to upgrade Junos OS, and the upgrade and downgrade policies for Junos OS for the MX Series. Upgrading or downgrading Junos OS might take several minutes, depending on the size and configuration of the network.

Starting in Junos OS 17.4R1 release, FreeBSD 11.x is the underlying OS for all Junos OS platforms which were previously running on FreeBSD 10.x based Junos OS. FreeBSD 11.x does not introduce any new Junos OS related modifications or features but is the latest version of FreeBSD.

The following table shows detailed information about which Junos OS can be used on which products:

Platform	FreeBSD 6.x-based Junos OS	FreeBSD 11.x-based Junos OS
MX5, MX10, MX40, MX80, MX104	YES	NO
MX240, MX480, MX960, MX2010, MX2020	NO	YES

Basic Procedure for Upgrading to Release 20.4R3

NOTE: Before upgrading, back up the file system and the currently active Junos OS configuration so that you can recover to a known, stable environment in case the upgrade is unsuccessful. Issue the following command:

```
user@host> request system snapshot
```

The installation process rebuilds the file system and completely reinstalls Junos OS. Configuration information from the previous software installation is retained, but the contents of log files might be erased. Stored files on the routing platform, such as configuration templates and shell scripts (the only exceptions are the `juniper.conf` and `ssh` files might be removed. To preserve the stored files, copy them to another system before upgrading or downgrading the routing platform. For more information, see the [Installation and Upgrade Guide](#).

For more information about the installation process, see [Installation and Upgrade Guide](#) and [Upgrading Junos OS with Upgraded FreeBSD](#).

Procedure to Upgrade to FreeBSD 11.x-Based Junos OS

Products impacted: MX240, MX480, MX960, MX2010, and MX2020.

To download and install FreeBSD 11.x-based Junos OS:

1. Using a Web browser, navigate to the All Junos Platforms software download URL on the Juniper Networks webpage:
<https://www.juniper.net/support/downloads/>
2. Select the name of the Junos OS platform for the software that you want to download.
3. Select the release number (the number of the software version that you want to download) from the Release drop-down list to the right of the Download Software page.
4. Select the Software tab.
5. In the Install Package section of the Software tab, select the software package for the release.
6. Log in to the Juniper Networks authentication system using the username (generally your e-mail address) and password supplied by a Juniper Networks representative.
7. Review and accept the End User License Agreement.
8. Download the software to a local host.
9. Copy the software to the routing platform or to your internal software distribution site.
10. Install the new jinstall package on the routing platform.

NOTE: We recommend that you upgrade all software packages out of band using the console because in-band connections are lost during the upgrade process.

All customers except the customers in the Eurasian Customs Union (currently composed of Armenia, Belarus, Kazakhstan, Kyrgyzstan, and Russia) can use the following package:

- For 32-bit Routing Engine version:

```
user@host> request system software add no-validate reboot  
source/junos-install-mx-x86-32-20.4R3.9-signed.tgz
```

- For 64-bit Routing Engine version:

```
user@host> request system software add no-validate reboot  
source/junos-install-mx-x86-64-20.4R3.9-signed.tgz
```

Customers in the Eurasian Customs Union (currently composed of Armenia, Belarus, Kazakhstan, Kyrgyzstan, and Russia) can use the following package (Limited encryption Junos package):

- For 32-bit Routing Engine version:

```
user@host> request system software add no-validate reboot
source/junos-install-mx-x86-32-20.4R3.x-limited.tgz
```

- For 64-bit Routing Engine version:

```
user@host> request system software add no-validate reboot
source/junos-install-mx-x86-64-20.4R3.9-limited.tgz
```

Replace source with one of the following values:

- **/pathname**—For a software package that is installed from a local directory on the router.
- For software packages that are downloaded and installed from a remote location:
 - **ftp://hostname/pathname**
 - **http://hostname/pathname**
 - **scp://hostname/pathname**

Do not use the **validate** option while upgrading from Junos OS (FreeBSD 6.x) to Junos OS (FreeBSD 11.x). This is because programs in the **junos-upgrade-x** package are built based on FreeBSD 11.x, and Junos OS (FreeBSD 6.x) would not be able to run these programs. You must run the **no-validate** option. The **no-validate** statement disables the validation procedure and allows you to use an import policy instead.

Use the **reboot** command to reboot the router after the upgrade is validated and installed. When the reboot is complete, the router displays the login prompt. The loading process might take 5 to 10 minutes.

Rebooting occurs only if the upgrade is successful.

NOTE:

- You need to install the Junos OS software package and host software package on the routers with the RE-MX-X6 and RE-MX-X8 Routing Engines. For upgrading the host OS on these routers with VM Host support, use the `junos-vmhost-install-x.tgz` image and specify the name of the regular package in the **request vmhost software add** command. For more information, see the VM Host Installation topic in the [Installation and Upgrade Guide](#).
- Starting in Junos OS Release 20.4R3, in order to install a VM host image based on Wind River Linux 9, you must upgrade the i40e NVM firmware on the following MX Series routers:
 - MX240, MX480, MX960, MX2010, MX2020, MX2008, MX10016, and MX10008

[See <https://kb.juniper.net/TSB17603>.]

NOTE: After you install a Junos OS Release 20.4R3 jinstall package, you cannot return to the previously installed Junos OS (FreeBSD 6.x) software by issuing the **request system software rollback** command. Instead, you must issue the **request system software add no-validate** command and specify the jinstall package that corresponds to the previously installed software.

NOTE: Most of the existing **request system** commands are not supported on routers with the RE-MX-X6 and RE-MX-X8 Routing Engines. See the VM Host Software Administrative Commands in the [Installation and Upgrade Guide](#).

Procedure to Upgrade to FreeBSD 6.x-Based Junos OS

Products impacted: MX5, MX10, MX40, MX80, MX104.

To download and install FreeBSD 6.x-based Junos OS:

1. Using a Web browser, navigate to the All Junos Platforms software download URL on the Juniper Networks webpage:
<https://www.juniper.net/support/downloads/>
2. Select the name of the Junos OS platform for the software that you want to download.
3. Select the release number (the number of the software version that you want to download) from the Release drop-down list to the right of the Download Software page.

4. Select the Software tab.
5. In the Install Package section of the Software tab, select the software package for the release.
6. Log in to the Juniper Networks authentication system using the username (generally your e-mail address) and password supplied by a Juniper Networks representative.
7. Review and accept the End User License Agreement.
8. Download the software to a local host.
9. Copy the software to the routing platform or to your internal software distribution site.
10. Install the new jinstall package on the routing platform.

NOTE: We recommend that you upgrade all software packages out of band using the console because in-band connections are lost during the upgrade process.

- All customers except the customers in the Eurasian Customs Union (currently composed of Armenia, Belarus, Kazakhstan, Kyrgyzstan, and Russia) can use the following package:

```
user@host> request system software add validate reboot source/jinstall-ppc-20.4R3.9-signed.tgz
```

- Customers in the Eurasian Customs Union (currently composed of Armenia, Belarus, Kazakhstan, Kyrgyzstan, and Russia) can use the following package (Limited encryption Junos OS package):

```
user@host> request system software add validate reboot  
source/jinstall-ppc-20.4R3.9-limited-signed.tgz
```

Replace source with one of the following values:

- **/pathname**—For a software package that is installed from a local directory on the router.
- For software packages that are downloaded and installed from a remote location:
 - **ftp://hostname/pathname**
 - **http://hostname/pathname**
 - **scp://hostname/pathname**

The validate option validates the software package against the current configuration as a prerequisite to adding the software package to ensure that the router reboots successfully. This is the default behavior when the software package being added is a different release.

Use the **reboot** command to reboot the router after the upgrade is validated and installed. When the reboot is complete, the router displays the login prompt. The loading process might take 5 to 10 minutes.

Rebooting occurs only if the upgrade is successful.

NOTE: After you install a Junos OS Release 20.4R3 jinstall package, you cannot return to the previously installed software by issuing the **request system software rollback** command. Instead, you must issue the **request system software add validate** command and specify the jinstall package that corresponds to the previously installed software.

Upgrade and Downgrade Support Policy for Junos OS Releases

We have two types of releases, standard EOL and EEOL:

- Standard End of Life (EOL) releases have engineering support for twenty four months after the first general availability date and customer support for an additional six more months.
- Extended End of Life (EEOL) releases have engineering support for thirty six months after the first general availability date and customer support for an additional six more months.

For both standard EOL and EEOL releases, you can upgrade to the next three subsequent releases or downgrade to the previous three releases. For example, you can upgrade from 19.2 to the next three releases – 19.3, 19.4 and 20.1 or downgrade to the previous three releases – 19.1, 18.4 and 18.3.

For EEOL releases only, you have an additional option - you can upgrade directly from one EEOL release to the next two subsequent EEOL releases, even if the target release is beyond the next three releases. Likewise, you can downgrade directly from one EEOL release to the previous two EEOL releases, even if the target release is beyond the previous three releases. For example, 19.2 is an EEOL release. Hence, you can upgrade from 19.2 to the next two EEOL releases – 19.3 and 19.4 or downgrade to the previous two EEOL releases – 19.1 and 18.4.

Release Type	End of Engineering (EOE)	End of Support (EOS)	Upgrade and Downgrade to subsequent 3 releases	Upgrade and Downgrade to subsequent 2 EEOL releases
Standard End of Life (EOL)	24 months	End of Engineering + 6 months	Yes	No
Extended End of Life (EEOL)	36 months	End of Engineering + 6 months	Yes	Yes

For more information about standard EOL and EEOL releases, see <https://www.juniper.net/support/eol/junos.html>.

For information about software installation and upgrade, see the [Installation and Upgrade Guide](#).

Upgrading a Router with Redundant Routing Engines

If the router has two Routing Engines, perform the following Junos OS installation on each Routing Engine separately to avoid disrupting network operation:

1. Disable graceful Routing Engine switchover (GRES) on the master Routing Engine, and save the configuration change to both Routing Engines.
2. Install the new Junos OS release on the backup Routing Engine while keeping the currently running software version on the master Routing Engine.
3. After making sure that the new software version is running correctly on the backup Routing Engine, switch over to the backup Routing Engine to activate the new software.
4. Install the new software on the original master Routing Engine that is now active as the backup Routing Engine.

For the detailed procedure, see the [Installation and Upgrade Guide](#).

Downgrading from Release 20.4R3

To downgrade from Release 20.4R3 to another supported release, follow the procedure for upgrading, but replace the 20.4R3 jinstall package with one that corresponds to the appropriate release.

NOTE: You cannot downgrade more than three releases.

For more information, see the [Installation and Upgrade Guide](#).

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Junos OS Release Notes for NFX Series

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These release notes accompany Junos OS Release 20.4R3 for the NFX Series Network Services Platforms. They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

You can also find these release notes on the Juniper Networks Junos OS Documentation webpage, located at https://www.juniper.net/documentation/product/en_US/junos-os.

What's New

IN THIS SECTION

- [What's New in Release 20.4R2 | 203](#)
- [What's New in Release 20.4R1 | 203](#)

Learn about new features introduced in the Junos OS main and maintenance releases for NFX Series.

NOTE: For information about NFX product compatibility, see [NFX Product Compatibility](#).

What's New in Release 20.4R2

There are no new features or enhancements to existing features for NFX Series devices in Junos OS Release 20.4R2.

What's New in Release 20.4R1

Application Security

- **AppQoE support for SaaS applications (NFX Series and SRX Series)**—Starting in Junos OS Release 20.4R1, we've extended application quality of experience (AppQoE) support for Software as a Service (SaaS) applications.

AppQoE performs service-level agreement (SLA) measurements across the available WAN -links such as underlay, GRE, IPsec or MPLS over GRE. It then sends SaaS application data over the most SLA-compliant link to provide a consistent service.

To configure AppQoE for SaaS applications:

1. Define the SLA rule type as SaaS (**set security advance-policy-based-routing sla-rule sla1 type saas**).
2. Include SaaS server details in the address book (**set security address-book global address address-book-name dns-name saas-server-url ipv4-only**).
3. Disable midstream switching to disengage advanced policy-based routing (APBR) and prevent further rule matching.
4. Attach the SLA rule to the policy-based APBR profile.

[See [Application Quality of Experience](#).]

- **Granular control over DNS-over-HTTP and DNS-over-TLS application traffic (NFX Series and SRX Series)**—In Junos OS Release 20.4R1, we introduce a new micro-application, DNS-ENCRYPTED, to enhance the application signature package. By configuring this micro-application in a security policy, you can have granular control for DNS-over-HTTP and DNS-over-TLS application traffic.

The DNS-ENCRYPTED application is enabled by default. You can disable it using the **request services application-identification application disable DNS-ENCRYPTED** command.

You can view the details of the micro-applications using the **show services show services application-identification application detail** command.

[See [Application Identification Support for Micro-Applications](#).]

High Availability

- **High availability on NFX350 devices**—Starting in Junos OS Release 20.4R1, NFX350 devices support the Chassis Cluster feature. You can configure a cluster of two NFX350 devices in active/passive or active/active mode to act as primary and secondary devices for protection against device failures. The high availability feature supports Layer 2 and Layer 3 features in dual CPE deployments.

[See [Chassis Cluster on NFX350 Devices](#) and [Upgrading or Disabling a Chassis Cluster on NFX350 Devices](#).]

Flow-Based and Packet-Based Processing

- **Support of IPFIX formatting and Chassis Cluster for NFX J-Flow functionality (NFX150, NFX250 NextGen, and NFX350)**—Starting with Junos OS Release 20.4R1, you can configure Chassis Cluster and define an IPFIX flow record template suitable for IPv4 traffic or IPv6 traffic. IPFIX is an enhanced version of J-flow version 9 template. Using IPFIX, you can collect a set of sampled flows and send the record to a specified host.

See [[Configuring Inline J-Flow to Use IPFIX Flow Templates on MX, vMX and T Series Routers, EX Series Switches, NFX Series Devices, and SRX Devices](#), [show services accounting flow](#), [show services accounting errors](#), [show services accounting status](#), and [clear services accounting statistics inline-jflow](#).]

Logical Systems and Tenant Systems

- **Support for MAP-E confidentiality CLI statement (NFX150, NFX250, NFX350, and SRX1500)**—Starting in Junos OS Release 20.4R1, we've introduced a global MAP-E **confidentiality** CLI statement to hide MAP-E rule parameters in CLI show commands and logs. To enable this configuration, include the **confidentiality** statement at the **[edit security softwires map-e]** hierarchy level. You need to have administrator privileges to enable or disable this configuration. This feature is supported for all domains of MAP-E.

[See [confidentiality](#) and [show security softwires map-e confidentiality status](#).]

Security

- **MACsec on NFX350 devices**—Starting in Junos OS Release 20.4R1, you can configure Media Access Control Security (MACsec) on NFX350 devices for secure communication for almost all types of traffic on Ethernet links. MACsec provides point-to-point security on Ethernet links between directly connected nodes and is capable of identifying and preventing most security threats, including denial of service, intrusion, man-in-the-middle, masquerading, passive wiretapping, and playback attacks.

[See [Configuring MACsec on NFX350 Devices](#) and [macsec](#).]

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What's Changed

IN THIS SECTION

- [Junos OS XML API and Scripting | 205](#)

Learn about what changed in the Junos OS main and maintenance releases for NFX Series devices.

Junos OS XML API and Scripting

- **The <get-interface-information/> RPC reply includes an <error-severity> element when execution fails (NFX Series)**—If the <get-interface-information/> RPC fails to execute, the device's RPC reply includes the <error-severity> element. In earlier releases, the RPC reply does not include the <error-severity> element.

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Known Limitations

IN THIS SECTION

- [Interfaces | 206](#)

Learn about known limitations in this release for NFX Series devices. For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Interfaces

- The link disable option puts the analyzer interface in an inconsistent state, with link state as DOWN and administrator state as UP. [PR1442224](#)
- On NFX250 devices, the LACP subsystem is not started automatically when dc-pfe process is restarted. [PR1583054](#)
- LACP port channel members are seen in detached state after configuring because of dc-pfe core. [PR1579647](#)

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Open Issues

IN THIS SECTION

- [High Availability | 207](#)
- [Virtual Network Functions \(VNFs\) | 207](#)

Learn about open issues in this release for NFX Series devices.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

High Availability

- On an NFX350 chassis cluster, when FPC0 (when node0 is primary) or FPC7 (when node1 is primary) is restarted by either using the **request chassis fpc slot slot restart node local** command or due to dcpfe core files on the primary, it restarts FPC1 or FPC8. This might break the pre-existing TCP sessions and fail to restart the TCP sessions. The TCP sessions might require a manual restart. [PR1557607](#)

Virtual Network Functions (VNFs)

- On NFX Series devices, while configuring **vmhost vlans** using **vlan-id-list**, the system allows duplicate VLAN IDs in the VLAN ID list. [PR1438907](#).
- On NFX Series devices, the OVS Analyzer configuration is lost after VNF reboot on non-dpdk SKUs. As a workaround, for non-dpdk SKUs, restart the VNF, de-activate and activate the analyzer configuration for the analyzer to function properly. [PR1480462](#).

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Resolved Issues

IN THIS SECTION

- [Resolved Issues: 20.4R2 | 208](#)
- [Resolved Issues: 20.4R1 | 209](#)

Learn which issues were resolved in the Junos OS Release 20.4R2 for NFX Series devices.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Resolved Issues: 20.4R2

General Routing

- The l2cpd core files might be seen on reboot. [PR1561235](#)

Interfaces

- On NFX Series devices, the following error message for interfaces might be seen: **FAILED(-1) read of SFP eeprom**. [PR1529939](#)

Platform and Infrastructure

- On NFX150 devices, when J-Flow v5 is configured and the J-Flow v5 server is reachable through an IPsec tunnel, and the MTU size of this IPsec tunnel is configured as 1500, the J-Flow packets are not generated on NFX Series devices. As a workaround, use J-Flow v9 or IPFIX version, instead of J-Flow v5, to enable the J-Flow functionality on NFX Series devices. [PR1539964](#)
- On NFX150, NFX250 NextGen, and NFX350 devices, the **EmulatorPin CPUSet** option does not get configured, which might result in vCPU running on a higher level up to 100%. [PR1540564](#)

Resolved Issues: 20.4R1

High Availability

- On NFX150 devices, upgrade from Junos OS Release 19.4 to Junos OS Release 20.2 fails and the `/usr/sbin/boot_mgmt_fsm: line 40: echo: write error: No space left on device` message is displayed. [PR1532334](#)

Interfaces

- When you configure analyzers on VNF interfaces with output port as other VNF interfaces, all the incoming and outgoing packets can be mirrored on to the designated analyzer port. However, after a system reboot, this functionality stops working and no packets are mirrored on the output analyzer port. [PR1480290](#)

Platform and Infrastructure

- On NFX150 devices, ZTP over LTE configuration commit fails for **operation=create** in XML operations configuration. [PR1511306](#)
- The device reads the board ID from eeprom directly using I2C upon power cycle. [PR1529667](#)

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Documentation Updates

There are no errata or changes in Junos OS Release 20.4R2 documentation for NFX Series devices.

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Migration, Upgrade, and Downgrade Instructions

IN THIS SECTION

- [Upgrade and Downgrade Support Policy for Junos OS Releases | 210](#)
- [Basic Procedure for Upgrading to Release 20.4 | 211](#)

This section contains the procedure to upgrade Junos OS, and the upgrade and downgrade policies for Junos OS for the NFX Series. Upgrading or downgrading Junos OS might take several hours, depending on the size and configuration of the network.

NOTE: For information about NFX product compatibility, see [NFX Product Compatibility](#).

Upgrade and Downgrade Support Policy for Junos OS Releases

We have two types of releases, standard EOL and EEOL:

- Standard End of Life (EOL) releases have engineering support for twenty four months after the first general availability date and customer support for an additional six more months.
- Extended End of Life (EEOL) releases have engineering support for thirty six months after the first general availability date and customer support for an additional six more months.

For both standard EOL and EEOL releases, you can upgrade to the next three subsequent releases or downgrade to the previous three releases. For example, you can upgrade from 19.2 to the next three releases – 19.3, 19.4 and 20.1 or downgrade to the previous three releases – 19.1, 18.4 and 18.3.

For EEOL releases only, you have an additional option - you can upgrade directly from one EEOL release to the next two subsequent EEOL releases, even if the target release is beyond the next three releases. Likewise, you can downgrade directly from one EEOL release to the previous two EEOL releases, even if the target release is beyond the previous three releases. For example, 19.2 is an EEOL release. Hence, you

can upgrade from 19.2 to the next two EEOL releases – 19.3 and 19.4 or downgrade to the previous two EEOL releases – 19.1 and 18.4.

Release Type	End of Engineering (EOE)	End of Support (EOS)	Upgrade and Downgrade to subsequent 3 releases	Upgrade and Downgrade to subsequent 2 EEOL releases
Standard End of Life (EOL)	24 months	End of Engineering + 6 months	Yes	No
Extended End of Life (EEOL)	36 months	End of Engineering + 6 months	Yes	Yes

For more information about standard EOL and EEOL releases, see <https://www.juniper.net/support/eol/junos.html>.

For information about software installation and upgrade, see the [Installation and Upgrade Guide](#).

Basic Procedure for Upgrading to Release 20.4

When upgrading or downgrading Junos OS, use the **jinstall** package. For information about the contents of the **jinstall** package and details of the installation process, see the [Installation and Upgrade Guide](#). Use other packages, such as the **jbundle** package, only when so instructed by a Juniper Networks support representative.

NOTE: The installation process rebuilds the file system and completely reinstalls Junos OS. Configuration information from the previous software installation is retained, but the contents of log files might be erased. Stored files on the device, such as configuration templates and shell scripts (the only exceptions are the `juniper.conf` and `ssh` files), might be removed. To preserve the stored files, copy them to another system before upgrading or downgrading the device. For more information, see the [Software Installation and Upgrade Guide](#).

NOTE: We recommend that you upgrade all software packages out of band using the console because in-band connections are lost during the upgrade process.

To download and install Junos OS Release 20.4R2:

1. Using a Web browser, navigate to the **All Junos Platforms** software download URL on the Juniper Networks webpage:
<https://www.juniper.net/support/downloads/>
2. Select the name of the Junos OS platform for the software that you want to download.
3. Select the **Software** tab.
4. Select the release number (the number of the software version that you want to download) from the Version drop-down list to the right of the Download Software page.
5. In the Install Package section of the Software tab, select the software package for the release.
6. Log in to the Juniper Networks authentication system using the username (generally your e-mail address) and password supplied by Juniper Networks representatives.
7. Review and accept the End User License Agreement.
8. Download the software to a local host.
9. Copy the software to the device or to your internal software distribution site.
10. Install the new package on the device.

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These release notes accompany Junos OS Release 20.4R3 for the PTX Series. They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

You can also find these release notes on the Juniper Networks Junos OS Documentation webpage, located at https://www.juniper.net/documentation/product/en_US/junos-os.

What's New

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- What's New in Release 20.4R3 | 214
- What's New in Release 20.4R2 | 214
- What's New in Release 20.4R1 | 214

Learn about new features introduced in Junos OS Release 20.4R3 for the PTX Series.

What's New in Release 20.4R3

There are no new features or enhancements to existing features for PTX Series routers in Junos OS Release 20.4R3.

What's New in Release 20.4R2

There are no new features or enhancements to existing features for PTX Series routers in Junos OS Release 20.4R2.

What's New in Release 20.4R1

Junos OS XML, API, and Scripting

- **Support for Certificate Authority Chain Profile (EX2300, EX3400, EX4300, MX240, MX480, MX960, PTX-5000, VMX, vSRX and QFX5200)**—Starting in Junos OS Release 20.4R1, you can configure intermediate Certificate Authority (CA) chain profile certificate and perform https REST API request using mutual and server authentications.

To configure intermediate ca-chain certificate, configure **ca-chain ca-chain** statement at the [edit system services rest https] hierarchy level.

- **Start time option for interval-based internal events that trigger event policies (EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—Starting in Junos OS Release 20.4R1, when you create an interval-based internal event for triggering event policies, you can specify the start date and time for the initial event. To specify a start time, configure the **start-time** option along with the **time-interval** option at the [edit event-options generate-event] hierarchy level.

[See [Generating Internal Events to Trigger Event Policies.](#)]

Junos Telemetry Interface

- **JTI support for inline Junos Traffic Vision sensors with gRPC services (MX Series and PTX Series)**—Junos OS Release 20.4R1 supports inline Jflow sensors for FPC3 and MPC 1 through 9. This feature enables you to monitor inline Junos Traffic Vision (previously known as Jflow) service statistics on a router and to export statistics to an outside collector at configurable intervals using remote procedure call (gRPC) services.

Use the resource path `/junos/system/linecard/services/inline-jflow/` in a subscription to export statistics.

You can view statistics in the collector output under `/components/`. The collector component ID in the statistics output will include the FPC slot number for which inline Junos Traffic Vision statistics are exported. For example, inline Jflow statistics for FPC 0 will be under **component id 0**, and inline Jflow statistics for FPC 1 will be under **component id 1**.

Inline Junos Traffic Vision statistics are slightly different, depending on the routing platform.

[See [Guidelines for gRPC and gNMI Sensors \(Junos Telemetry Interface\)](#).]

- **Enhancing debug information for JTI (PTX5000)**—Starting in Junos OS Release 20.4R1, debug commands supporting Junos telemetry interface (JTI) are enhanced to better support JTI.

The **show network-agent statistics (brief | detail)** command output now includes:

- Average latency values per sensor, which helps to check the latency of any given sensor on the device.
- Approximate circular buffer usage per sensor, which provides an early alert if drops are likely for any specific sensor.
- Time of subscription, which helps to correlate statistics information from the provisioning logs that are taken over a period of multiple subscriptions.

The **show extension-service request-response clients (brief | detail)** command output now includes:

- The username for which the session was authenticated in a remote procedure call (gRPC) session. If not authenticated, the **username** field displays as no authentication. This helps to identify which users have requested programmable operations.
- Login time of the gRPC client, which helps determine how long this client has been active.

[See [show network-agent statistics](#) and [show extension-service request-response clients](#).]

- **JTI support for persistent active gRPC sessions between collector and server during an SSL certificate update (ACX Series, MX Series, and PTX Series)**—Junos OS Release 20.4R1 supports persistent active remote procedure call (gRPC) sessions between the collector (client) and server during an SSL certificate update.

For secure channel authentication, the TLS protocol is used to maintain a secure channel between the collector and the server. TLS uses the server certificate and the client certificate to authenticate each other and send encrypted messages over the network. When an SSL certificate is updated, existing gRPC sessions are abruptly terminated, forcing the collector to initiate a new gRPC connection and subscribe to sensors again.

To avoid this problem, you can enable persistent active gRPC sessions by configuring **hot-reloading** at the `[edit system services extension-service request-response grpc ssl]` hierarchy level. After you enable this feature, gRPC sessions will remain active even when authentication certificates are updated.

After the certificate is updated, any new gRPC session will use the updated certificate.

[See [gRPC Services for Junos Telemetry Interface](#) and [ssl](#).]

- **BGP neighbor telemetry with sharding (MX Series, PTX Series, and QFX Series)**—Starting in Junos OS Release 20.4R1, BGP neighbor telemetry with sharding (multi-threading) is supported.

[See [Guidelines for gRPC and gNMI Sensors \(Junos Telemetry Interface\)](#).]

- **LACP sensors for actor partner states on JTI (MX Series and PTX Series)**—Starting in Junos OS Release 20.4R1, you can use Junos telemetry interface (JTI) and gRPC Network Management Interface (gNMI) services to export LACP actor partner states (also known as LACP port states). When a subscription is configured, ON_CHANGE or periodic streaming statistics are sent from devices to an outside collector.

You can subscribe to `/lacpd/` to collect all statistics or include the following resource paths individually in a subscription:

- `/lacpd/ae/member/partner_collecting`
- `/lacpd/ae/member/partner_synchronization`
- `/lacpd/ae/member/partner_timeout`
- `/lacpd/ae/member/partner_aggregatable`
- `/lacpd/ae/member/partner_distributing`
- `/junos/system/linecard/interface/traffic/`

[See [Guidelines for gRPC and gNMI Sensors \(Junos Telemetry Interface\)](#).]

- **Juniper Resiliency Interface for exception reporting and null route detection (ACX Series, PTX Series, and MX Series)**—Starting in Junos OS Release 20.4R1, you can use Juniper Resiliency Interface to detect and reduce Mean Time to Repair (MTTR) first-order network issues. Juniper Resiliency Interface uses a push model for data reporting from the entities in the system which encounter packet drops. This automates the workflow for detecting, reporting, and mitigating adverse exceptions.

To collect kernel routing table and routing protocol process exceptions, configure the **set system resiliency exceptions** statement at the `[edit]` hierarchy level to specify exception reporting based on kernel exceptions, and routing exceptions.

You can display exceptions from a remote collector by means of remote procedure call (gRPC) services or gRPC network management interface (gNMI) services. Display on-box exceptions by accessing the `/var/log` file or the database at `/var/db/ResiliencyExceptions.db`. No Junos operational mode commands display these exceptions.

MPLS

- **Support for optimizing auto-bandwidth adjustments for MPLS LSPs (MX Series and PTX Series)**—Starting in Junos OS Release 20.4R1, you can configure faster auto-bandwidth adjustment for MPLS LSPs under overflow or underflow conditions. This feature decreases the minimum allowed **adjust-threshold-overflow-limit** and **adjust-interval** to 150 seconds when **adjust-threshold-overflow-limit** and **adjust-threshold-underflow-limit** cross the configured threshold values. In releases earlier than Junos OS Evolved Release 20.4R1, the **adjust-interval** is 300 seconds under overflow or underflow conditions.

You can configure faster in-place LSP bandwidth update that avoids signaling of a new LSP instance as part of make-before-break. To configure faster in-place LSP bandwidth update, include the **in-place-lsp-bandwidth-update** configuration statement at the **[edit protocols mpls label-switched-path *lsp-name*]** hierarchy level.

You can also configure RSVP interfaces to support subscription percentage per priority. To configure subscription percentage per priority, include the **subscription priority *priority* percent *value*** configuration statement at the **[edit protocols rsvp interface *interface-name*]** hierarchy level.

[See [Configuring Optimized Auto-bandwidth Adjustments for MPLS LSPs](#).]

- **Re-engineering of SR-TE (MX Series, PTX Series)**—Starting with Junos OS Release 20.4R1, you can incorporate the following features to enhance the debugging capability of segment routing traffic-engineering (SR-TE):
 - rib-group import functionality.
 - Display of SR-TE routes installed from various tunnel sources using the **show spring-traffic-engineering** command.
 - Template map for BGP SR-TE tunnels.
 - Compute profile in template with distributed Constrained Shortest Path First (CSPF) for dynamic SR-TE tunnels.
 - 6PE (IPv6 over IPv4 SR-TE tunnel)
 - no-chained-composite-next-hop option

[See [source-packet-routing](#) and [show spring-traffic-engineering](#).]

- **Support for express segments to establish end-to-end segment routing path (MX Series and PTX Series)**—Starting in Junos OS Release 20.4R1, express segments can be used to establish end-to-end TE paths between interconnected TE networks. Express segments (also known as virtual TE links) are generated dynamically through policies matching the underlay LSPs. Express segments and the corresponding abstracted topology (required by RFC7926) is generated with policies.

To apply a policy, include the **policy *policy-name*** statement at the **[edit protocols express-segment traffic-engineering]** hierarchy level.

To configure express segment, include the **express-segment** statement under the **[edit protocols]** hierarchy level.

[See *How to Establish End-to-End Segment Routing Paths Using Express Segments*.]

Network Management and Monitoring

- **Configuration retrieval using the configuration revision identifier (EX3400, EX4300, MX204, MX240, MX480, MX960, MX2020, PTX3000, PTX10008, QFX5100, QFX10002-60C, SRX5800, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, you can use the configuration revision identifier feature to view the configuration for a specific revision. This configuration database revision can be viewed with the CLI command **show system configuration revision**.

[See [show system configuration revision](#).]

- **Junos XML protocol operations support loading and comparing configurations using the configuration revision identifier (EX3400, EX4300, MX204, MX240, MX480, MX960, MX2020, PTX3000, PTX10008, QFX5100, QFX10002-60C, SRX5800, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, the Junos XML management protocol operations support loading and comparing configurations by referencing the configuration revision identifier of a committed configuration. You can execute the **<load-configuration>** operation with the **configuration-revision** attribute to load the configuration with the given revision identifier into the candidate configuration. Additionally, you can compare the candidate or active configuration to a previously committed configuration by referencing the configuration revision identifier for the comparison configuration. The **<get-configuration>** operation supports the **compare="configuration-revision"** and **configuration-revision** attributes to perform the comparison.

[See [<get-configuration>](#) and [<load-configuration>](#).]

Routing Policy and Firewall Filters

- **Support for route's next-hop weight in policy match condition (MX Series, PTX Series, and QFX Series)**—Starting in Junos OS Release 20.4R1, a route with multiple next-hop paths can use the weight associated with a path to identify primary and backup paths. The path with the lowest weight is used as the primary path, and any paths with higher weights are treated as backup paths. You can use the next-hop weight as a match condition in export policies to redistribute IGP and BGP routes based on whether the primary or backup paths are active.

Configure this match condition using the **[edit policy-options policy-statement *policy-name* term *term-name* from]** statement.

[See [policy-statement](#) and [show policy](#).]

- **Unicast RPF support for IPv4 and IPv6 (PTX10004)**—Starting in Junos OS Release 20.4R1, PTX10004 devices support unicast reverse-path-forwarding (uRPF) for both IPv4 and IPv6 traffic flows. uRPF helps protect against DoS and DDoS attacks by verifying the unicast source address of packets arriving on a protected interface. Packets that are not from a valid path can be discarded. You can enable RPF checking for a given interface from the **[edit interfaces *name* unit *number* family *inet* | *inet6* rpf-check]** hierarchy level, and create a discard rule at the **[edit firewall filter *name* term default then reject]** hierarchy level.

[See [Example: Configuring Unicast Reverse-Path-Forwarding Check](#).]

Routing Protocols

- **Support for multiple single-hop EBGp sessions on different links using the same IPv6 link-local address (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, you are no longer required to have unique peer addresses for Juniper devices for every EBGp session. You can now enable single-hop EBGp sessions on different links over multiple directly connected peers that use the same IPv6 link-local address.

In earlier Junos OS Releases, BGP peers could be configured with link-local addresses, but multiple BGP peers could not be configured to use the same link-local address on different interfaces.

[See [Configure Multiple Single-Hop EBGp Sessions on Different Links Using the Same Link-Local Address \(IPv6\)](#).]

- **Support for IS-IS flood-reflector interfaces (PTX1000, QFX10002, QFX10008)**—Starting in Junos OS Release 20.4R1, we support the IS-IS flood reflector feature that offers better scalability for a Level 2 topology. Flood reflectors enable the creation of topologies where Level 1 areas provide transit forwarding for Level 2 destinations within a Level 2 topology.

The flexible tunnel interfaces (FTI) are designated as flood-reflector interfaces. To enable the flood reflector on an FTI, include the **flood-reflector** statement at the **[edit protocols isis interface *interface name* level *level number*]** hierarchy level.

You can configure the interface to be either the reflector or the client. To enable the reflector, you can use the **flood-reflector reflector *cluster-id*** statement at the **[edit protocols isis level *level number*]** hierarchy level.

To enable the flood reflector client, include the **flood-reflector client** statement at the **[edit protocols isis level *level number*]** hierarchy level.

NOTE: You can configure the flood reflector feature on FTIs at Level 2 only.

[See [How to Configure Flood-Reflector Interfaces in IS-IS Networks](#).]

- **Support for BGP Labeled Unicast prefix SID (MX Series and PTX Series)**—Starting in Junos OS 20.4R1, BGP labeled unicast can carry segment routing global block label range and index information through the prefix segment attribute. With this feature we support segment routing using the BGP labeled unicast prefix segments and the MPLS data plane in medium to large scaled data centers. The controller directs the server to assign a stack- of labels to an incoming packet based on the available network state information. The assigned label stack avoids congested paths and steers the packet through a best available path.

To configure and advertise the SRGB label range specifically for BGP include the **source-packet-routing srgb start-label *start-label* index-range *index-range* advertise-srgb** configuration statements at the **[edit protocols bgp]** hierarchy level.

To advertise prefix SIDs to external BGP peers, include the **advertise-prefix-sid** configuration statement at the **[edit protocols bgp]** hierarchy level. You can configure this statement globally or for specific BGP groups or BGP neighbors.

[See [srgb](#).]

- **Support for relaxing BGP router ID format from /32 to a nonzero ID per RFC6286 (EX9208, EX9214, MX204, PTX-5000, VRR, and QFX5120-32C)()**—Starting in Junos OS Release 20.4R1, you can establish a BGP connection using a BGP identifier that is a 4-octet, unsigned, nonzero integer and it needs to be unique only within the autonomous system (AS) per RFC 6286. In earlier releases, the BGP ID of a BGP speaker was required to be a valid IPv4 host address assigned to the BGP speaker.

To enable this feature, use the **bgp-identifier identifier group bgp group name bgp-identifier identifier neighbor peer address bgp-identifier identifier** configuration statement at the **[edit protocols bgp]** hierarchy level.

[See [router-id](#)]

- **IPv6 support in TED (MX Series, PTX Series)**—Starting in Junos OS Release 20.4R1, you can configure IS-IS traffic engineering to store IPv6 information in the traffic engineering database (TED) in addition to IPv4 addresses. BGP-LS distributes this information as routes from the TED to the Isdist.0 routing table using the TED import policies. These routes are advertised to BGP-TE peers as network layer reachability information (NLRI) with IPv6 router ID type, length, and value (TLV).

With this enhancement, you can benefit from obtaining the complete network topology in the TED.

[See [Link-State Distribution Using BGP Overview](#).]

Software Installation and Upgrade

- **ZTP with DHCPv6 client support (EX3400, EX4300, PTX1000, PTX5000, PTX10002-60C, PTX10008, QFX5100, QFX5200, QFX10002, and QFX10002-60C)**—Starting in Junos OS Release 20.4R1, zero touch supports the DHCPv6 client. During the bootstrap process, the device first uses the DHCPv4 client to request for information regarding image and configuration file from the DHCP server. The device checks the DHCPv4 bindings sequentially. If one of the DHCPv4 bindings fails, the device continues to check for bindings until provisioning is successful. However, if there are no DHCPv4 bindings, the device checks for DHCPv6 bindings and follows the same process as for DHCPv4 until the device can be provisioned successfully. Both DHCPv4 and DHCPv6 clients are included as part of the default configuration on the device.

The DHCP server uses DHCPv6 options 59 and 17 and applicable suboptions to exchange ZTP-related information between itself and the DHCP client.

NOTE: ZTP supports only HTTP and HTTPS transport protocols.

[See [Zero Touch Provisioning](#).]

System Logging

- **Support for time averaged watermark (MX Series, PTX Series, and QFX Series)**—Starting in Junos OS Release 20.4R1, you can capture steady state data of routing and forwarding (RIB/FIB) table routes using the **time-averaged-watermark-interval** configuration statement at the **[edit routing-options]** hierarchy level. Time averaged watermark is calculated whenever the time averaged interval is changed from CLI. Time averaged watermark is logged in syslog if the logs are enabled in the system at **LOG_NOTICE** level. The default time averaged watermark interval is 1 day. You can see the timed averaged watermark using the existing **show route summary** command.

[See [routing-options](#) and [show route summary](#).]

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What's Changed

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Learn about what changed in this release for PTX Series.

What's Changed in Release 20.4R3

General Routing

- **SSH session connection limit and rate limit per connection (PTX Series and QFX Series)**—We have introduced SSH **connection-limit** and **rate-limit** options at the **edit system services ssh** hierarchy levels to enable SSH connection limit and rate limit per connection. The default connection limit value is 75 connections and there is no default value associated with rate limit.

Network Management and Monitoring

- **Enhancement to the snmp mib walk command (PTX Series, QFX Series, EX Series, MX Series, SRX Series)**—The `ipv6IfOperStatus` field displays the current operational state of the interface. The `noIfIdentifier(3)` state indicates that no valid Interface Identifier is assigned to the interface. This state usually indicates that the link-local interface address failed Duplicate Address Detection. When you specify the 'Duplicate Address Detected' error flag on the interface, the new value (`noIfIdentifier(3)`) is displayed. Previously, the `snmp mib walk` command did not display the new value (`noIfIdentifier(3)`).
- **Changes in contextEngineID for SNMPv3 INFORMS (PTX Series, QFX Series, ACX Series, EX Series, MX Series, and SRX Series)**—Now the `contextEngineID` of SNMPv3 INFORMS is set to the local engine-id of Junos devices. In earlier releases, the `contextEngineID` of SNMPv3 INFORMS was set to remote engine-id.

[See [SNMP MIBs and Traps Supported by Junos OS](#).]

- **The configuration accepts only defined identity values for nodes of type identityref in YANG data models (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—If you configure a statement that has type `identityref` in the corresponding YANG data model, the device accepts only defined identity values (as defined by an identity statement) as valid input. In earlier releases, the device also accepts values that are not defined identity values.

What's Changed in Release 20.4R2

General Routing

- **Secure boot disabled alarm is raised (PTX10008)**—The **Secure boot disabled** alarm is raised when the system boots with secure boot disabled in bios.

Interfaces and Chassis

- **Warning message when taking an FPC offline**—PTX10003-80C and PTX10003-160C devices do not support the **request chassis fpc slot *slot-number* online** command. The only way to bring up an FPC (MPC) that is offline is by rebooting the chassis. So, when you take an FPC offline by using the **request chassis fpc slot *slot-number* offline** command, the screen displays the following message. 'Warning : FPC slot cannot be made online using a CLI command. You need to perform router reboot using "request system reboot" to online the FPC slot Do you wish to continue ? yes,no (no).'

[See [request chassis fpc.](#)]

Junos XML API and Scripting

- **Refreshing scripts from an HTTPS server requires a certificate (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—When you refresh a local commit, event, op, SNMP, or Juniper Extension Toolkit (JET) script from an HTTPS server, you must specify the certificate (Root CA or self-signed) that the device uses to validate the server's certificate, thus ensuring that the server is authentic. In earlier releases, when you refresh scripts from an HTTPS server, the device does not perform certificate validation.

When you refresh a script using the **request system scripts refresh-from** operational mode command, include the **cert-file** option and specify the certificate path. Before you refresh a script using the **set refresh** or **set refresh-from** configuration mode command, first configure the **cert-file** statement under the hierarchy level where you configure the script. The certificate must be in Privacy-Enhanced Mail (PEM) format.

[See [request system scripts refresh-from](#) and [cert-file.](#)]

- **The jcs:invoke() function supports suppression of root login and logout events in system log files for SLAX commit scripts (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—The **jcs:invoke()** extension function supports the **no-login-logout** parameter in SLAX commit scripts. If you include the parameter, the function does not generate and log UI_LOGIN_EVENT and UI_LOGOUT_EVENT messages when the script logs in as root to execute the specified remote procedure call (RPC). If you omit the parameter, the function behaves as in earlier releases in which the root UI_LOGIN_EVENT and UI_LOGOUT_EVENT messages are included in system log files.

[See [invoke\(\) Function \(SLAX and XSLT\).](#)]

- **The jcs:invoke() function supports suppression of root login and logout events in system log files for SLAX event scripts (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—The **jcs:invoke()** extension function supports the **no-login-logout** parameter in SLAX event scripts. If you include the parameter, the function does not generate and log UI_LOGIN_EVENT and UI_LOGOUT_EVENT

messages when the script logs in as root to execute the specified remote procedure call (RPC). If you omit the parameter, the function behaves as in earlier releases in which the root UI_LOGIN_EVENT and UI_LOGOUT_EVENT messages are included in system log files.

[See [invoke\(\) Function \(SLAX and XSLT\)](#).]

What's Changed in Release 20.4R1

Class of Service (CoS)

- We've corrected the output of the **show class-of-service interface | display xml** command. The output is of the following sort: `<container> <leaf-1> data </leaf-1><leaf-2>data </leaf-2> <leaf-3> data</leaf-3> <leaf-1> data </leaf-1> <leaf-2> data </leaf-2> <leaf-3> data </leaf-3> </container>` will now appear correctly as `<container> <leaf-1> data </leaf-1><leaf-2>data </leaf-2> <leaf-3> data</leaf-3></container> <container> <leaf-1> data </leaf-1> <leaf-2> data </leaf-2> <leaf-3> data </leaf-3> </container>`.

General Routing

- **Control plane DDoS protection packet type option for ARP traffic (PTX Series and QFX Series)**— Starting in this release, we've renamed the **arp-snoop** packet type option in the **edit system ddos-protection protocols arp** protocol group to **arp**. This packet type option enables you to change the default control plane distributed denial of service (DDoS) protection policer parameters for ARP traffic.

[See [protocols \(DDoS\) \(PTX Series and QFX Series\)](#).]

- **Support for unicast ARP request on table entry expiration**—You can configure the device to send a unicast ARP request instead of the default broadcast request when an ARP table entry is about to expire. The retry requests are unicast at intervals of 5 seconds. Without this option, the retry requests are broadcast at intervals of 800 milliseconds. This behavior reduces ARP overall broadcast traffic. It also supports the use case where access nodes are configured not to forward broadcast ARP requests toward customer CPEs for security reasons and instead translate ARP broadcasts to unicast requests. To confirm whether this is configured, you can issue the following command: **show configuration system arp | grep unicast-mode-on-expire**.

[See [arp](#).]

High Availability (HA) and Resiliency

- The **ntp boot-server** command is deprecated — The **boot-server** option for the **ntp** command is no longer needed, and has been deprecated.

MPLS

- The **show mpls lsp extensivel** and **show mpls lsp detail** commands display next hop gateway LSPid—When you use the **show mpls lsp extensivel** and **show mpls lsp detail** commands, you'll see next hop gateway LSPid in the output as well.

Network Management and Monitoring

- **Warning changed for configuration statements that correspond to deviate not-supported nodes in YANG data models (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—If you configure a statement corresponding to a YANG data model node that defines the **deviate not-supported** statement, the Junos OS configuration annotates that statement with the comment **Warning: statement ignored: unsupported platform**. In earlier releases, the warning is **Warning: 'statement' is deprecated**.

User Interface and Configuration

- **Verbose format option for exporting JSON configuration data (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—The Junos OS CLI exposes the **verbose** statement at the **edit system export-format json** hierarchy level. The default format for exporting configuration data in JSON changed from **verbose** format to **ietf** format starting in Junos OS Release 16.1R1. You can explicitly specify the default export format for JSON configuration data by configuring the appropriate statement at the **edit system export-format json** hierarchy level. Although the **verbose** statement is exposed in the Junos OS CLI as of the current release, you can configure this statement starting in Junos OS Release 16.1R1.

[See [export-format](#).]

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Known Limitations

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Learn about known limitations in this release for PTX Series Routers.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

General Routing

- During reconfigurations or link events at the physical interface level, `pe.ipw.misc_int.status:iq_disabled` interrupts can be seen. These do not indicate impact to traffic. [PR1476553](#)
- When counter sample is enabled, it's trying to fetch the IFD stats for sflow enabled interfaces using `rtsock` msgs to kernel. This blocks call and wait for the reply of earlier request and sends a new request only after receive the reply of first one. So, FPC is occupied when this request was made and couldn't reply on time and hence the scheduler slip will occur. So, if you set the pooling interval 0, then this message will not come like the below command:

set protocols sflow polling-interval 0

set protocols sflow interfaces <intf-name> polling-interval 0 This is only required if the polling interval is set at the interface level. [PR1517076](#)

- This is about the address selection for IPv6 trace-route. In case of IPv4, the address marked with primary flag gets selected. In case of IPv6, the address from the head of the address list gets selected as per the design. Changing it similar to IPV4 could cause other behavioural changes which is not suggested. [PR1518978](#)
- Route resolution over BGP-LU labels with default composite-next-hop knob is not supported on PTX1K and QFX10K. Workaround: `set routing-options forwarding-table chained-composite-next-hop transit no-labeled-bgp`, commit [PR1544277](#)
- On PTX10008, end-to-end traffic is not flowing for ethernet-switching in enterprise style. [PR1583219](#)

Routing Protocols

- Due to a race condition between route re-convergence and the BGP-PIC version up message to the Packet Forwarding Engine, after a remote transit router reboot, certain BGP routes might reuse stale LDP next hops and cause packet discard at the transit router during the route re-convergence window.

[PR1495435](#)

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Learn about open issues in this release 20.4R3 for PTX Series Routers.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

General Routing

- On PTX platforms, some non-fatal interrupts (for example, CM cache or AQD interrupts) are logged as fatal interrupts. The following log messages will be shown on CM parity interrupt: fpc0 TQCHIP 0: CM parity Fatal interrupt, Interrupt status: 0x10 fpc0 CMSNG: Fatal ASIC error, chip TQ fpc0 TQCHIP 0: CM cache parity Fatal interrupt has occurred 181 time(s) in 180010 msecs TQCHIP 0: CM cache parity Fatal interrupt has occurred 181 time(s) in 180005 msecs. [PR1089955](#)
- On the PTX Platform with FPC-PTX-P1-A or FPC2-PTX-P1A, you might encounter a single event upset (SEU) event that might cause a linked-list corruption of the TQCHIP. The following syslog message gets reported: Jan 9 08:16:47.295 router fpc0 TQCHIP1: Fatal error pqt_min_free_cnt is zero Jan 9 08:16:47.295 router fpc0 CMSNG: Fatal ASIC error, chip TQ Jan 9 08:16:47.295 router fpc0 TQ Chip::FATAL ERROR!! from PQT free count is zero Jan 9 08:16:47.380 router alarmd[2427]: Alarm set: FPC color=RED, class=CHASSIS, reason=FPC 0 Fatal Errors - TQ Chip Error code: 0x50002 Jan 9 08:16:47.380 router craft[2051]: Fatal alarm set, FPC 0 Fatal Errors - TQ Chip Error code: 0x50002 The Junos OS Chassis Management error handling detects such a condition, raises an alarm, and disables the affected Packet Forwarding Engine entity. To recover this Packet Forwarding Engine entity, restart the FPC. Contact your Juniper support representative if the issue persists even after the FPC restarts. [PR1254415](#)
- When CFP2-DCO-T-WDM-1 is plugged in a PTX Series PIC, after FPC restarts, the carrier frequency offset TCA is raised even when TCA is not enabled. [PR1301471](#)
- On 30-port MACsec-enabled line card (LC1101-M-30C, LC1101-M-30Q, and LC1101-M-96X) of the PTX10008 chassis, when the exclude-protocol lacp statement configured at the [edit security macsec connectivity-association connectivity-association-name] hierarchy level is deleted or deactivated, the LACP protocol's Mux State shown under the output of CLI command **show lacp interface**, might remain as attached or detached and might not change to distributing state. [PR1331412](#)
- The log of **SMART ATA Error Log Structure error: invalid SMART checksum** might be seen on FPC with WINTEC mSata SSD. [PR1354070](#)
- Due to transient hardware condition, single-bit error (SBE) events are corrected and have no operational impact. Reporting of those events had been disabled to prevent alarms and possibly unnecessary hardware replacements. This change applies to all platforms using Hybrid Memory Controller (HMC). [PR1384435](#)
- The telemetry statistics might not account correctly for the traffic on SR-TE policies (both byte count and packet count) on PTX series devices. This is a sensor-related issue. [PR1413680](#)
- On the PTX3000 routers, the firewall counter for lo0 does not increment. [PR1420560](#)
- On PTX series platforms, if a firewall filter with multiple match conditions is configured on interfaces which are Up and the firewall filter is modified (either a new action is added or the condition is added or removed etc.), the FPC might crash and restart. It might affect the service and traffic. [PR1432116](#)
- Memory leaks are expected in this release. [PR1438358](#)
- This is a timing issue during the sxe interface bring up (w.r.t i40e driver). This can be recovered by rebooting the complete board. [PR1442249](#)

- On the PTX10001 with NGRE platforms, the vmhost disk usage might keep increasing due to no log rotation for resild log and temperature sensor information was incorrectly written into resild log, resulting in resild log file size continuously increasing. [PR1480217](#)
- The SNMP index for bundle interface might become zero in Packet Forwarding Engine after restarting the FPC. This could cause the sflow records to have either **input interface value (IIF)** or **output interface value (OIF)** as 0 value. [PR1484322](#)
- When executing **show interfaces ae<> extensive** CLI on PTX1000, sometimes the aggregated Ethernet member link stats might be missing in the output. The issue is momentary issue and does not happen all the time. If run into the issue, please execute **show interfaces ae<> extensive** CLI again. [PR1519218](#)
- Flapping might be observed on channelized ports of PTX Series routers during ZTP, when one of the port is disabled on the supporting device. [PR1534614](#)
- SFlow reports incorrect "Extended Router Data" for traffic going over non-default VRF. [PR1537190](#)
- The output VLAN is not reported correctly in the extended switch data for IPIP transit traffic when you configure both dynamic tunnel and FTI as backups. [PR1537648](#)
- The **socket to sflowd closed** error comes up when the ukern socket to sflowd daemon (server) is closed. The error is rectified by itself as the client successfully reestablishes the connection in the subsequent attempts. When these errors are consistent, it indicates a communication issue between sflowd and the sFlow running on the FPC. [PR1538863](#)
- When we run continuous sync (show interfaces aex extensive) and async(SNMP polling) queries on aggregated Ethernet interface in parallel, we may observe spikes in aggregated Ethernet interface framing errors counter in between correct values. [PR1539537](#)
- On PTX platforms, when Inline Jflow is configured and high sampling rate (more than 4000 per second) is set, high CPU utilization may be observed and this might result in relevant impacts on traffic analysis and billing. [PR1569229](#)
- When firewall is configured with both discard and port-mirror as actions in the same term, mirrored packet will be corrupted (will have two Layer 2 headers). [PR1576914](#)
- On PTX Series routers and QFX Series switches, the traffic from TACACS port 49 might not be classified into a proper DDoS queue. When the issue happens, it might cause the unclassified traffic to get dropped when the CPU utilization is very high. [PR1578579](#)
- On PTX platforms with **vlan-ccc** configured, if it acts as a provider edge device and forwards the ISIS packet between CEs over the layer 2 circuit tunnel, the ISIS packet might be corrupted. In this case, the ISIS adjacency might not be formed. [PR1580047](#)
- On PTX10008, end-to-end traffic is not flowing for ethernet-switching in EP Style. [PR1583219](#)
The command **show chassis clocks** is handled by chassisd daemon on legacy platforms. On PTX10008, however, chassisd is not a running daemon. Hence the previously observed output: **error: the chassis-control subsystem is not running**. [PR1583715](#)
- ISIS over Layer 2 circuit will not come up if the encapsulation is translation cross-connect. [PR1590387](#)

- When LLDP is used and interfaces are flapping on PTX platforms, memory leaks might be observed and it will cause the l2cpd process to crash. [PR1608699](#)
- On PTX1000 platforms, ARP resolution failure with IRB configuration results in packet drop. [PR1612205](#)
- In the PTX5000 with behavior aggregate (BA) classification scenario, the COS queue-num 3 is applied to TTL packets by default. If the IEEE-802.1ad classifier is configured for the packets which are mapped to the CoS output queue-num 3 (The Packet Forwarding Engine internal hostbound queue classification code-points=1100), in a very rare case, if some TTL expired packets are passed via the other interfaces which are also mapped to the same hostbound queue (in this case, queue 3), the packets will be dropped by Packet Forwarding Engine because of some TTL expiry packets in the same queue. [PR1584042](#)
- **Failed to get pechip handle for chip 0 and prds_encap_sample_flood_lpbk_desc_install: Egress NH descriptor install OK for Flabel 7808** errors are seen during restart. [PR1585594](#)
- On PTX5000 and PTX10003 platforms, higher latency than configured or default value might be observed in the traffic passing through the device. This issue is seen as the VOQ (Virtual output queue) size is not correctly set in ASIC. [PR1588514](#)
- On Junos platforms with Telemetry implemented, the jsd (JET service process) crashes might occur if tracing is enabled for libgrpc while the collector continues to flap. [PR1589103](#)
- On PTX3000 and PTX5000 platforms, if a 40G or 100G interface flaps, the interface might stay down even after disabling or enabling it. [PR1589170](#)
- Upon frequent inetflow route changes, Heap Memory of the FPC will get leaked on PTX platforms. Once running out of memory resources, FPC might restart along with a core file. Syslog emergency messages will get recorded once crossing the 90 percent threshold. The current FPC Heap Utilization can be validated via the **show chassis fpc** command. [PR1589133](#)

Layer 2 Ethernet Services

- It was observed rarely that issuing a **request system zeroize** did not trigger ZTP. A simple workaround is to reinitiate ZTP. [PR1529246](#)

MPLS

- At high scale, LSP setup rate will be relatively slower in IPinIP networks. [PR1457992](#)
- In the case of the LDP route with multiple next-hops, the last NH weight in table mpls.0 is not set properly when the total number of LDP NHs is multiple of 8 + 1, e.g., 9, 17. This might lead to some backup route active as the primary path, which might result in a traffic loop. [PR1582037](#)

Network Management and Monitoring

- On PTX10008 platforms, syslog does not log information on IPv4 after upgrade. [PR1611504](#)

Platform and Infrastructure

- A buffer overflow vulnerability in the TCP/IP stack of Juniper Networks Junos OS allows an attacker to send specific sequences of packets to the device thereby causing a Denial of Service (DoS). Please refer to [JSA11200](#) for more information. [PR1557881](#)
- On PTX series platforms running Junos, traffic loss might be observed in a scaled firewall filter configuration setup due to FPC crash. When the issue occurs, a core file is generated which could be checked using the CLI command **show system core-dumps**. `user@router> show system core-dumps`
`-rw-r--r-- 1 root wheel 89322187 /var/crash/core-NGMPC0.gz.core.0 ----> Core file` [PR1586817](#)

Routing Protocols

- The ssh connection limit on PTX5000 platform will be ignored. `== ssh {#### Warning: statement ignored: unsupported platform (PTX5000)##connection-limit 3; } ==.` [PR1559305](#)
- In BGP Resource Public Key Infrastructure (RPKI) scenario, if the session record-lifetime is configured less than the hold-time, the record-lifetime for route validation (RV) might expire while the session is still up, which will cause the rpd crash. [PR1585321](#)
- In BGP multipath scenario, if an interface for a single hop EBGP peer goes down, the rpd might crash on the backup Routing Engine. If NSR switchover is performed, the rpd crash might be observed on the newly primary Routing Engine, hence there may be traffic impact. [PR1589141](#)

User Interface and Configuration

- When a user tries to deactivate the mpls related configuration, The commit fails on backup Routing Engine. [PR1519367](#)
- On PTX platforms, the default routing policy might not be changed back after it is changed to network-services enhanced mode. [PR1587174](#)

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This section lists the issues fixed in the Junos OS main release and the maintenance releases for the PTX Series.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Resolved Issues: 20.4R3

General Routing

- The kernel crash might happen if NSR is enabled. [PR1545143](#)
- Packet drop might happen on the aggregated Ethernet bundle which have the single child member only. [PR1551736](#)
- The device might run out of service post GRES or unified ISSU. [PR1558958](#)
- Traffic drop might be seen in 128 or more way ECMP paths after FPC restart. [PR1559528](#)
- The **request system software delete** CLI command is improved to add new option **archived** to delete all old software versions except current and rollback. [PR1566173](#)
- Another port will also shutdown after shutting down one port on PTX10002-60C or QFX10002-60C. [PR1568294](#)
- LLDP out-of-bounds read vulnerability in l2cpd. [PR1569312](#)
- Upgrading PTX1000 with unified SSDs (2x32G SSD) might result in boot loop in certain scenario. [PR1571275](#)

- The gRPC sessions hang in CLOSED state. [PR1571999](#)
- Channelized ports on PTX10002 platforms might drop traffic. [PR1575742](#)
- On the PTX5000 platform, traffic loss might be observed. [PR1578511](#)
- TACACS traffic might be dropped. [PR1578579](#)
- BFD sessions might flap during traffic spikes on PTX platforms. [PR1578599](#)
- Authentication might fail if the password contains special characters. [PR1580003](#)
- The IS-IS packet might be corrupted on the provider edge device over the layer 2 circuit tunnel. [PR1580047](#)
- PTX Series routers might drop traffic. [PR1580211](#)
- The FEC91 mode might not get enabled automatically for QSFP28-SR4 SFP used on WAN interface. [PR1582200](#)
- Junos telemetry Interfaces: Missing Leaves - Transceiver/state. [PR1583076](#)
- The packets might be dropped by Packet Forwarding Engine of PTX5000 after changing the queue of IEEE-802.1ad classifier on FPC-PTX-P1-A or FPC2-PTX-P1A. [PR1584042](#)
- On Junos OS PTX Series platforms, FPC resource usage increases when certain packets are processed which are being VXLAN encapsulated (CVE-2021-31361). [PR1584197](#)
- JDI-RCT: T/PTX: **Failed to get pechip handle for chip 0 and prds_encap_sample_flood_lpbk_desc_install: Egress NH descriptor install OK for Flabel 7808** errors are seen bringup. [PR1585594](#)
- The na-grpc process crash might be seen and existing telemetry connections will be disconnected. [PR1587956](#)
- There might be higher latency in traffic flow than configured or default value. [PR1588514](#)
- The jsd process crash might be seen in a rare condition in a telemetry scenario. [PR1589103](#)
- On Junos OS PTX Series platforms, an FPC heap memory leak will be triggered by certain Flowspec route operations which might lead to an FPC crash (CVE-2021-31367). [PR1589133](#)
- The 40G and 100G interface might stay down after link flaps on PTX3000 and PTX5000. [PR1589170](#)
- The L2cpd-agent might go unresponsive after starting telemetry service. [PR1592473](#)
- CRC errors increase continuously after interface flap. [PR1600768](#)
- Traffic blackhole might be seen due to the **RS Fatal** error on FPC-PTX-P1-A/FPC2-PTX-P1A/FPC-SFF-PTX-P1-A/FPC-SFF-PTX-T. [PR1600935](#)
- The layer 2 circuit packets with destination MAC address 01:00:0c:cc:cc:cd may get punted. [PR1601360](#)
- On PTX platforms, link flaps might be observed momentarily. [PR1606008](#)

Forwarding and Sampling

- Junos OS: User-defined ARP Policer isn't applied on aggregated Ethernet interface until firewall process is restarted (CVE-2021-0289). [PR1528403](#)

General Routing

- Node name should not be attached to the system hostname under LLDP. [PR1593991](#)

MPLS

- Sub-optimal routing issues might be seen in the case of LDP route with multiple next-hops. [PR1582037](#)
- The LDP replication session might not get synchronized when dual-transport is enabled. [PR1598174](#)
- VPLS connection might get down if configuration statement **dual-transport** is configured. [PR1601854](#)

Multicast

- Multicast traffic in MVPN setup might be blackholed on some PTX platforms acting as transit LSR. [PR1555274](#)
- FPC might crash in a multicast scenario. [PR1569957](#)

Platform and Infrastructure

- Junos OS: Upon receipt of specific sequences of genuine packets destined to the device the kernel will crash and restart (vmcore) (CVE-2021-0283, CVE-2021-0284). [PR1557881](#)
- FPC might crash in a scaled firewall configuration. [PR1586817](#)

Routing Protocols

- Route validation states might flip between VALID/INVALID/UNKNOWN in some corner case. [PR1556656](#)
- BGP session carrying VPNv4 prefix with IPv6 next-hop might be dropped. [PR1580578](#)
- Process rpd crash might be seen in certain IS-IS scenario. [PR1583484](#)
- The rpd crash might be seen when BGP RPKI session record-lifetime is configured less than the hold-time. [PR1585321](#)
- The rpd might crash in BGP multipath scenario if interface for a single hop EBGP peer goes down. [PR1589141](#)

Resolved Issues: 20.4R2

Forwarding and Sampling

- The l2ald process might crash due to next-hop issue in the EVPN-MPLS. [PR1548124](#)

General Routing

- Aggregate Ethernet interfaces do not display member links' statistic. [PR1505596](#)
- Error messages "t6e_dfe_tuning_state:et-6/0/0 - Failed to dfe tuning count 10" might be seen after links flap [PR1512919](#)
- The chassisd memory leak might cause traffic loss [PR1537194](#)
- The kernel crash might happen if NSR is enabled. [PR1545143](#)
- On the PTX10K platforms, traffic might get dropped when the set routing-options forwarding-table no-ecmp-fast-reroute configuration is changed to 128 ECMP entries. [PR1547457](#)
- The rpd crash might be seen when BGP service route is resolved over color-only SRTE policy [PR1550736](#)
- The interface filter with source-port 0 matches everything instead of port 0. [PR1551305](#)
- Packet drop might happen on the AE bundle which have the single child member only. [PR1551736](#)
- There might be traffic drop when default EXP classifier maps traffic to FC with no schedulers. [PR1554266](#)
- The LCMD process might consume memory until all of the free memory available to VMHOST gets exhausted. [PR1555386](#)
- The micro BFD session might flap with DDoS policer. [PR1557782](#)
- EVO System - after recovering from restart routing immediately, object-info anomalies is observed on rpdagent [PR1561812](#)
- An enhancement to enable watchdog petting log on PTX10K Line Cards [PR1561980](#)
- Junos OS: Upon receipt of specific packets BFD sessions might flap due to DDoS policer implementation in Packet Forwarding Engine. [PR1564807](#)
- Improve 'request system software delete' CLI command to add new option "archived" to delete all old software versions except current and rollback [PR1566173](#)
- gRPC session hanging in CLOSED state. [PR1571999](#)
- [LDP] [MPLS] JUNOS:JDI_FT_REGRESSION:PLATFORM:ROUTING:LDP: Higher Traffic loss observed after Link Flap with LDP. [PR1578511](#)
- BFD sessions might flap during traffic spikes on PTX platforms. [PR1578599](#)
- [BGP] [URPF] JUNOS:JDI_FT_REGRESSION:PLATFORM:ROUTING:URPF: Traffic dropped at PFE with uRPF config [PR1580211](#)

Infrastructure

- The kernel crash with core file might be seen if churn happens for a flood composite next hop. [PR1548545](#)

Layer 2 Ethernet Services

- The copying of files to the RCB over WAN ports is slow [PR1496895](#)

MPLS

- Traffic loss might be observed due to rpd crash in MPLS scenario [PR1528460](#)
- MPLS-LIB memory leak might be seen in SR scenario. [PR1556495](#)
- Traffic sent over an LSP may be dropped if two consecutive PLRs along the LSP perform local repair and bypass protecting the second PLR fails. [PR1566101](#)

MPLS

- FPC might crash in a multicast scenario. [PR1569957](#)

Network Management and Monitoring

- The mib2d process crashes and generates a core file on backup Routing Engine. [PR1557384](#)

Platform and Infrastructure

- The BGP session replication might fail to start after the session crashes on the backup Routing Engine. [PR1552603](#)

Routing Policy and Firewall Filters

- Generate route goes to hidden state when protect core knob is enabled [PR1562867](#)

Routing Protocols

- Traffic might be silently discarded when the BGP route getting deleted which is part of multipath [PR1514966](#)
- The rpd memory leak might be seen in the BGP scenario. [PR1547273](#)
- The BGP session might not come up if extended-nexthop is enabled by default on the other vendor remote peer. [PR1555288](#)
- The rpd might restart after interface flap if layer2-map. [PR1557710](#)
- BGP LU session flap might be seen with AIGP used scenario [PR1558102](#)
- Traffic loss might occur for stitched traffic from SR towards LDP if no-eligible-backup is configured. [PR1558565](#)
- The pppd memory leak may cause traffic loss. [PR1561850](#)

VPNs

- The rpd might crash during a race condition under BGP multipath scenario. [PR1567918](#)

Resolved Issues: 20.4R1

General Routing

- On PTX10008 and PTX5000 routers, the output of the **show filter index number counter** command shows value as zero. [PR1420057](#)
- On PTX10016 routers, after device reboot, the FPC takes a long time to come up and hence MKA sessions establishment is delayed. The error message **Frame 08: sp = 0x48d222b8, pc = 0x10fad3bc , blaze fpc2 SCHED: Thread 59 (PFE Manager) ran for 2177 ms without yielding** is observed. [PR1477585](#)
- On PTX10016 routers, if aggregated Ethernet member or interface flow control is in disabled state, then it does not enable its own. [PR1478715](#)
- The Layer 2 VPN might flap and the CE device facing interface cannot restore the TX optical laser power even if the Layer 2 VPN is in the **Up** status under the asynchronous-notification. [PR1486181](#)
- In IP-in-IP, end-to-end (CE device to CE device) traceroute is not working as expected. [PR1488379](#)
- Dynamic tunnels traceoptions does not offer state tracing and causes JTASK_SCHED_SLIP with single underlay route bounce. [PR1493236](#)
- FPC ukern core file is not transferred to Routing Engine in a scaled setup. [PR1500418](#)
- The error message **mpls_extra NULL** might be seen during add, change, and delete of MPLS route. [PR1502385](#)
- The packetio crashes during the initialization and this might result in a second reboot. [PR1505150](#)
- ERO update by the controller for branch LSP might cause issues. [PR1508412](#)
- BIND does not sufficiently limit the number of fetches performed when processing referrals. [PR1512212](#)
- The routes update might fail upon the HMC memory issue and traffic impact might be seen. [PR1515092](#)
- On PTX5000 and PTX3000 routers, the FPC E might get stuck when the packet is switched internally between FPC connected port towards Routing Engine connected ports. [PR1519673](#)
- Sampling with the rate limiter statement enabled crosses the sample rate of 65535. [PR1525589](#)
- Running SNMP MIB walk and executing **show interfaces** command might cause the picd to crash. [PR1533766](#)
- The error message **expr_dfw_action_topo_connect_anh:1434 expr_dfw_action_topo_connect_anh:eda_anh_discard is FALSE for nh-id 568 - return** is observed in PTX1000 routers. [PR1540064](#)

- The Packet Forwarding Engine might crash in MPLS IPv6-tunneling scenario when the next hop changes. [PR1540793](#)
- Traffic might drop silently after swapping an FPC type 3 card with an FPC type 1 card in the same slot on a PTX3000 router. [PR1547790](#)

Infrastructure

- Interface drop counters might display 0 during a race condition and voq statistics are also polled simultaneously. [PR1537960](#)

Interfaces and Chassis

- The error message **Request failed: OID not increasing: ieee8021CfmStackServiceSelectorType** is observed. [PR1517046](#)
- EOAM IEEE802.3ah link discovery state is **Down** instead of **Active Send Local** after deactivating interfaces. [PR1532979](#)
- Logs are not being written in `/var/log/messages` on certain PTX Series platforms. [PR1551374](#)

MPLS

- The SNMP trap is sent with incorrect OID `jnxSpSvcSetZoneEntered`. [PR1517667](#)

Network Management and Monitoring

- The SNMP MIB `ifInErrors` [OID 1.3.6.1.2.1.2.2.1.14] reports wrong values. [PR1534286](#)
- The syslog messages might not be sent with correct port. [PR1545829](#)

Routing Protocols

- The **show dynamic-tunnels database** command does not show the statistics for the first time and fetches the traffic statistics the second time. [PR1445705](#)
- The `ppmd` process crashes after configuring S-BFD responder on the PTX Series routers with the RE-DUO-2600 Routing Engine. [PR1477525](#)
- The `rpd` process might report 100 percent CPU usage with the BGP route damping enabled. [PR1514635](#)

SEE ALSO

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Documentation Updates

There are no errata or changes in Junos OS Release 20.4R3 documentation for PTX Series Routers.

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Migration, Upgrade, and Downgrade Instructions

IN THIS SECTION

- [Basic Procedure for Upgrading to Release 20.4 | 239](#)
- [Upgrade and Downgrade Support Policy for Junos OS Releases | 242](#)
- [Upgrading a Router with Redundant Routing Engines | 243](#)

This section contains the procedure to upgrade Junos OS, and the upgrade and downgrade policies for Junos OS for the PTX Series. Upgrading or downgrading Junos OS might take several hours, depending on the size and configuration of the network.

Basic Procedure for Upgrading to Release 20.4

When upgrading or downgrading Junos OS, use the **jinstall** package. For information about the contents of the **jinstall** package and details of the installation process, see the [Installation and Upgrade Guide](#). Use

other packages, such as the **jbundle** package, only when so instructed by a Juniper Networks support representative.

NOTE: Back up the file system and the currently active Junos OS configuration before upgrading Junos OS. This allows you to recover to a known, stable environment if the upgrade is unsuccessful. Issue the following command:

```
user@host>request system snapshot
```

NOTE: The installation process rebuilds the file system and completely reinstalls Junos OS. Configuration information from the previous software installation is retained, but the contents of log files might be erased. Stored files on the router, such as configuration templates and shell scripts (the only exceptions are the `juniper.conf` and `ssh` files), might be removed. To preserve the stored files, copy them to another system before upgrading or downgrading the routing platform. For more information, see the [Installation and Upgrade Guide](#).

NOTE: We recommend that you upgrade all software packages out of band using the console because in-band connections are lost during the upgrade process.

To download and install Junos OS Release 20.4R3:

1. Using a Web browser, navigate to the All Junos Platforms software download URL on the Juniper Networks webpage:
<https://support.juniper.net/support/downloads/>
2. Select the name of the Junos OS platform for the software that you want to download.
3. Select the release number (the number of the software version that you want to download) from the Release drop-down list to the right of the Download Software page.
4. Select the **Software** tab.
5. In the Install Package section of the Software tab, select the software package for the release.
6. Log in to the Juniper Networks authentication system by using the username (generally your e-mail address) and password supplied by Juniper Networks representatives.

7. Review and accept the End User License Agreement.
8. Download the software to a local host.
9. Copy the software to the routing platform or to your internal software distribution site.
10. Install the new jinstall package on the router.

NOTE: We recommend that you upgrade all software packages out of band using the console because in-band connections are lost during the upgrade process.

All customers except the customers in the Eurasian Customs Union (currently composed of Armenia, Belarus, Kazakhstan, Kyrgyzstan, and Russia) can use the following package:

```
user@host> request system software add validate reboot
source/junos-install-ptx-x86-64-20.4R2.9.tgz
```

Customers in the Eurasian Customs Union (currently composed of Armenia, Belarus, Kazakhstan, Kyrgyzstan, and Russia) can use the following package (limited encryption Junos OS package):

```
user@host> request system software add validate reboot
source/junos-install-ptx-x86-64-20.4R2.9-limited.tgz
```

Replace the source with one of the following values:

- **/pathname**—For a software package that is installed from a local directory on the router.
- For software packages that are downloaded and installed from a remote location:
 - **ftp://hostname/pathname**
 - **http://hostname/pathname**
 - **scp://hostname/pathname**

The **validate** option validates the software package against the current configuration as a prerequisite to adding the software package to ensure that the router reboots successfully. This is the default behavior when the software package being added is a different release.

Adding the **reboot** command reboots the router after the upgrade is validated and installed. When the reboot is complete, the router displays the login prompt. The loading process might take 5 to 10 minutes.

Rebooting occurs only if the upgrade is successful.

NOTE: You need to install the Junos OS software package and host software package on the routers with the RE-PTX-X8 Routing Engine. For upgrading the host OS on this router with VM Host support, use the `junos-vmhost-install-x.tgz` image and specify the name of the regular package in the **`request vmhost software add`** command. For more information, see the VM Host Installation topic in the [Installation and Upgrade Guide](#).

NOTE: After you install a Junos OS Release 20.4 jinstall package, you cannot return to the previously installed software by issuing the **`request system software rollback`** command. Instead, you must issue the **`request system software add validate`** command and specify the jinstall package that corresponds to the previously installed software.

NOTE: Most of the existing **`request system`** commands are not supported on routers with RE-PTX-X8 Routing Engines. See the VM Host Software Administrative Commands in the [Installation and Upgrade Guide](#).

Upgrade and Downgrade Support Policy for Junos OS Releases

We have two types of releases, standard EOL and EEOL:

- Standard End of Life (EOL) releases have engineering support for twenty four months after the first general availability date and customer support for an additional six more months.
- Extended End of Life (EEOL) releases have engineering support for thirty six months after the first general availability date and customer support for an additional six more months.

For both standard EOL and EEOL releases, you can upgrade to the next three subsequent releases or downgrade to the previous three releases. For example, you can upgrade from 19.2 to the next three releases – 19.3, 19.4 and 20.1 or downgrade to the previous three releases – 19.1, 18.4 and 18.3.

For EEOL releases only, you have an additional option - you can upgrade directly from one EEOL release to the next two subsequent EEOL releases, even if the target release is beyond the next three releases. Likewise, you can downgrade directly from one EEOL release to the previous two EEOL releases, even if the target release is beyond the previous three releases. For example, 19.2 is an EEOL release. Hence, you

can upgrade from 19.2 to the next two EEOL releases – 19.3 and 19.4 or downgrade to the previous two EEOL releases – 19.1 and 18.4.

Release Type	End of Engineering (EOE)	End of Support (EOS)	Upgrade and Downgrade to subsequent 3 releases	Upgrade and Downgrade to subsequent 2 EEOL releases
Standard End of Life (EOL)	24 months	End of Engineering + 6 months	Yes	No
Extended End of Life (EEOL)	36 months	End of Engineering + 6 months	Yes	Yes

For more information about standard EOL and EEOL releases, see <https://www.juniper.net/support/eol/junos.html>.

For information about software installation and upgrade, see the [Installation and Upgrade Guide](#).

Upgrading a Router with Redundant Routing Engines

If the router has two Routing Engines, perform a Junos OS installation on each Routing Engine separately to avoid disrupting network operation as follows:

1. Disable graceful Routing Engine switchover (GRES) on the master Routing Engine and save the configuration change to both Routing Engines.
2. Install the new Junos OS release on the backup Routing Engine while keeping the currently running software version on the master Routing Engine.
3. After making sure that the new software version is running correctly on the backup Routing Engine, switch over to the backup Routing Engine to activate the new software.
4. Install the new software on the original master Routing Engine that is now active as the backup Routing Engine.

For the detailed procedure, see the [Installation and Upgrade Guide](#).

SEE ALSO

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Junos OS Release Notes for the QFX Series

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These release notes accompany Junos OS Release 20.4R3 for the QFX Series. They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

You can also find these release notes on the Juniper Networks Junos OS Documentation webpage, located at https://www.juniper.net/documentation/product/en_US/junos-os.

What's New

IN THIS SECTION

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- What's New in Release 20.4R2 | 245
- What's New in Release 20.4R1 | 246

Learn about new features introduced in the Junos OS main and maintenance releases for QFX Series Switches.

NOTE: The following QFX Series platforms are supported in Release 20.4R3: QFX5100, QFX5110-32Q, QFX5110-48S, QFX5120, QFX5200, QFX5210, QFX10002, QFX10002-60C, QFX10008, and QFX10016.

Junos on White Box runs on Accton Edgecore AS7816-64X switches in this release. The software is based on Junos OS running on QFX5210 switches, so release-note items that apply to QFX5210 switches also apply to Junos on White Box.

What's New in Release 20.4R3

MPLS

- **Support for optimizing auto-bandwidth adjustments for MPLS LSPs (QFX10008)**—Starting in Junos OS Release 20.4R3, you can configure faster auto-bandwidth adjustment for MPLS LSPs under overflow or underflow conditions. This feature decreases the minimum allowed **adjust-threshold-overflow-limit** and **adjust-interval** to 150 seconds when **adjust-threshold-overflow-limit** and **adjust-threshold-underflow-limit** cross the configured threshold values. In releases earlier than Junos OS Release 20.4R3, the **adjust-interval** is 300 seconds under overflow or underflow conditions.

You can configure faster in-place LSP bandwidth update that avoids signaling of a new LSP instance as part of make-before-break. To configure faster in-place LSP bandwidth update, include the **in-place-lsp-bandwidth-update** configuration statement at the **[edit protocols mpls label-switched-path lsp-name]** hierarchy level.

You can also configure RSVP interfaces to support subscription percentage per priority. To configure subscription percentage per priority, include the **subscription priority priority percent value** configuration statement at the **[edit protocols rsvp interface interface-name]** hierarchy level.

[See [Configuring Optimized Auto-bandwidth Adjustments for MPLS LSPs](#).]

When you enable the configuration statement **route-age-bgp-view** at the **[edit protocols bgp]** or the **[edit protocols bgp family inet unicast]** hierarchy level, the **route-age** of BGP routes reflects protocol updates events. However, it does not reflect internal events causing any route change, such as route-resolution, bgp-multipath.

What's New in Release 20.4R2

There are no new features or enhancements to existing features for QFX Series Switches in Junos OS Release 20.4R2.

What's New in Release 20.4R1

Hardware

- **Support for QSA Adapter (QFX10002-60C)**—Starting in Junos OS Release 20.4R1, you can use the QSA Adapter to support 1GbE and 10GbE connections on the QFX10002-60C switches.

[See the [Hardware Compatibility Tool \(HCT\)](#) for details.]

- **Support for transceivers, AOCs, and DACs (QFX5120-48T)**—Starting in Junos OS Release 20.4R1, transceivers, AOCs, and DACs are supported on the QFX5120-48T switches.

[See the [Hardware Compatibility Tool \(HCT\)](#) for details.]

- **New QFX5120-48YM Switch (QFX Series)**—In Junos OS Release 20.4R1, we introduce the QFX5120-48YM, an ideal switch for data center top-of-rack, leaf-and-spine deployments, enterprise multicloud deployments, and campus distribution or core deployments. The QFX5120-48YM is a 25GbE/100GbE switch that offers 48 SFP28 ports and 8 QSFP28 ports. The 48 SFP28 ports support 1-Gbps, 10-Gbps, and 25-Gbps speeds and the 8 QSFP28 ports support 40-Gbps and 100-Gbps speeds. QFX5120-48YM switches support both manual and auto-channelization, but manual CLI channelization always takes precedence. [See [Port Settings](#).]

To install the QFX5120-48YM switch hardware and perform initial software configuration, routine maintenance, and troubleshooting, see [QFX5120 Switch Hardware Guide](#). See [Feature Explorer](#) for the complete list of features for any platform.

[Table 3 on page 246](#) summarizes the QFX5120-48YM features in Junos OS Release 20.4R1.

Table 3: Features Supported by the QFX5120-48YM

Feature	Description
Access security	• DHCP and DHCPv6 snooping. [See DHCP Snooping.] • IP and IPv6 source guard. [See Understanding IP Source Guard for Port Security on Switches.] • Dynamic ARP inspection (DAI). [See Understanding and Using Dynamic ARP Inspection (DAI).] • IPv6 neighbor discovery inspection. [See IPv6 Neighbor Discovery Inspection.]
Authentication and access control	• IEEE 802.1X authentication. [See User Access and Authentication User Guide.] • RADIUS and TACACS+ authentication and accounting. [See Authentication Order for RADIUS, TACACS+, and Local Password.] • Authentication bypass access (based on host MAC address) and fallback. [See Static MAC Bypass of 802.1X and MAC RADIUS.] • Captive portal authentication for Layer 2 and Layer 3 interfaces. [See Captive Portal Authentication.]

Table 3: Features Supported by the QFX5120-48YM (*continued*)

Feature	Description
Class of service (CoS)	• QFX5120-48YM switches support all class-of-service (CoS) features except the following: shared unicast and multi-destination classifiers, forwarding classes, and output queues; CoS flexible hierarchical scheduling (ETS); virtual output queue (VOQ) architecture; and CoS command to detect the source of RED-dropped packets. [See CoS Support on QFX Series Switches, EX4600 Line of Switches, and QFabric Systems.]
DHCP	• DHCP server. [See DHCP Server.] • DHCP relay agent and DHCP smart relay. [See DHCP Relay Agent.] • DHCP server and client in separate routing instances. [See DHCP Message Exchange Between DHCP Clients and DHCP Server in Different VRFs.] • DHCP relay with option 82 for Layer 2 VLANs and Layer 3 interface. [See DHCP Relay Agent Information Option (Option 82).] • DHCP option 82 with textual interface description. [See DHCP Relay Agent Information Option (Option 82).] • DHCPv6 option 79. [See relay-agent-option-79.] • DHCP static addresses. [See Configuring Static DHCP IP Addresses.] • Extended DHCP (also referred to as virtual router (VR) aware DHCP). [See Legacy DHCP and Extended DHCP.]

Table 3: Features Supported by the QFX5120-48YM (continued)

Feature	Description
EVPN-VXLAN	

Table 3: Features Supported by the QFX5120-48YM (*continued*)

Feature	Description
	• In a centrally routed bridging overlay, the QFX5120-48YM switch can act as a leaf device (Layer 2 VXLAN gateway). [See Example: Configuring an EVPN Control Plane and VXLAN Data Plane.] • In an edge-routed bridging overlay, the QFX5120-48YM switch can act as a leaf device (Layer 2 and 3 VXLAN gateways). [See Example: Configuring a QFX5110 Switch as Layer 2 and 3 VXLAN Gateways in an EVPN-VXLAN Edge-Routed Bridging Overlay.] • In a campus EVPN multihoming environment, you can deploy two QFX5120-48YM switches as distribution switches with ESI-LAG (Layer 2 and 3 VXLAN gateways) to eliminate STP. • In a campus fabric environment, you can deploy QFX5120-48YM switches as distribution or core switches in a centrally routed bridging overlay or an edge routed bridging overlay. • In the spine and leaf roles, the QFX5120-48YM switches support the following features: • Firewall filtering and policing of EVPN-VXLAN traffic. [See Overview of Firewall Filters (QFX Series).] • IGMPv2 snooping in a centrally routed bridging overlay. Supported use cases include intra-VLAN, inter-VLAN with IRB interfaces and PIM, and inter-VLAN with a PIM gateway and Layer 2 connectivity. [See Overview of Multicast Forwarding with IGMP Snooping or MLD Snooping in an EVPN-VXLAN Environment.] • Support for IPv6 data traffic. [See Routing IPv6 Data Traffic through an EVPN-VXLAN Network with an IPv4 Underlay.] • MAC filtering, storm control, and port mirroring. [See MAC Filtering, Storm Control, and Port Mirroring Support in an EVPN-VXLAN Environment.] • EVPN pure Type 5 routes. [See EVPN Type-5 Route with VXLAN Encapsulation for EVPN-VXLAN.] • EVPN proxy ARP and ARP suppression, and proxy Neighbor Discovery Protocol (NDP) and NDP suppression. [See EVPN Proxy ARP and ARP Suppression, and Proxy NDP and NDP Suppression.] • Support for OSPF, IS-IS, BGP, and static routing on IRB interfaces. [See Supported Protocols on an IRB Interface in EVPN-VXLAN.] • Virtual machine traffic optimization for ingress interfaces. [See Ingress Virtual Machine Traffic Optimization.] • Layer 2 and 3 families, encapsulation types, and VXLAN on the same physical interface. [See Understanding Flexible Ethernet Services Support With EVPN-VXLAN.]

Table 3: Features Supported by the QFX5120-48YM (continued)

Feature	Description
	• Selective multicast forwarding. [See Overview of Selective Multicast Forwarding.] • MAC mobility and duplicate MAC address detection and suppression. [See Overview of MAC Mobility.]
Firewalls and policers	• Firewall filters provide rules that define whether to permit, deny, or forward packets that are transiting an interface on the device from a source address to a destination address. The supported firewall filter and policer features include: • Policer mark down action and policing/rate limiting • Single-rate two-color marking (ingress), single-rate tricolor marking (color aware, color blind), and two-rate tricolor marking • Filter-based forwarding (FBF) and FBF with destination and source prefix list on IPv6 interfaces • Dynamic allocation of TCAM memory to firewall filters and error message displayed when TCAM is full • Enhanced filter classification of CPU-generated packets • Firewall filter actions: assign forwarding class, counters; logging, syslog, reject; mirroring to an interface; and permit, drop, police, and mark • Firewall filter flexible match conditions • Firewall filters on loopback interface and management interface • IPv6 fields for ingress port and VLAN firewall filters • Policer action for MPLS firewall filters • Port firewall filters (egress and ingress), routed firewall filters (egress and ingress) and VLAN firewall filters (egress and ingress) • TCP/UDP port ranges in classification • Loopback filter optimization • Firewall filtering and policing on EVPN-VXLAN traffic • Filter-based GRE de-encapsulation • Firewall filter support on Layer 3 interfaces. [See Firewall Filter Match Conditions and Actions (QFX5220).]
High availability	• Nonstop bridging (NSB), and nonstop active routing (NSR) for IPv6 and OSPFv2.

Table 3: Features Supported by the QFX5120-48YM (*continued*)

Feature	Description
Interfaces and chassis	• Support for the following resiliency features: • Operating system resiliency to recover the Junos OS software by using the recovery mode option on the Grub menu, which is visible after BIOS has booted up. • Partial resiliency for DIMM errors, machine-check exception (MCE), and PCI Express advanced error reporting (AER). If required, you can take assistance from Juniper Networks Technical Assistance Center (JTAC) to manually debug these type of errors when they occur. [See Channelizing Interfaces on QFX5120-48YM Switches.] • Support for channelizing interfaces. The QFX5120-48YM contains a total of 56 ports, of which 8 are QSFP28 ports and 48 are SFP28 ports. To channelize speed, you use the channel-speed statement. For setting speed, you use the set chassis fpc 0 pic 0 port <25g 1g 10g 50g 100g 40g> command. The speeds supported are: • 1 Gbps, 10 Gbps, and 25 Gbps on SFP28 • 40 Gbps, 100 Gbps, 4x25 Gbps, 4x10 Gbps, 2x50 Gbps on QSFP28 • 2x50 Gbps, 4x25 Gbps, or 4x10 Gbps channelization is supported on ports 50 and 52. [See Port Settings.]
Junos OS XML API and scripting	• Python, SLAX, and XSLT scripting languages, commit scripts and macros, event policy and event scripts, op scripts, and SNMP scripts. [See Automation Scripting User Guide.]

Table 3: Features Supported by the QFX5120-48YM (continued)

Feature	Description
Layer 2 features	

Table 3: Features Supported by the QFX5120-48YM (*continued*)

Feature	Description
	• Layer 2 protocol tunneling (L2PT) support to tunnel any of the following Layer 2 protocols: CDP, E-LMI, GVRP, IEEE 802.1X, IEEE 802.3AH, LACP, LLDP, MMRP, MVRP, STP (including RSTP and MSTP), UDLD, VSTP, and VTP. [See Layer 2 Protocol Tunneling.] • Support for the following Layer 2 multicast features: • IGMP snooping with IGMPv1, IGMPv2, and IGMPv3 • IGMP proxy • IGMP querier • Virtual router (VRF-lite) IGMP snooping [See IGMP Snooping Overview.] • Support for the following Layer 2 unicast features: • 802.1D • 802.1w (RSTP) • 802.1s (MST) • BPDU protection • Loop protection • Root protection • VSTP • 802.1Q VLAN trunking • 802.1p • IRB (Integrated routing and bridging Interface) • Layer 3 vlan-tagged sub-interfaces • 4096 VLANs • Multiple VLAN Registration Protocol (802.1ak) • MAC address filtering • MAC address aging configuration • Static MAC address assignment for interface • Pe-VLAN MAC learning (limit) • Pe-VLAN MAC learning (limit) • MAC Learning Disable • Persistent MAC (Sticky MAC) • Link Aggregation (Static and Dynamic) with LACP (Fast and Slow LACP) • LLDP • MC-LAG with configuration sync

Table 3: Features Supported by the QFX5120-48YM (continued)

Feature	Description
	• Uplink Failure Detection (UFD) • VxLAN L2 Gateway (Static, OVSDb, EVPN) • QinQ Tag manipulation • 802.1x (Access control) [See Layer 2 Networking.]

Table 3: Features Supported by the QFX5120-48YM (continued)

Feature	Description
Layer 3 features	

Table 3: Features Supported by the QFX5120-48YM (*continued*)

Feature	Description
	• Traceroute over Layer 3 VPN. • Virtual routing and forwarding (VRF) support in IRB interfaces in a Layer 3 VPN. • VRF-lite, BGP, IGMP, IS-IS, OSPF, PIM, and RIP. • Support for the following Layer 3 multicast features: • IGMP version 1 (IGMPv1), version 2 (IGMPv2), and version 3 (IGMPv3) • IGMP filtering • PIM sparse mode (PIM-SM) • PIM dense mode (PIM-DM) • PIM source-specific multicast (PIM-SSM) Multicast Source Discovery Protocol (MSDP) IGMP and PIM are also supported on virtual routers. [See Multicast Overview.] • Support for the following Layer 3 unicast features: • Virtual Router Redundancy Protocol (VRRP) • Static routing • OSPFv2 • IPv4 BGP • IPv4 MBGP • BGP 4-byte ASN support • BGP Add Path (BGP-AP) • IS-IS • BFD (for RIP, OSPF, ISIS, BGP, PIM) • Filter based forwarding (FBF) • Unicast reverse path forwarding (unicast RPF) • IP directed broadcast traffic forwarding • IPv4 over GRE • IPv6 neighbor discovery protocol • Path MTU discovery • IPv6 CoS (BA, classification and rewrite, scheduling based on TC) • IPv6 ping • IPv6 static routing • IPv6 traceroute • IPv6 stateless auto-configuration

Table 3: Features Supported by the QFX5120-48YM (*continued*)

Feature	Description
	• IPv6 OSPFv3 • IPv6 IS-IS • IPv6 BGP • VRRPv3 • 32-way equal-cost multipath (ECMP) • VXLAN Layer 3 Gateway • MPLS over UDP • DHCP snooping • IPv6 Ready Logo certification
MPLS	• MPLS support for label edge routers (LERs) and label switch routers (LSR). [See MPLS Overview for Switches.] • MPLS signaling protocols LDP and RSVP. [See LDP Overview and RSVP Overview.] • Fast reroute (FRR) support (a component of MPLS local protection for both one-to-one and many-to-one local protection). • Static LSPs. [See LSP Overview.] • MPLS node protection, link protection, and statistics for static LSPs. • MPLS OAM (LSP ping). • MPLS statistics. [See statistics (Protocols MPLS).] • MPLS automatic bandwidth allocation and dynamic count sizing. • MPLS with RSVP-based LSPs. • IRB interfaces over an MPLS core network. [See Example: Configuring IRB Interfaces on QFX5100 Switches over an MPLS Core Network.] • MPLS stitching for virtual machine connections. [See Using MPLS Stitching with BGP to Connect Virtual Machines.] • MPLS over Layer 3 subinterfaces. [See MPLS Limitations on QFX Series and EX4600 Switches.] • RSVP-traffic engineering (RSVP-TE), traffic engineering extensions (OSPF-TE, IS-IS-TE), Path Computation Element Protocol (PCEP), and PCE-initiated LSPs for the PCEP implementation. [See MPLS Applications User Guide.] • Equal-cost multipath (ECMP) operation on MPLS using firewall filters.

Table 3: Features Supported by the QFX5120-48YM (*continued*)

Feature	Description
OVSDB-VXLAN	• In an OVSDB-VXLAN environment, the QFX5120-48YM switch can act as a Layer 2 VXLAN gateway. [See Understanding the OVSDB Protocol Running on Juniper Networks Devices.] • In a manual (PIM-based) VXLAN environment, the QFX5120-48YM switch can act as: • A Layer 2 VXLAN gateway. • A transit Layer 3 switch for downstream VTEPs. • A Layer 2 VXLAN gateway. [See Examples: Manually Configuring VXLANs on QFX Series and EX4600 Switches.]
Network management and monitoring	• Support for the following services: • sFlow networking monitoring technology—Collects samples of network packets and sends them in a UDP datagram to a monitoring station called a <i>collector</i>. You can configure sFlow technology on a device to monitor traffic continuously at wire speed on all interfaces simultaneously. The inline-sampling configuration option is available. • Local, remote, and extended port mirroring—Copies packets entering or exiting a port or entering a VLAN and sends the copies to a local interface (local port mirroring), to a VLAN (remote port mirroring), or to the IP address of a device running an analyzer application on a remote network (extended port mirroring). When you use extended port mirroring, the mirrored packets are GRE-encapsulated. • Storm control—Causes a device to monitor traffic levels and take a specified action when a specified traffic level—called the <i>storm control level</i>—is exceeded, thus preventing packets from proliferating and degrading service. You can configure devices to drop broadcast and unknown unicast packets, shut down interfaces, or temporarily disable interfaces when the storm control level is exceeded. [See Overview of sFlow Technology, Understanding Port Mirroring, and Understanding Storm Control.] • Support for adding nonnative YANG modules to the Junos OS schema. [See Understanding the Management of Nonnative YANG Modules on Devices Running Junos OS.] • Puppet for Junos OS support. [See Puppet for Junos OS Administration Guide.]
Port Security	• Support for Media Access Control security with 256-bit cipher suite. GCM-AES-256 has a maximum key length of 256 bits and is also available with extended packet numbering (GCM-AES-XPB-256). [See Understanding Media Access Control Security (MACsec).]

Table 3: Features Supported by the QFX5120-48YM (*continued*)

Feature	Description
Software installation and upgrade	- Secure boot—The implementation is based on the UEFI 2.4 standard. [See Software Installation and Upgrade Guide.] - You need a license to use the features on the QFX5120-48YM. To find out what features are supported on this device, see QFX Switch Support for the Juniper Flex Program. To add, delete, and manage licenses, see Managing Licenses. - Zero-touch provisioning (ZTP). [See Zero Touch Provisioning Overview.] - Virtualization enables the switch to support multiple instances of Junos OS and other operating systems on the same Routing Engine. One instance of Junos OS, which runs as a guest operating system, is launched by default. You need to log in to this instance for operations and management. The Routing Engines on the QFX5120-48YM switches support the Wind River Linux 9 (WRL9) kernel version. [See What Are VMHosts?.]

To view the hardware compatibility matrix for optical interfaces and transceivers supported on the QFX5120-48YM, see the [Hardware Compatibility Tool](#).

Class of Service (CoS)

- **Priority-based flow control (PFC) using Differentiated Services code points (DSCP) at Layer 3 for untagged traffic (QFX5210 switches)**—Starting in Junos OS Release 20.4R1, to support lossless traffic across Layer 3 connections to Layer 2 subnetworks on QFX5210 switches, you can configure priority-based flow control (PFC) to operate using 6-bit DSCP values from Layer 3 headers of untagged VLAN traffic. You can do this rather than use IEEE 802.1p priority values in Layer 2 VLAN-tagged packet headers. You need DSCP-based PFC to support remote direct memory access (RDMA) over converged Ethernet version 2 (RoCEv2).

To enable DSCP-based PFC:

1. Map a forwarding class to a PFC priority using the **pfc-priority** statement.
2. Define a congestion notification profile to enable PFC on traffic specified by a 6-bit DSCP value.
3. Set up a classifier for the DSCP value and the PFC-mapped forwarding class.

[See [Understanding PFC Using DSCP at Layer 3 for Untagged Traffic](#).]

EVPN

- **Multicast with IGMPv3 in EVPN-VXLAN centrally-routed bridging overlay fabrics (QFX10000, QFX5110, and QFX5120)**—Starting in Junos OS Release 20.4R1, you can configure IGMPv3 multicast in an EVPN-VXLAN centrally-routed bridging overlay fabric with multihoming for the following IPv4 multicast traffic use cases:

- Intra-VLAN forwarding
- Inter-VLAN routing using:
 - IRB interfaces configured with PIM
 - A PIM gateway router (for Layer 2 or Layer 3 connectivity)
 - An external multicast router

IGMPv3 multicast works with these multicast optimizations:

- IGMP snooping
- Selective multicast (SMET) forwarding
- Assisted replication (AR)

These devices process IGMPv3 reports in one of two modes:

- As any-source multicast (ASM) (*,G) reports by default
- As source-specific multicast (SSM) (S,G) reports only (if you explicitly configure this mode)

[See [Overview of Multicast Forwarding with IGMP Snooping in an EVPN-VXLAN Environment](#), [Overview of Selective Multicast Forwarding](#), [Assisted Replication Multicast Optimization in EVPN Networks](#), and [evpn-ssm-reports-only](#).]

- **MAC-VRF with EVPN-VXLAN (MX Series and vMX routers; QFX5100, QFX5110, QFX5120, QFX5200, QFX5210, QFX10002-60C, QFX10008, and QFX10016 switches)**—Data center service providers must support multiple customers with their own routing and bridging policies in the same physical network. To accommodate this requirement, you can now configure multiple customer-specific EVPN instances (EVIs) of type **mac-vrf**, each of which can support a different EVPN service type. This configuration results in customer-specific virtual routing and forwarding (VRF) tables with MAC addresses on each Juniper Networks device that serves as a virtual tunnel endpoint (VTEP) in the EVPN-VXLAN network.

NOTE: We support MAC-VRF routing instances for EVPN unicast routes only.

To support this feature, we introduce a uniform routing instance configuration, which complies with RFC 7432, *BGP MPLS-Based Ethernet VPN*. The uniform configuration eliminates hardware restrictions that limit the number of EVIs and combinations of EVIs with their respective policies that can simultaneously exist. The common configuration includes the following new CLI elements:

- The **mac-vrf** keyword at the **[edit routing-instances name instance-type]** hierarchy level.
- The **service-type** configuration statement at the **[edit routing-instances name]** hierarchy level. We support VLAN-based, VLAN-aware, and VLAN-bundle service types.
- (QFX10000 line of switches only) The **forwarding-instance** configuration statement at the **[edit routing-instances name]** hierarchy level. With this optional configuration statement, you can map

multiple routing instances to a single forwarding instance. If you don't include this configuration statement, the default forwarding instance is used.

We continue to support the existing method of routing instance configuration along with the new uniform routing instance configuration.

[See [EVPN User Guide](#).]

- **Filter-based forwarding in EVPN-VXLAN networks (QFX10002-36Q, QFX10002-72Q, QFX10002-60C, QFX10008, and QFX10016)**—Instead of using the routing functionality typically provided by routing protocols, you can now use firewall filter-based forwarding. With filter-based forwarding, you can use firewall filter match conditions and actions to better control how traffic is routed in your EVPN-VXLAN network.

We support the following filter-based forwarding use cases:

- Redirecting traffic received on IRB interfaces. To set up a firewall filter:
 - Create an input filter.

NOTE: When specifying an IP address for this filter, you can use either IPv4 or IPv6 addresses.

- Specify the following match criteria:
 - The source and destination IP addresses in the inner header after a packet is de-encapsulated.
 - The source and destination ports in the inner header.
- Specify an action that directs matching packets to one of the following:
 - A routing instance
 - A next hop
 - A next hop and a routing instance
- Apply the filter to an IRB interface with or without a virtual gateway address or an anycast address.
- Handling transit traffic that matches a VXLAN network identifier (VNI). To set up a firewall filter:
 - Create an input filter.

NOTE: When specifying an IP address for this filter, you must use an IPv4 address.

- Specify the following match criteria:
 - VNI.

- The source and destination IP addresses in the outer header.
- For the action, specify **count** or any other firewall filter action that is supported by the QFX10000 line of switches.
- Apply the filter to a Layer 3 interface.

[See [Understanding Filter-Based Forwarding](#).]

- **Loop detection for EVPN-VXLAN fabrics (QFX5120, QFX5200, QFX5210, and QFX5220)**—You can configure loop detection on the server-facing Layer 2 interfaces of the leaf devices in an EVPN-VXLAN fabric. This feature can detect the following types of Ethernet loops:
 - A loop between two interfaces with different Ethernet segment identifiers (ESIs). This loop is typically caused by miswiring fabric components.
 - A loop between two interfaces with the same ESI. This loop is typically caused by miswiring a third-party switch to the fabric.

After you've enabled loop detection, the interfaces periodically send multicast loop-detection protocol data units (PDUs). If a loop detection-enabled interface receives a PDU, a loop is detected, which triggers the configured action to break the loop. For example, if the configured action is interface-down, the interface is brought down. After the revert-interval timer expires, the configured action is reverted, and the interface is brought back up again.

[See [loop-detect](#).]

- **IPv6 multicast with MLDv1 and MLDv2 in EVPN-VXLAN centrally-routed bridging overlay fabrics (QFX10000, QFX5110, and QFX5120)**—Starting in Junos OS Release 20.4R1, you can configure MLDv1 and MLDv2 multicast in an EVPN-VXLAN centrally-routed bridging overlay fabric with multihoming for the following IPv6 multicast traffic use cases:
 - Intra-VLAN forwarding
 - Inter-VLAN routing using:
 - IRB interfaces configured with PIM
 - A PIM gateway router (for Layer 2 or Layer 3 connectivity)
 - An external multicast router

MLD multicast works with these multicast optimizations:

- MLD snooping
- Selective multicast (SMET) forwarding
- Assisted replication (AR)

These devices process MLD reports as follows:

- MLDv1 reports as any-source multicast (ASM) (*,G) reports

- MLDv2 reports in one of two modes:
 - As any-source multicast (ASM) (*,G) reports by default
 - As source-specific multicast (SSM) (S,G) reports only (if you explicitly configure this mode)

[See [Overview of Multicast Forwarding with IGMP Snooping or MLD Snooping in an EVPN-VXLAN Environment](#), [Overview of Selective Multicast Forwarding](#), [Assisted Replication Multicast Optimization in EVPN Networks](#), and [evpn-ssm-reports-only](#).]

- **Seamless EVPN-VXLAN stitching with multicast support (QFX10002-36Q, QFX10002-72Q, QFX10008, and QFX10016)**—Starting in Junos OS Release 20.4R1, we support the following multicast features with the seamless EVPN-VXLAN stitching data center interconnect (DCI) use case:
 - Protocol Independent Multicast (PIM)
 - Internet Group Management Protocol version 2 (IGMPv2) snooping
 - Assisted replication (AR) with the following use cases:
 - The super spine device, which interconnects the data centers, and the AR replicator are two separate devices.
 - The super spine device and the AR replicator are the same device.
 - Selective multicast Ethernet tag (SMET)

[See [interconnect](#).]

Flow-Based and Packet-Based Processing

- **Support for user-defined flex hashing for MPLS traffic flows (QFX5210; Accton Edgecore AS7816 running Junos OS on White Box)**—Starting in Junos OS Release 20.4R1, you can configure user-defined flex hashing to load-balance MPLS traffic based on TCP or UDP source and destination port information. User-defined flex hashing, which supports protocol versions IPv4 and IPv6, enables you to set byte offsets in packet headers to influence hashing computation. You specify two offsets, each 2 bytes in length, from the first 128 bytes of a packet. You can configure the selected bytes to be directly used for hashing or to be used only when the data pattern in these bytes matches specific values (conditional match). To provide load balancing in spine layers, configure flex hashing and encapsulate the traffic in VXLAN, thus enabling entropy at UDP source ports. At de-encapsulation, configure the **no-inner-payload** statement to load-balance based on the outer UDP header.

To configure user-defined flex hashing:

```
set forwarding-options enhanced-hash-key flex-hashing name ethernet mpls num-labels num_labels hash-offset
offset1 base-offset1 offset1-value offset1_value offset1-mask offset1_mask offset2 base-offset2 offset2-value
offset2_value offset2-mask offset2_mask
```

To configure a conditional match (repeat the following command with values for offsets and match data 2-4):

```
set forwarding-options enhanced-hash-key conditional-match name offset1 base-offset1 offset1-value
offset1_value offset1-mask offset1_mask matchdata1 matchdata1 matchdata1-mask matchdata1-mask
```

To enable load balancing on VXLAN transit traffic based on the outer UDP header:

```
set forwarding-options enhanced-hash-key vxlan no-inner-payload
```

To troubleshoot, use the **show forwarding-options enhanced-hash-key** command.

Limitations:

- Use a maximum of two MPLS labels.
- Use only even values for **offset1** and **offset2**.
- If you are using conditional matches, configure the conditions before you attach them to the flex-hashing entry.
- An aggregated Ethernet (ae) or LAG interface is not supported as an input interface. You *can* configure input interfaces on LAGs by configuring the same user-defined flex-hashing data and the same conditional-match data on all *member* interfaces of a LAG interface.
- Apply a similar set of commands to the various member interfaces. For example, if the members of a particular LAG are xe-0/0/2, xe-0/0/3, and xe-0/0/4, configure three slightly different flex-hashing rules on those individual member interfaces—the rules are identical except that they have different names for the incoming traffic:
 - **set forwarding-options enhanced-hash-key flex-hashing FLEX_L2_V6_TCP_4 ethtype mpls interface xe-0/0/2**
set forwarding-options enhanced-hash-key flex-hashing FLEX_L2_V6_TCP_4 ethtype mpls num-labels 2
set forwarding-options enhanced-hash-key flex-hashing FLEX_L2_V6_TCP_4 ethtype mpls conditional-match COND_L2_V6_TCP_4
set forwarding-options enhanced-hash-key flex-hashing FLEX_L2_V6_TCP_4 ethtype mpls hash-offset offset1 base-offset1 start-of-L3-OuterHeader
set forwarding-options enhanced-hash-key flex-hashing FLEX_L2_V6_TCP_4 ethtype mpls hash-offset offset1 offset2 offset2-mask ffff
 [...configuration commands truncated...]
 - **set forwarding-options enhanced-hash-key flex-hashing FLEX_L2_V6_TCP_5 ethtype mpls interface xe-0/0/3**
 [...configuration commands truncated...]

- **set forwarding-options enhanced-hash-key flex-hashing FLEX_L2_V6_TCP_6 ethtype mpls interface xe-0/0/4**

[...configuration commands truncated...]

- Use unique flex-data profile names and unique conditional-data profile names for each member interface—for example, in the following conditional-data profile names, the port number is unique in each instance:
 - ...enhanced-hash-key conditional-match COND_L1_V6_UDP_SRC_PORT_1...
 - ...enhanced-hash-key conditional-match COND_L1_V6_UDP_SRC_PORT_2...

[See [flex-hashing](#).]

High Availability (HA) and Resiliency

- **Disk health monitoring (QFX5100 and QFX5120-48Y)**—Starting in Junos OS Release 20.4R1, the QFX5100 and QFX5120-48Y switches support disk-health monitoring. This feature detects solid-state drive (SSD) failures and reboots the device gracefully to handle those failures. With this feature in place, you need not manually intervene to recover the system from an SSD failure condition. The QFX5100 switches used in a Virtual Chassis as well support this feature.

[See [show chassis routing-engine](#)]

Interfaces and Chassis

- **Retrieve an ECMP or trunk interface hardware hash result for a given input for load balancing (QFX5120)**—Starting in Junos OS Release 20.4R1, you can view the hash parameters that are used by the hashing algorithm and the final egress interface for the traffic you are interested in. Use the CLI command **show forwarding-options load-balance ecmp|trunk** to retrieve this information. The command output provides you information to troubleshoot issues for which you need to know the packet path.

[See [Configuring the Fields in the Algorithm Used To Hash LAG Bundle and ECMP Traffic \(CLI Procedure\)](#) and [show forwarding-options load-balance ecmp|trunk](#).]

IP Tunneling

- **IPIP encapsulation for flexible tunnel interfaces (FTIs) (QFX5100, QFX5110, QFX5120, QFX5200, and QFX5210)**—We've extended flexible tunnel interfaces (FTIs) and existing forwarding constructs to support configuring static IPv4 IP-in-IP tunnels and RIB APIs. To configure an IP-in-IP tunnel on a FTI, use the **ipip** option at the **[edit interfaces interface-name unit logical-unit-number tunnel encapsulation]** hierarchy level.

[See [Configuring Flexible Tunnel Interfaces](#) and [ipip](#).]

Juniper Extension Toolkit

- **Support for static backup paths with IP-in-IP tunnel encapsulation and provisioning APIs (QFX5100, QFX5110, QFX5120, QFX5200, and QFX5210)**—We've enhanced Juniper Extension Toolkit (JET) APIs to enable a controller to set up underlay network backup paths that use IP-in-IP tunnels with IPv4

encapsulation. JET APIs notify the controller of active paths, interfaces, and changes to the interface state. The loop-free backup paths help quickly restore failed core transport networks built with only IP protocols.

[See [JET APIs on Juniper EngNet](#).]

- **Support for policy match condition to match programmed routes (QFX5100, QFX5110, QFX5120, QFX5200, and QFX5210)**—We've introduced a new option programmed that allows policy matches for routes injected by JET APIs. To allow policy matches for routes injected by JET APIs, use the **programmed** option at the **[edit policy-options policy-statement policy-name term term-name from]** hierarchy level. To view details about programmed routes, use the **show route programmed (detail | extensive)** command.

[See [policy-statement](#) and [show route](#).]

- **RIB service API option to control route distribution (QFX5100, QFX5110, QFX5120, QFX5200, and QFX5210)**—We've added a no-advertise flag to the RIB service API per-route RouteAttributes object to limit re-advertisement of the provisioned route. You can set this flag to TRUE to prevent the route from being redistributed to routing protocols and advertised to peers.

[See [JET APIs on Juniper EngNet](#).]

Junos OS XML, API, and Scripting

- **Support for Certificate Authority Chain Profile (EX2300, EX3400, EX4300, MX240, MX480, MX960, PTX-5000, VMX, vSRX and QFX5200)**—Starting in Junos OS Release 20.4R1, you can configure intermediate Certificate Authority (CA) chain profile certificate and perform https REST API request using mutual and server authentications.

To configure intermediate ca-chain certificate, configure **ca-chain ca-chain** statement at the **[edit system services rest https]** hierarchy level.

- **Start time option for interval-based internal events that trigger event policies (EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—Starting in Junos OS Release 20.4R1, when you create an interval-based internal event for triggering event policies, you can specify the start date and time for the initial event. To specify a start time, configure the **start-time** option along with the **time-interval** option at the **[edit event-options generate-event]** hierarchy level.

[See [Generating Internal Events to Trigger Event Policies](#).]

Junos Telemetry Interface

- **BGP neighbor telemetry with sharding (MX Series, PTX Series, and QFX Series)**—Starting in Junos OS Release 20.4R1, BGP neighbor telemetry with sharding (multi-threading) is supported.

[See [Guidelines for gRPC and gNMI Sensors \(Junos Telemetry Interface\)](#).]

Network Management and Monitoring

- **Support for sFlow technology with VXLAN (QFX5110, QFX5120-32C, QFX5120-48Y, and QFX5120-48T-6C)**—Starting in Junos OS Release 20.4R1, we support sFlow technology with VXLAN. sFlow is a monitoring technology for high-speed switched or routed networks. The sFlow agent performs

packet sampling and gathers interface statistics, and then combines the information into UDP datagrams that are sent to sFlow collectors.

Keep the following points in mind:

- True egress sampling is not supported on the switches. Egress sampling is done at the end of the ingress pipeline, and so the egress samples do not contain modifications that are made to the packet in the egress pipeline. When the packet flow is from the access port to the network port, sFlow is configured on the egress network port, and thus packets are captured without VXLAN encapsulation at the collector.
- Egress sampling for BUM traffic is not supported.
- Extended router data with next-hop information is not supported on the switches.
- Sampling on ingress interfaces does not capture CPU-bound traffic.
- You cannot configure sFlow on a (LAG), but you can configure it individually on a LAG member interface.
- You must not configure sFlow for more than one logical interface on a physical interface.

[See [Overview of sFlow Technology](#).]

- **Configuration retrieval using the configuration revision identifier (EX3400, EX4300, MX204, MX240, MX480, MX960, MX2020, PTX3000, PTX10008, QFX5100, QFX10002-60C, SRX5800, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, you can use the configuration revision identifier feature to view the configuration for a specific revision. This configuration database revision can be viewed with the CLI command **show system configuration revision**.

[See [show system configuration revision](#).]

- **sFlow sampling support for IP-IP traffic (QFX5100 and QFX5200)**—Starting in Junos OS Release 20.4R1, you can use sFlow technology to sample IP over IP (IP-IP) traffic at a physical port. This feature is supported for IP-IP tunnels with an IPv4 outer header that carry IPv4 or IPv6 traffic. You can use sFlow monitoring technology to randomly sample network packets from IP-IP tunnels to send the samples to a destination collector for monitoring. Devices that act as an IP-IP tunnel entry point, transit device, or tunnel endpoint support sFlow sampling.

[See [Overview of sFlow Technology](#).]

- **Junos XML protocol operations support loading and comparing configurations using the configuration revision identifier (EX3400, EX4300, MX204, MX240, MX480, MX960, MX2020, PTX3000, PTX10008, QFX5100, QFX10002-60C, SRX5800, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, the Junos XML management protocol operations support loading and comparing configurations by referencing the configuration revision identifier of a committed configuration. You can execute the **<load-configuration>** operation with the **configuration-revision** attribute to load the configuration with the given revision identifier into the candidate configuration. Additionally, you can compare the candidate or active configuration to a previously committed configuration by referencing the configuration revision identifier for the comparison configuration. The **<get-configuration>** operation supports the **compare="configuration-revision"** and **configuration-revision** attributes to perform the comparison.

[See [<get-configuration>](#) and [<load-configuration>](#).]

Platform and Infrastructure

- **Flooding bridge protocol data units (BPDUs) using existing ingress port-based firewall filters (QFX5100, QFX5110, QFX5120-32C, QFX5200, and QFX5210)**—Starting in Junos OS Release 20.4R1, you can configure a new firewall filter CLI action to flood BPDUs using the **set firewall family ethernet-switching filter f1 term t1 then flood** statement. The flexibility to flood BPDUs on a per port basis for the QFX5000 line of switches can be achieved by using the existing ingress port-based firewall filters.

[See [Configuring a Firewall Filter](#).]

Routing Policy and Firewall Filters

- **Support for route's next-hop weight in policy match condition (MX Series, PTX Series, and QFX Series)**—Starting in Junos OS Release 20.4R1, a route with multiple next-hop paths can use the weight associated with a path to identify primary and backup paths. The path with the lowest weight is used as the primary path, and any paths with higher weights are treated as backup paths. You can use the next-hop weight as a match condition in export policies to redistribute IGP and BGP routes based on whether the primary or backup paths are active.

Configure this match condition using the **[edit policy-options policy-statement policy-name term term-name from]** statement.

[See [policy-statement](#) and [show policy](#).]

- **IPv6 support for firewall filtering and policing on EVPN-VXLAN traffic (QFX5100, QFX5110, QFX5120, QFX5200, and QFX5210)**—Starting with Junos OS Release 20.4R1, you can use IPv6 for firewall filters and policers on VXLAN traffic in an EVPN topology. Configure firewall filters at the **[edit firewall]** hierarchy level. For each firewall filter that you apply to a VXLAN, you can specify **family ethernet-switching** to filter Layer 2 (Ethernet) packets or **family inet** or **family inet6** to filter on IRB interfaces. You can apply firewall filters and policers on CE-facing interfaces in the ingress direction only. For IRB interfaces, you can apply filtering only at the ingress point of a non-encapsulated frame routed through the IRB interface.

[See [Understanding VXLANs](#) and [Overview of Firewall Filters](#).]

- **Support for matching IPv6 source addresses from an inet6 egress interface (QFX5110)**—Starting in Junos OS Release 20.4R1, you can configure a firewall filter on an IPv6 egress interface to match specified IPv6 source or destination addresses, for example, to protect a third-party device connected to the switch.

[See [eracl-ip6-match](#) and [Example: Configuring an Egress Filter Based on IPv6 Source or Destination IP Addresses](#).]

Routing Protocols

- **Support for multiple single-hop EBGP sessions on different links using the same IPv6 link-local address (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, you are no longer required to have unique peer addresses for Juniper devices for every EBGP session. You can now enable single-hop EBGP sessions on different links over multiple directly connected peers that use the same IPv6 link-local address.

In earlier Junos OS Releases, BGP peers could be configured with link-local addresses, but multiple BGP peers could not be configured to use the same link-local address on different interfaces.

[See [Configure Multiple Single-Hop EBGP Sessions on Different Links Using the Same Link-Local Address \(IPv6\)](#).]

- **Support for IS-IS flood-reflector interfaces (PTX1000, QFX10002, QFX10008)**—Starting in Junos OS Release 20.4R1, we support the IS-IS flood reflector feature, which allows creation of IS-IS flood reflection topologies. Flood reflection allows the creation of topologies where Level 1 areas provide transit forwarding for Level 2 destinations within a Level 2 topology that provides better scalability.

We designate flexible tunnel interfaces (FTI) as flood-reflector interfaces. To enable the flood reflector on an FTI, include the **flood-reflector** statement at the **[edit protocols isis interface *interface name* level *level number*** hierarchy level.

You can configure the interface to be either the reflector or the client. To enable the reflector, you can use the **flood-reflector reflector *cluster-id*** statement at the **[edit protocols isis level *level number*** hierarchy level.

To enable the flood reflector client, include the **flood-reflector client** statement at the **[edit protocols isis level *level number*** hierarchy level.

NOTE: You can configure the flood reflector feature on FTIs at Level 2 only.

[See [How to Configure Flood Reflector Interfaces in IS-IS Networks](#).]

- **Support for relaxing BGP router ID format from /32 to a nonzero ID per RFC6286 (EX9208, EX9214, MX204, PTX5000, VRR, and QFX5120-32C)**—Starting in Junos OS Release 20.4R1, you can establish a BGP connection using a BGP identifier that is a 4-octet, unsigned, nonzero integer and it needs to be unique only within the autonomous system (AS) per RFC 6286. In earlier releases, the BGP ID of a BGP speaker was required to be a valid IPv4 host address assigned to the BGP speaker.

To enable this feature, use the **bgp-identifier *identifier* group *bgp group name* bgp-identifier *identifier* neighbor *peer address* bgp-identifier *identifier*** configuration statement at the **[edit protocols bgp]** hierarchy level.

[See [router-id](#).]

- **Support for IPv4 VPN unicast and IPv6 VPN unicast address families in BGP (QFX10002-60C, QFX10002, QFX10008, and QFX10016)**—Starting in Junos OS Release 20.4R1, on QFX Series switches, the following

address families are supported to enable advertisement or reception, or both, of multiple paths to a destination to and from the same BGP peer, instead of advertising and receiving only one path to and from the same BGP peer, under the `[edit protocols bgp group group-name]` hierarchy. You can configure the **add-path** statement at the BGP global, group level, and peer level.

- IPv4 VPN unicast (family inet-vpn)
- IPv6 VPN unicast (family inet6-vpn)

[See [Understanding the Advertisement of Multiple Paths to a Single Destination in BGP.](#)]

Software Defined Networking (SDN)

- **PCEP support for color (MX480, QFX5200)**—Starting in Junos OS Release 20.4R1, the Path Computation Element Protocol (PCEP) supports color for colored segment routing LSPs. This includes Path Computation Element (PCE)-initiated, Path Computation Client (PCC)-controlled, and PCC-delegated segment routing LSPs. With this PCEP extension, you can configure candidate paths based on color and endpoints, where the active candidate path is the path with the highest segment routing preference, or based on source priority.

[See [Understanding Static Segment Routing LSP in MPLS Networks.](#)]

- **Static VXLAN at VLAN or bridge domain level (MX5, MX10, MX40, MX80, MX150, MX240, MX480, MX960, MX2008, MX2010, MX2020, MX10003, MX10008, MX10016 routers and QFX5120-32C, QFX5120-48T, and QFX5120-48Y switches)**—In Junos OS Release 20.3R1 and earlier, we supported the configuration of static VXLAN at the global level only. By including the **remote-vtep-list** configuration statement at the `[edit switch-options]` or `[edit routing-instances name]` hierarchy level, you can map all local VLANs or bridge domains to the remote virtual tunnel endpoints (VTEPs) in the list.

Starting in Junos OS Release 20.4R1, you can also configure static VXLAN at the VLAN or bridge domain level using the **static-remote-vtep-list** configuration statement at the `[edit vlans name vxlan]`, `[edit bridge-domains name vxlan]`, or `[edit routing-instances name bridge-domains name vxlan]` hierarchy level.

When specifying remote VTEPs at the VLAN level in the default switching instance, you must also specify the same VTEPs at the global level in the default switching instance. Or when specifying remote VTEPs at the bridge domain level in a routing instance, you must also specify the same VTEPs at the global level in the same routing instance. For example, if you specify a VTEP in the **static-remote-vtep-list** at the `[edit routing-instances name bridge-domains name vxlan]` hierarchy level, you must also specify the VTEP in the **remote-vtep-list** at the `[edit routing-instances name]` hierarchy level.

To replicate and flood BUM traffic, you must specify the **ingress-node-replication** configuration statement at the `[edit vlans name vxlan]`, `[edit bridge-domains name vxlan]`, or `[edit routing-instances name bridge-domains name vxlan]` hierarchy level. This configuration restricts the BUM traffic flood domain to only those VTEPs mapped to a particular bridge domain or VLAN.

[See [Static VXLAN](#) and [static-remote-vtep-list](#).]

Software Installation and Upgrade

- **Phone-home client (EX4600, EX4650, EX9200, QFX5110, QFX5200, QFX5210, QFX5120-32C, and QFX5120-48Y)**—Starting with Junos OS Release 20.4R1, you can use either the legacy DHCP-options-based ZTP or the phone-home client (PHC) to provision software for the switch. When the switch boots up, if there are DHCP options that have been received from the DHCP server for ZTP, ZTP resumes. If DHCP options are not present, PHC is attempted. PHC enables the switch to securely obtain bootstrapping data, such as a configuration or software image, with no user intervention other than having to physically connect the switch to the network. When the switch first boots up, PHC connects to a redirect server, which redirects to a phone home server to obtain the configuration or software image.

To initiate either DHCP-options-based ZTP or PHC, the switch must be in a factory-default state, or you can issue the **request system zeroize** command.

[See [Understanding the Phone-Home Client](#)

- **ZTP with DHCPv6 client support (EX3400, EX4300, PTX1000, PTX5000, PTX10002-60C, PTX10008, QFX5100, QFX5200, QFX10002, and QFX10002-60C)**—Starting in Junos OS Release 20.4R1, zero touch supports the DHCPv6 client. During the bootstrap process, the device first uses the DHCPv4 client to request for information regarding image and configuration file from the DHCP server. The device checks the DHCPv4 bindings sequentially. If one of the DHCPv4 bindings fails, the device continues to check for bindings until provisioning is successful. However, if there are no DHCPv4 bindings, the device checks for DHCPv6 bindings and follows the same process as for DHCPv4 until the device can be provisioned successfully. Both DHCPv4 and DHCPv6 clients are included as part of the default configuration on the device.

The DHCP server uses DHCPv6 options 59 and 17 and applicable suboptions to exchange ZTP-related information between itself and the DHCP client.

NOTE: ZTP supports only HTTP and HTTPS transport protocols.

[See [Zero Touch Provisioning](#).]

System Management

- **PTP transparent clock (QFX5120-32C)**—Starting in Junos OS Release 20.4R1, you can use a transparent clock to update Precision Time Protocol (PTP) packets with the residence time as the packets pass through QFX5120-32C switches. The PTP Transparent Clock (PTP TC) is defined in IEEE 1588-2008 (PTPv2). QFX5120-32C switches support end-to-end transparent clocks, which include only the residence time. To use a transparent clock, enable the **e2e-transparent** statement at the **[edit protocols ptp]** hierarchy level.

[See [Understanding Transparent Clocks in Precision Time Protocol](#).]

System Logging

- **Support for time averaged watermark (MX Series, PTX Series, and QFX Series)**—Starting in Junos OS Release 20.4R1, you can capture steady state data of routing and forwarding (RIB/FIB) table routes using the **time-averaged-watermark-interval** configuration statement at the **[edit routing-options]** hierarchy level. Time averaged watermark is calculated whenever the time averaged interval is changed from CLI. Time averaged watermark is logged in syslog if the logs are enabled in the system at **LOG_NOTICE** level. The default time averaged watermark interval is 1 day. You can see the timed averaged watermark using the existing **show route summary** command.

[See [routing-options](#) and [show route summary](#).]

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What's Changed

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Learn about what changed in Junos OS main and maintenance releases for QFX Series Switches.

What's Changed in Release 20.4R3

Class of Service (CoS)

- On a Layer 2 interface, use unit * to apply a classifier or rewrite rule to all of the logical units on that interface.

EVPN

- **Community information no longer included in VRF routing table**— The QFX series switches will no longer include the inherited advertised route target communities, EVPN extended communities, or vxlan encapsulation communities for EVPN Type 2 and EVPN Type 5 routes when an IP host is added in the VRF routing table.

Interfaces and Chassis

- **Blocking duplicate IP detection in the same routing instance (ACX Series, EX Series, MX Series, NFX Series, PTX Series, QFX Series, and SRX Series)**—Junos will no longer accept duplicate IPs between different logical interfaces in the same routing instance. Refer to the table mentioned in the topic [inet \(interfaces\)](#). When you try to configure same IP on two logical interfaces inside same routing instance, the commit will be blocked with the error displayed as shown below: `edit user@host# set interfaces ge-0/0/1 unit 0 family inet address 2.2.2.2/24 edit user@host# commit commit complete edit user@host# set interfaces ge-0/0/2 unit 0 family inet address 2.2.2.2/24 edit user@host# commit edit interfaces ge-0/0/2 unit 0 family inet 'address 2.2.2.2/24' identical local address found on rt_inst default, intfs ge-0/0/2.0 and ge-0/0/1.0, family inet. error: configuration check-out failed`

[See [inet\(interfaces\)](#).]

Layer 2 Ethernet Services

- **Link selection support for DHCP**—We have introduced the `link-selection` statement at the `edit forwarding-options dhcp-relay relay-option-82` hierarchy level, which allows DHCP relay to add suboption 5 to option 82. Suboption 5 allows DHCP proxy clients and relay agents to request an IP address for a specific subnet from a specific IP address range and scope. Prior to this release, the DHCP relay dropped packets during the renewal DHCP process and the DHCP server used the leaf's address as a destination to acknowledge the DHCP renewal message.

[See [relay-option-82](#).]

Network Management and Monitoring

- **Enhancement to the `snmp mib walk` command (EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**— The `ipv6IfOperStatus` field displays the current operational state of the interface. The `noIfIdentifier(3)` state indicates that no valid Interface Identifier is assigned to the interface. This state usually indicates that the link-local interface address failed Duplicate Address Detection. When you specify the 'Duplicate Address Detected' error flag on the interface, the new value (`noIfIdentifier(3)`) is displayed. Previously, the `snmp mib walk` command did not display the new value (`noIfIdentifier(3)`).
- **Changes in contextEngineID for SNMPv3 INFORMS (ACX Series, EX Series, MX Series, PTX Series,**

QFX Series, and SRX Series)—Now the contextEngineID of SNMPv3 INFORMS is set to the local engine-id of Junos devices. In earlier releases, the contextEngineID of SNMPv3 INFORMS was set to remote engine-id.

[See [SNMP MIBs and Traps Supported by Junos OS.](#)]

- **The configuration accepts only defined identity values for nodes of type identityref in YANG data models (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—If you configure a statement that has type identityref in the corresponding YANG data model, the device accepts only defined identity values (as defined by an identity statement) as valid input. In earlier releases, the device also accepts values that are not defined identity values.
- **Change in OID ifHighSpeed**—Now, the object identifier (OID) ifHighSpeed displays the negotiated speed once negotiation is completed. If the speed is not negotiated, ifHighSpeed displays the actual maximum speed of the interface. In earlier releases, ifHighSpeed always displayed the actual speed of the interface.

[See [SNMP MIBs and Traps Supported by Junos OS.](#)]

What's Changed in Release 20.4R2

EVPN

- **Support for displaying SVLBNH information**—You can now view shared VXLAN load balancing next hop (SVLBNH) information when you display the VXLAN tunnel endpoint information for a specified ESI and routing instance by using the **show ethernet-switching vxlan-tunnel-end-point esi esi-identifier esi-identifier instance instance svlbnh** command.

General Routing

- **Support only for manual channelization on QSFP-100G-SR4-T2 optics (QFX5120-48T and QFX5120-32C)**— We recommend that you use the active optical cable (AOC) for auto-channelization. The QSFP-100G-SR4-T2 cables do not support auto-channelization. To use the QSFP-100G-SR4-T2 optics with an external breakout cable, you must configure the channelization manually by running the **channel-speed** statement at the **edit chassis fpc slot-number pic pic-number (port port-number | port-range port-range-low port-range-high)** hierarchy level.

[See [channel-speed.](#)]

Junos XML API and Scripting

- **Refreshing scripts from an HTTPS server requires a certificate (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—When you refresh a local commit, event, op, SNMP, or Juniper Extension Toolkit (JET) script from an HTTPS server, you must specify the certificate (Root CA or self-signed) that the device uses to validate the server's certificate, thus ensuring that the server is authentic. In earlier releases, when you refresh scripts from an HTTPS server, the device does not perform certificate validation.

When you refresh a script using the **request system scripts refresh-from** operational mode command, include the **cert-file** option and specify the certificate path. Before you refresh a script using the **set**

refresh or set **refresh-from** configuration mode command, first configure the **cert-file** statement under the hierarchy level where you configure the script. The certificate must be in Privacy-Enhanced Mail (PEM) format.

[See [request system scripts refresh-from](#) and [cert-file](#).]

- The **jcs:invoke()** function supports suppression of root login and logout events in system log files for **SLAX commit scripts (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—The **jcs:invoke()** extension function supports the **no-login-logout** parameter in SLAX commit scripts. If you include the parameter, the function does not generate and log UI_LOGIN_EVENT and UI_LOGOUT_EVENT messages when the script logs in as root to execute the specified remote procedure call (RPC). If you omit the parameter, the function behaves as in earlier releases in which the root UI_LOGIN_EVENT and UI_LOGOUT_EVENT messages are included in system log files.

[See [invoke\(\) Function \(SLAX and XSLT\)](#).]

- The **jcs:invoke()** function supports suppression of root login and logout events in system log files for **SLAX event scripts (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—The **jcs:invoke()** extension function supports the **no-login-logout** parameter in SLAX event scripts. If you include the parameter, the function does not generate and log UI_LOGIN_EVENT and UI_LOGOUT_EVENT messages when the script logs in as root to execute the specified remote procedure call (RPC). If you omit the parameter, the function behaves as in earlier releases in which the root UI_LOGIN_EVENT and UI_LOGOUT_EVENT messages are included in system log files.

[See [invoke\(\) Function \(SLAX and XSLT\)](#).]

Software Licensing

- **License key format (QFX5120-32C, QFX5120-48Y, and QFX5200)**—When you are upgrading from Junos OS release 20.4R1 to Junos OS release 20.4R2 or later releases, you need new license keys to use the features on the listed devices. Contact [Customer Care](#) to exchange license keys for Junos OS releases 20.4R2 or later.

What's Changed in Release 20.4R1

Class of Service (CoS)

- We've corrected the output of the "show class-of-service interface | display xml" command. Output of the following sort: <container> <leaf-1> data <leaf-2> data <leaf-3> data <leaf-1> data <leaf-2> data <leaf-3> data will now appear correctly as: <container> <leaf-1> data <leaf-2> data <leaf-3> data <container> <leaf-1> data <leaf-2> data <leaf-3> data.

General Routing

- **Control plane DDoS protection packet type option for ARP traffic (PTX Series and QFX Series)**— Starting in this release, we've renamed the **arp-snoop** packet type option in the **edit system ddos-protection protocols arp** protocol group to **arp**. This packet type option enables you to change the default control plane distributed denial-of-service (DDoS) protection policer parameters for ARP traffic.

[See [protocols \(DDoS\) \(PTX Series and QFX Series\)](#).]

- **Change in license bandwidth command on vMX virtual routers**—Starting in Junos OS, to use the available license bandwidth, explicitly set the license bandwidth using the **set chassis license bandwidth <ln-mbps>** command.

[See [Configuring Licenses on vMX Virtual Routers](#).]

High Availability (HA) and Resiliency

- **The ntp boot-server command is deprecated** — The **boot-server** option for the **ntp** command is no longer needed, and has been deprecated.

MPLS

- **The show mpls lsp extensivel and show mpls lsp detail commands display next hop gateway LSPid**—When you use the **show mpls lsp extensivel** and **show mpls lsp detail** commands, you'll see next hop gateway LSPid in the output as well.

Network Management and Monitoring

- **Warning changed for configuration statements that correspond to deviate not-supported nodes in YANG data models (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—If you configure a statement corresponding to a YANG data model node that defines the **deviate not-supported** statement, the Junos OS configuration annotates that statement with the comment **Warning: statement ignored: unsupported platform**. In earlier releases, the warning is **Warning: 'statement' is deprecated**.

User Interface and Configuration

- **Verbose format option for exporting JSON configuration data (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—The Junos OS CLI exposes the **verbose** statement at the **edit system export-format json** hierarchy level. The default format for exporting configuration data in JSON changed from **verbose** format to **ietf** format starting in Junos OS Release 16.1R1. You can explicitly specify the default export format for JSON configuration data by configuring the appropriate statement at the **edit system export-format json** hierarchy level. Although the **verbose** statement is exposed in the Junos OS CLI as of the current release, you can configure this statement starting in Junos OS Release 16.1R1.

[See [export-format](#).]

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Known Limitations

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Learn about known limitations in Junos OS Release 20.4R3 for QFX Series Switches. For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Class of Service (CoS)

- Traffic might be dropped by the destination device. [PR1568333](#)

EVPN

- VTEP interface displays output packets underlying the physical interface. [PR1549820](#)
- The TYPE-5 decapsulation nexthop does not get installed if the peer does not advertising the TYPE-5 route. [PR1599717](#)

Layer 2 Features

- On the QFX5000 line of switches with SP-style aggregated Ethernet interface, the child member link changes leads to momentary CRC error and drops traffic when there is a scale of logical interfaces. [PR1532342](#)

Platform and Infrastructure

- After you configure and delete the Ethernet loopback configuration, the interface goes down and does not come up. [PR1353734](#)
- On QFX10002-60C line of switches, continuous FPC might crash and the dcpfe process might generate the core file. [PR1612871](#)
- Reroute counter log events occurs sometimes when you change the routes pointed by the unilist next hop. [PR1380350](#)
- Router might become non responsive when you reboot the router with the **request system reboot** statement. [PR1385970](#)
- On the QFX10000 line of switches, the analyzer does not mirror after you add the child member to an aggregated Ethernet interface. [PR1417694](#)
- When you tag the spine underlay and untag again, the inner packet comes and goes over the TYPE 2 tunnel, resulting in IPv4 discarding the traffic silently on PECHIP. [PR1435864](#)
- After you rename VLAN on the trunk interface, the local host MAC learning halts for more than 30 seconds. [PR1454274](#)
- Changing the scaled firewall profiles on the fly does not release the TCAM resources as expected. [PR1512242](#)
- On the QFX5110-48S line of switches, the **Number of Datagrams** and **Number of Flow Samples** values displays their value as zero in the output of the **show sflow collector detail** command, . [PR1525356](#)
- On the QFX5000 line of switches with storm control, significant difference between the configured rate and actual rate is observed. [PR1526906](#)
- The following notification appears during the reboot or powering off the device: **[FAILED] Failed unmounting /var**. [PR1527581](#)
- The **show chassis environment** and **show chassis fan** commands have different string status for fan failure. [PR1527628](#)
- The vmcore process might generate a core file during the aggressive sensor subscribe and unsubscribe. [PR1528432](#)
- When you disable the auto-channelization with the 4x10G breakout cable, the dut interface goes in the **Down** state while the 10G interfaces on the peer goes in the **Up** state. [PR1531850](#)

- On the QFX10002 line of switches in a dynamic IP-IP tunnel transit scenario, the sFlow export data does not include the **nexthop** field. [PR1533307](#)
- The input runs error counter does not increment when the packets are sent with the size less than 64 bytes. [PR1533322](#)
- On the QFX5100 line of switches that does not run the QFX-5E codes (non TVP architecture), when you install the image with Broadcom SDK upgrade (6.5.X), the CPU utilization might go up by around 5 percent. [PR1534234](#)
- ECMP over GRE does not work for the BGP routes. [PR1537924](#)
- The QFX5100 line of switches sends maximum samples of around 700 per second to the sFlow collector. [PR1539815](#)
- Power interruption during firmware upgrade is not recommended. [PR1543192](#)
- The QFX10000 line of switches do not keep the inner vlan-id on the egress EP interface. [PR1546840](#)
- On the QFX5120 and QFX5200 line of switches, the EVPN VXLAN loop detect and CFM must not be configured on the same VLAN. [PR1553384](#)

Routing Protocols

- Node protection for the RSVP LSP on FTI interfaces does not work. [PR1456350](#)
- On the QFX5000 line of switches, the PIP decapsulation in the forward filter does not work when you commit the **NO from matching conditions AND when subnet masks < 32** statement. [PR1511893](#)
- On the QFX5210 line of switches, when you configure two Flex Hash rules, deactivating the first Flex Hash rules, the second Flex Hash rules do not get programmed in the hardware. [PR1521306](#)
- On the QFX5100-24Q line of switches, error messages occurs when you deactivate and activate interfaces. [PR1522701](#)
- On the QFX5000 line of switches, error messages on the Packet Forwarding Engine console occurs when you bring up the 512 IP-IP tunnels. [PR1525270](#)
- On the QFX5120-48YM line of switches, when the scale of IPv4 and IPv6 routes are present in the LPM profile, few of the IPv6 routes do not get installed when the ports on which the routes are learnt flaps due to the LPM table full error. [PR1557655](#)

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Learn about open issues in Junos OS Release 20.4R3 for QFX Series Switches. For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Class of Service (CoS)

- The buffer allocation for the VCP ports might not get released in the Packet Forwarding Engine after physically moving the port location. [PR1581187](#)

EVPN

- On the QFX5120 line of switches, traffic received on the VTEP, which is not configured in the static-remote-vtep-list does not gets dropped. [PR1543779](#)
- MAC address of the end-host gets wrongly programmed in the forwarding table after the ESI failover. [PR1584595](#)
- I-ESI modification workflow on DC-GW occurs. [PR1600600](#)
- The MAC address of the end-host gets wrongly programmed in the forwarding table after the ESI failovers. [PR1584595](#)

High Availability (HA) and Resiliency

- On the QFX5200-32C line of switches, the reboot time gets degraded from 205 seconds in Junos OS Release 20.2R1 to 260 seconds in Junos OS Release 20.3. [PR1511607](#)

Infrastructure

- The following error message occurs during FTP: `ftpd[14105]: bl_init: connect failed for /var/run/blacklistd.sock(No such file or directory)`. [PR1315605](#)

Interfaces and Chassis

- Newly added MC-LAGs do not come up after the Routing Engine switchovers. [PR1583547](#)

Junos Fusion Satellite Software

- The output of the od temperature sensor 2 for opus SD106 does not appear. [PR1582981](#)

Layer 2 Features

- On QFX5110 and QFX5120 line of switches, changing lo0 IP address might sometimes either result in stale entry of IP in `mpls_entry` table or missing IP entry, which results in traffic drop for VXLAN traffic. [PR1472333](#)
- The MAC addresses learnt from the MC-LAG client device might keep flapping between the ICL interface and mc-ae interface after you disable one child link in the mc-ae interface. [PR1582473](#)
- Traffic drop might occur on the aggregated Ethernet interface. [PR1585320](#)
- On the QFX5000 line of switches, the DF might not forward BUM traffic. [PR1575976](#)
- On the QFX5100 line of switches, traffic might be dropped in the Packet Forwarding Engine after you commit changes related to TPID in the dcd process. [PR1477156](#)
- On the QFX5100 line of switches, traffic might be dropped in the Packet Forwarding Engine after making changes related to TPID in the dcd process. [PR1477156](#)

Layer 2 Ethernet Services

- ZTP does not get activated after all the values of the device becomes zero. [PR1529246](#)

Platform and Infrastructure

- The sFlow sampling fails when you configure the egress interface with more than 8 in a line card. [PR1202870](#)
- On the QFX10000 line of switches, the source MAC and TTL values do not get updated for the routed multicast packets in EVPN-VXLAN. [PR1346894](#)
- The backup Routing Engine might crash after GRES occurs continuously for more than 10 times. [PR1348806](#)
- On QFX5110 line of switches, the VCP ports goes down after the members split and merge in the Virtual Chassis. [PR1606705](#)
- Upon receipt of specific sequences of genuine packets destined to the device, the kernel crashes and restarts. [PR1557881](#)
- The VCF might become not stable. [PR1559172](#)
- The subscriber management infrastructure daemon (smid) process might become nonresponsive at hundred percent. [PR1559402](#)
- The dcpfe process might crash after you commit the EVPN-VXLAN profile configuration and ARP resolution might fail leading to traffic issues. [PR1561588](#)
- On the QFX5110-32Q line of switches, LACP does not come up in the non-oversubscribed mode for a set of ports. [PR1563171](#)
- On QFX10002 line of switches, continuous Layer 3 traffic drop occurs with the MC-LAG configuration. [PR1610173](#)
- On QFX5000 Virtual Chassis, MAC move or MAC flap might be triggered. [PR1610295](#)
- The QFX5000 and QFX10000 line of switches might become nonresponsive after reboot for sometime. [PR1584902](#)
- Due to the transient hardware condition, the single-bit error (SBE) events gets corrected and have no operational impact. Those reported events had been disabled to prevent alarms and possibly unnecessary hardware replacements. [PR1384435](#)
- On QFX10002-60C line of switches, complete packet loss occurs while testing the vacl ingress terms scale configuration. [PR1581767](#)
- On QFX10000 line of switches, the dcpfe and fpc process might crash. [PR1597479](#)
- The IS-IS adjacency might fail to be formed if you configure the MTU size of an IRB interface with a value great than 1496 bytes. [PR1595823](#)

- The fpc process might crash when you modify a firewall filter. [PR1432116](#)
- On the QFX5200 line of switches, the ISSU might fail. [PR1438690](#)
- On the QFX5000 line of switches, the port qualifier is not supported. [PR1440980](#)
- Storm-control does not rate-limit the ARP packets. [PR1461958](#)
- On the QFX5110 line of switches, the VXLAN VNI (mcast) scaling causes traffic issue. [PR1462548](#)
- On the QFX5000 line of switches, analyzer does not work. [PR1576327](#)
- The OSPF session over IRB might not come up in the EVPN-VXLAN scenario. [PR1577183](#)
- On the QFX10000 line of switches, the port might not go in to the **Down** state immediately during some abnormal type of line card reboots. [PR1577315](#)
- The unexpected next hop might occur after the route gets deleted. [PR1477603](#)
- The SNMP index in the Packet Forwarding Engine reports as 0 that causes sFlow to report either IIF or OIF (not both) as 0 in the sFlow record data at the collector. [PR1484322](#)
- Interface on platforms using the Broadcom chipset might have abnormal status. [PR1495564](#)
- On the QFX10000 line of switches, the **show pfe filter hw filter-name** command does not retrieve the Packet Forwarding Engine program. [PR1495712](#)
- On the QFX5100 line of switches, degradation occurs during the system reboot time and FPC online time. [PR1513540](#)
- SNMP trap of power failure might not be sent out. [PR1520144](#)
- The Layer 3 classifier takes effect though the configured Layer 2 classifier. [PR1520570](#)
- The FIPS mode is not supported. [PR1530951](#)
- FPC might crash in a scaled firewall configuration. [PR1586817](#)
- On the QFX5100 line of switches that does not run the QFX-5E codes (non TVP architecture), when you install image with the Broadcom SDK upgrade (6.5.X), the CPU utilization might go up by around 5 percent. [PR1534234](#)
- The MAC addresses might not be relearned successfully after the MAC address age timeouts. [PR1567723](#)
- DCI traffic loss of the hundred percent occurs in the transit spine devices. [PR1572238](#)
- On the QFX10000 line of switches, the dcpfe or fpc process might crash if the ARP MAC occurs. [PR1572876](#)
- On the QFX5100 Virtual Chassis, the firewall counter does not get updated as expected with PACL applied. [PR1535825](#)
- On the QFX10002-72Q line of switches, the sFlow egress samples at PHP router do not include explicit-null (IPv4 and IPv6). [PR1537946](#)
- The dcpfe process might crash when any interface flaps. [PR1579736](#)

- On the QFX5000 line of switches, traffic loss might occur in the EVPN-VxLAN scenario. [PR1580005](#)
- The IS-IS packet might be corrupted on the provider edge device over the Layer 2 circuit. tunnel [PR1580047](#)
- The DHCP packets might be dropped if you apply the **dyn-dhcpv4_v6_trap** dynamic filter on the interface. [PR1580352](#)
- On the QFX10002 line of switches, the sFlow log error message occurs when you enable the egress sampling on the dynamic IP-IP tunnel encapsulation scenario. [PR1538863](#)
- On the QFX10008 line of switches with the Layer 2 and Layer 3 multicast configurations, the vmcore process generates the core file on the primary and backup Routing Engines. [PR1539259](#)
- The following error message occurs when you reboot the device with the enterprise base configurations: **Error BCMX: Failed to add lport 0x0 (unit , port) . -8: Entry exists.** [PR1541159](#)
- On the QFX5000 Virtual Chassis fan, traffic loss might occur after you swap the primary and backup Routing Engines. [PR1544353](#)
- On the QFX5000 line of switches, the installed next hop local-bias filter fails on the Packet Forwarding Engine with MAC-VRF. [PR1544850](#)
- The QFX10000 line of switches do not keep the inner vlan-id on the egress EP interface. [PR1546840](#)
- Need to move WRL7 to RCPL31 for the QFX-10-M and QFX-10-F line of switches. [PR1547565](#)
- On the QFX10000 line of switches, there are no local MAC-IP entry for vlan-bundle service. [PR1548456](#)
- During power cycle, 100G port down issue occurs occasionally on et-0/0/54 and et-0/0/55 with INNOLIGHT 100G-AOC cables. [PR1548525](#)
- Host shell gets unreachable after the image installation. [PR1548710](#)
- On the QFX10000 line of switches, only untagged traffic flows through the ethernet-bridge interface. [PR1550700](#)
- On the QFX10002-60C line of switches, ethernet-bridge is not supported. [PR1551037](#)
- In Junos OS Release 20.2, some features appears as a licensed feature. While using the features, the alarms and commit warnings are displayed. However, there are no functional impact. [PR1558017](#)
- pic_create_ifname: 0/0/0 pic type F050 is not supported. The log messages generated under chassisd and other messages in logs appers. [PR1566440](#)
- Unexpected multicast traffic streams occurs after you enable EVPN. [PR1570689](#)
- BUM traffic from AR-LEAF does not display the correct count in the output of the **show interfaces vtep extensive** command. [PR1579614](#)
- On the QFX5210-64C line of switches, PSU firmware upgrades through JUNOS. [PR1589572](#)
- On QFX10008 line of switches, the system reboot takes approximately 9 minutes for FPCs to come online after system reboot command is issued. It is about 2 minutes more than 20.4 releases. [PR1605002](#)

- IRB ARP/ND entries missing in GW nodes and traffic black-hole after modify WAN-RD or WAN-RT of EVPN VXLAN DCI routing instance. [PR1611154](#)
- Traffic received on the Type-5 tunnel on Spine are not forwarded out to leaf after modifying lo0 interface IP on QFX1000. [PR1615253](#)
- On QFX10008 and QFX10016 line of switches with EVPN-VXLAN configuration and dual GRES enabled, the dcpfe process generates core file after the below sequence of events. [PR1615430](#)
- On QFX5210-64C line of switches after reboot, the carrier transition counter does not increment when the link flaps. [PR1605037](#)
- On the EVPN VXLAN environment, VNI for the EAD/ES route sets to 1. [PR1594981](#)
- The mcsnoopd process generates the core files at `snmp_token_db_gencfg_handler,krt_decode_gencfg,krt_ifstate_resync_read,krt_async_rcv_ifstate_resync_phase`. [PR1596483](#)
- During an image upgrade, sometimes the system goes for NMI or auto vmcore generation. [PR1601867](#)
- BUM traffic might be looped for a short time when you add the interface as the CE interface. [PR1493650](#)
- Additional firewall policy must be added to allow the client DNS queries from the QF directors. [PR1509383](#)
- Offer message from the server reaching the relay agent appears. ,However, the messages does not get forwarded to IRB on which clients are connected. [PR1530160](#)
- On the QFX10000 line of switches, the firewall filter logs are incorrectly populated the protocol 8847 entries. [PR1582780](#)
- The DHCP offer packets might be dropped on the Spine device in the VXLAN multi-homing setup. [PR1585715](#)
- Traffic drop occurs while verifying the traffic status on the fec interface after reboot. [PR1586036](#)
- Packet DMA memory leak might occur in the EVPN-VXLAN scenario after receiving some packets. [PR1587609](#)
- The LLDP packets drop on the SP style interface. [PR1589702](#)
- Multiple crashes with the `toe_interrupt_errors` error message might occur. [PR1593025](#)

Routing Policy and Firewall Filters

- When you upgrade Junos OS to a specific version, the configuration validation might fail and the rpd process might crash. [PR1538172](#)
- The rpd process might crash when you delete the routing table. [PR1565629](#)

Routing Protocols

- On the QFX-5100 Virtual Chassis or Virtual Chassis Fan, the following error occurs in the hardware with the mini-PDT base configurations: **BRCM_NH-,brcm_nh_bdvlan_ucast_uninstall(), 128:l3 nh 6594 unintsall failed.** [PR1407175](#)
- On the QFX5100 line of switches, the fxpc process might generate the core filr when you cleanup the configurations after ISSU. [PR1578115](#)
- The multi-hop BFD session might flap if you execute the **RSI (Request Support Information) collection** command. [PR1589765](#)
- IPv4 static route might still forward traffic unexpectedly even when the static route configuration has already been deleted. [PR1599084](#)
- Filter with destination-port-range-optimize or source-port-range-optimize fails. [PR1582452](#)
- Encapsulation and decapsulation of VXLAN traffic fails leading to traffic drop. [PR1582713](#)
- With IGMP snooping implemented, unexpected jitter issue might occur leading to traffic loss. [PR1583207](#)

VPNs

- On the QFX10000, the auto-RP functionality does not work in the NGMVPN scenario. [PR1604014](#)

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Resolved Issues

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Learn which issues were resolved in Junos OS main and maintenance releases for QFX Series Switches.

For the most complete and latest information about known Junos OS defects, use the Juniper online [Junos Problem Report Search](#) application.

Resolved Issues: 20.4R3

Class of Service (CoS)

- The buffer allocation for the VCP ports might not get released in the Packet Forwarding Engine after you physically move the port location. [PR1581187](#)
- On the QFX5000 line of switches, the DSCP classifier might not work properly. [PR1585361](#)
- The TCP-ECN traffic might not be forwarded with high priority. [PR1585854](#)
- Not able to configure policer with bandwidth-limit greater than 50G. [PR1575049](#)

EVPN

- The l2ald process might crash and restart with a l2ald core file created when you enable the global level telemetry sensor. [PR1570757](#)
- Configuring static-mac and no-mac-learning simultaneously on the VXLAN interface causes to stale the MAC or IP entry in the EVPN database. [PR1576147](#)
- Policy with mac-filter-list might not work if you commit changes in the policy unrelated to that policy in the EVPN scenario. [PR1567623](#)
- After device reboots in an EVPN-VXLAN setup with graceful restart, the EVPN routes does not get advertised to the EVPN peers until the rpd process gets up for 180 seconds. [PR1586246](#)
- Traffic loss might occur under the EVPN-VXLAN scenario when MAC-IP moves from one CE interface to another. [PR1591264](#)

- The device announces router-mac, target, and EVPN-VXLAN community to the BGP IPv4 NLRI. [PR1600653](#)
- Traffic sent by the QFX5000 switch leaf to remote leaf with link down. [PR1605375](#)

Interfaces and Chassis

- New added MC-LAGs do not come up after the Routing Engine switchovers. [PR1583547](#)
- Removing the configuration from interface stanza might cause the dcpfe process to crash. [PR1594356](#)
- Traffic loss might occur when you deactivate and activate member links of the logical child or ICCP interface. [PR1542840](#)

Layer 2 Features

- On the QFX5110-32Q line of switches, LACP does not come up in the non-oversubscribed mode for a set of ports. [PR1563171](#)
- Traffic forwarding for VLAN 2 might not be correct when a VLAN member is removed from the ESI interface. [PR1570446](#)
- On the QFX5000 line of switches, software with forwarded VXLAN decapsulated packets contains illegal length. [PR1574435](#)
- On the QFX5000 line of switches, DF might not forward BUM traffic. [PR1575976](#)
- MAC addresses learnt from the MC-LAG client device might keep flapping between the ICL interface and MC-AE interface after you disable one child link in the MC-AE interface. [PR1582473](#)
- Traffic might drop on the aggregated Ethernet interface. [PR1585320](#)
- In the OVSDB VXLAN scenario, the inner VLAN tag 8 gets added unexpectedly into the encapsulated Ethernet header. [PR1531319](#)
- Packets received on a port that is in the **LACP Detached** and **Broadcom STP Blocked** might get forwarded. [PR1553570](#)
- LACP gets into the **Detached** state when you delete VLAN on the aggregate Ethernet interface configured on the SP style. [PR1555862](#)

Layer 2 Ethernet Services

- The DHCP client becomes offline for 120 seconds after sending the **DHCPINFORM** message in the DHCP relay scenario. [PR1575740](#)
- The DHCP relay drops packets during the renewal DHCP process. [PR1576417](#)

Network Management and Monitoring

- Slow memory leak might occur in the snmpd process. [PR1575790](#)

Platform and Infrastructure

- Kernel crash might occur after NSSU when you perform GRES. [PR1533874](#)
- The dcpfe process might crash with auto-channelization enabled. [PR1484336](#)
- On the QFX10000 line of switches, need to enhancement to enable watchdog petting log on line cards. [PR1527535](#)
- On the QFX5110-32Q line of switches, the ports from 20 to 27 might flap when you insert the QSPF-40G transceiver into the port 29 to 31. [PR1535216](#)
- On the QFX10000 line of switches, Denial of Service (DoS) upon receipt of the DVMRP packets gets received on the multi-homing ESI in VXLAN. [PR1539194](#)
- The Layer 3 static license is required even though it is included in the base license. [PR1557631](#)
- On QFX5120-48T line of switches after removing 1G speed on interface, the speed does not come back as 10g. [PR1591038](#)
- The socket connection drops due to keepalive timer expiration with the port 33015. [PR1598019](#)
- The egress interface of the GRE tunnel does not dynamically get updated when the destination to tunnel changes. [PR1602391](#)
- Sampled memory might leak when the analyzer is in the **Down** state. [PR1564790](#)
- Traffic loss might occur in the MC-LAG scenario. [PR1565287](#)
- On the QFX5000 line of switches, the dcpfe process might crash after deleting the VXLAN configuration [PR1562692](#)
- The received encapsulated VXLAN packets might be dropped if you load the EVPN-VXLAN configuration. [PR1562814](#)
- The VXLAN queue DDoS violation and RARP packets flood might occur if you receive the RARP packets more than the supported DDoS bandwidth. [PR1560243](#)
- On the QFX10000-60S-6Q line of switches, line card takes more than 15 minutes to boot up after you trigger the panic or watchdog reboot. [PR1559725](#)
- Timestamp discrepancy might occur in the IPFIX packet flows exported. [PR1558131](#)
- The DF (Designated Forwarder) might not forward traffic. [PR1567752](#)
- The 100G port with module QSFP 100G-SR4-T2 converts to two channelized interfaces without any channelized configuration. [PR1567937](#)
- The BFD session flaps between leaf and core during the Spine reboot causing other protocols to flap. [PR1568615](#)
- Multicast streams might stop flooding in the VXLAN setup. [PR1606256](#)

- The LLDP packets received on the VXLAN enabled port might get flooded unexpectedly. [PR1607249](#)
- Ping to lo0/IRB over Type-5 fails. [PR1610093](#)
- The dcpfe process might crash and cause FPC to restart due to the traffic burst. [PR1534340](#)
- The BFD neighborship fails with the EVPN-VXLAN configuration after the Layer 2 learning restarts. [PR1538600](#)
- The traffic does not get load-balanced properly in an EVPN overlay-ecmp setup. [PR1550020](#)
- The interface might not come up with 1G optics. [PR1554098](#)
- The dcpfe process might crash and restart with a dcpfe core file created when you run the Type5 EVPN-VXLAN with 2000 VLANs. [PR1556561](#)
- The MAC addresses learned in a Virtual Chassis might fail due to aging out in the MAC scaling environment. [PR1558128](#)
- On the QFX5000 line of switches, incorrect ARP reply might be sent through the aggregated Ethernet interface. [PR1554389](#)
- Traffic loss might occur on a VXLAN enabled VLAN. [PR1554600](#)
- The Virtual Chassis Fabric might become unstable. [PR1559172](#)
- The subscriber management infrastructure daemon (smid) process might become nonresponsive at 100 percent. [PR1559402](#)
- On the QFX5110 line of switches, the untagged traffic routed over native-vlan might be dropped. [PR1560038](#)
- VXLAN queue DDos violation and RARP packets flood might occur if the RARP packets are received more than the supported DDos bandwidth. [PR1560243](#)
- The tunable optics SFP+-10G-T-DWDM-ZR does not work. [PR1561181](#)
- The dcpfe process might crash on after committing the EVPN-VXLAN profile configuration and ARP resolution might fail causing traffic issues. [PR1561588](#)
- On the QFX5000 line of switches, the dcpfe process might crash after you delete the VXLAN configuration. [PR1562692](#)
- MAC addresses might not be relearned successfully after the MAC address age timeouts. [PR1567723](#)
- Another port gets shutdown after one port on the QFX10002-60C device shuts down. [PR1568294](#)
- The dcpfe process might crash if the TYPE-5 tunnel fails to get installed for EVPN-VXLAN. [PR1570136](#)
- Unexpected packet loss might occur if you delete the subunit of the physical interface. [PR1571286](#)
- DCI traffic loss of 100 percent occurs in the transit spine devices. [PR1572238](#)
- The EVPN VXLAN CE interface with RSTP configured might cause LACP or BFD issues. [PR1572504](#)
- On the QFX10000 line of switches, the dcpfe or fpc process might crash if you move the ARP MAC. [PR1572876](#)

- On the QFX10008 chassis, the dcpfe process generates a core file. [PR1572889](#)
- On the QFX10008 and QFX10016 line of switches, traffic loss might be observed due to faulty FPC. [PR1574779](#)
- BFD might flap occasionally during the spine reboot. [PR1575296](#)
- On the QFX5000 line of switches, analyzer does not work. [PR1576327](#)
- The OSPF session over IRB might not come up in the EVPN-VXLAN scenario. [PR1577183](#)
- On the QFX10000 line of switches, the WAN port links might not come down immediately during some abnormal type of line card reboot. [PR1577315](#)
- TACACS traffic might be dropped. [PR1578579](#)
- The dcpfe process might crash when any interface flaps. [PR1579736](#)
- The IS-IS packet might be corrupted on the provider edge device over the l2circuit tunnel. [PR1580047](#)
- The dcpfe process crashes when you check the virtual tunnel-nexthop packet status. [PR1580114](#)
- The DHCP packets might be dropped if you apply the **dyn-dhcpv4_v6_trap** dynamic filter on the interface. [PR1580352](#)
- When you apply mapping analyzers to the channelized port, mirror might not work properly. [PR1580473](#)
- On the QFX5120-32C line of switches, the following error appears: **kern.ipc.maxpipekva exceeded; see tuning error**. [PR1581192](#)
- The switchover might get affected with the shared VXLAN tunnel. [PR1581524](#)
- The traffic might not be load-balanced properly in an EVPN overlay-ecmp setup. [PR1582017](#)
- On the QFX5100 line of switches, few 40G ports might not be channelized successfully. [PR1582105](#)
- On the QFX10000 line of switches, the firewall filter logs incorrectly populates the protocol 8847 entries. [PR1582780](#)
- On the QFX5000 line of switches, the firewall filter do not get programmed after you delete a large filter and add a new one in a single commit. [PR1583440](#)
- The ZTP process might cause the traffic to discard silently. [PR1585057](#)
- The DHCP offer packets might be dropped on the spine device in the VXLAN multi-homing setup. [PR1585715](#)
- Inter and intra VNI traffic drop might occur in the spine device with the EVPN-VXLAN CRB configuration. [PR1586537](#)
- The na-grpc process might crash and existing telemetry connections gets disconnected. [PR1587956](#)
- When you remove the member interface from the aggregated Ethernet interface, the member interface do not get removed from mirroring in the analyzer. [PR1589579](#)
- Traffic loop occurs when you add the logical child interface in the case of the multihomed SP style in the EVPN-VXLAN. [PR1543966](#)

- The LLDP packets drop on the SP style interface. [PR1589702](#)
- On the QFX5120 line of switches, the MPLS traffic might not be forwarded after the aggregate interface flaps. [PR1589840](#)
- VXLAN DDoS violation might occur when you disable the port mirror analyzer output interface. [PR1590150](#)
- The Routing Engine kernel might crash due to the logical child interface of the aggregated interface. [PR1592456](#)
- The IPv4 fragmented packets might be broken if you configure the PTP transparent clock. [PR1592463](#)
- On the QFX10002, QFX10008, and QFX10016 line of switches, the MPLS traffic might get discarded on passive monitoring interface. [PR1592693](#)
- Multiple crashes with the **toe_interrupt_errors** error message occurs. [PR1593025](#)
- the BFD session might flap during the Routing Engine switchover. [PR1593244](#)
- The dcpfe process might crash in an EVPN-VXLAN scenario. [PR1593950](#)
- Packet drop might occur in an ECMP next-hop flap scenario. [PR1594030](#)
- The existing ECMP route traffic might be dropped if you configure a static ECMP route with the same number of next-hops as the existing ECMP route. [PR1594573](#)
- In the QFX10002-72Q line of switches, SNMP walk jnxOperatingEntry displays only two PSU even if you install four PSU. [PR1555852](#)
- The VGA might be down when you configure the IRB interface with multi VGA addresses. [PR1555338](#)
- The lcmd process might consume memory until all of the free memory available to VMHOST gets exhausted. [PR1555386](#)
- The re-installation of the Type-5 tunnels might fail in an EVPN-VXLAN scenario. [PR1595197](#)
- Traffic might be dropped after you reboot the backup FPC in a Virtual Chassis scenario. [PR1596773](#)
- The logical child interface does not come up in a specific condition. [PR1552938](#)
- The Layer 2 multicast traffic received on the VCP (Virtual Chassis port) ports might be dropped if you enable the igmp-snooping and STP/VSTP. [PR1553159](#)
- LACP timeout issue might occur while you poll for the QSFP diagnostics. [PR1549121](#)
- The dcpfe process might crash due to the chip SDK fault. [PR1552645](#)
- On the QFX10008 and QFX10016 line of switches, traffic loss might occur due to faulty FPC. [PR1574779](#)
- Port mirroring might not work when the analyzer output gets in a trunk interface. [PR1575129](#)
- On the QFX10000 line of switches, a high rate of 802.3x pause frames gets sent out of the interfaces. [PR1575280](#)
- The BFD session might flap occasionally during the Spine reboot. [PR1575296](#)

- The traffic does not get load-balanced properly in the EVPN overlay-ecmp setup. [PR1550020](#)
- The interface might not be brought up when you configure QinQ. [PR1597261](#)
- On the QFX5000 line of switches, the sFlow impacts on the ICMP traffic. [PR1598239](#)
- The dcpfe process might crash if the TYPE-5 tunnel fails to be installed for EVPN-VXLAN. [PR1570136](#)
- Traffic forwarding for VLAN 2 might not be correct when you remove a VLAN member from the ESI interface. [PR1570446](#)
- The dcpfe process crashes in the VXLAN scenario. [PR1571170](#)
- Unexpected packet loss might occur if you delete the subunit of the physical interface. [PR1571286](#)
- On the QFX5100, QFX5110, QFX5120, QFX5200, and QFX5210 line of switches, the DDoS violations might be reported for the IP multicast miss traffic incorrectly. [PR1598678](#)
- File permissions gets changed for the `/var/db/scripts` files after reboot. [PR1599365](#)
- On the QFX10002-60C line of switches with IRB interface, the Layer 3 traffic might be silently discarded. [PR1599692](#)
- Unable to disable the configuration mode command for the management port em1. [PR1600905](#)
- On the QFX10000 line of switches, the dcpfe process might crash. [PR1546572](#)
- On the QFX5000 line of switches, the static MAC on an interface might not work. [PR1546655](#)
- On the QFX5120-48y-8c line of switches, the dcpfe process generates core file when you issue the `show pfe vxlan nh-usage` command in an ERB EMC scenario with around 6000 ARP entries. [PR1601949](#)
- The status of FPC becomes down and the dcpfe process might generate core file dump in some cases. [PR1602583](#)
- Traffic loss might occur in the MC-LAG scenario. [PR1602811](#)
- On the QFX5000 line of switches, traffic drop might occur in the Virtual Chassis scenario when you configure the firewall filter. [PR1602914](#)
- Duplicate packets might appear when you bring up all the interfaces on the spine switch. [PR1604393](#)
- MAC movement might occur between the ICL and MC-LAG interface if you add or remove the VLANs on the ICL interface. [PR1605234](#)
- Upon the receipt of the specific sequences of genuine packets destined to the device, the kernel crashes and restarts. [PR1557881](#)
- FPC might crash in a scaled-firewall configuration. [PR1586817](#)

Routing Policy and Firewall Filters

- The rpd might crash when deletion of routing table occurs. [PR1565629](#)

Routing Protocols

- The remaining BFD sessions of the aggregated Ethernet interface flaps continuously if one of the BFD sessions gets deleted. [PR1516556](#)
- The fpc and dcpfe process might crash when you insert and remove the channelized interface of QSFP. [PR1547231](#)
- Memory leak might occur in the MSDP scenario. [PR1571906](#)
- A filter could not be installed if the filter has large scaled number of terms. [PR1555337](#)
- The rpd memory might leak in the BGP scenario. [PR1547273](#)
- The BFD sessions over IRB interface becomes nonresponsive in the **Init** state with the incremental FRR errors. [PR1541851](#)
- The BFD sessions on the Layer 3 sub-interface of the ESI aggregated Ethernet interface might keep flapping when the upstream underlay or overlay BGP flaps. [PR1544982](#)
- There might be traffic loss when the GRE interface flaps. [PR1566428](#)
- The GRE egress traffic might not be forwarded between different routing instances. [PR1573411](#)
- On the QFX5000 line of switches, continuous traffic destined to a device configured with MC-LAG leading to nodes losing their control connection, which can impact traffic. [PR1552877](#)
- The QFX5000 line of switches might drop the DHCP packets in the static VXLAN scenario. [PR1576168](#)
- Multicast Packets with TTL=1 gets dropped on the VXLAN enabled interface when you enable the igmp-snooping or MLD-snooping. [PR1576775](#)
- On the QFX5000 line of switches, traffic loss might occur in the EVPN-VxLAN scenario. [PR1580005](#)
- The BGP session carrying the VPNv4 prefix with IPv6 next-hop might be dropped. [PR1580578](#)
- Traffic loss might occur when IPv6 traffic forwarded by IPv4 GRE tunnel. [PR1582408](#)
- With IGMP snooping implemented, unexpected jitter issue might cause traffic loss. [PR1583207](#)
- The rpd process might crash after you commit the configured static group 224.0.0.0. [PR1586631](#)
- The multi-hop BFD session might flap if you commit the **RSI (Request Support Information)** collection command. [PR1589765](#)
- The traffic might get silently discarded or forwarded through not-best-path in the BGP setup. [PR1592550](#)
- There might be traffic loss when the GRE interface flaps. [PR1566428](#)
- On the QFX5000 line of switches, memory leak might occur. [PR1566483](#)

Resolved Issues: 20.4R2

EVPN

- The l2ald process might crash under a VLAN-based EVPN-VXLAN scenario. [PR1550109](#)
- On the QFX10000 devices, the l2ald process generates the core file at `l2ald_VXLAN_ifl_create_event_handler` at `/src/junos/usr/sbin/l2ald/platform/junos/l2ald_rtsock_VXLAN.c:477`. [PR1560068](#)
- `global-mac-ip-table-aging-time`; change from a high to low value might not take effect. [PR1562925](#)

Forwarding and Sampling

- The l2ald process might crash due to next-hop issue in the EVPN-MPLS. [PR1548124](#)
- Configuration archive transfer-on-commit fails while running Junos OS Release 18.2R3-S6.5. [PR1563641](#)

General Routing

- The DHCP relay-reply packets are dropped in the DHCPv6 relay scenario. [PR1352613](#)

Interfaces and Chassis

- MAC address entry issue might be seen after MC-LAG interface failover or failback. [PR1562535](#)

Layer 2 Features

- On the QFX5120 devices, packets with VLAN ID 0 are dropped. [PR1566850](#)
- On the QFX5000, software-forwarded VXLAN de-encapsulated packets have illegal length. [PR1574435](#)

Layer 2 Ethernet Services

- DHCP packet drop might be seen when the DHCP relay is configured on a leaf device. [PR1554992](#)

Platform and Infrastructure

- On the QFX5000 line of switches, the number of egress ACL filter entries is only 512 in Junos OS Release 19.4R1. [PR1472206](#)
- On the QFX10000 device, the chassisd process might generate core files on the backup Routing Engine after commit for 200 seconds due to the following error message: `CHASSISD_MAIN_THREAD_STALLED`. [PR1481143](#)
- Channelized interfaces might fail to come up. [PR1512203](#)
- Some inter-VLAN traffic flows do not converge after rebooting a spine (QFX10002) device in an EVPN-VXLAN non-collapsed scaled scenario. [PR1522585](#)
- Traffic loss might be observed on interfaces in a VXLAN environment. [PR1524955](#)
- On the QFX10002, the firewall logs are incorrectly populating from the Packet Forwarding Engine. [PR1533814](#)

- The following Packet Forwarding Engine error message is observed in BRCM-VIRTUAL:
brcm_virtual_tunnel_port_create() ,489: Failed NW VXLAN port token(45) hw-id(7026) status(Entry not found). [PR1535555](#)
- The BFD sessions might not come up in a VXLAN scenario. [PR1538600](#)
- The rpd memory leak might be observed on the backup Routing Engine due to link flaps. [PR1539601](#)
- Unable to take RSI properly due to the authentication error. [PR1539654](#)
- FPC might not be recognized after power cycle (hard reboot). [PR1540107](#)
- OSPFv3 session might keep flapping and OSPFv3 hellos might be dropped in the host path. [PR1547032](#)
- On the QFX10000 device, traffic might get dropped when the **set routing-options forwarding-table no-ecmp-fast-reroute** configuration is changed to 128 ECMP entries. [PR1547457](#)
- On the QFX5100 Virtual Chassis, the backup Routing Engines clear the reporting alarm for a PEM failure intermittently for a missing power source. [PR1548079](#)
- The 40GbE interface might be channelized after restarting the Virtual Chassis member. [PR1548267](#)
- Neighbor Solicitation might be dropped from the peer device. [PR1550632](#)
- The interface filter with source-port 0 matches everything instead of port 0. [PR1551305](#)
- On the QFX5110 and QFX5120 devices, the DHCPv6 traffic received over a VTEP might not be forwarded. [PR1551710](#)
- On the QFX5000 devices, the ARP resolution might fail. [PR1552671](#)
- The dcpfe process might crash and the non-channelization interfaces might not come up. [PR1552798](#)
- The **action-shutdown** command of storm control does not work for the ARP broadcast packets. [PR1552815](#)
- Traffic might not pass due to the addition of the VLAN tag 2 while passing through the Virtual Chassis port. [PR1555835](#)
- Traffic might be dropped when a firewall filter rule uses the then VLAN action. [PR1556198](#)
- The dcpfe process might crash and restart with the dcpfe process generated core file created while running the type-5 EVPN-VXLAN with 2000 VLANs. [PR1556561](#)
- On the QFX5120-48YM device, the **Multiple License Warning Messages** are observed. [PR1556816](#)
- Traffic storm might be caused by the analyzer due to the flapping of the link. [PR1557274](#)
- On the QFX5000 devices, the firewall filter might fail. [PR1558320](#)
- On the QFX5120 device, amber LEDs are displayed for the fan modules after upgrading to Junos OS Release 20.2R1. [PR1558407](#)
- PRBS (Pseudo Random Binary Sequence) test on the QFX5200 devices fails for 100GbE interfaces with the default settings. [PR1560086](#)

- When configuring the static MAC and static ARP on the EVPN core aggregate interface, the underlay next-hop programming might not be updated in the Packet Forwarding Engine. [PR1561084](#)
- PTP BC with G.8275.2.enh profile_2 512 clients does not come up. [PR1561348](#)
- PTP lock status gets stuck in the **Acquiring** state instead of the **Phase Aligned** state. [PR1561372](#)
- Firewall filters might not work after ISSU. [PR1561690](#)
- On the QFX10000 devices, the dcpfe process might crash during the configuration changes. [PR1561746](#)
- Traffic loss might occur in a large-scaled EVPN scenario when the next-hop type changes between discard and unicast. [PR1562425](#)
- On the QFX5000 devices, port mirroring might not work as expected. [PR1562607](#)
- On the QFX5120 devices, storm control with IRB interface might not work correctly. [PR1564020](#)
- On the QFX5100 Virtual Chassis, continuous message about agentd-pfe-proxy_telemetry_publisher is observed. [PR1566528](#)
- On the QFX5100 devices, the following internal comment is displayed in the output of the **show configuration** command: **Placeholder for QFX platform configuration**. [PR1567037](#)
- On the QFX10002 devices, discrepancy in inet.1 versus Packet Forwarding Engine reports multicast routes. [PR1567353](#)
- PTP management message with SMTLV is sent only to the first port number to go active in the member **multicast-mode l2-ifl**. [PR1571283](#)
- Issue is observed in telemetry when the **set services analytics streaming-server <> <>** configuration is present and server is not reachable. [PR1581192](#)
- The dcpfe process might crash and cause FPC to restart due to the traffic burst. [PR1534340](#)
- On the QFX10000 devices, the dcpfe process might crash in the specific MAC move cases and traffic loss might be observed in the EVPN-VXLAN scenario. [PR1542709](#)
- The switchover might be affected with the shared VXLAN tunnel. [PR1581524](#)

Routing Policy and Firewall Filters

- The policy configuration might be mismatched between the rpd and mgd process when **deactivate policy-options prefix-list** is involved in the configuration sequence. [PR1523891](#)

Routing Protocols

- Traffic might be silently discarded when the BGP route gets deleted which is part of multipath. [PR1514966](#)
- The OSPF neighborhood gets stuck in the **Start** state after configuring EVPN-VXLAN. [PR1519244](#)
- BGP LU session flap might be seen with the AIGP used scenario. [PR1558102](#)

- On the QFX5110-32Q device, the following syslog error message is observed after loading the NC type-5 EVPN-VXLAN configuration: **BCM-L2,pfe_bcm_l2_sp_bridge_port_tpid_set() Config TPID New/Old (8100:8100) Other-Tpid's ba49, 4aa0, 80f.** [PR1558189](#)
- On the QFX5120-48Y devices, the Layer 3 IPv4 traffic issue is observed after loading the non-collapsed type 5 EVPN-VXLAN configuration. [PR1560173](#)
- On the QFX5110 devices, the ARP resolution might fail if **native-vlan-id** is configured on the VXLAN interface. [PR1563569](#)
- The dcpfe process might crash when the size of the Local Bias Filter Bitmap string exceeds 256 characters. [PR1568159](#)
- On the QFX5000 devices, the untagged packets might not work. [PR1568533](#)
- The GRE egress traffic might not be forwarded between the different routing-instances. [PR1573411](#)

Resolved Issues: 20.4R1

EVPN

- EVPN-VXLAN core isolation do not work when the system is rebooted or the routing is restarted. [PR1461795](#)
- Unable to create a new VTEP interface. [PR1520078](#)
- ARP table might not be updated after performing VMotion or a network loop. [PR1521526](#)
- All the ARP reply packets towards to some address are flooded across the entire fabric. [PR1535515](#)
- EVPN-VXLAN registers mac-move counters under "system statistics bridge" even though there is no actual mac-move for multi-home (MH) clients. [PR1538117](#)
- Observed Layer 2 core file when system is rebooted when shared-tunnels are configured. [PR1548502](#)
- The l2ald process crashes and generates a core file l2ald_iff_rtm_delete_subintf_ifbds during the datacenter interconnect (dci) fusion run. [PR1550109](#)

General Routing

- Port LEDs do not work on the QFX5100-48T-6Q platforms. [PR1317750](#)
- On the QFX5100 switches, the interface output counter is double counted for self-generated traffic. [PR1462748](#)
- IRB MAC is not programmed in hardware when the MAC persistence timer expires. [PR1484440](#)
- Virtual Chassis is not stable with 100-Gigabit Ethernet and 40-Gigabit Ethernet interfaces. [PR1497563](#)
- BFD sessions flap after deactivating or activating the aggregated Ethernet interface or executing GRES. [PR1500798](#)
- The error message "mpls_extra NULL" might be seen when you add, change, or delete MPLS route. [PR1502385](#)

- The interface becomes physically down after changing to the FEC none mode. [PR1502959](#)
- LLDP is not acquired when native-vlan-id and tagged VLAN-ID are the same on a port. [PR1504354](#)
- "Media type" in show interface command is displayed as "Fiber" for SFP-10G-T. [PR1504630](#)
- The archival function might fail in certain conditions. [PR1507044](#)
- The fxpc might crash and restart with a fxpc core file created while installing image through ZTP. [PR1508611](#)
- Traffic might be affected on QFX10002, QFX10008, and QFX10016 line of switches. [PR1509220](#)
- The output VLAN push might not work. [PR1510629](#)
- Multicast traffic loss is observed because of the few multicast routes missing in the spine node. [PR1510794](#)
- The QFX10000-36Q line card used on QFX10008 and QFX10016 line of switches might fail to detect any QSFP. [PR1511155](#)
- Display issue, Virtual Chassis environment, Configured num-65-127-prefix value is shown incorrect for the command O/P "show chassis forwarding-options" [PR1512712](#)
- In the VXLAN configuration, the firewall filters might not be loaded into the TCAM with the following message due to TCAM overflow after upgrading to Junos OS Release 18.1R3-S1, 18.2R1, and later. [PR1514710](#)
- The routes update might fail upon the HMC memory issue and traffic impact might be seen. [PR1515092](#)
- The 100-Gigabit Ethernet AOC non-breakout port might be auto-channelized to other speed. [PR1515487](#)
- The MAC learning might not work properly after multiple MTU changes on the access port in the VXLAN scenario. [PR1516653](#)
- The dcpfe (PFE) process might crash due to memory leak. [PR1517030](#)
- The vgd process might generate a core file when the OVSDDB server restarts. [PR1518807](#)
- Traffic forwarding might be affected when adding, removing, or modifying the VLAN or VNI configurations such as VLAN-ID, VNI-ID, and Ingress-Replication command. [PR1519019](#)
- QFX5100: cprod timeout triggers high CPU. [PR1520956](#)
- The interfaces on the EX4600-EM-8F expansion module do not come up on the QFX5100-24Q with the non-QFX5E image. [PR1521523](#)
- Output interface index in SFLOW packet is zero when transit traffic is observed on the IRB interface with VRRP enabled. [PR1521732](#)
- On the QFX10002, QFX10008, and QFX10016 line of switches, the following error message is observed during specific steps while clearing and loading the scaled configuration again:
`PRDS_SLU_SAL:jprds_sl_u_sal_update_lrnent(),1379:jprds_sl_u_sal_update_lrnent call failed.` [PR1522852](#)

- The ECMP and LAG hash polarization might occur if the "hash-parameters" statement is not configured. [PR1525387](#)
- Sampling with the rate limiter command enabled, crosses the sample rate 65535. [PR1525589](#)
- Traffic loss might be observed when traffic is locally routed between the two VXLANs on the QFX5120 switch. [PR1527939](#)
- The MPLS EXP classifier might not work on QFX10000 line of switches. [PR1531095](#)
- Running SNMP MIB walk and executing 'show interfaces' command may cause the picd to crash. [PR1533766](#)
- High rate of ARP or NS packets might be observed between a device that runs Junos OS and host when the device that runs Junos OS receives an ARP or NS packet on an interface in transition. [PR1534796](#)
- The filter instance do not get removed from Packet Forwarding Engine after deactivating VLAN and IRB. [PR1537108](#)
- Interfaces are not created after channel-speed 10G is applied across ports 48 to 53 on QFX5100-48T. [PR1538340](#)
- Management Ethernet link down alarm seen while verifying system alarms in Virtual Chassis setup. [PR1538674](#)
- Traffic loss might be seen in OVSDB VXLAN scenario. [PR1540208](#)
- Inter VLAN traffic drop might be observed in EVPN-VXLAN scenario. [PR1541406](#)
- On QFX10002-60C switch, the "show pfe filter" CLI command is unavailable. [PR1545019](#)
- The neighbor solicitation might be dropped from the peer device. [PR1550632](#)
- DHCP IPv6 is not working for QFX5110-48s-4c. [PR1551710](#)
- On QFX10000 and PTX10000 line of devices with Junos OS Releases 20.1R1 and later, cannot collect RSI properly because of the authentication error. [PR1556816](#)

Infrastructure

- The kernel might crash if a file or a directory is accessed for the first time and is not created locally. [PR1518898](#)
- OID ifOutDiscards reports zero and sometimes shows valid value. [PR1522561](#)

Interfaces and Chassis

- The dcpfe might crash when the ICL is disabled and then enabled. [PR1525234](#)
- The logical interface might flap after adding or deleting native VLAN configuration. [PR1539991](#)

Layer 2 Features

- Flow control is enabled in Packet Forwarding Engine irrespective of the interface configuration and the fix causes small amount of packet loss when a parameter related to an interface such as "interface description" on any port is changed. [PR1496766](#)
- The dcpfe or FPC might crash generate a core file because of the memory leak after the VLAN add and VLAN delete operation. [PR1505239](#)
- On the QFX5000 line of switches, traffic imbalance might be observed if hash-params is not configured. [PR1514793](#)
- The MAC address in the hardware table might not synchronize between the master and the member in Virtual Chassis after the MAC flap. [PR1521324](#)
- On QFX5110 switch, the EVPN-VXLAN check traffic when VXLAN encap header fails. [PR1541316](#)

Platform and Infrastructure

- On QFX5110 and QFX5120 platforms, unicast RPF check in strict mode might not work properly. [PR1417546](#)

Routing Policy and Firewall Filters

- The policy configuration might be mismatched between rpd and mgd when "deactivate policy-options prefix-list" is involved in configuration sequence. [PR1523891](#)

Routing Protocols

- System upgrade or installation might fail on QFX5100-48T-6Q VC/VCF. [PR1486632](#)
- The IPv6 traffic might drop when falling back from IP-in-IP tunnel to inet.0/inet6.0. [PR1508631](#)
- Scale of filters with egress-to-ingress command is enabled. [PR1514570](#)
- The rpd process might report 100 percent CPU usage with the BGP route damping enabled. [PR1514635](#)
- The remaining BFD sessions of the aggregated Ethernet interface flap continuously if one of the BFD sessions is deleted. [PR1516556](#)
- Stale tunnel entries are seen after negative triggers. [PR1516818](#)
- The BFD sessions might flap continuously after disruptive switchover followed by GRES. [PR1518106](#)
- On QFX5210-64C, enabling IPv6 flow-based Packet Forwarding Engine hashing gives commit error. [PR1519018](#)
- Firewall "sample" configuration gives the warning as unsupported on QFX10002-36q and will not work. [PR1521763](#)
- Errors are seen during script run with negative triggers at scale[LOG: Err]
BRMCM_NH-,brcm_nh_ipntunnel_unilist_install(),5752:IPoIP < src: 1.1.1.1, dst: 1.1.3.1> NH id 2537, Tunnel id: 512 failed to create decap obj Table full: vrf 1 vid 4082 intf 4058 of nh 131074(3)] when the tunnel color attribute is deleted for all the tunnels at scale. [PR1526405](#)

- On QFX5000 line of switches, the IPIP firewall filter term with decapsulate action need to be duplicated for each from protocol. [PR1527755](#)
- On QFX5000 line of switches, the fxpc process might crash if the VXLAN interface flap. [PR1528490](#)

Virtual Chassis

- On the QFX5000 Virtual Chassis, the DDoS violations that occur on the backup are not reported to the Routing Engine. [PR1490552](#)
- On QFX5120 and QFX5210 platforms unexpected storm control events might occur. [PR1519893](#)

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Documentation Updates

There are no errata or changes in Junos OS Release 20.4R3 documentation for the QFX Series Switches.

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Migration, Upgrade, and Downgrade Instructions

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This section contains the procedure to upgrade Junos OS, and the upgrade and downgrade policies for Junos OS. Upgrading or downgrading Junos OS can take several hours, depending on the size and configuration of the network.

Upgrading Software on QFX Series Switches

When upgrading or downgrading Junos OS, always use the jinstall package. Use other packages (such as the jbundle package) only when so instructed by a Juniper Networks support representative. For information about the contents of the jinstall package and details of the installation process, see the [Installation and Upgrade Guide](#) and [Junos OS Basics](#) in the QFX Series documentation.

If you are not familiar with the download and installation process, follow these steps:

1. In a browser, go to <https://www.juniper.net/support/downloads/junos.html>.

The Junos Platforms Download Software page appears.

2. In the QFX Series section of the Junos Platforms Download Software page, select the QFX Series platform for which you want to download the software.
3. Select **20.4** in the Release pull-down list to the right of the Software tab on the Download Software page.

4. In the Install Package section of the Software tab, select the QFX Series Install Package for the 20.4 release.

An Alert box appears.

5. In the Alert box, click the link to the PSN document for details about the software, and click the link to download it.

A login screen appears.

6. Log in to the Juniper Networks authentication system using the username (generally your e-mail address) and password supplied by Juniper Networks representatives.
7. Download the software to a local host.
8. Copy the software to the device or to your internal software distribution site.
9. Install the new jinstall package on the device.

NOTE: We recommend that you upgrade all software packages out of band using the console, because in-band connections are lost during the upgrade process.

Customers in the United States and Canada use the following command:

```
user@host> request system software add
source/jinstall-host-qfx-5-x86-64-20.4-R1.n-secure-signed.tgz reboot
```

Replace **source** with one of the following values:

- **/pathname**—For a software package that is installed from a local directory on the switch.
- For software packages that are downloaded and installed from a remote location:
 - **ftp://hostname/pathname**
 - **http://hostname/pathname**
 - **scp://hostname/pathname** (available only for Canada and U.S. version)

Adding the **reboot** command reboots the switch after the upgrade is installed. When the reboot is complete, the switch displays the login prompt. The loading process can take 5 to 10 minutes.

Rebooting occurs only if the upgrade is successful.



NOTE: After you install a Junos OS Release 20.4 **jinstall** package, you can issue the **request system software rollback** command to return to the previously installed software.

Installing the Software on QFX10002-60C Switches

This section explains how to upgrade the software, which includes both the host OS and the Junos OS. This upgrade requires that you use a VM host package—for example, a **junos-vmhost-install-x.tgz**.

During a software upgrade, the alternate partition of the SSD is upgraded, which will become primary partition after a reboot. If there is a boot failure on the primary SSD, the switch can boot using the snapshot available on the alternate SSD.

NOTE: The QFX10002-60C switch supports only the 64-bit version of Junos OS.

NOTE: If you have important files in directories other than /config and /var, copy the files to a secure location before upgrading. The files under /config and /var (except /var/etc) are preserved after the upgrade.

To upgrade the software, you can use the following methods:

If the installation package resides locally on the switch, execute the **request vmhost software add <pathname><source>** command.

For example:

```
user@switch> request vmhost software add /var/tmp/junos-vmhost-install-qfx-x86-64-20.4R3.9.tgz
```

If the Install Package resides remotely from the switch, execute the **request vmhost software add <pathname><source>** command.

For example:

```
user@switch> request vmhost software add
ftp://ftpserver/directory/junos-vmhost-install-qfx-x86-64-20.4R3.9.tgz
```

After the reboot has finished, verify that the new version of software has been properly installed by executing the **show version** command.

```
user@switch> show version
```

Installing the Software on QFX10002 Switches

NOTE: If you are upgrading from a version of software that does not have the FreeBSD 10 kernel (15.1X53-D30, for example), you will need to upgrade from Junos OS Release 15.1X53-D30 to Junos OS Release 15.1X53-D32. After you have installed Junos OS Release 15.1X53-D32, you can upgrade to Junos OS Release 15.1X53-D60 or Junos OS Release 18.3R1.

NOTE: On the switch, use the **force-host** option to force-install the latest version of the Host OS. However, by default, if the Host OS version is different from the one that is already installed on the switch, the latest version is installed without using the **force-host** option.

If the installation package resides locally on the switch, execute the **request system software add <pathname><source> reboot** command.

For example:

```
user@switch> request system software add
/var/tmp/jinstall-host-qfx-10-f-x86-64-20.4R3.n-secure-signed.tgz reboot
```

If the Install Package resides remotely from the switch, execute the **request system software add <pathname><source> reboot** command.

For example:

```
user@switch> request system software add
ftp://ftpserver/directory/jinstall-host-qfx-10-f-x86-64-20.4R3.n-secure-signed.tgz reboot
```

After the reboot has finished, verify that the new version of software has been properly installed by executing the **show version** command.

```
user@switch> show version
```

Upgrading Software from Junos OS Release 15.1X53-D3X to Junos OS Release 15.1X53-D60, 15.1X53-D61.7, 15.1X53-D62, and 15.1X53-D63 on QFX10008 and QFX10016 Switches

NOTE: Before you install the software, back up any critical files in `/var/home`. For more information regarding how to back up critical files, contact Customer Support at <https://www.juniper.net/support>.

The switch contains two Routing Engines, so you will need to install the software on each Routing Engine (re0 and re1).

If the installation package resides locally on the switch, execute the **request system software add <pathname><source>** command.

To install the software on re0:

```
user@switch> request system software add
/var/tmp/jinstall-host-qfx-10-m-15.1X53-D60.n-secure-domestic-signed.tgz re0
```

If the Install Package resides remotely from the switch, execute the **request system software add <pathname><source> re0** command.

For example:

```
user@switch> request system software add
ftp://ftpserver/directory/jinstall-host-qfx-10-m-15.1X53-D60.n-secure-domestic-signed.tgz re0
```

To install the software on re1:

```
user@switch> request system software add
/var/tmp/jinstall-host-qfx-10-m-15.1X53-D60.n-secure-domestic-signed.tgz re1
```

If the Install Package resides remotely from the switch, execute the **request system software add <pathname><source> re1** command.

For example:

```
user@switch> request system software add
ftp://ftpserver/directory/jinstall-host-qfx-10-m-15.1X53-D60.n-secure-domestic-signed.tgz re1
```

Reboot both Routing Engines.

For example:

```
user@switch> request system reboot both-routing-engines
```

After the reboot has finished, verify that the new version of software has been properly installed by executing the **show version** command.

```
user@switch> show version
```

Installing the Software on QFX10008 and QFX10016 Switches

Because the switch has two Routing Engines, perform a Junos OS installation on each Routing Engine separately to avoid disrupting network operation.

NOTE: Before you install the software, back up any critical files in **/var/home**. For more information regarding how to back up critical files, contact Customer Support at <https://www.juniper.net/support>.



WARNING: If graceful Routing Engine switchover (GRES), nonstop bridging (NSB), or nonstop active routing (NSR) is enabled when you initiate a software installation, the software does not install properly. Make sure you issue the CLI **delete chassis redundancy** command when prompted. If GRES is enabled, it will be removed with the **redundancy** command. By default, NSR is disabled. If NSR is enabled, remove the nonstop-routing statement from the **[edit routing-options]** hierarchy level to disable it.

1. Log in to the master Routing Engine's console.

For more information about logging in to the Routing Engine through the console port, see the specific hardware guide for your switch.

2. From the command line, enter configuration mode:

```
user@switch> configure
```

3. Disable Routing Engine redundancy:

```
user@switch# delete chassis redundancy
```

4. Disable nonstop-bridging:

```
user@switch# delete protocols layer2-control nonstop-bridging
```

5. Save the configuration change on both Routing Engines:

```
user@switch# commit synchronize
```

6. Exit the CLI configuration mode:

```
user@switch# exit
```


After the switch has been prepared, you first install the new Junos OS release on the backup Routing Engine, while keeping the currently running software version on the master Routing Engine. This enables the master Routing Engine to continue operations, minimizing disruption to your network.

After making sure that the new software version is running correctly on the backup Routing Engine, you are ready to switch routing control to the backup Routing Engine, and then upgrade or downgrade the software version on the other Routing Engine.

7. Log in to the console port on the other Routing Engine (currently the backup).

For more information about logging in to the Routing Engine through the console port, see the specific hardware guide for your switch.

8. Install the new software package using the **request system software add** command:

```
user@switch> request system software add validate
/var/tmp/jinstall-host-qfx-10-f-x86-64-20.4R3.n-secure-signed.tgz
```

For more information about the **request system software add** command, see the [CLI Explorer](#).

9. Reboot the switch to start the new software using the **request system reboot** command:

```
user@switch> request system reboot
```

NOTE: You must reboot the switch to load the new installation of Junos OS on the switch.

To abort the installation, do not reboot your switch. Instead, finish the installation and then issue the **request system software delete <package-name>** command. This is your last chance to stop the installation.

All the software is loaded when you reboot the switch. Installation can take between 5 and 10 minutes. The switch then reboots from the boot device on which the software was just installed. When the reboot is complete, the switch displays the login prompt.

While the software is being upgraded, the Routing Engine on which you are performing the installation is not sending traffic.

10. Log in and issue the **show version** command to verify the version of the software installed.

```
user@switch> show version
```

Once the software is installed on the backup Routing Engine, you are ready to switch routing control to the backup Routing Engine, and then upgrade or downgrade the master Routing Engine software.

11. Log in to the master Routing Engine console port.

For more information about logging in to the Routing Engine through the console port, see the specific hardware guide for your switch.

12. Transfer routing control to the backup Routing Engine:

```
user@switch> request chassis routing-engine master switch
```

For more information about the **request chassis routing-engine master** command, see the [CLI Explorer](#).

13. Verify that the backup Routing Engine (slot 1) is the master Routing Engine:

```
user@switch> show chassis routing-engine
```

```
Routing Engine status:
Slot 0:
  Current state           Backup
  Election priority       Master (default)
Routing Engine status:
Slot 1:
  Current state           Master
  Election priority       Backup (default)
```

14. Install the new software package using the **request system software add** command:

```
user@switch> request system software add validate
/var/tmp/jinstall-host-qfx-10-f-x86-64-20.4R3.n-secure-signed.tgz
```

For more information about the **request system software add** command, see the [CLI Explorer](#).

15. Reboot the Routing Engine using the **request system reboot** command:

```
user@switch> request system reboot
```

NOTE: You must reboot to load the new installation of Junos OS on the switch.

To abort the installation, do not reboot your system. Instead, finish the installation and then issue the **request system software delete jinstall <package-name>** command. This is your last chance to stop the installation.

The software is loaded when you reboot the system. Installation can take between 5 and 10 minutes. The switch then reboots from the boot device on which the software was just installed. When the reboot is complete, the switch displays the login prompt.

While the software is being upgraded, the Routing Engine on which you are performing the installation does not send traffic.

16. Log in and issue the **show version** command to verify the version of the software installed.

17. Transfer routing control back to the master Routing Engine:

```
user@switch> request chassis routing-engine master switch
```

For more information about the **request chassis routing-engine master** command, see the [CLI Explorer](#).

18. Verify that the master Routing Engine (slot 0) is indeed the master Routing Engine:

```
user@switch> show chassis routing-engine
```

```
Routing Engine status:
  Slot 0:
    Current state           Master
    Election priority       Master (default)
Routing Engine status:
  Slot 1:
    Current state           Backup
    Election priority       Backup (default)
```

Performing a Unified ISSU

You can use unified ISSU to upgrade the software running on the switch with minimal traffic disruption during the upgrade.

NOTE: Unified ISSU is supported in Junos OS Release 13.2X51-D15 and later.

Perform the following tasks:

- [Preparing the Switch for Software Installation on page 316](#)
- [Upgrading the Software Using Unified ISSU on page 316](#)

Preparing the Switch for Software Installation

Before you begin software installation using unified ISSU:

- Ensure that nonstop active routing (NSR), nonstop bridging (NSB), and graceful Routing Engine switchover (GRES) are enabled. NSB and GRES enable NSB-supported Layer 2 protocols to synchronize protocol information between the master and backup Routing Engines.

To verify that nonstop active routing is enabled:

NOTE: If nonstop active routing is enabled, then graceful Routing Engine switchover is enabled.

```
user@switch> show task replication
Stateful Replication: Enabled
RE mode: Master
```

If nonstop active routing is not enabled (**Stateful Replication** is **Disabled**), see *Configuring Nonstop Active Routing on Switches* for information about how to enable it.

- Enable nonstop bridging (NSB). See *Configuring Nonstop Bridging on Switches (CLI Procedure)* for information on how to enable it.
- (Optional) Back up the system software—Junos OS, the active configuration, and log files—on the switch to an external storage device with the **request system snapshot** command.

Upgrading the Software Using Unified ISSU

This procedure describes how to upgrade the software running on a standalone switch.

To upgrade the switch using unified ISSU:

1. Download the software package by following the procedure in the Downloading Software Files with a Browser section in *Installing Software Packages on QFX Series Devices*.
2. Copy the software package or packages to the switch. We recommend that you copy the file to the `/var/tmp` directory.
3. Log in to the console connection. Using a console connection allows you to monitor the progress of the upgrade.
4. Start the ISSU:
 - On the switch, enter:

```
user@switch> request system software in-service-upgrade /var/tmp/package-name.tgz
```

where *package-name.tgz* is, for example, *jinstall-host-qfx-10-f-x86-64-20.4R3.n-secure-signed.tgz*.

NOTE: During the upgrade, you cannot access the Junos OS CLI.

The switch displays status messages similar to the following messages as the upgrade executes:

```
warning: Do NOT use /user during ISSU. Changes to /user during ISSU may get
lost!
ISSU: Validating Image
ISSU: Preparing Backup RE
Prepare for ISSU
ISSU: Backup RE Prepare Done
Extracting jinstall-host-qfx-5-f-x86-64-18.3R1.n-secure-signed.tgz ...
Install jinstall-host-qfx-5-f-x86-64-19.2R1.n-secure-signed.tgz completed
Spawning the backup RE
Spawn backup RE, index 0 successful
GRES in progress
GRES done in 0 seconds
Waiting for backup RE switchover ready
GRES operational
Copying home directories
Copying home directories successful
Initiating Chassis In-Service-Upgrade
Chassis ISSU Started
ISSU: Preparing Daemons
ISSU: Daemons Ready for ISSU
```

```

ISSU: Starting Upgrade for FRUs
ISSU: FPC Warm Booting
ISSU: FPC Warm Booted
ISSU: Preparing for Switchover
ISSU: Ready for Switchover
Checking In-Service-Upgrade status
  Item          Status          Reason
  FPC 0         Online (ISSU)
Send ISSU done to chassisd on backup RE
Chassis ISSU Completed
ISSU: IDLE
Initiate em0 device handoff

```

NOTE: A unified ISSU might stop, instead of abort, if the FPC is at the warm boot stage. Also, any links that go down and up will not be detected during a warm boot of the Packet Forwarding Engine (PFE).

NOTE: If the unified ISSU process stops, you can look at the log files to diagnose the problem. The log files are located at `/var/log/vjunos-log.tgz`.

5. Log in after the reboot of the switch completes. To verify that the software has been upgraded, enter the following command:

```
user@switch> show version
```

6. Ensure that the resilient dual-root partitions feature operates correctly, by copying the new Junos OS image into the alternate root partitions of all of the switches:

```
user@switch> request system snapshot slice alternate
```

Resilient dual-root partitions allow the switch to boot transparently from the alternate root partition if the system fails to boot from the primary root partition.

Upgrade and Downgrade Support Policy for Junos OS Releases

We have two types of releases, EOL and EEOL:

- End of Life (EOL) releases have engineering support for twenty four months after the first general availability date and customer support for an additional six more months.
- Extended End of Life (EEOL) releases have engineering support for thirty six months after the first general availability date and customer support for an additional six more months.

For both EOL and EEOL releases, you can upgrade to the next three subsequent releases or downgrade to the previous three releases. For example, you can upgrade from 19.2 to the next three releases – 19.3, 19.4 and 20.1 or downgrade to the previous three releases – 19.1, 18.4 and 18.3.

For EEOL releases only, you have an additional option - you can upgrade directly from one EEOL release to the next two subsequent EEOL releases, even if the target release is beyond the next three releases. Likewise, you can downgrade directly from one EEOL release to the previous two EEOL releases, even if the target release is beyond the previous three releases. For example, 19.2 is an EEOL release. Hence, you can upgrade from 19.2 to the next two EEOL releases – 19.3 and 19.4 or downgrade to the previous two EEOL releases – 19.1 and 18.4.

Release Type	End of Engineering (EOE)	End of Support (EOS)	Upgrade and Downgrade to subsequent 3 releases	Upgrade and Downgrade to subsequent 2 EEOL releases
End of Life (EOL)	24 months	End of Engineering + 6 months	Yes	No
Extended End of Life (EEOL)	36 months	End of Engineering + 6 months	Yes	Yes

For more information about EOL and EEOL releases, see <https://www.juniper.net/support/eol/junos.html>.

For information about software installation and upgrade, see the [Installation and Upgrade Guide](#).

SEE ALSO

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Junos OS Release Notes for SRX Series

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These release notes accompany Junos OS Release 20.4R3 for the SRX Series Services Gateways. They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

You can also find these release notes on the Juniper Networks Junos OS Documentation webpage, located at https://www.juniper.net/documentation/product/en_US/junos-os.

What's New

IN THIS SECTION

- [What's New in Release 20.4R3 | 321](#)
- [What's New in Release 20.4R2 | 321](#)
- [What's New in Release 20.4R1 | 321](#)

Learn about new features introduced in the Junos OS main and maintenance releases for the SRX Series.

What's New in Release 20.4R3

There are no new features for SRX Series Services Gateways in Junos OS Release 20.4R3.

What's New in Release 20.4R2

There are no new features for SRX Series Services Gateways in Junos OS Release 20.4R2.

What's New in Release 20.4R1

Application Layer Gateways (ALGs)

- **SIP ALG load-balancing enhancement (SRX5400, SRX5600, and SRX5800)**—Starting in Junos OS Release 20.4R1, you can configure two new CLI commands to distribute the SIP load among all the available SPUs to better balance SIP traffic. The new commands are **set security alg sip bulk-call-mode** and **set security alg sip enable-call-distribution**.

[See [bulk-call-mode](#) and [enable-call-distribution](#).]

Application Security

- **AppQoE support for SaaS applications (NFX Series and SRX Series)**—Starting in Junos OS Release 20.4R1, we've extended application quality of experience (AppQoE) support for Software as a Service (SaaS) applications.

AppQoE performs service-level agreement (SLA) measurements across the available WAN links such as underlay, GRE, IPsec or MPLS over GRE. It then sends SaaS application data over the most SLA-compliant link to provide a consistent service.

To configure AppQoE for SaaS applications:

1. Define the SLA rule type as SaaS (**set security advance-policy-based-routing sla-rule sla1 type saas**).
2. Include SaaS server details in the address book (**set security address-book global address *address-book* dns-name *saas-server-url* ipv4-only**).
3. Attach the SLA rule to the policy-based APBR profile.

[See [Application Quality of Experience](#).]

- **SNI-based dynamic application information for SSL proxy profile (SRX Series)**—Starting in Junos OS Release 20.4R1, we've enhanced the selection mechanism for SSL proxy profiles. An SSL proxy can now use Server Name Indication (SNI) TLS extensions to identify dynamic applications.

SSL proxy defers SSL profile selection until the dynamic application is detected in a client hello message based on the SNI. Next, the SSL proxy does a firewall rule lookup based on the identified application and selects an appropriate SSL proxy profile.

Using SNI-based dynamic application information results in more accurate SSL proxy profile selection for the session. By default, this feature is enabled on SRX Series devices.

[See [Unified Policies for SSL Proxy](#) and [global-config \(Services\)](#).]

- **Granular control over DNS-over-HTTP and DNS-over-TLS application traffic (NFX Series, SRX Series and vSRX)**—In Junos OS Release 20.4R1, we introduce a new micro-application, DNS-ENCRYPTED, to enhance the application signature package. By configuring this micro-application in a security policy, you can have granular control for DNS-over-HTTP and DNS-over-TLS application traffic.

The DNS-ENCRYPTED application is enabled by default. You can disable it using the **request services application-identification application disable DNS-ENCRYPTED** command.

You can view the details of the micro-applications using the **show services show services application-identification application detail** command.

[See [Application Identification Support for Micro-Applications](#).]

- **Support for tunneling applications in unified policies (SRX Series and vSRX)**—In Junos OS Release 20.4R1, we've enhanced unified policy functionality on security devices to manage tunneling applications. You can now block a specific tunneling application by using a unified policy.

For example, to block tunneling applications such as QUIC or SOCKS, you can configure a unified policy with the deny or reject action for these applications.

[See [Application Identification Support for Unified Policies](#).]

ATP Cloud

- **Support for filtering DNS requests for disallowed domains (SRX4100, SRX4200, SRX4600, and vSRX)**—Starting in Junos OS Release 20.4R1, you can configure DNS filtering to identify DNS requests for disallowed domains. You can either:
 - Block access to the disallowed domain by sending a DNS response that contains the IP address or fully qualified domain name (FQDN) of a DNS sinkhole server. This ensures that when the client attempts to send traffic to the disallowed domain, the traffic instead goes to the sinkhole server.
 - Log the DNS request and reject access.

The DNS sinkhole must be configured only for DNS profile category.

[See [dns-filtering](#), [security-intelligence](#), [clear services security-intelligence dns-statistics](#), and [show services security-intelligence dns-statistics](#).]

Authentication and Access Control

- **Logical domain support for device identity authentication (SRX1500, SRX4100, SRX4200, SRX4600, SRX5400, SRX5600, and SRX5800)**—Starting in Junos OS Release 20.4R1, we've enabled logical system and tenant system support for policy search that uses device information from JIMS. The user firewall uses the logical system name or the tenant name as a differentiator. The logical system name or tenant name must be globally unique and consistent in the JIMS server and SRX Series device. The JIMS server forwards the differentiator to be included in the device identification authentication entries. The device authentication entries are distributed into the root logical system.

[See [Understanding Integrated User Firewall support in a Logical System](#), [end-user-profile](#).]

Chassis Clustering

- **Enabling and disabling control link (SRX1500)**—Starting in Junos OS Release 20.4R1, you can enable or disable the control links on SRX1500 using the operational and configuration mode CLI commands listed below.

In earlier Junos OS releases, if you wanted to disable the control links and fabric links, you had to manually unplug the cables for control links and fabric links, which was very inconvenient.

Using these commands helps you control the status of the cluster nodes and provides protection against version mismatch during a cluster upgrade, and minimizes failovers.

Table 4: Configuration and Operational Mode Commands

Configuration Mode	Operational Mode
To disable the control link, run the set chassis cluster control-interface (node0 node1) disable command on node 0 or node 1. NOTE: If you disable the links using the configuration command, then the links remain disabled even after system reboot.	To disable the control link from the local node, run the request chassis cluster control-interface (node0 node1) disable command. NOTE: If you have disabled control link using the operational mode CLI commands, the links will be enabled after system reboot.

Table 4: Configuration and Operational Mode Commands (*continued*)

Configuration Mode	Operational Mode
To enable the control link, run the delete chassis cluster control-interface (node0 node1) disable on both the nodes. You need to ensure that you disable the control and fabric interfaces on both nodes using CLI configuration, to keep control and fabric links disabled when you reboot the nodes. Use the set interfaces (fab0 fab1) disable and delete interfaces (fab0 fab1) disable CLI commands to disable or enable the fabric interfaces.	To enable the control link from the local node, run the request chassis cluster control-interface (node0 node1) enable command.

[See [cluster \(Chassis\)](#).]

- **In-service software upgrade (ISSU) (SRX1500, SRX4100, SRX4200, SRX4600, SRX5400, SRX5600, and SRX5800)**—Starting in Junos OS Release 20.4R1, you can use the new **status** option with the **request system software in-service-upgrade** command to display ISSU status during upgrade.

[See [request system software in-service-upgrade](#).]

- **Support for single PSU operation without alarms (SRX1500 and SRX4600)**—Starting in Junos OS Release 20.4R1, a new argument, **pem-absence**, is available at the **[edit chassis alarm]** hierarchy level. You can use **set chassis alarm pem-absence ignore** to ignore the power supply unit (PSU) alarm. By default, the PSU alarm is raised when any PSU is missing or not energized.

[See [Understanding Chassis Alarms](#), [show chassis alarms](#), and [pem-absence](#).]

Flow-Based and Packet-Based Processing

- **Express Path+ support for packet-based services in traditional and unified policies (SRX4600, SRX5400, SRX5600, and SRX5800)**—Starting in Junos OS Release 20.4R1, Express Path+ (formerly known as services offloading) is supported for packet-based services in traditional and unified policies.

[Table 5 on page 325](#) lists the features that are supported by Express Path+.

Table 5: Features Supported by Express Path+

Supported Features
ALG
APBR (application based routing)
The Express Path+ works after APBR ignores subsequent traffic.
Web Filtering
GPRS (GTP and SCTP)
IDP
Screens
See Understanding Screens Options on SRX Series Devices .
Juniper Security Intelligence (SecIntel)
Unified Policies with AppID and URL category matching
UserFW

[See [Express Path Overview](#).]

- **New resource-manager commands (SRX5400)**—Starting in Junos OS 20.4R1 Release, you can configure memory load and CPU resources using three new resource-manager configuration statements. The new configuration statements **service-memory**, **session-memory**, and **cpu** are available at the **[edit security resource-manager]** hierarchy level.

[See [services-memory \(resource-manager\)](#), [session-memory \(resource-manager\)](#), [cpu \(resource-manager\)](#), [show resource-manager memory](#), [show resource-manager](#), [show resource-manager cpu](#), and [show security flow session](#).]

- **Support for trace and debug of data packets (SRX1500, SRX4100, SRX4200, SRX4600, SRX5400, SRX5600, SRX5800, and vSRX)**—Starting in Junos OS Release 20.4R1, you can trace packet footprints. To enable tracing of packet footprints, use the **traceoptions flag jexec** command at the **[edit security flow]** or **[edit logical-systems logical system name security flow]** hierarchy level.

The packet trace logs are captured in a sequential time order. The sequential trace enhances flow debuggability for packet processing with multiple logical systems and tenant systems, tunnel piping, multiple reinjection, and so on.

[See [traceoptions \(Security Flow\)](#) and [show security flow status](#).]

- **Pass-through authentication of IP-IP and GRE tunnel traffic in TAP mode (SRX300, SRX320, SRX340, SRX345, SRX380, SRX550 HM, SRX1500, and vSRX)**—Starting in Junos OS Release 20.4R1, SRX Series devices perform pass-through authentication of IP-IP and GRE tunnel traffic when in TAP mode. To use TAP mode, connect the SRX Series device to the mirror port of the connected switch, which provides a copy of the traffic traversing the switch. In TAP mode, the SRX Series device processes incoming traffic from the TAP interface and generates a security log or report containing with information about threats detected, application usage, and user details.

[See [Configure User Authentication Methods](#).]

- **Enhancement in Resource Management (SRX Series Devices)**—Starting in Junos OS Release 20.4R1, when the Layer7 packets such as ALG or User Firewall create flow sessions, you can control whether to drop the packet or forward the packet if resource is busy.

Configure the **security-service** under the **edit security forwarding-options** hierarchy to implement the resource management. When you configure the **security service** as **fail-open**, the session skips the application level and forwards the packet.

By default, the **security service** is **fail-closed**, and allows the session to drop at the application level. When you use **fail-closed** option, make sure **set security forwarding-options security-service fail-open** is not configured.

[See [Traffic Processing on SRX Series Devices Overview](#)]

- **Enhancement of Flow Reroute in Multiple Routing Table (SRX Series Devices)**—Starting in Junos OS Release 20.4R1, the flow reroutes the traffic using multiple routing table. Earlier to this release, flow reroute was supported with only one routing table.

If there is one routing table involved in route lookup, there is no change in the implementation of reroute. If there are more than one routing table involved in the route lookup and when there is a route change in any one of the routing table, you can mark all the affected flows for reroute. We support 16 routing tables.

[See [Traffic Processing on SRX Series Devices Overview](#)]

Interfaces and Chassis

- **CPU load monitoring (SRX1500, SRX4100, SRX4200, SRX4600, SRX5400, SRX5600, and SRX5800)**—Starting in Junos OS Release 20.4R1, the reswatch process is used to monitor Routing Engine CPU load and Junos kernel usage.

[See [resource-watch](#).]

- **Support for captive portal on Wi-Fi Mini-Physical Interface Module (SRX320, SRX340, SRX345, SRX380, and SRX550HM)**—Starting in Junos OS Release 20.4R1, we support captive portal for integrated guest

access management on the Wi-Fi Mini-PIM card. You can set up captive portal authentication to do either of the following:

- Redirect Web browser requests to a login page that requires the user to input a simple password.
- Use username and password authentication with the RADIUS server.

To configure captive portal, you must enable a DHCP server and configure two address pools for the Wi-Fi interface. After you configure captive portal, you can customize the login page display settings.

You can enable or disable captive portal on different virtual access points (VAPs), VAPs under the same access point use the same captive portal authentication type.

[See [Wi-Fi Mini-Physical Interface Module Overview](#).]

- **Support for Annex J and G.Fast with specialized SFP (SRX380, SRX300, SRX320, SRX340, and SRX345)**—Starting in Junos OS Release 20.4R1, we support G.Fast and Annex J specification with SFP xDSL for ADSL2/ADSL2+ and all VDSL2 profiles on SRX Series devices. Annex J is a specification in ITU-T recommendations G.992.3 and G.992.5 for all digital mode ADSL with improved spectral compatibility with ADSL over ISDN. You can configure Annex J by using the **dsl-sfp-options** cli command.

[See [Configuring ADSL Interfaces](#).]

Intrusion Detection and Prevention

- **IDP utility to read PCAP and generate protocol (SRX1500, SRX4200, SRX4600, vSRX, and vSRX3.0)**—Starting in Junos OS Release 20.4R1, you can use the `pcap-analysis` operational command to display the generated IDP context.

Execute the operational command **request security idp pcap-analysis /var/tmp/http.pcap from-zone trust to-zone untrust** by providing the zone details for selecting the corresponding interface. The output displays as a list of IDP contexts associated with the PCAP, and the data matched for that IDP context and attacks. However, configure the rulebase, IDP active policy, interfaces with necessary configuration like IPv4 addresses, zones, and policies for pcap utility to detect contexts and attacks.

You can process upto 3MB files and save a maximum length of 8K of context data to generate a unique context.

```

Attack Detected:  Yes
Rulebase          Rule Id      Attack Name
IPS-Rulebase-1    Rule-1      HTTP:AUDIT:URL
IDP context statistics:
Context name          #Hits      #Data
http-url              1          /
http-get-url          1          /
http-header-host      1          7.0.0.1
http-header-user-agent 1          lwp-request/5.827
                     libwww-perl/5.833
http-header           2          te:

```

		deflate,gzip;q=0.3
		&&
		connection:
		TE, close
http-request	1	GET /
		HTTP/1.1
http-request-method	1	GET

[See [IDP Utility for PCAP](#)]

Juniper Extension Toolkit (JET)

- **Juniper Extension Toolkit (JET) support for 64-bit applications (MX5, MX10, MX40, MX80, MX104, MX150, MX204, MX240, MX480, MX960, MX2008, MX2010, MX2020, MX10003, MX10008, MX ELM, JunosV Firefly, cSRX, SRX100, SRX110, SRX210, SRX220, SRX240, SRX300, SRX320, SRX340, SRX345, SRX550, SRX550HM, SRX650, SRX720E, SRX750E, SRX1400, SRX1500,SRX3400, SRX3600, SRX4100, SRX4200, SRX4400, SRX4600, SRX4800, SRX5400, SRX5600, SRX5800, SRX7X0E, SRX-ES7, SRX-ES8, VMX, and VSRX)**—Starting in Junos OS Release 20.4R1, JET supports 64-bit applications. Use the following commands to compile 64-bit applications for use with the AMD64 or ARM64 64-bit processor architecture.
 - **mk-amd64:** Compiles the application for use with AMD64 and Junos OS with FreeBSD.
 - **mk-amd64,bsd:** Compiles the application for use with AMD64 and Junos OS with upgraded FreeBSD.
 - **mk-arm64,bsd:** Compiles the application for use with ARM64 and Junos OS with upgraded FreeBSD.

[See [Develop On-Device JET Applications.](#)]

Junos OS XML and API Scripting

- **Start time option for interval-based internal events that trigger event policies (EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—Starting in Junos OS Release 20.4R1, when you create an interval-based internal event for triggering event policies, you can specify the start date and time for the initial event. To specify a start time, configure the **start-time** option along with the **time-interval** option at the **[edit event-options generate-event]** hierarchy level.

[See [Generating Internal Events to Trigger Event Policies.](#)]

J-Web

- **Enhanced Setup Wizard (SRX Series)**—Starting in Junos OS Release 20.4R1, we’ve refreshed the Setup Wizard settings for better experience. You can:
 - Access the wizard with the changed menu by selecting Device Administration>Reset Configuration.
 - View the new look of the wizard page.
 - Create one admin user with super user permissions.

- Add and search NTP servers inline.
- Synchronize device time with your computer time.
- Configure interfaces directly under Zones & Interfaces instead of using a pop-up page.
- Configure Tap settings with reorganized options.
- View improved statements in the commit-success page.

In Standalone mode, the Setup Wizard does not support advanced services configuration.

[See [Start J-Web](#).]

- **Support for captive portal (SRX Series)**—Starting in Junos OS Release 20.4R1, you can configure captive portal while:

- Creating a rule on the **Security Policies & Objects > Security Policies** page.
- Adding a logical interface on the **Network > Connectivity > Ports** page. The IP address configured for web authentication is also used for captive portal.
- Uploading the logo on the **Security Services > Firewall Authentication > Authentication Settings** page. The selected logo is also used for captive portal.

The captive portal authenticates the user request to access an SRX Series protected resource using HTTPS browser.

[See [Add a Rule](#), [Add a Logical Interface](#), and [About the Authentication Settings Page](#).]

- **Change in the Monitor tab menus (SRX Series)**—Starting in Junos OS Release 20.4R1, we've reorganized the Monitor tab into the following menus for enhanced experience:

- Interfaces
- Logs
- Maps and Charts
- Statistics
- Reports

Additionally:

- Support for Adobe Flash Player will end on December 31, 2020. Therefore, J-Web will support only the Monitor tab submenus that don't require flash components.
- A new Traffic Map page is added under Monitor>Maps and Charts. Use this page to visualize inbound and outbound traffic between geographic regions.

[See [Monitor Interfaces](#) and [Monitor Traffic Map](#).]

- **Enhanced Source NAT feature (SRX Series)**—Starting in Junos OS Release 20.4R1, we've refreshed the Source NAT page to improve user experience. You can configure:

- Source NAT inline on the NAT Policies page (Create>Source NAT).
- Source NAT pool and destination NAT pool on the NAT Pools page (Create>Source NAT Pool or Create>Destination NAT Pool).

[See [About the NAT Policies Page](#).]

Layer 2 Features

- **LLDP on routed and reth interfaces (SRX1500)**—Starting in Junos OS Release 20.4R1, Link Layer Discovery Protocol (LLDP) is supported on routed interfaces and redundant Ethernet (reth) interfaces. LLDP is a link-layer protocol used by network devices to advertise capabilities, identity, and other information to a LAN.

[See [LLDP Overview](#).]

Logical Systems and Tenant Systems

- **Support for MAP-E confidentiality CLI statement (NFX150, NFX250, NFX350, and SRX1500)**—Starting in Junos OS Release 20.4R1, we've introduced a global MAP-E **confidentiality** CLI statement to hide MAP-E rule parameters in CLI show commands and logs. To enable this configuration, include the **confidentiality** statement at the **[edit security softwires map-e]** hierarchy level. You need administrator privileges to enable or disable this configuration. This feature is supported for all domains of MAP-E.

[See [confidentiality](#) and [show security softwires map-e confidentiality status](#).]

Multinode High Availability

- **Multinode high availability solution (SRX5400, SRX5600, and SRX5800 with SPC3 card)**—In Junos OS Release 20.4R1, we introduce the multinode high availability solution, where two SRX Series devices can be either co-located or spread across geographies. This solution also provides redundancy across service levels.

When you configure multinode high availability in the SRX Series device (in the active-backup mode), one node acts as the active device and the other acts as the backup device, ensuring failover of services to the backup device in the event of software, hardware, or path monitoring failure. Traffic is then routed toward the active node by upstream and downstream routers.

The active and backup nodes are interconnected with an IP-based link called interchassis link (ICL). The active and backup nodes synchronize data plane based session states. They also synchronize control plane states for certain services.

[Table 6 on page 332](#) lists the multinode high availability features that we support.

Table 6: Feature Support for Multinode High Availability

Feature	Description
Active and Backup modes	We've introduced active and backup states on SRX Series devices that operate in the multinode HA mode. [See Multinode High Availability.]
IPsec VPN support	IPsec feature is supported on multinode HA. IPsec runs actively on one node (or active node). It can fail over to the secondary node (or backup node). IKE negotiations occur from the active node and the states are synchronized with the backup node. After synchronization, the backup node takes over the primary role and continues without bringing down the tunnels after switchover. You can run the show command(s) on both active and backup nodes to display the status of IKE and IPsec security associations. You can delete the IKE and IPsec security associations only on the active node. [See Support for VPN on HA Nodes in Multinode High Availability Solution.] When a packet enters a flow session which is on IPsec VPN and on the backup node, the packet is dropped. When the packet enters a clear text session, irrespective of the control plane transition state the clear text session moves to the active state. When you configure the set chassis high-availability services-redundancy-group 1 process-packet-on-backup command, the IPsec VPN related packet is not dropped in the backup node.

- **Configuration retrieval using the configuration revision identifier (EX3400, EX4300, MX204, MX240, MX480, MX960, MX2020, PTX3000, PTX10008, QFX5100, QFX10002-60C, SRX5800, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, you can use the configuration revision identifier feature to view the configuration for a specific revision. This configuration database revision can be viewed with the CLI command **show system configuration revision**.

[See [show system configuration revision](#).]

- **Junos XML protocol operations support loading and comparing configurations using the configuration revision identifier (EX3400, EX4300, MX204, MX240, MX480, MX960, MX2020, PTX3000, PTX10008, QFX5100, QFX10002-60C, SRX5800, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, the Junos XML management protocol operations support loading and comparing configurations by referencing the configuration revision identifier of a committed configuration. You can execute the **<load-configuration>** operation with the **configuration-revision** attribute to load the configuration with the given revision identifier into the candidate configuration. Additionally, you can compare the candidate or active configuration to a previously committed configuration by referencing the configuration revision identifier for the comparison configuration. The **<get-configuration>** operation supports the **compare="configuration-revision"** and **configuration-revision** attributes to perform the comparison.

[See [<get-configuration>](#) and [<load-configuration>](#).]

Securing GTP and SCTP Traffic

- **Support for listening mode, syslog identity information, and rate-limit configuration enhancement (SRX1500, SRX4100, SRX4200, SRX4600, SRX5400, SRX5600, SRX5800, and vSRX)**—Starting in Junos OS Release 20.4R1, we support:
 - **GTP listening mode**—The SRX Series firewall enables GPRS tunneling protocol (GTP) listening mode on the S5, S8, or S11 interface. In listening mode, the firewall can perform sanity check and stateful inspection on GTP packets per GTP configuration. To configure, include the **listening-mode** statement at the **[edit security gtp profile profile-name]** hierarchy level.
 - **Syslog identity information**—This feature adds information such as the user equipment (UE) IP address, International Mobile Station Identity (IMSI), and access point name (APN) to existing syslog messages during GTP management. This helps in mapping identities between a user ID and an IP address.
 - **rate-limit configuration enhancement**—We've added an alarm threshold option and a drop threshold option to the existing **rate-limit** configuration. This enhancement reduces the duplicate drop logs for the destination GPRS support node (GSN).

[See [listening-mode](#), and [rate-limit \(Security GTP\)](#).]

Security

- **Unified policies support for zone-context and global-level policies (SRX Series and vSRX)**—Starting in Junos OS Release 20.4R1, unified policies support both zone-context and global-level policies at the same time. In previous releases, unified policies supported only zone-context policies.

If there is any unified policy match, either in a zone-context or in a global context, then it is added to potential match list.

If there is no match in the zone-context, policy search occurs in the global context.

[See [Global Security Policies](#).]

- **Tunnel inspection for VXLAN passthrough (SRX4100, SRX4200, SRX4600, and vSRX)**—Starting in Junos OS Release 20.4R1, you can allow L4 or L7 services to perform an inspection against the inner ethernet frame. VXLAN is one of the supported protocols and is designed to accommodate most overlay or underlay protocols which require inner inspection. VXLAN traffic is only inspected if there is a security policy configured to perform the inspection.

[See [tunnel-inspection](#) and [show security flow session](#).]

- **Security policy support for security inspection on VXLAN tunnels (SRX4100, SRX4200, SRX4600, and vSRX)**—Starting in Junos OS Release 20.4R1, you can perform security inspection on VXLAN tunnels by performing policy control twice. Configure an outer policy for the outer header and an inner policy for the inner header.

Configure a tunnel inspection profile to connect the outer policy and inner policy. The tunnel inspection profile is attached to the outer policy and it points to a group of inner policies (policy set). When the packet matches the outer policy, the SRX device decapsulates the packet to get the inner header. Using inner packet content along with the attached tunnel inspection profile of outer policy, the second policy lookup gets the desired inner policy applies the security services to inner packet.

[See [tunnel-inspection](#).]

- **Support for unidirectional session refreshing (SRX Series)**—Starting in Junos OS Release 20.4R1, SRX Series device support unidirectional session refreshing. You can do either of the following:
 - Refresh a session by any packet from any direction. This is an existing session-refreshing mechanism and the default behavior.
 - Refresh a session by only the packets in the initial direction (unidirectional refreshing).

By default, unidirectional session refreshing is disabled. To enable the feature, include the **unidirectional-session-refreshing** statement at the **[edit security zones security-zone zone-name]** hierarchy level.

[See [unidirectional-session-refreshing](#).]

Unified Threat Management (UTM)

- **Custom response page in UTM Web filtering profile (SRX Series and vSRX)**—Starting in Junos OS Release 20.4R1, you can configure a custom response page for a URL that is configured with the block or quarantine actions in the UTM Web filtering profile. The custom response page can include predefined page variables, your corporate branding, acceptable use policies, and links to your internal resources. You can enable the **custom-page** statement at the **[edit security utm custom-objects custom-message name type]** hierarchy level and configure the customized HTML file at the **[edit security utm custom-objects custom-message name custom-page-file file-name]** hierarchy level.

[See [custom-page](#), [custom-page-file](#), and [custom-message \(Security Web Filtering\)](#).]

- **URL pattern wildcard enhancement (SRX Series and vSRX)**—Starting in Junos OS Release 20.4R1, the URL pattern supports new regular expressions and defines new pattern matching rules for the domain name and URL path. This enhancement allows you to configure better and user-friendly URL pattern matching in the Web filtering function. You can use the asterisk (*), caret (^), and question mark (?) wildcards for a domain name match. The URL match supports the prefix match and keyword match. You can use the asterisk (*) wildcard for the URL match.

[See [url-pattern](#).]

- **Dynamic-address group rescan enhancement (SRX Series and vSRX)**—In the current dynamic-address implementation, when you add a host address to the dynamic-address group, the system does not terminate and rescan the existing sessions of the host. Starting in Junos OS Release 20.4R1, when you add a host address to the dynamic-address group, the system rescans the sessions including the existing sessions to ensure that the traffic matches the updated policy. The **session-scan** option is disabled by default. You can enable the **session-scan** option at the **[edit security dynamic-address address-name name session-scan]** or **[edit security dynamic-address session-scan]** hierarchy level.

[See [session-scan](#) and [hold-interval](#).]

VPNs

- **Support for load redistribution (SRX5400, SRX5600, and SRX5800)**—Starting in Junos OS Release 20.4R1, you can redistribute or migrate a tunnel that belongs to a site-to-site or Auto VPN gateway to a new processing unit. Load distribution of tunnels tears down the tunnels associated with a peer and anchors them to the user-specified processing unit the next time a tunnel for the same peer is established. Use the **request security re-distribution ipsec-vpn** command to specify the processing unit to which you want to distribute or migrate the tunnels.

[See [request security re-distribution ipsec-vpn](#), [show security re-distribution ipsec-vpn](#), and [show security ipsec tunnel-distribution](#).]

- **PowerMode IPsec support (SRX4600)**—PowerMode IPsec (PMI) is supported on SRX5400, SRX5600, SRX5800, and vSRX. Starting in Junos OS Release 20.4R1, you can configure PMI on the SRX4600 also. PMI provides IPsec performance improvements using Vector Packet Processing (VPP) and Intel Advanced Encryption Standard New Instructions (AES-NI). SRX4600 also supports the fat core feature that increases single tunnel performance. If one of the tunnels loaded with high traffic and other tunnels have less traffic, the resources share within the fat group and result in an even CPU utilization of the resources.

[See [Improving IPsec Performance with PowerMode IPsec](#) and [power-mode-ipsec](#).]

- **PowerMode IPsec performance improvement (SRX5400, SRX5600, and SRX5800 with SPC3 cards)**—Starting in Junos OS Release 20.4R1, we've improved PowerMode IPsec (PMI) performance by distributing load between the AES-NI instructions on the SPUs and the on-board Intel QuickAssist Technology (QAT), Hardware-based cryptographic acceleration for symmetric fat tunnels in SPC3 cards provides higher performance. Load balancing helps to provide higher throughput for IPsec. PMI uses AES-NI and QAT for encryption and FPGA for decryption of cryptographic operation. To enable QAT with AES-NI, include **power-mode-ipsec-qat** at the **[edit security flow]** hierarchy level.

[See [power-mode-ipsec-qat](#) and [inline-fpga-crypto](#).]

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What's Changed

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Learn about what changed in the Junos OS main and maintenance releases for SRX Series.

What's Changed in Release 20.4R3

Network Management and Monitoring

- **The configuration accepts only defined identity values for nodes of type identityref in YANG data models (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—If you configure a statement that has type identityref in the corresponding YANG data model, the device accepts only defined identity values (as defined by an identity statement) as valid input. In earlier releases, the device also accepts values that are not defined identity values.

What's Changed in Release 20.4R2

Flow-Based and Packet-Based Processing

- **Default MKA transmit interval (SRX380)**—On SRX380 devices, the default MACsec Key Agreement (MKA) transmit interval is 2000 milliseconds. If you deploy an SRX380 device with other security peer device with MACsec secure link, you must change the MKA transmit interval on the peer device to 2000 milliseconds to match the new default MKA transmit interval of the SRX380 device.

[See [transmit-interval \(MACsec\)](#).]

- **New output field added in show pfe statistics traffic command (SRX380)**—Starting in Junos OS Release, you'll see Unicast EAPOL in the output of the show pfe statistics traffic command.

[See [show pfe statistics traffic](#).]

- **Self-generated IKE packets chooses outgoing interface matching source IP Address (SRX Series)** — A self-generated Internet Key Exchange (IKE) packet always select the ECMP outgoing interface that matches source IP address. Note that filter-based forwarding for self-generated traffic with rerouting is not supported.

Junos XML API and Scripting

- **Refreshing scripts from an HTTPS server requires a certificate (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—When you refresh a local commit, event, op, SNMP, or Juniper Extension Toolkit (JET) script from an HTTPS server, you must specify the certificate (Root CA or self-signed) that the device uses to validate the server's certificate, thus ensuring that the server is authentic. In earlier releases, when you refresh scripts from an HTTPS server, the device does not perform certificate validation.

When you refresh a script using the **request system scripts refresh-from** operational mode command, include the **cert-file** option and specify the certificate path. Before you refresh a script using the **set refresh** or **set refresh-from** configuration mode command, first configure the **cert-file** statement under the hierarchy level where you configure the script. The certificate must be in Privacy-Enhanced Mail (PEM) format.

[See [request system scripts refresh-from](#) and [cert-file](#).]

- The `jcs:invoke()` function supports suppression of root login and logout events in system log files for SLAX commit scripts (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)—The `jcs:invoke()` extension function supports the **no-login-logout** parameter in SLAX commit scripts. If you include the parameter, the function does not generate and log `UI_LOGIN_EVENT` and `UI_LOGOUT_EVENT` messages when the script logs in as root to execute the specified remote procedure call (RPC). If you omit the parameter, the function behaves as in earlier releases in which the root `UI_LOGIN_EVENT` and `UI_LOGOUT_EVENT` messages are included in system log files.

[See [invoke\(\) Function \(SLAX and XSLT\)](#).]

- The `jcs:invoke()` function supports suppression of root login and logout events in system log files for SLAX event scripts (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)—The `jcs:invoke()` extension function supports the **no-login-logout** parameter in SLAX event scripts. If you include the parameter, the function does not generate and log `UI_LOGIN_EVENT` and `UI_LOGOUT_EVENT` messages when the script logs in as root to execute the specified remote procedure call (RPC). If you omit the parameter, the function behaves as in earlier releases in which the root `UI_LOGIN_EVENT` and `UI_LOGOUT_EVENT` messages are included in system log files.

[See [invoke\(\) Function \(SLAX and XSLT\)](#).]

VPNs

- **View the traffic selector type for an IPsec tunnel (SRX Series and MX Series)**—You can run the `show security ipsec security-associations detail` command to display the traffic selector type for a VPN. The `show security ipsec security-associations detail` command displays **proxy-id** or **traffic-selector** as a value for the **TS Type** output field based on your configuration.

[See [show security ipsec security-associations](#).]

What's Changed in Release 20.4R1

Class of Service (CoS)

- We've corrected the output of the "show class-of-service interface | display xml" command. Output of the following sort: `<container> <leaf-1> data </leaf-1> <leaf-2> data </leaf-2> <leaf-3> data </leaf-3> <leaf-1> data </leaf-1> <leaf-2> data </leaf-2> <leaf-3> data </leaf-3> </container>` will now appear correctly as: `<container> <leaf-1> data </leaf-1> <leaf-2> data </leaf-2> <leaf-3> data </leaf-3> </container> <container> <leaf-1> data </leaf-1> <leaf-2> data </leaf-2> <leaf-3> data </leaf-3> </container>`

Flow-Based and Packet-Based Processing

- On SRX Series devices in earlier releases, when the session table was full there was no alarm set to indicate this. Starting from this release, when the percent of flow session table utilization is 95% on FPC and PIC, an alarm message `Flow session table is almost full on FPC <number> PIC <number> is set`. Similarly, when the percent of DCP session table utilization is 95% on FPC and PIC, an alarm message `DCP session table is almost full on FPC <number> PIC <number> is set`.

[See [Understanding Session Cache](#).]

- **Default MKA transmit interval (SRX380)**—On SRX380 devices, the default MACsec Key Agreement (MKA) transmit interval is 2000 milliseconds. If you deploy an SRX380 device with other security peer device with MACsec secure link, you must change the MKA transmit interval on the peer device to 2000 milliseconds to match the new default MKA transmit interval of the SRX380 device.

[See [transmit-interval \(MACsec\)](#)~~transmit-interval (MACsec)~~.]

Intrusion Detection and Prevention (IDP)

- **Intelligent Offload State (SRX Series)**— We have introduced a new field in the **show security idp status** command to see the status of the IDP Intelligent offload.

[See [show security idp status](#).]

Interfaces and Chassis

- **g mode supported on radio 2.4GHz of Wi-Fi MPIM (SRX320, SRX340, SRX345, and SRX550M)**—Starting in Junos OS Release 20.4R1, radio 2 with frequency 2.4 GHz supports mode g on SRX Wi-Fi MPIM.

[See [Wi-Fi Mini Physical Interface Module \(MPIM\)](#)].

J-Web

- **Adobe Flash Player support (SRX Series)**—Adobe Flash Player support will end on December 31, 2020. Due to this, the Flash dependent J-Web monitor pages will not load correctly for Junos OS Release 20.3R1 and earlier releases.
- **Change in the J-Web browser tab title (SRX Series)**—The J-Web browser tab title displays the device model and hostname. These details are also displayed when you hover over the J-Web browser tab.

For example, when you access J-Web for an SRX320 device with the hostname srx320-xyz, the J-Web browser tab displays the title as *J-Web (srx320 - srx320-xyz)*.

If the hostname isn't configured, the J-Web browser tab title displays the host URL or IP address; for example, *J-Web (srx320 - <device IP address>)*.

Network Address Translation (NAT)

- **Port block allocation support (SRX300, SRX320, SRX340, SRX345, SRX380, SRX550HM, SRX1500, SRX4100, SRX4200, and SRX4600)**—You can configure the port block allocation size 1 through 64512. To save system memory, the recommended port block allocation size is 64. If you configure the port block allocation size lesser than 64, the system displays the warning message warning: To save system memory, the block size is recommended to be no less than 64. In the earlier releases, you can configure port block allocation size 1 through 64512 on SRX5400, SRX5600, and SRX5800 only.

[See [Configure Port Block Allocation Size](#).]

Network Management and Monitoring

- **Warning changed for configuration statements that correspond to deviate not-supported nodes in YANG data models (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—If you configure a statement corresponding to a YANG data model node that defines the **deviate not-supported** statement, the Junos OS configuration annotates that statement with the comment **Warning: statement ignored: unsupported platform**. In earlier releases, the warning is **Warning: 'statement' is deprecated**.

Platform and Infrastructure

- **Support for fully qualified domain name (FQDN) for log server (SRX Series)**—Starting in Junos OS Release, you can configure TTL value for a DNS server cache with hostname or IP address.

[See [Configuring the TTL Value for DNS Server Caching](#).]

Securing GTP and SCTP Traffic

- **Deprecated CLI configuration statements and operational commands for GTP and SCTP (SRX1500, SRX4100, SRX4200, SRX4600, SRX5400, SRX5600, SRX5800, and vSRX)**—Starting in Junos OS Release 20.4R1, we've removed the term `gprs` from all the configuration statements and operational commands for GTP and SCTP. As a part of this change:
 - We've deprecated the `[set security gprs]` hierarchy-level and all the configuration options under this hierarchy-level.
 - All the configuration statements previously available under the `[set security gprs gtp]` hierarchy-level are now available under the `[set security gtp]` hierarchy-level.
 - All the configuration statements previously available under the `[set security gprs sctp]` hierarchy-level are now available under the `[set security sctp]` hierarchy-level.
 - Replace the `show security gprs gtp configuration` command by `show security gtp profile` command.
 - Replace the `identifier` option by `profile-name` in the `show security gtp profile` command.
 - For default applications like `junos-gprs-gtp` and `junos-gprs-sctp`, you need not remove the term `gprs`.

[See [Configuration Statements and Operational Commands](#).]

User Interface and Configuration

- **Verbose format option for exporting JSON configuration data (ACX Series, EX Series, MX Series, PTX Series, QFX Series, and SRX Series)**—The Junos OS CLI exposes the **verbose** statement at the **edit system export-format json** hierarchy level. The default format for exporting configuration data in JSON changed from **verbose** format to **ietf** format starting in Junos OS Release 16.1R1. You can explicitly specify the default export format for JSON configuration data by configuring the appropriate statement at the **edit system export-format json** hierarchy level. Although the **verbose** statement is exposed in the Junos OS CLI as of the current release, you can configure this statement starting in Junos OS Release 16.1R1.

[See [export-format](#).]

VPNs

- **Dynamic CA profiles loaded only on active nodes (SRX5400, SRX5600, and SRX5800)**—When you enable the multinode high availability feature, the dynamic CA profiles are loaded only on the node during the IKE negotiation. If a failover occurs, the new active node undergoes a new IKE negotiation and loads the dynamic CA certificates as part of that negotiation. When PKID restarts, dynamic CA certificates are deleted only from the node where PKID was restarted.
- **Public key infrastructure warning message (SRX5400, SRX5600, SRX5800)**—When you generate a public key infrastructure (PKI) public/private key pair for a local digital certificate, with key pair size of 4096 bits and DSA encryption, a warning message is displayed. **root@hostname> request security pki generate-key-pair certificate-id test type dsa size 4096** Generating a key-pair with a large modulus is very time-consuming. Progress is reported to the trace log, and a log message is generated upon completion. Because generating a local digital certificate with large key pair size is time consuming, we recommend you to check the trace log for the progress of generating a key pair.

[See [request security pki generate-key-pair \(Security\)](#).request security pki generate-key-pair (Security).]

- **Delay in VPN tunnel establishment negotiation (SRX5400, SRX5600, and SRX5800)** —In an IPsec VPN configuration, if you configure the **establish-tunnels immediately** option under the **[edit security ipsec vpn <vpn-name>]** hierarchy, it may take up to five seconds to start the negotiation for VPN tunnel establishment. In the earlier Junos OS releases, the negotiation for VPN tunnel establishment starts immediately.

[See [vpn \(Security\)](#).]

- **The junos-ike package installed by default (SRX5000 line of devices)**— For the SRX5000 line of devices with RE3 installed, the junos-ike package is installed by default. As a result, the iked and ikemd processes run on the Routing Engine by default instead of the IPsec key management daemon (kmd). In earlier Junos OS releases, the junos-ike package is an optional package for SRX5000 devices with RE3, and IPsec Key Management Daemon (KMD) runs by default.

[See [Enabling IPsec VPN Feature Set on SRX5K-SPC3 Services Processing Card.](#)]

- **IKE index displayed in show security ipsec security-associations detail output (SRX5400,SRX5600, and SRX5800)**—When you execute the **show security ipsec security-associations detail** command, a new output field, **IKE SA Index**, corresponding to every IPsec Security Association (SA) within a tunnel is displayed under each IPsec SA information.

[See [show security ipsec security-associations.](#)]

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Known Limitations

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Learn about known limitations in this release for SRX Series devices.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Flow-Based and Packet-Based Processing

- For accelerated flows such as Express Path, the packet or byte counters in the session close log and show session output take into account only the values that accumulated while traversing the NP. [PR1546430](#)

General Routing

- IPC cannot afford pushing 50,000 NAT rules with complicated matching criteria from Routing Engine to Packet Forwarding Engine. It is hard to simply provide how much rules with complicated matching criteria can be supported, because memory size consumption depends on different rule content. [PR1481845](#)
- Junos OS DHCP server pool change br-lan will wait for the lease timeout before getting a new IP in the new DHCP pool. [PR1516241](#)
- In SRX380, MACsec show security macsec statistics command, when encryption-offset is enabled, the encrypted bytes and encrypted packets will include both encrypted and protected bytes. [PR1534840](#)
- On SRX300, SRX320, SRX340, SRX345, SRX550M, and SRX1500 devices, if CoS code points with values other than 000 are configured, packet loss might be seen for certain traffic patterns due to the firewall not having enough buffer. [PR1544709](#)
- Due to enhancements in AppID starting Junos OS Release 21.1R1, database files are not compatible with earlier releases. Hence, this issue is expected to be seen during downgrade from Junos OS Release 21.1R1 to earlier releases. [PR1554490](#)
- On all Junos platforms, the l2ald crash could be observed on changing the routing-instance from VPLS to non-L2 routing-instance, with same routing-instance name is being used for both VPLS and non-L2 routing-instance. [PR1586516](#)

J-Web

- The Threat Map page available under the Monitor menu does not support a redirection to the Logs page from top destination countries, top source countries, and specific country threat count. [PR1542392](#)
- HTTP client application wget version 1.12 or lower version uses HTTP/1.0 protocol version for their requests will be facing request timeout issue. [PR1552063](#)
- In the NAT Policies page: White space when reordering multiple rules: When you try to reorder more than one rule at a time using drag and drop, a white space is seen in the rule dropped area. It will disappear when the grid is scrolled up or down. Hit count value zero when searching for a rule: The hit count field in the search results will always show zero when searching for rules. But in the normal grid view when not performing the search operation, the Hit count is updated correctly. Policy grid is greyed out while adding a new rule with an existing context or ruleset expanded and scrolled down almost more than 20 rules. The page loads normally if you refresh the menu. Select all checkbox and delete: When Select all

checkbox is used repeatedly or when deleting multiple rules using Select all checkbox, you may receive a browser warning: A Web page is slowing down your browser. [PR1558757](#)

- The Firefox browser displays an unsaved changes error message in the J-Web Basic Settings page if the Autofill logins and passwords option is selected under the Browser Privacy and security settings. [PR1560549](#)

Routing Policy and Firewall Filters

- On all SRX platforms with TCP proxy is enabled, firewall TCP proxy does not send the window scale TCP option if the dynamic application is configured in the security policies, This might cause the throughput to reduce. [PR1492738](#)

VPNs

- In some scenario(e.g configuring firewall filter) sometimes srx5K might show obsolete IPsec SA and NHTB entry even when the peer tear down the tunnel. [PR1432925](#)
- On the SRX5000 line of devices with an SPC3 card, sometimes IKE SA is not seen on the device when the st0 binding on the VPN configuration object is changed from one interface to another (for example, st0.x to st0.y). [PR1441411](#)
- In SPC2 and SPC3 mixed-mode HA deployments, tunnel per second (TPS) is getting affected while dead peer detection (DCD) is being served on existing tunnels. This limitation is due to a large chunk of CPU being occupied by infrastructure (gencfg) used by IKED to synchronize its DPD state to the backup nodes. [PR1473482](#)

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Open Issues

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Learn about open issues in this release for SRX Series devices.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Flow-Based and Packet-Based Processing

- Use an antireplay window size of 512 for IPv4 or IPv6 in fat-tunnel. The ESP sequence check might otherwise report out-of-order packets if the fat-tunnel parallel encryption is within 384 packets (12 cores * 32 packets in one batch). Hence, there are no out-of-order packets with 512 antireplay window size. [PR1470637](#)

General Routing

- The LLDP protocol can be configured on SRX4000 and SRX5000 lines of devices, although it is not actually supported on those platforms. [PR1540797](#)
- The PKI CMPv2 (RFC 4210) client certificate enrolment does not properly work on SRX Series devices when using root-CA. [PR1549954](#)
- PKID core might occur during cert signature validation . This core is not very frequent and occurs due to memory corruption . [PR1573892](#)
- With ssl-proxy configured along with web-proxy, the client session might not closed on the device even though proxy session ends gracefully. [PR1580526](#)

- SRX platforms using SkyATP with security-intelligence configured, might disconnect from the cloud after several days with the error "Connection status: Request client certificate failed". The issue can be recovered by the CLI command "restart pki-service". [PR1585362](#)
- Web-proxy: Getting UNKNOWN instead of HTTP-PROXY for application and UNKNOWN instead of GOOGLE-GEN in RT-FLOW close messages These messages can be seen in the RT-flow close log and these are due to JDPI not engaged for the session. This may affect the app identification for the web-proxy session traffic. [PR1588139](#)
- On SRX345, icmp checksum error and packet drops are observed while doing rapid ping on vdsl interface with MTU 1514. [PR1591230](#)
- In 20.3 R3, 20.4R3 and 21.1R2, Sometimes on reboot schedule-report are not getting generated. [PR1594377](#)
- For release 20.3R3 & 20.4R3 & 21.1R2 & 21.2R1, phone-home ZTP is failing on branch srx as phone home client is unable to connect to Phone Home Server/Redirect Server. [PR1598462](#)
- With Network Address Translation (NAT) and Endpoint-independent Mapping (EIM) enabled, traffic unsupported by EIM might not be translated due to packets injected back to NAT gateway. When this issue happens, EIM unsupported traffic could be dropped. Also, the issue could cause looping at NAT gateway. In the end, looping occurred at NAT gateway affects device performance. [PR1601890](#)

Infrastructure

- 21.2 latest is in stable12 bsd release. there is a limitation where image validation is not supported across different bsd versions. image validation will fail from stable11 to stable12. for upgrade between different bsd releases, we have to use "no-validate" option. however, currently ISSU doesn't support no-validate option. PR 1569992 tracks the ISSU infra change to include no-validate option. [PR1568757](#)

Intrusion Detection and Prevention (IDP)

- Starting from Junos OS Release 21.1, either greater-than or less-than are allowed for age-of-attack filter of dynamic attack group configuration. The age-of-attack field in signatures will be changed to CVE dates from activation dates. Anomalies and generic attacks will be part of all groups created. [PR1397599](#)

Platform and Infrastructure

- If authentication (e.g., tacplus-server, radius-server) is configured on a device, it may fail to open files in a rare case, which may cause the process mgd to crash. [PR1600615](#)

Routing Policy and Firewall Filters

- Policy Report out is not coming as expected due to this issue. From functionality point of view there is no impact on services running on data and policy lookup. So data-path services are not impacted. [PR1582020](#)
- The issue is related to output of one of the CLI command where it display some additional then expected data. However it will not cause any issue with data path functionality on PFE. It's more like display issue. [PR1582344](#)

VPNs

- When multiple traffic selectors are configured on a particular VPN, the iked process checks for a maximum of 1 DPD probe that is sent to the peer for the configured DPD interval. The DPD probe is sent to the peer if traffic flows over even one of the tunnels for the given VPN object. [PR1366585](#)
- In the output of the show security ipsec inactive-tunnels command, Tunnel Down Reason is not displayed as this functionality is not supported in Junos OS Release 18.2R2 and later. [PR1383329](#)
- On SRX5400, SRX5600, and SRX5800 devices with an SPC3 card, a new behavior has been introduced that differs from the behavior on the older SPC2 card. The SRX Series device with AutoVPN configuration can now accept multiple IPsec tunnels from a peer device (with the same source IP address and port number) using different IKE IDs. [PR1407356](#)

- On SRX5400, SRX5600, and SRX5800 devices, during in-service software upgrade (ISSU), the IPsec tunnels flap, causing a disruption of traffic. The IPsec tunnels recover automatically after the ISSU process is completed. [PR1416334](#)
- In some scenario(e.g configuring firewall filter) sometimes srx5K might show obsolete IPsec SA and NHTB entry even when the peer tear down the tunnel. [PR1432925](#)
- On the SRX5000 line of devices with SPC3 and SPC2 mixed mode, with a very large number of IKE peers (60,000) with dead peer detection (DPD) enabled, IPsec tunnels might flap in some cases when IKE and IPsec rekeys are happening at the same time. [PR1473523](#)
- Dev to Provide content [PR1519830](#)
- IPsec protocol change for an existing VPN tunnel will not work. User must not configure an IPsec policy with proposals using both ESP and AH protocols. [A] IPsec protocol change for an existing VPN tunnel will not work. For example in the following VPN configuration: set security ipsec proposal IPSEC_PROP_AH protocol ah set security ipsec proposal IPSEC_PROP_AH authentication-algorithm hmac-sha1-96 set security ipsec proposal IPSEC_PROP_ESP protocol esp <+== IPsec protocol cannot change set security ipsec proposal IPSEC_PROP_ESP authentication-algorithm hmac-sha1-96 set security ipsec proposal IPSEC_PROP_ESP encryption-algorithm 3des-cbc set security ipsec policy IPSEC_POL proposals IPSEC_PROP_ESP <+== cannot change 'proposals' to use a proposal using a different IPsec protocol set security ipsec vpn IPSEC_VPN bind-interface st0.1 set security ipsec vpn IPSEC_VPN ike gateway IKE_GW set security ipsec vpn IPSEC_VPN ike ipsec-policy IPSEC_POL <+== cannot change 'ipsec-policy' to a policy using a different IPsec protocol If there is a need to change the IPsec protocol (esp or ah), the 'security ipsec vpn IPSEC_VPN' needs to be deactivated/committed first. [B] IPsec policy with both ESP and AH IPsec proposals will no longer work. For example: set security ipsec proposal IPSEC_PROP_AH protocol ah set security ipsec proposal IPSEC_PROP_AH authentication-algorithm hmac-sha1-96 set security ipsec proposal IPSEC_PROP_ESP protocol esp set security ipsec proposal IPSEC_PROP_ESP authentication-algorithm hmac-sha1-96 set security ipsec proposal IPSEC_PROP_ESP encryption-algorithm 3des-cbc set security ipsec policy IPSEC_POL proposals IPSEC_PROP_ESP set security ipsec policy IPSEC_POL proposals IPSEC_PROP_AH <+== ipsec policy IPSEC_POL has both a proposal using esp and a proposal using esp This configuration will continue be allowed to commit, but the IPsec traffic will not work. User must not configure an IPsec policy with proposals using both ESP and AH protocols. [PR1552701](#)
- On SRX5000 Series devices with SRX5K-SPC3 used or vSRXs with junos-ike package loaded, when IPsec VPN is used and SNMP get is performed on jnxIpSecTunnelMonTable, a memory corruption might be triggered, which results in the ikemd process crash. [PR1582036](#)

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Resolved Issues

Learn which issues were resolved in the Junos OS main and maintenance releases for SRX Series devices.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Resolved Issues: 20.4R3

Authentication and Access Control

- Unified-access-control(UAC) authentication might not work post system reboot [PR1585158](#)

Chassis Clustering

- Security policies might not be synced to all PFEs (Packet Forwarding Engine) post upgrade [PR1591559](#)

Class of Service (CoS)

- L2 CoS rewrite (802.1p) on egress interface might not work as expected [PR1570143](#)

EVPN

- The mustd.core process generates core file during upgrading or while committing a configuration [PR1577548](#)

Flow-Based and Packet-Based Processing

- The srxpfe process might crash during route churn [PR1572240](#)
- The flowd/srxpfe process might crash when clearing the TCP-Proxy session [PR1573842](#)
- Session using custom application might timeout when RG1 failover occurs [PR1580444](#)

General Routing

- SSL-FP Logging for non SNI session [PR1442391](#)
- The flowd might coredump frequently on SRX340 [PR1463689](#)
- The spcd process might crash during early initialization [PR1535536](#)
- The kmd process might crash when the interface flaps [PR1544800](#)
- SRX1500 reports fans running at over speed. [PR1546132](#)

- Certain SRX4100, SRX4200 and JRR200 devices SSD may encounter "buffer I/O error" leading to drive failure [PR1554716](#)
- The PIC in SRX5K-SPC3/MX-SPC3 card might get stuck in offline status after flowd crash occurs on it [PR1560305](#)
- SPC3 is not supported on MX in 21.1R1 and 20.4R2 for deployment. [PR1561188](#)
- Fabric probe packets might be processed incorrectly when power-mode-ipsec (PMI) is enabled [PR1564117](#)
- Wi-Fi mPIM on SRX Series devices is reaching out to NTP and DNS servers. [PR1569680](#)
- Packets with the MAC address of eth0 and macvlan0@eth0 interface might be sent out to the management interface on VMHOST platform with NG-RE [PR1571753](#)
- MACsec not using network-control queue [PR1571977](#)
- The srxpfe process might be crashed if BFD session is deleted from the PFE [PR1578946](#)
- Traffic is dropped to/through VRRP virtual IP on SRX380 [PR1581554](#)
- The ipfd process might crash with a coredump when SecProfiling thread feeds are fetched from Policy Enforcer(PE) [PR1582454](#)
- The srxpfe process might crash on SRX1500 [PR1582989](#)
- Packet drop or srxpfe coredump might be observed due to Glacis FPGA limitation [PR1583127](#)
- The application-identification process may crash with a core if multiple commands are run simultaneously [PR1583606](#)
- Secure Web proxy continue sending DNS query for unresolved DNS entry even after the entry was removed [PR1585542](#)
- On SRX-Series devices, significant performance improvements for JDPI's micro-application identification were included in this release [PR1585683](#)
- The 1G interfaces might not come up after device reboot [PR1585698](#)
- The l2ald process might crash on issuing ethernet-switching commands [PR1586426](#)
- The l2ald process might crash on changing the routing-instance [PR1586516](#)
- On SRX-Series devices, the protocol-version command which controls TLS-versions (1.1, 1.2, 1.3, etc) within SSL-Proxy has been unhidden [PR1587149](#)
- Garbage characters might be received in quarantine notification [PR1587962](#)
- IP packets might be dropped on SRX platforms [PR1588627](#)
- The jsqsyncd process files generation might cause device to panic crash after upgrade [PR1589108](#)
- Pass-through traffic might fail post reboot when Secure Web Proxy is configured [PR1589957](#)
- Traffic loss might be observed for interface configured in subnet 137.63.0.0/16 [PR1590040](#)
- The REST API does not work for SRX380 [PR1590810](#)

- 20.1R3 & 20.3R3: SecIntel: The issue (empty feed-name) starts with the hit returned from cache which points to the node with the parameter of feed-ID (2) inconsistent with the feeds-update (when it's 1). As a result the incorrect feed-ID points to the empty entry in the array of the feed-names. [PR1591236](#)
- Jweb Deny log nested-application="UNKNOWN" instead of specific application [PR1593560](#)
- Node1 fpc0(SPM) goes down after ISSU and RG0 failover [PR1595462](#)
- Jflow V9 application-id record: Network based application recognition value for IPv4 application-id are not as expected [PR1595787](#)
- Delay might be observed between Services Processing Card(SPC) failing and failover to other node [PR1596118](#)
- The flowd might core dump if application-services security policy is configured [PR1597111](#)
- The SRXPFE process might crash and generate a core file post "targeted-broadcast forward-only" interface-config commit [PR1597863](#)
- The flowd core may be seen if the AppQOS module receiving two packets of a session [PR1597875](#)
- The flowd process might crash in AppQoS scenarios [PR1599191](#)
- httpd-gk core might be observed when ipsec vpn is configured [PR1599398](#)
- On SRX-Series devices, a core file may be generated when using DNS-Inspection features such as Secintel or DGA/Tunnel Detection [PR1604773](#)

Interfaces and Chassis

- 20.4R2: Tenant: Facing configuration check-out failed with error message: identical local address found on rt_inst [default], intfs [PR1581877](#)

Intrusion Detection and Prevention (IDP)

- The IDP policy process might become unresponsive and fail to compile the IDP policy after an IDP automatic update. [PR1577684](#)
- IDP policy compilation failure for over 1000 custom signatures [PR1589399](#)
- IDP signature DB update fails [PR1594283](#)
- Custom attack IDP policies might fail to compile [PR1598867](#)
- IDP policy compilation is not happening when a commit check is issued prior to a commit [PR1599954](#)
- The srxpfe might crash while the IDP security package contains a new detector [PR1601380](#)

J-Web

- Junos OS: J-Web allows a locally authenticated attacker to escalate their privileges to root. (CVE-2021-0278) [PR1511853](#)
- To improve performance in Monitoring > Network > Interfaces page, Admin Status is removed, Services and Protocols data merged into one Host inbound traffic. [PR1574895](#)

- [J-Web] zone info disappears when functional zone is configured [PR1594366](#)
- [J-Web] a custom application name contains "any" is listed under Pre-defined Applications [PR1597221](#)
- J-Web may not display customer defined application services if one new policy is created [PR1599434](#)
- J-web application might crash with httpd core-dumps [PR1602228](#)
- Radius users might not be able to view/modify configuration via J-web [PR1603993](#)
- On all SRX platforms, some widgets in JWeb might not load properly for logical systems users [PR1604929](#)

MPLS

- Incorrect EXP bit change might be seen in certain conditions under MPLS scenario. [PR1555797](#)

Network Address Translation (NAT)

- Incorrect IPv6 UDP checksum inserted after translation of packet from IPv4 to IPv6 [PR1596952](#)

Platform and Infrastructure

- Junos OS: Upon receipt of specific sequences of genuine packets destined to the device the kernel will crash and restart (vmcore) (CVE-2021-0283, CVE-2021-0284) [PR1557881](#)
- On SRX5k, the alarm "Power Budget:Insufficient Power" may be raised incorrectly when the second SCB does not contain an RE [PR1568183](#)
- Due to a nstraced memory leak, e2e debug does not get disabled from the IOC card [PR1574759](#)

Routing Policy and Firewall Filters

- The dns-name can't be resolved if customer-defined routing instance is configured under name-server [PR1539980](#)

Routing Protocols

- Short multicast packets drop using PIM when multicast traffic received at a non-RPT/SPT interface [PR1579452](#)
- BGP session carrying VPNv4 prefix with IPv6 next-hop might be dropped [PR1580578](#)

Services Applications

- Extra data-plane CPU cycles for processing GTP traffic on SRX5000 Series device [PR1586367](#)

User Interface and Configuration

- After image upgrade device might fail to come up due to certain configurations [PR1585479](#)

VPNs

- The iked process might crash by operational commands on the SRX5000 line of devices with SRX5000-SPC3 card installed. [PR1566649](#)
- The SRXPFE process might crash and generate a core file when IPsec VPN is used [PR1574409](#)

- IKEv2 soft-lifetime timer might expire later than expected time [PR1574717](#)
- The iked process may crash when IKEv2 negotiation fails on MX/SRX devices [PR1577484](#)
- The from-self packet might be dropped when it forwards through an IPsec VPN tunnel [PR1577550](#)
- The flow sessions on SPC3 may get wrong Tunnel ID when "establish-tunnels on-traffic" configured under "ipsec vpn" [PR1581648](#)
- Memory leaks on the iked process on SRX5000 Series with SRX5K-SPC3 installed [PR1586324](#)
- The IPSec tunnel might not come up if configured with configuration payload in a certain scenario [PR1593408](#)
- The kmd process might crash when VPN peer initiates using source-port other than 500 [PR1596103](#)

Resolved Issues: 20.4R2

Chassis Clustering

- Disabled node on chassis cluster sent out ARP request packets. [PR1548173](#)
- SPU pause might be seen under GPRS tunneling protocol scenario. [PR1559802](#)

Flow-Based and Packet-Based Processing

- The THR capacity update on SRX Series devices. [PR1538058](#)
- Adjust the default route change timeout value. [PR1553621](#)
- The usp_max_tcplib_connection is not expected on SRX1500, SRX4100, and SRX4200 devices. [PR1563881](#)

Forwarding and Sampling

- The configuration archive transfer-on-commit fails when running Junos OS Release 18.2R3-S6.5. [PR1563641](#)

General Routing

- The JNH memory leak could be observed on MPCs or MICs. [PR1542882](#)
- The output of the command show services application-identification group detail incorrectly included Micro-Applications (Micro-Apps) in the output of every group. [PR1544727](#)
- On SRX4100 and SRX4200 devices, if PEM0 is removed, the output of jnxOperatingDescr.2 command might be incomplete. [PR1547053](#)
- Advanced anti-malware file/email statistics does not increment with latest PB version. [PR1547094](#)
- On vSRX2.0, vSRX3.0, SRX1500, SRX4100, SRX4200, SRX4600 running chassis cluster in Junos OS Release 18.3 or later, multiple messages of "LCC: ch_cluster_lcc_set_context:564: failed to lock chassis_vmx mutex 11" are generated in the chassisd log file. These messages may recur after every few seconds and they do not have any impact on system operation. [PR1547953](#)

- LCMD log "gw_cb_presence:136: PEM(slot = 0): error detecting presence (fruid = 15, drv_id = 30, status = -11)" is generated every second on the SRX4100 and SRX4200 devices. [PR1550249](#)
- The dcpfe process might stop and the non-channelization interfaces might not come up. [PR1552798](#)
- On SRX1500, SRX-SFP-1GE-T(Part#740-013111) for a copper cable might be corrupted after reboot. [PR1552820](#)
- The re-define volume is displayed in traffic map. [PR1553066](#)
- The speed mismatch error is seen while trying to commit reth0 with gigether-options. [PR1553888](#)
- An ipfd core file might be generated when using adaptive threat profiling. [PR1554556](#)
- On the SRX550M device, the dumpdisklabel command fails with message "ERROR: Unknown platform srx550m". [PR1557311](#)
- Application identity unknown packet capture utility does not function on SRX Series devices when enhanced-services mode is enabled. [PR1558812](#)
- The show security log report top idp group-by threat-severity order-by count top-number 5 where-attack command display changes. [PR1560027](#)
- The PKID process runs at 100 percent when the device is unable to connect to a particular URL. [PR1560374](#)
- The DNS commands may not be executed and any new configuration may not take effect on connecting the SRX Series device to Juniper Sky ATP. [PR1561169](#)
- The idpd process might stop when committing IDP configuration under logical systems and tenant systems during RGs failover. [PR1561298](#)
- The flowd process might pause and generates a core dump if JFlow version 9 is configured. [PR1567871](#)
- Traffic going through the VRRP interface might be dropped when VRRP enabled IRB interface goes down. [PR1572920](#)

Interfaces and Chassis

- Backup Routing Engine or backup node may get stuck in bad status with improper backup-router configuration. [PR1530935](#)

Intrusion Detection and Prevention (IDP)

- IDP now supports the ability to create dynamic-attack-groups based on attack-prefix wildcards. [PR1537195](#)
- IDP policy load might fail post image upgrade for Junos OS 15.1X49 releases. [PR1546542](#)
- The IDP policy process might become unresponsive and fail to compile the IDP policy after an IDP automatic update. [PR1577684](#)

J-Web

- J-Web GUI doesn't allow to save the rules with more than 2,500 cumulative shared objects. [PR1540047](#)

- After commit pending changes message is shown, the contents of other messages, landing page, or pop-ups will not be visible completely. [PR1554024](#)

Platform and Infrastructure

- Commit is not failing as expected after removing the reth interface. [PR1538273](#)
- The show chassis errors command is not supported on SRX5000 Series devices with RE3 and SCB3 installed anymore. [PR1560562](#)
- The show chassis ethernet-switch errors command unexpectedly shows error counters for port 14 on the SRX5800 device. [PR1563978](#)

Routing Policy and Firewall Filters

- Global policies working with multi-zones cause high CPU utilization. [PR1549366](#)
- Policy configured with route-active-on condition might incorrectly work for local routes. [PR1549592](#)
- The junos-defaults construct within a unified-policies application match criteria now restricts the ports and protocols of a flow on a per-dynamic-application basis. [PR1551984](#)
- Unified policies in global zone contexts do not work when from-zone or to-zone is defined. [PR1558009](#)
- On the SRX5000 line of devices, the secondary node might get stuck in performing ColdSync after a reboot, upgrade, or if ISSU is performed. [PR1558382](#)
- The traffic might be dropped due to inserting one global policy above others on SRX Series devices. [PR1558827](#)

Unified Threat Management (UTM)

- Stream buffer memory leak might happen when UTM is configured under unified policies. [PR1557278](#)
- UTM license expiry event lost may cause the device can't quit in advance service mode and the maximum-sessions is decreased by half. [PR1563874](#)

User Interface and Configuration

- The outbound-ssh routing-instance command is shown as unsupported. [PR1558808](#)

VPNs

- On the SRX5000 line of devices with SPC3 card, when the encryption algorithm is not configured in IPsec proposal, the output of show security ipsec security-associations command might display empty space instead of keyword null for encryption algorithm. [PR1507270](#)
- After IPsec tunnel using policy-based VPN is overwritten by another VPN client, traffic using this IPsec tunnel will be dropped. [PR1546537](#)
- Traffic that goes through policy-based IPsec tunnel might be dropped after RGO failover. [PR1550232](#)
- The iked process might stop with Multinode High Availability setup. [PR1559121](#)
- When there are multiple IPsec SAs, backup SA starts IPsec rekey. [PR1565132](#)

Resolved Issues: 20.4R1

Application Layer Gateways (ALGs)

- The SCCP ALG does not work on SRX Series devices running with Junos OS Release 17.3R1 and later. [PR1535356](#)

Flow-Based and Packet-Based Processing

- CLI autocomplete is now available for both SecIntel and advanced anti-malware products. [PR1487419](#)
- A condition within TCP proxy could result in downloads becoming permanently stuck or not completing. TCP proxy is used by multiple services, including Juniper ATP Cloud in block mode, ICAP, SSL proxy, antivirus, content filtering, and antispam. [PR1502977](#)
- In a dual CPE scenario, if the rule match is completed before application identification is done, AppQoS moves the session to the other node. [PR1514973](#)
- VRRP does not work on the redundant Ethernet interface with a VLAN ID greater than 1023. [PR1515046](#)
- PCAP file generated using packet capture was improper on the SRX5000 line of devices. [PR1515691](#)
- A logic issue was corrected in SSL proxy that could lead to an srxpfe or flowd core file under load. [PR1516903](#)
- The PPPoE session does not come up after return to zero on SRX Series devices. [PR1518709](#)
- FQDN-based security log stream does not dynamically update the IP address. [PR1520071](#)
- TAP mode behavior has been improved and the configuration has been greatly simplified. [PR1521066](#)
- The TCP packet might be dropped if syn-proxy protection is enabled. [PR1521325](#)
- Hide routing-instance under edit system name-server for SRX Series devices starts from Junos OS Release 20.4. [PR1521666](#)
- On SRX Series devices with chassis cluster, high CPU usage might be seen due to the llmd process. [PR1521794](#)
- Adaptive threat profiling would stop submitting new IP addresses to the feed after a limit of 10,000 has been reached. [PR1524284](#)
- On the SRX1500 device, the traffic rate shown in the CLI command is not accurate. [PR1527511](#)
- The MAC table is null in Layer 2 mode after one pass-through session is created successfully. [PR1528286](#)
- On SRX Series devices, a node of chassis cluster might stop passing traffic. The traffic forwarding can be restored by a manual failover to node 1. [PR1528898](#)
- When no LSYS or TSYS flow trace is configured and no root-override is configured, the latest behavior is to not log any flow trace for that LSYS or TSYS, instead of dumping all to root flow trace as before. [PR1530904](#)
- On SRX4100 and SRX4200 devices, four out of eight fans might not work. [PR1534706](#)

- The firewall filter SA and DA tags are not in the log messages as expected in port details. [PR1539338](#)
- Packet drop might be seen when a packet with destination port 0 is received on the SRX380 device. [PR1540414](#)
- The rst-invalidate-session command does not work if configured together with the no-sequence-check command. [PR1541954](#)
- The nsd process might crash when DNS-based allowlisting is configured under SSL proxy. [PR1542942](#)
- Application fragmented traffic might get dropped on SRX Series devices. [PR1543044](#)
- The Wi-Fi Mini-Physical Interface Module (Mini-PIM) does not support pure g mode with radio 2.4 GHz. [PR1543824](#)
- Need syslog to indicate signature download completion. [PR1545580](#)
- The flowd process might generate core files when the user changes the flow mode configuration to packet mode. [PR1546653](#)

Interfaces and Chassis

- Fabric interface might be monitored down after chassis cluster reboot. [PR1503075](#)
- On SRX320, SRX340, SRX345, SRX380, and SRX550M devices with an LTE Mini-Physical Interface Module (Mini-PIM), the LTE connection might drop and fail to automatically recover because of firmware issue. [PR1520879](#)
- When SRX Series devices receive proxy ARP requests on VRRP interfaces, SRX Series devices send ARP replies with the underlying interface MAC address. [PR1526851](#)

Intrusion Detection and Prevention (IDP)

- The flowd or srxpfe process might generate core files during the idpd process commit on SRX Series devices. [PR1521682](#)
- Adaptive threat profiling incorrectly classifies hosts when Server-to-Client (S2C) IDP signatures are used. [PR1533116](#)
- SOF support for partial packet plugins on traditional or unified policy. [PR1542497](#)

J-Web

- The parameters show another LSYS at J-Web in a multiple LSYS scenario. [PR1518675](#)
- Sometimes, when you edit the local gateway in the remote access VPN workflow under VPN>IPsec VPN, J-web might not display one or more drop-down values. [PR1521788](#)
- In the SRX5000 line of devices, J-Web can take up to 60 seconds to 90 seconds to load 60000 security policies. [PR1521841](#)

Layer 2 Ethernet Services

- DHCP might not work after performing request system zeroize or load factory-default on SRX Series devices. [PR1521704](#)

Network Address Translation (NAT)

- NAT PBA size 1 on SRX Series devices. [PR1525822](#)

Platform and Infrastructure

- Syslog reporting "PFE_FLOWD_SELFPING_PACKET_LOSS: Traffic impact: Selfping packets loss/err: 300 within 600 second" error messages in node 0 and node 1 control panel. [PR1522130](#)

Routing Policy and Firewall Filters

- Junos OS upgrade may encounter failure in certain conditions when enabling ATP. [PR1519222](#)
- Traffic might be dropped when policies are changed in SRX Series devices. [PR1527570](#)
- The show security dynamic-address feed-name command could not list secprofiling feed. [PR1537714](#)
- The flowd or srxpfe process might crash when an SRX Series or NFX Series device running Junos OS Release 18.2R1 or later supports the unified policy feature. [PR1544554](#)
- Traffic might be dropped unexpectedly when the URL category match condition is used on a security policy. [PR1546120](#)
- NSD process stops when the secprofiling feed name is 64 bytes. [PR1549676](#)

Routing Protocols

- The rpd process might report 100 percent CPU usage with the BGP route damping enabled. [PR1514635](#)

Subscriber Access Management

- Incorrect counter type (counter instead of gauge) specified for some values in MIB jnxUserAAAMib. [PR1533900](#)

Unified Threat Management (UTM)

- UTM causes e-mails from outside to inside to not be received. [PR1523222](#)
- Stream buffer memory leak might happen when UTM is configured under unified policies. [PR1557278](#)

VPNs

- The IKE tunnel negotiation might fail if IKE_INIT request is re-transmitted. [PR1460907](#)
- IPsec traffic may get dropped after RGO failover. [PR1522931](#)
- On all SRX Series devices using IPsec with NAT traversal, MTU size for the external interface might be changed after IPsec SA is re-established. [PR1530684](#)
- After IPsec tunnel using policy-based VPN is overwritten by another VPN client, traffic using this IPsec tunnel will be dropped. [PR1546537](#)

SEE ALSO

[What's New | 320](#)[What's Changed | 336](#)[Known Limitations | 342](#)[Open Issues | 345](#)[Documentation Updates | 359](#)[Migration, Upgrade, and Downgrade Instructions | 359](#)

Documentation Updates

There are no errata or changes in Junos OS Release 20.4R3 documentation for the SRX Series.

SEE ALSO

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Migration, Upgrade, and Downgrade Instructions

This section contains the procedure to upgrade Junos OS, and the upgrade and downgrade policies for Junos OS. Upgrading or downgrading Junos OS can take several hours, depending on the size and configuration of the network.

Upgrade and Downgrade Support Policy for Junos OS Releases

We have two types of releases, standard EOL and EEOL:

- Standard End of Life (EOL) releases have engineering support for twenty four months after the first general availability date and customer support for an additional six more months.

- Extended End of Life (EOL) releases have engineering support for thirty six months after the first general availability date and customer support for an additional six more months.

For both standard EOL and EEOL releases, you can upgrade to the next three subsequent releases or downgrade to the previous three releases. For example, you can upgrade from 19.2 to the next three releases – 19.3, 19.4 and 20.1 or downgrade to the previous three releases – 19.1, 18.4 and 18.3.

For EEOL releases only, you have an additional option - you can upgrade directly from one EEOL release to the next two subsequent EEOL releases, even if the target release is beyond the next three releases. Likewise, you can downgrade directly from one EEOL release to the previous two EEOL releases, even if the target release is beyond the previous three releases. For example, 19.2 is an EEOL release. Hence, you can upgrade from 19.2 to the next two EEOL releases – 19.3 and 19.4 or downgrade to the previous two EEOL releases – 19.1 and 18.4.

Release Type	End of Engineering (EOE)	End of Support (EOS)	Upgrade and Downgrade to subsequent 3 releases	Upgrade and Downgrade to subsequent 2 EEOL releases
Standard End of Life (EOL)	24 months	End of Engineering + 6 months	Yes	No
Extended End of Life (EEOL)	36 months	End of Engineering + 6 months	Yes	Yes

For more information about standard EOL and EEOL releases, see <https://www.juniper.net/support/eol/junos.html>.

For information about software installation and upgrade, see the [Installation and Upgrade Guide](#).

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Junos OS Release Notes for vMX

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These release notes accompany Junos OS Release 20.4R3 for vMX. They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

You can also find these release notes on the Juniper Networks Junos OS Documentation webpage, located at https://www.juniper.net/documentation/product/en_US/junos-os.

What's New

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- [What's New in Release 20.4R2 | 362](#)
- [What's New in Release 20.4R1 | 362](#)

Learn about new features introduced in the Junos OS main and maintenance releases for vMX.

What's New in Release 20.4R3

There are no new features for vMX in Junos OS Release 20.4R3.

What's New in Release 20.4R2

There are no new features for vMX in Junos OS Release 20.4R2.

What's New in Release 20.4R1

EVPN

- **MC-LAG emulation in an EVPN deployment (EX Series, MX Series, and vMX)**—Starting in Junos OS Release 20.4R1, you can emulate the function of an MC-LAG in active-standby mode in an EVPN configuration without having to configure an ICCP or ICL interface. In a standard EVPN configuration, logical interfaces configured on an aggregated Ethernet interface can have different designated forwarder election roles. To emulate an MC-LAG configuration, the designated forwarder (DF) takes on the role of the aggregated Ethernet interface. The provider edge (PE) that is the non-DF will send LACP out-of-sync packets to the CE. This causes LACP to go down on the CE device, and the CE device does not use the links connected to the non-DF for sending traffic. If the connection between a CE and a DF PE fails, the PE is re-elected as a DF. If the connection between a CE and a non-DF PE fails, the current DF PE is not changed.

To enable this functionality, configure the **lACP-oos-on-ndf** statement at the **[edit interfaces interface name esi df-election-granularity per-esi]** hierarchy.

Juniper Extension Toolkit (JET)

- **Juniper Extension Toolkit (JET) support for 64-bit applications (MX5, MX10, MX40, MX80, MX104, MX150, MX204, MX240, MX480, MX960, MX2008, MX2010, MX2020, MX10003, MX10008, MX ELM, JunosV Firefly, cSRX, SRX100, SRX110, SRX210, SRX220, SRX240, SRX300, SRX320, SRX340, SRX345, SRX550, SRX550HM, SRX650, SRX720E, SRX750E, SRX1400, SRX1500, SRX3400, SRX3600, SRX4100, SRX4200, SRX4400, SRX4600, SRX4800, SRX5400, SRX5600, SRX5800, SRX7X0E, SRX-ES7, SRX-ES8, VMX, and VSRX)**—Starting in Junos OS Release 20.4R1, JET supports 64-bit applications. Use the following commands to compile 64-bit applications for use with the AMD64 or ARM64 64-bit processor architecture.
 - **mk-amd64**: Compiles the application for use with AMD64 and Junos OS with FreeBSD.
 - **mk-amd64,bsd**: Compiles the application for use with AMD64 and Junos OS with upgraded FreeBSD.
 - **mk-arm64,bsd**: Compiles the application for use with ARM64 and Junos OS with upgraded FreeBSD.

[See [Develop On-Device JET Applications](#).]

- **Configure inner source MAC address for flexible VXLAN tunnels (MX Series and vMX with MPC1-MPC9E or LC2101)**—Starting in Junos OS Release 20.4R1, you can use the Juniper Extension Toolkit (JET) RIB Service API to configure the source MAC address used in IPv4 and IPv6 flexible VXLAN tunnel encapsulation profiles. The source MAC addresses is stored in the inner Ethernet header of VXLAN encapsulation. If you don't specify a source MAC address, the default source MAC address 00:00:5e:00:52:01 is used to encapsulate IPv4 and IPv6 flexible VXLAN tunnels.

Use the **show route detail**, **show route extensive**, and **show flexible-tunnels profiles** CLI commands or the **get-route-information** and **get-flexible-tunnels-profiles** RPC/NETCONF commands to view the source MAC address that is specified in the flexible tunnel profile.

[See [Understanding Programmable Flexible VXLAN Tunnels](#) and [JET APIs on Juniper EngNet](#).]

Junos OS XML ,API, and Scripting

- **Support for Certificate Authority Chain Profile (EX2300, EX3400, EX4300, MX240, MX480, MX960, PTX-5000, VMX, vSRX and QFX5200)**—Starting in Junos OS Release 20.4R1, you can configure intermediate Certificate Authority (CA) chain profile certificate and perform https REST API request using mutual and server authentications.

To configure intermediate ca-chain certificate, configure **ca-chain ca-chain** statement at the **[edit system services rest https]** hierarchy level.

Network Management and Monitoring

- **Configuration retrieval using the configuration revision identifier (EX3400, EX4300, MX204, MX240, MX480, MX960, MX2020, PTX3000, PTX10008, QFX5100, QFX10002-60C, SRX5800, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, you can use the configuration revision identifier feature to view the configuration for a specific revision. This configuration database revision can be viewed with the CLI command **show system configuration revision**.

[See [show system configuration revision](#).]

- **Junos XML protocol operations support loading and comparing configurations using the configuration revision identifier (EX3400, EX4300, MX204, MX240, MX480, MX960, MX2020, PTX3000, PTX10008, QFX5100, QFX10002-60C, SRX5800, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, the Junos XML management protocol operations support loading and comparing configurations by referencing the configuration revision identifier of a committed configuration. You can execute the **<load-configuration>** operation with the **configuration-revision** attribute to load the configuration with the given revision identifier into the candidate configuration. Additionally, you can compare the candidate or active configuration to a previously committed configuration by referencing the configuration revision identifier for the comparison configuration. The **<get-configuration>** operation supports the **compare="configuration-revision"** and **configuration-revision** attributes to perform the comparison.

[See [<get-configuration>](#) and [<load-configuration>](#).]

Routing Protocols

- **Support for multiple single-hop EBGP sessions on different links using the same IPv6 link-local address (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, you are no longer required to have unique peer addresses for Juniper devices for every EBGP session. You can now enable single-hop EBGP sessions on different links over multiple directly connected peers that use the same IPv6 link-local address.

In earlier Junos OS Releases, BGP peers could be configured with link-local addresses, but multiple BGP peers could not be configured to use the same link-local address on different interfaces.

[See [Configure Multiple Single-Hop EBGP Sessions on Different Links Using the Same Link-Local Address \(IPv6\)](#).]

What's Changed

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- [What's Changed in Release 20.4R2](#) | 365
- [What's Changed in Release 20.4R1](#) | 365

Learn about what changed in the Junos OS main and maintenance releases for vMX.

What's Changed in Release 20.4R3

Network Management and Monitoring

- The configuration accepts only defined identity values for nodes of type `identityref` in YANG data models (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)—If you configure a statement that has type `identityref` in the corresponding YANG data model, the device accepts only defined identity values (as defined by an identity statement) as valid input. In earlier releases, the device also accepts values that are not defined identity values.

Platform and Infrastructure

- We are discontinuing support for vMX on Microsoft Azure starting in Junos OS Release 18.1R1.

What's Changed in Release 20.4R2

Junos OS XML API and Scripting

- Refreshing scripts from an HTTPS server requires a certificate (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)—When you refresh a local commit, event, op, SNMP, or Juniper Extension Toolkit (JET) script from an HTTPS server, you must specify the certificate (Root CA or self-signed) that the device uses to validate the server's certificate, thus ensuring that the server is authentic. In earlier releases, when you refresh scripts from an HTTPS server, the device does not perform certificate validation.

When you refresh a script using the **request system scripts refresh-from** operational mode command, include the **cert-file** option and specify the certificate path. Before you refresh a script using the **set refresh** or **set refresh-from** configuration mode command, first configure the **cert-file** statement under the hierarchy level where you configure the script. The certificate must be in Privacy-Enhanced Mail (PEM) format.

[See [request system scripts refresh-from](#) and [cert-file](#).]

What's Changed in Release 20.4R1

Licensing

- Change in license bandwidth command on vMX virtual routers—Starting in Junos OS, to use the available license bandwidth, explicitly set the license bandwidth use the **set chassis license bandwidth <In Mbps>** command

[See [Configuring Licenses on vMX Virtual Routers](#).]

Known Limitations

IN THIS SECTION

- [Platform and Infrastructure | 366](#)

Learn about known limitations in this release for vMX.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Platform and Infrastructure

- The eODN feature supports up to 1K SRTE LSPs with 32K express segment links in TED. [PR1561947](#)
- RPM is a Juniper Networks proprietary feature. In the case of Junos OS, the RPM client never set the DF bit. Hence, we don't see this issue between a Junos OS RPM client and a Junos OS RPM server. Whereas in the case of Junos OS Evolved, the RPM client is setting the DF bit while sending the RPM probes to RPM server. In case of Junos OS TVP-based platforms, the RPM server is not able to decode the DF bit properly. This causes an interwork failure between Junos OS Evolved as the RPM client and TVP platforms as the RPM server (and vice versa). This issue is not applicable for non-TVP-based Junos OS platforms acting as the RPM server. [PR1508127](#)

Open Issues

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- [General Routing | 367](#)

Learn about open issues in this release for vMX.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

General Routing

- On vMX, the blockpointer in the ktree is getting corrupted leading to core-file generation. There is no function impact such as fpc restart or system down and the issue is not seen in hardware setups. [PR1525594](#)

Resolved Issues

Learn which issues were resolved in the Junos OS main and maintenance releases for vMX.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Resolved Issues: 20.4R3

Platform and Infrastructure

- Communication between two CEs might be failed when BGP rib-sharding is enabled. [PR1582210](#)
- Interface hold-time up does not work on vMX and MX150. [PR1604554](#)
- In the VXLAN scenario, the locally originated packets have UDP source port 0. [PR1571970](#)
- The l2ald process might crash on changing the routing-instance. [PR1584737](#)
- Interwork failure between Junos OS Evolved as RPM client and TVP platforms as RPM server and vice versa. [PR1508127](#)
- Traffic loss might be observed due to FPC crash on MX and vMX platforms. [PR1563144](#)

Resolved Issues: 20.4R2

General Routing

- Multiple vmxt processes might generate core files. [PR1534641](#)
- Configuring HFRR (link-protection) on an interface might cause the rpd process to crash. [PR1555866](#)

Resolved Issues: 20.4R1

Interfaces and Chassis

- Random packet drop with flow cache disabled, when NIC mapped to NUMA node 1. [PR1458742](#)

- On vMX instances, configuring the statement ranges for autosensed VLANs (either stacked VLANs or single-tag VLANs) might not work. This is because the VLANs are not programmed on the NIC drivers. [PR1503538](#)
- After the peer is moved out of the protection group, the path protection is not removed from the PE device. Multipath route is still present. [PR1538956](#)

Network Management and Monitoring

- After l2cpd service is restarted, the context of registration from l2cpd to snmpd was failing due to incorrect reinitialization. Because of this, if an NMS polls the dot1dStp objects by prefixing the context might fail. As a workaround, restart snmpd or reconfigure protocols hierarchy. [PR1561736](#)

Licensing

Starting in Junos OS Release 19.2R1, Juniper Agile Licensing introduces a new capability that significantly improves the ease of license management network wide. The Juniper Agile License Manager is a software application that runs on your network and provides an on-premise repository of licenses that are dynamically consumed by Juniper Networks devices and applications as required. Integration with Juniper's Entitlement Management System and Portal provides an intuitive extension of the existing user experience that enables you to manage all your licenses.

- The Agile License Manager is a new option that provides more efficient management of licenses, but you can continue to use individual license keys for each device if required.
- To use vMX or vBNG feature licenses in the Junos OS Release 19.2R1 version, you need new license keys. Previous license keys will continue to be supported for previous Junos OS releases, but for the Junos OS Release 19.2R1 and later you need to carry out a one-time migration of existing licenses. Contact [Customer Care](#) to exchange previous licenses. Note that you can choose to use individual license keys for each device, or to deploy Agile License Manager for more efficient management of licenses.
- For more information about Agile Licensing keys and capabilities, see [Juniper Agile Licensing portal FAQ](#).

See [Juniper Agile Licensing Guide](#) for more details on how to obtain, install, and use the License Manager.

Upgrade Instructions

You cannot upgrade Junos OS for the vMX router from earlier releases using the **request system software add** command.

You must deploy a new vMX instance using the downloaded software package.

Remember to prepare for upgrades with new license keys and/or deploying Agile License Manager.

Junos OS Release Notes for vRR

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These release notes accompany Junos OS Release 20.4R3 for vRR. They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

You can also find these release notes on the Juniper Networks Junos OS Documentation webpage, located at https://www.juniper.net/documentation/product/en_US/junos-os.

What's New

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- [What's New in Release 20.4R3 | 369](#)
- [What's New in Release 20.4R2 | 370](#)
- [What's New in Release 20.4R1 | 370](#)

Learn about new features introduced in the Junos OS main and maintenance releases for vRR.

What's New in Release 20.4R3

There are no new features for vRR in Junos OS Release 20.4R3.

To learn about common BGP or routing Junos OS features supported on vRR for Junos OS Release 20.4R3, see [What's New](#) for MX Series routers.

What's New in Release 20.4R2

There are no new features for vRR in Junos OS Release 20.4R2.

To learn about common BGP or routing Junos OS features supported on vRR for Junos OS Release 20.4R2, see [What's New](#) for MX Series routers.

What's New in Release 20.4R1

To learn about common BGP or routing Junos OS features supported on vRR for Junos OS Release 20.4R1, see [What's New](#) for MX Series routers.

Routing Protocols

- **Support for relaxing BGP router ID format from /32 to a nonzero ID per RFC 6286 (EX9208, EX9214, MX204, PTX5000, VRR, and QFX5120-32C)**—Starting in Junos OS Release 20.4R1, you can establish a BGP connection using a BGP identifier that is a 4-octet, unsigned, nonzero integer and it needs to be unique only within the autonomous system (AS) per RFC 6286. In earlier releases, the BGP ID of a BGP speaker was required to be a valid IPv4 host address assigned to the BGP speaker.

To enable this feature, use the **bgp-identifier *identifier* group *bgp group name* bgp-identifier *identifier* neighbor peer address bgp-identifier *identifier*** configuration statement at the [edit protocols bgp] hierarchy level.

[See [router-id](#)]

What's Changed

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- [What's Changed in Release 20.4R2 | 371](#)
- [What's Changed in Release 20.4R1 | 371](#)

Learn about what changed in the Junos OS main and maintenance releases for vRR.

What's Changed in Release 20.4R3

There are no changes in behavior or syntax for vRR in Junos OS Release 20.4R3.

To learn more about common BGP or routing changes in behavior or syntax in Junos OS Release 20.4R3, see [What's Changed](#) for MX Series routers.

What's Changed in Release 20.4R2

There are no changes in behavior or syntax for vRR in Junos OS Release 20.4R2.

To learn more about common BGP or routing changes in behavior or syntax in Junos OS Release 20.4R2, see [What's Changed](#) for MX Series routers.

What's Changed in Release 20.4R1

There are no changes in behavior or syntax for vRR in Junos OS Release 20.4R1.

To learn more about common BGP or routing changes in behavior or syntax in Junos OS Release 20.4R1, see [What's Changed](#) for MX Series routers.

Known Limitations

There are no known limitations for vRR in Junos OS Release 20.4R3.

To learn more about common BGP or routing known limitation in Junos OS Release 20.4R3, see [Known Limitations](#) for MX Series routers.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Open Issues

There are no known issues for vRR in Junos OS Release 20.4R3.

To learn more about common BGP or routing open issues in Junos OS Release 20.4R3, see [Open Issues](#) for MX Series routers.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Resolved Issues

Learn which issues were resolved in the Junos OS main and maintenance releases for vRR.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Resolved Issues: 20.4R3

To learn more about common BGP or routing resolved issues in Junos OS Release 20.4R3, see [Resolved Issues](#) for MX Series routers.

General Routing

- Option-60 (Vendor-Class-Identifier) is not sent during ZTP. [PR1582038](#)

Resolved Issues: 20.4R2

To learn more about common BGP or routing resolved issues in Junos OS Release 20.4R2, see [Resolved Issues](#) for MX Series routers.

General Routing

- 6PE prefixes might not be removed from RIB upon reception of withdrawal from a BGP neighbor when RIB sharding is enabled. [PR1556271](#)

Resolved Issues: 20.4R1

There are no resolved issues for vRR in Junos OS Release 20.4R1.

To learn more about common BGP or routing resolved issues in Junos OS Release 20.4R1, see [Resolved Issues](#) for MX Series routers.

Junos OS Release Notes for vSRX

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These release notes accompany Junos OS Release 20.4R3 for vSRX. They describe new and changed features, limitations, and known and resolved problems in the hardware and software.

You can also find these release notes on the Juniper Networks Junos OS Documentation webpage, located at https://www.juniper.net/documentation/product/en_US/junos-os.

What's New

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- [What's New in Release 20.4R1 | 374](#)

Learn about new features introduced in the Junos OS main and maintenance releases for vSRX.

What's New in Release 20.4R3

There are no new features for vSRX in Junos OS Release 20.4R3.

What's New in Release 20.4R2

Platform and Infrastructure

- **vSRX 3.0 support in Oracle Cloud Infrastructure (vSRX 3.0)**—Starting in Junos OS Release 20.4R2, you can deploy vSRX 3.0 in Oracle Cloud Infrastructure (OCI).

You must download the vSRX 3.0 software from the [Juniper Support Downloads](#) page and upload the software into an OCI compartment. vSRX 3.0 software download is not available in the OCI Marketplace.

vSRX 3.0 is available with built-in 60 days free trial eval license. The eval license expires after 60 days. OCI supports Bring Your Own License (BYOL) licensing model. The BYOL license model allows you to customize your license, subscription, and support to fit your needs. You can purchase BYOL from Juniper Networks or Juniper Networks authorized reseller.

[See [Licenses for vSRX](#) and [vSRX Deployment Guide for Private and Public Cloud Platforms](#).]

What's New in Release 20.4R1

ATP Cloud

- **Support for filtering DNS requests for disallowed domains (SRX4100, SRX4200, SRX4600, and vSRX)**—Starting in Junos OS Release 20.4R1, you can configure DNS filtering to identify DNS requests for disallowed domains. You can either:
 - Block access to the disallowed domain by sending a DNS response that contains the IP address or fully qualified domain name (FQDN) of a DNS sinkhole server. This ensures that when the client attempts to send traffic to the disallowed domain, the traffic instead goes to the sinkhole server.
 - Log the DNS request and reject access.

The DNS sinkhole must be configured only for DNS profile category.

[See [dns-filtering](#), [security-intelligence](#), [clear services security-intelligence dns-statistics](#), and [show services security-intelligence dns-statistics](#).]

Flow-Based Packet-Based Processing

- **Pass-through authentication of IP-IP and GRE tunnel traffic in TAP mode (SRX300, SRX320, SRX340, SRX345, SRX380, SRX550 HM, SRX1500, and vSRX)**—Starting in Junos OS Release 20.4R1, SRX Series devices perform pass-through authentication of IP-IP and GRE tunnel traffic when in TAP mode. To use TAP mode, connect the SRX Series device to the mirror port of the connected switch, which provides a copy of the traffic traversing the switch. In TAP mode, the SRX Series device processes incoming traffic from the TAP interface and generates a security log or report containing with information about threats detected, application usage, and user details.
- [See [Configure User Authentication Methods](#).]
- **Support for trace and debug of data packets (SRX1500, SRX4100, SRX4200, SRX4600, SRX5400, SRX5600, SRX5800, and vSRX)**—Starting in Junos OS Release 20.4R1, you can trace packet footprints.

To enable tracing of packet footprints, use the **traceoptions flag jexec** command at the **[edit security flow]** or **[edit logical-systems *logical system name* security flow]** hierarchy level.

The packet trace logs are captured in a sequential time order. The sequential trace enhances flow debuggability for packet processing with multiple logical systems and tenant systems, tunnel piping, multiple reinjection, and so on.

[See [traceoptions \(Security Flow\)](#) and [show security flow status](#).]

High Availability

- **SR-IOV 10GbE high availability support (vSRX 3.0)**—Starting in Junos OS Release 20.4R1, vSRX 3.0 supports high availability (HA) single-root I/O virtualization (SR-IOV) deployment.

If you have a physical network interface card (NIC) that supports SR-IOV, you can attach SR-IOV-enabled vNICs or virtual functions to the vSRX 3.0 instance.

With this feature, you can access the hardware directly from a virtual machines environment and efficiently share the PCIe devices to optimize performance and capacity. Also, this feature allows you to create many VFs associated with a single physical function (PF) extending the capacity of a device and lowering hardware costs.

We recommend that you configure all revenue ports of vSRX 3.0 as SR-IOV. On KVM, you can configure SR-IOV high availability on management port: -fxp0/ control port- em0 / fabric port-ge-0/0/*.

SR-IOV high availability Layer 2 function is not supported. Also, SR-IOV high availability with the vSRX 3.0 on VMWare and Mellanox NICs is not supported.

[See [Configuring SR-IOV 10-Gigabit High Availability on vSRX 3.0](#).]

Juniper Extension Toolkit (JET)

- **Juniper Extension Toolkit (JET) support for 64-bit applications (MX5, MX10, MX40, MX80, MX104, MX150, MX204, MX240, MX480, MX960, MX2008, MX2010, MX2020, MX10003, MX10008, MX ELM, JunosV Firefly, cSRX, SRX100, SRX110, SRX210, SRX220, SRX240, SRX300, SRX320, SRX340, SRX345, SRX550, SRX550HM, SRX650, SRX720E, SRX750E, SRX1400, SRX1500, SRX3400, SRX3600, SRX4100, SRX4200, SRX4400, SRX4600, SRX4800, SRX5400, SRX5600, SRX5800, SRX7X0E, SRX-ES7, SRX-ES8, VMX, and VSRX)**—Starting in Junos OS Release 20.4R1, JET supports 64-bit applications. Use the following commands to compile 64-bit applications for use with the AMD64 or ARM64 64-bit processor architecture.
 - **mk-amd64:** Compiles the application for use with AMD64 and Junos OS with FreeBSD.
 - **mk-amd64,bsd:** Compiles the application for use with AMD64 and Junos OS with upgraded FreeBSD.
 - **mk-arm64,bsd:** Compiles the application for use with ARM64 and Junos OS with upgraded FreeBSD.

[See [Develop On-Device JET Applications](#).]

Junos OS XML ,API, and Scripting

- **Support for Certificate Authority Chain Profile (EX2300, EX3400, EX4300, MX240, MX480, MX960, PTX-5000, VMX, vSRX and QFX5200)**—Starting in Junos OS Release 20.4R1, you can configure intermediate Certificate Authority (CA) chain profile certificate and perform https REST API request using mutual and server authentications.

To configure intermediate ca-chain certificate, configure **ca-chain ca-chain** statement at the **[edit system services rest https]** hierarchy level.

Network Management and Monitoring

- **Configuration retrieval using the configuration revision identifier (EX3400, EX4300, MX204, MX240, MX480, MX960, MX2020, PTX3000, PTX10008, QFX5100, QFX10002-60C, SRX5800, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, you can use the configuration revision identifier feature to view the configuration for a specific revision. This configuration database revision can be viewed with the CLI command **show system configuration revision**.

[See [show system configuration revision](#).]

- **Junos XML protocol operations support loading and comparing configurations using the configuration revision identifier (EX3400, EX4300, MX204, MX240, MX480, MX960, MX2020, PTX3000, PTX10008, QFX5100, QFX10002-60C, SRX5800, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, the Junos XML management protocol operations support loading and comparing configurations by referencing the configuration revision identifier of a committed configuration. You can execute the **<load-configuration>** operation with the **configuration-revision** attribute to load the configuration with the given revision identifier into the candidate configuration. Additionally, you can compare the candidate or active configuration to a previously committed configuration by referencing the configuration revision identifier for the comparison configuration. The **<get-configuration>** operation supports the **compare="configuration-revision"** and **configuration-revision** attributes to perform the comparison.

[See [<get-configuration>](#) and [<load-configuration>](#).]

Platform and Infrastructure

- **LiquidIO DPDK driver support (vSRX3.0)**—Starting in Junos OS Release 20.4R1, vSRX3.0 supports LiquidIO DPDK driver with KVM hypervisor. If you use the LiquidIO II smart NICs, then you can use vSRX3.0 by the virtual function of SR-IOV.

[See [Requirements for vSRX on KVM](#).]

Routing Protocols

- **Support for multiple single-hop EBGP sessions on different links using the same IPv6 link-local address (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)**—Starting in Junos OS Release 20.4R1, you are no longer required to have unique peer addresses for Juniper devices for every EBGP session. You can now enable single-hop EBGP sessions on different links over multiple directly connected peers that use the same IPv6 link-local address.

In earlier Junos OS Releases, BGP peers could be configured with link-local addresses, but multiple BGP peers could not be configured to use the same link-local address on different interfaces.

[See [Configure Multiple Single-Hop EBGP Sessions on Different Links Using the Same Link-Local Address \(IPv6\)](#).]

VPNs

- **AWS Key Management Service (KMS) Integration support (vSRX 3.0)**—Starting in Junos OS Release 20.4R1, you can safeguard the private keys used by the PKI daemon and IKED using AWS Key Management Service (KMS). You can establish a PKI daemon-based VPN tunnel using the keypairs generated at the KMS. The KMS server creates, stores, and performs the needed keypair operations. After you enable KMS, all the PKI daemons keypairs previously created are deleted.

[See [Deploying vSRX 3.0 for Securing Data using AWS KMS](#).]

What's Changed

IN THIS SECTION

- [What's Changed in Release 20.4R3 | 378](#)
- [What's Changed in Release 20.4R2 | 378](#)
- [What's Changed in Release 20.4R1 | 378](#)

Learn about what changed in the Junos OS main and maintenance releases for vSRX.

What's Changed in Release 20.4R3

Network Management and Monitoring

- The configuration accepts only defined identity values for nodes of type `identityref` in YANG data models (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)—If you configure a statement that has type `identityref` in the corresponding YANG data model, the device accepts only defined identity values (as defined by an identity statement) as valid input. In earlier releases, the device also accepts values that are not defined identity values.

What's Changed in Release 20.4R2

Junos OS XML API and Scripting

- Refreshing scripts from an HTTPS server requires a certificate (ACX Series, EX Series, MX Series, PTX Series, QFX Series, SRX Series, vMX, and vSRX)—When you refresh a local commit, event, op, SNMP, or Juniper Extension Toolkit (JET) script from an HTTPS server, you must specify the certificate (Root CA or self-signed) that the device uses to validate the server's certificate, thus ensuring that the server is authentic. In earlier releases, when you refresh scripts from an HTTPS server, the device does not perform certificate validation.

When you refresh a script using the **request system scripts refresh-from** operational mode command, include the **cert-file** option and specify the certificate path. Before you refresh a script using the **set refresh** or **set refresh-from** configuration mode command, first configure the **cert-file** statement under the hierarchy level where you configure the script. The certificate must be in Privacy-Enhanced Mail (PEM) format.

[See [request system scripts refresh-from](#) and [cert-file](#).]

What's Changed in Release 20.4R1

Platform and Infrastructure

- On vSRX 3.0 instances with AWS Key Management Service (KMS), if the MEK is changed, then the keypairs will be re-encrypted using the newly set Master Encryption Key (MEK).
- Repetition of WALinuxAgent logs causing file size increase (vSRX 3.0)—The Azure WALinuxAgent performs the provisioning job for the vSRX instances. When a new vSRX instance is deployed, the continued increasing size of the waagent log file might cause the vSRX to stop.

If the vSRX is still operating, then delete the `/var/log/waagent.log` directly or run the `clear log waagent.log` command to clear the log file. Or you can run the `set groups azure-provision system syslog file waagent.log archive size 1m` and `set groups azure-provision system syslog file waagent.log archive files 10` commands to prevent the growing of the waagent logs.

These configurations will cause the rotation of log of waagent with the size bigger than 1MB and set a maximum of 10 backups.

See [vSRX with Microsoft Azure](#).

Known Limitations

There are no known behaviors for vSRX in Junos OS Release 20.4R3.

Open Issues

IN THIS SECTION

- [Platform and Infrastructure](#) | 379

Learn about open issues in this release for vSRX.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Platform and Infrastructure

- IPV6 Traffic redirection by NSX-T edge infrastructure is not supported. To get it working check possible workaround. [PR1527130](#)
- Ensure the MTU on the host is large enough before setting the MTU in vSRX. [PR1537984](#)
- When upgrading to Junos OS Release 20.4R1 or later, any existing entries within the on-box logging database (security logs) are cleared. This is due to the high performance database design that is not forward-compatible from Junos OS Release 20.3 or earlier versions. These are the logs normally visible within J-Web under the Monitoring>Logs page. [PR1541674](#)
- With ssl-proxy configured along with web-proxy, the client session might not closed on the device even though proxy session ends gracefully. [PR1580526](#)
- SRX platforms using SkyATP with security-intelligence configured, might disconnect from the cloud after several days with the error "Connection status: Request client certificate failed". The issue can be recovered by the CLI command "restart pki-service". [PR1585362](#)
- Web-proxy: Getting UNKNOWN instead of HTTP-PROXY for application and UNKNOWN instead of GOOGLE-GEN in RT-FLOW close messages These messages can be seen in the RT-flow close log and

these are due to JDPI not engaged for the session. This may affect the app identification for the web-proxy session traffic. [PR1588139](#)

Resolved Issues

Learn which issues were resolved in the Junos OS main and maintenance releases for vSRX.

For the most complete and latest information about known Junos OS defects, use the Juniper Networks online [Junos Problem Report Search](#) application.

Resolved Issues: 20.4R3

Authentication and Access Control

- Unified-access-control(UAC) authentication might not work post system reboot [PR1585158](#)

Flow-Based and Packet-Based Processing

- The srxpfe process might crash during route churn [PR1572240](#)
- The flowd/srxpfe process might crash when clearing the TCP-Proxy session [PR1573842](#)
- Multicast traffic drop may occur on TAP interface on SRX devices [PR1583214](#)
- Unexpected traffic drop happens if both PMI and flexible-vlan-tagging are implemented [PR1584299](#)

General Routing

- IKE configure mode payload is not pushing secondary DNS and secondary WINS attributes to Xauth module with IKEv1. Hence, the client is not getting assigned with secondary DNS and secondary WINS with IKEv1. [PR1558831](#)
- Delay in vSRX CLI prompt might be observed [PR1559741](#)
- Fabric probe packets might be processed incorrectly when power-mode-ipsec (PMI) is enabled [PR1564117](#)
- The srxpfe/flowd process might crash when Sky-ATP is used [PR1573157](#)
- The srxpfe process might stop and generate a core file during the feed update process. [PR1579631](#)
- The incorrect DNS UDP checksums may be generated when vSRX3.0 performs DNS Sinkhole [PR1582827](#)
- vSRX unreachable over SSH after integration with KMS on AWS [PR1584415](#)
- Jflow V9 application-id record: Network based application recognition value for IPv4 application-id are not as expected [PR1595787](#)
- ALG traffic might be dropped [PR1598017](#)

Intrusion Detection and Prevention (IDP)

- Global data SHM utilization increase quickly and FTP traffic might impacted [PR1585485](#)
- Application-identification related signatures might not get triggered [PR1588450](#)

J-Web

- To improve performance in Monitoring > Network > Interfaces page, Admin Status is removed, Services and Protocols data merged into one Host inbound traffic. [PR1574895](#)

Platform and Infrastructure

- COS queue egress interface forwarding-class might not work as expected [PR1538286](#)

Resolved Issues: 20.4R2***J-Web***

- J-Web GUI does not allow you to save a rule if the cumulative shared objects are more than 2500 before the policy grid is saved. When there are several shared objects, there will be a noticeable delay in opening sources and destinations of a rule, and performing the rule action. [PR1540047](#)
- When the commit pending changes message is shown on the J-Web GUI, the contents of other messages, landing page, or pop-ups will not be clearly visible. [PR1554024](#)

Platform and Infrastructure

- Configuration integrity mismatch error in vSRX3.0 running on Azure with key-vault integrated. [PR1551419](#)
- The pkid process runs at 100 percent when the device is unable to connect to a particular URL. [PR1560374](#)
- The srpxfe process might stop and generate a core file during the feed update process. [PR1579631](#)

Routing Protocols

- Traffic might be lost during mirror data transmit from the primary ppmdd or bfdd. [PR1570228](#)

Resolved Issues: 20.4R1***Application Security***

- The flowd or srpxfe process might crash when SSL proxy and AppSecure process traffic simultaneously. [PR1516969](#)
- During rare circumstances, if the AppID unknown packet capture functionality is enabled, the srpxfe process might crash and generate a core file. [PR1538991](#)

Chassis Clustering

- The control link might be broken when there is excessive traffic load on the control link in a vSRX cluster deployment. [PR1524243](#)

CLI

- On Microsoft Azure deployments, SSH public key authentication is not supported for vSRX 3.0 CLI and portal deployment. [PR1402028](#)
- Commit is not successful when configuration committed without active probe settings options (all options under active probe settings are optional). [PR1533420](#)
- The master-password configuration is rejected if master-encryption-password (MEK) is not set. [PR1537251](#)

Flow-Based and Packet-Based Processing

- A chassis cluster node might stop passing traffic. [PR1528898](#)

Install and Upgrade

- Upgrading to Junos OS Release 20.4R1 or later releases with a large, pre-existing security-log database might result in LLMD consuming large amounts of CPU. [PR1548423](#)

Interfaces and Chassis

- LiquidIO SR-IOV configuring ge interface as DHCP client does not work; no IP address obtained. [PR1529228](#)

Intrusion Detection and Prevention (IDP)

- The flowd or srpxpfe process might generate core files during the idpd process commit on SRX Series devices. [PR1521682](#)
- When adaptive threat profiling is configured within an IDP rule base and logging is enabled, on the vSRX instances the Packet Forwarding Engine process might stop and generate a core file. [PR1532737](#)

Platform and Infrastructure

- The vSRX may restart unexpectedly. [PR1479156](#)
- In vSRX3.0 on Azure with key-vault enabled, change in MEK results in deletion of certificates. [PR1513456](#)
- With CSO SD-WAN configuration loaded, the flowd process generates core files while deleting the GRE IPsec configuration. [PR1513461](#)
- Configuration integrity mismatch error in vSRX3.0 running on Azure with key-vault integrated. [PR1551419](#)

Routing Policy and Firewall Filters

- Junos OS upgrade may encounter failure in certain conditions when enabling ATP. [PR1519222](#)

User Access and Authentication

- On vSRX 3.0 on Azure, with Microsoft Azure Hardware Security Module (HSM) enabled, keypair generation fails if you reuse the certificate ID for creating a new keypair—even if the previous keypair was deleted. [PR1490558](#)

VPNs

- The Ping-icmp test fails after configuring ECMP routes over multipoint tunnel interface VPNs. [PR1438311](#)
- The flowd process might stop in an IPsec VPN scenario. [PR1517262](#)

Migration, Upgrade, and Downgrade Instructions

IN THIS SECTION

- [Upgrading Software Packages | 384](#)
- [Validating the OVA Image | 389](#)

This section contains information about how to upgrade Junos OS for vSRX using the CLI. Upgrading or downgrading Junos OS can take several hours, depending on the size and configuration of the network.

You also can upgrade to Junos OS Release 20.4R3 for vSRX using J-Web (see [J-Web](#)) or the Junos Space Network Management Platform (see [Junos Space](#)).

Direct upgrade of vSRX from Junos OS 15.1X49 Releases to Junos OS Releases 17.4, 18.1, 18.2, 18.3, 18.4, 19.1, or 19.2 is supported.

The following limitations apply:

- Direct upgrade of vSRX from Junos OS 15.1X49 Releases to Junos OS Release 19.3 and higher is not supported. For upgrade between other combinations of Junos OS Releases in vSRX and vSRX 3.0, the general Junos OS upgrade policy applies.
- The file system mounted on /var usage must be below 14% of capacity.

Check this using the following command:

```
show system storage | match " /var$" /dev/vtbd1s1f
```

2.7G	82M	2.4G	3%	/var
------	-----	------	----	------

Using the **request system storage cleanup** command might help reach that percentage.

- The Junos OS upgrade image must be placed in the directory `/var/host-mnt/var/tmp/`. Use the **request system software add /var/host-mnt/var/tmp/<upgrade_image>**
- We recommend that you deploy a new vSRX virtual machine (VM) instead of performing a Junos OS upgrade. That also gives you the option to move from vSRX to the newer and more recommended vSRX 3.0.
- Ensure to back up valuable items such as configurations, license-keys, certificates, and other files that you would like to keep.

NOTE: For ESXi deployments, the firmware upgrade from Junos OS Release 15.1X49-Dxx to Junos OS releases 17.x, 18.x, or 19.x is not recommended if there are more than three network adapters on the 15.1X49-Dxx vSRX instance. If there are more than three network adapters and you want to upgrade, then we recommend that you either delete all the additional network adapters and add the network adapters after the upgrade or deploy a new vSRX instance on the targeted OS version.

Upgrading Software Packages

To upgrade the software using the CLI:

1. Download the **Junos OS Release 20.4R2 for vSRX .tgz** file from the [Juniper Networks website](#). Note the size of the software image.
2. Verify that you have enough free disk space on the vSRX instance to upload the new software image.

```
root@vsrx> show system storage
```

Filesystem	Size	Used	Avail	Capacity	Mounted on
/dev/vtbd0s1a	694M	433M	206M	68%	/
devfs	1.0K	1.0K	0B	100%	/dev
/dev/md0	1.3G	1.3G	0B	100%	/junos
/cf	694M	433M	206M	68%	/junos/cf
devfs	1.0K	1.0K	0B	100%	/junos/dev/
procfs	4.0K	4.0K	0B	100%	/proc
/dev/vtbd1s1e	302M	22K	278M	0%	/config

/dev/vtbd1s1f	2.7G	69M	2.4G	3%	/var
/dev/vtbd3s2	91M	782K	91M	1%	/var/host
/dev/md1	302M	1.9M	276M	1%	/mfs
/var/jail	2.7G	69M	2.4G	3%	/jail/var
/var/jails/rest-api	2.7G	69M	2.4G	3%	/web-api/var
/var/log	2.7G	69M	2.4G	3%	/jail/var/log
devfs	1.0K	1.0K	0B	100%	/jail/dev
192.168.1.1:/var/tmp/corefiles		4.5G	125M	4.1G	3%
/var/crash/corefiles					
192.168.1.1:/var/volatile		1.9G	4.0K	1.9G	0%
/var/log/host					
192.168.1.1:/var/log		4.5G	125M	4.1G	3%
/var/log/hostlogs					
192.168.1.1:/var/traffic-log		4.5G	125M	4.1G	3%
/var/traffic-log					
192.168.1.1:/var/local		4.5G	125M	4.1G	3%
/var/db/host					
192.168.1.1:/var/db/aamwd		4.5G	125M	4.1G	3%
/var/db/aamwd					
192.168.1.1:/var/db/secinteld		4.5G	125M	4.1G	3%
/var/db/secinteld					

3. Optionally, free up more disk space, if needed, to upload the image.

```

root@vsrx> request system storage cleanup
List of files to delete:
Size Date   Name
11B Sep 25 14:15 /var/jail/tmp/alarmd.ts
259.7K Sep 25 14:11 /var/log/hostlogs/vjunos0.log.1.gz
494B Sep 25 14:15 /var/log/interactive-commands.0.gz
20.4K Sep 25 14:15 /var/log/messages.0.gz
27B Sep 25 14:15 /var/log/wtmp.0.gz
27B Sep 25 14:14 /var/log/wtmp.1.gz
3027B Sep 25 14:13 /var/tmp/BSD.var.dist
0B Sep 25 14:14 /var/tmp/LOCK_FILE
666B Sep 25 14:14 /var/tmp/appidd_trace_debug
0B Sep 25 14:14 /var/tmp/eedebug_bin_file
34B Sep 25 14:14 /var/tmp/gksdchk.log
46B Sep 25 14:14 /var/tmp/kmdchk.log
57B Sep 25 14:14 /var/tmp/krt_rpf_filter.txt
42B Sep 25 14:13 /var/tmp/pfe_debug_commands

```

```

0B Sep 25 14:14 /var/tmp/pkg_cleanup.log.err
30B Sep 25 14:14 /var/tmp/policy_status
0B Sep 25 14:14 /var/tmp/rtssdb/if-rtssdb
Delete these files ? [yes,no] (no) yes
<
output omitted>

```

NOTE: If this command does not free up enough disk space, see [\[SRX\] Common and safe files to remove in order to increase available system storage](#) for details on safe files you can manually remove from vSRX to free up disk space.

4. Use FTP, SCP, or a similar utility to upload the Junos OS Release 20.4R2 for vSRX .tgz file to **/var/crash/corefiles/** on the local file system of your vSRX VM. For example:

```

root@vsrx> file copy ftp://username:prompt@ftp.hostname.net/pathname/
junos-vsr-x86-64-20.4-2021-09-04.0_RELEASE_20.4_THROTTLE.tgz /var/crash/corefiles/

```

5. From operational mode, install the software upgrade package.

```

root@vsrx> request system software add
/var/crash/corefiles/junos-vsr-x86-64-20.4-2021-09-04.0_RELEASE_20.4_THROTTLE.tgz
no-copy no-validate reboot
Verified junos-vsr-x86-64-20.4-2021-09-04.0_RELEASE_20.4_THROTTLE signed by
PackageDevelopmentEc_2017 method ECDSA256+SHA256
THIS IS A SIGNED PACKAGE
WARNING:      This package will load JUNOS 20.4 software.
WARNING:      It will save JUNOS configuration files, and SSH keys
WARNING:      (if configured), but erase all other files and information
WARNING:      stored on this machine. It will attempt to preserve dumps
WARNING:      and log files, but this can not be guaranteed. This is the
WARNING:      pre-installation stage and all the software is loaded when
WARNING:      you reboot the system.
Saving the config files ...
Pushing Junos image package to the host...
Installing
/var/tmp/install-media-srx-mr-vsr-20.4-2021-09-04.0_RELEASE_20.4_THROTTLE.tgz
Extracting the package ...
total 975372
-rw-r--r-- 1 30426 950 710337073 Oct 19 17:31
junos-srx-mr-vsr-20.4-2021-09-04.0_RELEASE_20.4_THROTTLE-app.tgz

```

```

-rw-r--r-- 1 30426 950 288433266 Oct 19 17:31
junos-srx-mr-vsrx-20.4-2021-09-04.0_RELEASE_20.4_THROTTLE-linux.tgz
Setting up Junos host applications for installation ...
=====
Host OS upgrade is FORCED
Current Host OS version: 3.0.4
New Host OS version: 3.0.4
Min host OS version required for applications: 0.2.4
=====
Installing Host OS ...
upgrade_platform: -----
upgrade_platform: Parameters passed:
upgrade_platform: silent=0
upgrade_platform:
package=/var/tmp/junos-srx-mr-vsrx-20.4-2021-09-04.0_RELEASE_20.4_THROTTLE-linux.tgz
upgrade_platform: clean install=0
upgrade_platform: clean upgrade=0
upgrade_platform: Need reboot after staging=0
upgrade_platform: -----
upgrade_platform:
upgrade_platform: Checking input
/var/tmp/junos-srx-mr-vsrx-20.4-2021-09-04.0_RELEASE_20.4_THROTTLE-linux.tgz ...
upgrade_platform: Input package
/var/tmp/junos-srx-mr-vsrx-20.4-2021-09-04.0_RELEASE_20.4_THROTTLE-linux.tgz is
valid.
upgrade_platform: Backing up boot assets..
cp: omitting directory '.'
bzImage-intel-x86-64.bin: OK
initramfs.cpio.gz: OK
version.txt: OK
initrd.cpio.gz: OK
upgrade_platform: Checksum verified and OK...
/boot
upgrade_platform: Backup completed
upgrade_platform: Staging the upgrade package -
/var/tmp/junos-srx-mr-vsrx-20.4-2021-09-04.0_RELEASE_20.4_THROTTLE-linux.tgz..
./
./bzImage-intel-x86-64.bin
./initramfs.cpio.gz
./upgrade_platform
./HOST_COMPAT_VERSION
./version.txt
./initrd.cpio.gz
./linux.checksum

```

```

./host-version
bzImage-intel-x86-64.bin: OK
initramfs.cpio.gz: OK
version.txt: OK
upgrade_platform: Checksum verified and OK...
upgrade_platform: Staging of
/var/tmp/junos-srx-mr-vsrx-20.4-2021-09-04.0_RELEASE_20.4_THROTTLE-linux.tgz
completed
upgrade_platform: System need *REBOOT* to complete the upgrade
upgrade_platform: Run upgrade_platform with option -r | --rollback to rollback
the upgrade
Host OS upgrade staged. Reboot the system to complete installation!
WARNING:      A REBOOT IS REQUIRED TO LOAD THIS SOFTWARE CORRECTLY. Use the
WARNING:      'request system reboot' command when software installation is
WARNING:      complete. To abort the installation, do not reboot your system,
WARNING:      instead use the 'request system software rollback'
WARNING:      command as soon as this operation completes.
NOTICE: 'pending' set will be activated at next reboot...
Rebooting. Please wait ...
shutdown: [pid 13050]
Shutdown NOW!
*** FINAL System shutdown message from root@ ***
System going down IMMEDIATELY
Shutdown NOW!
System shutdown time has arrived\x07\x07

```

If no errors occur, Junos OS reboots automatically to complete the upgrade process. You have successfully upgraded to Junos OS Release 20.4R2 for vSRX.

NOTE: Starting in Junos OS Release 17.4R1, upon completion of the vSRX image upgrade, the original image is removed by default as part of the upgrade process.

6. Log in and use the **show version** command to verify the upgrade.

```

--- JUNOS 20.4-2021-09-04.0_RELEASE_20.4_THROTTLE Kernel 64-bit
JNPR-11.0-20171012.170745_fbsd-
At least one package installed on this device has limited support.
Run 'file show /etc/notices/unsupported.txt' for details.
root@:~ # cli
root> show version
Model: vsrx

```

```

Junos: 20.4-2021-09-04.0_RELEASE_20.4_THROTTLE
JUNOS OS Kernel 64-bit [20171012.170745_fbsd-builder_stable_11]
JUNOS OS libs [20171012.170745_fbsd-builder_stable_11]
JUNOS OS runtime [20171012.170745_fbsd-builder_stable_11]
JUNOS OS time zone information [20171012.170745_fbsd-builder_stable_11]
JUNOS OS libs compat32 [20171012.170745_fbsd-builder_stable_11]
JUNOS OS 32-bit compatibility [20171012.170745_fbsd-builder_stable_11]
JUNOS py extensions [20171017.110007_ssd-builder_release_174_throttle]
JUNOS py base [20171017.110007_ssd-builder_release_174_throttle]
JUNOS OS vmguest [20171012.170745_fbsd-builder_stable_11]
JUNOS OS crypto [20171012.170745_fbsd-builder_stable_11]
JUNOS network stack and utilities
[20171017.110007_ssd-builder_release_174_throttle]
JUNOS libs [20171017.110007_ssd-builder_release_174_throttle]
JUNOS libs compat32 [20171017.110007_ssd-builder_release_174_throttle]
JUNOS runtime [20171017.110007_ssd-builder_release_174_throttle]
JUNOS Web Management Platform Package
[20171017.110007_ssd-builder_release_174_throttle]
JUNOS srx libs compat32 [20171017.110007_ssd-builder_release_174_throttle]
JUNOS srx runtime [20171017.110007_ssd-builder_release_174_throttle]
JUNOS common platform support [20171017.110007_ssd-builder_release_174_throttle]
JUNOS srx platform support [20171017.110007_ssd-builder_release_174_throttle]
JUNOS mtx network modules [20171017.110007_ssd-builder_release_174_throttle]
JUNOS modules [20171017.110007_ssd-builder_release_174_throttle]
JUNOS srxtvp modules [20171017.110007_ssd-builder_release_174_throttle]
JUNOS srxtvp libs [20171017.110007_ssd-builder_release_174_throttle]
JUNOS srx libs [20171017.110007_ssd-builder_release_174_throttle]
JUNOS srx Data Plane Crypto Support
[20171017.110007_ssd-builder_release_174_throttle]
JUNOS daemons [20171017.110007_ssd-builder_release_174_throttle]
JUNOS srx daemons [20171017.110007_ssd-builder_release_174_throttle]
JUNOS Online Documentation [20171017.110007_ssd-builder_release_174_throttle]
JUNOS jail runtime [20171012.170745_fbsd-builder_stable_11]
JUNOS FIPS mode utilities [20171017.110007_ssd-builder_release_174_throttle]

```

Validating the OVA Image

If you have downloaded a vSRX .ova image and need to validate it, see [Validating the vSRX .ova File for VMware](#).

Note that only .ova (VMware platform) vSRX images can be validated. The .qcow2 vSRX images for use with KVM cannot be validated the same way. File checksums for all software images are, however, available on the download page.

Upgrading Using ISSU

In-service software upgrade (ISSU) enables you to upgrade between two different Junos OS releases with no disruption on the control plane and with minimal disruption of traffic.

For additional information about using ISSU on routing and switching devices, see the [High Availability User Guide](#).

For additional information about using ISSU on security devices, see the [Chassis Cluster User Guide for SRX Series Devices](#).

For information about ISSU support across platforms and Junos OS releases, see the [In-Service Software Upgrade \(ISSU\)](#) Web application.

Licensing

Starting in 2020, Juniper Networks introduced a new software licensing model. The Juniper Flex Program comprises a framework, a set of policies, and various tools that help unify and thereby simplify the multiple product-driven licensing and packaging approaches that we've developed at Juniper Networks over the past several years.

The major components of the framework are:

- A focus on customer segments (enterprise, service provider, and cloud) and use cases for Juniper Networks hardware and software products.
- The introduction of a common three-tiered model (standard, advanced, and premium) for all Juniper Networks software products.
- The introduction of subscription licenses and subscription portability for all Juniper Networks products, including Junos OS and Contrail.

For information about the list of supported products, see [Juniper Flex Program](#).

Compliance Advisor

For regulatory compliance information about [Common Criteria](#), [FIPS](#), [Homologation](#), [RoHS2](#), and [USGv6](#) for Juniper Networks products, see the [Juniper Networks Compliance Advisor](#).

Finding More Information

- **Feature Explorer**—Juniper Networks Feature Explorer helps you in exploring software feature information to find the right software release and product for your network. <https://apps.juniper.net/feature-explorer/>
- **PR Search Tool**—Keep track of the latest and additional information about Junos OS open defects and issues resolved. prsearch.juniper.net.
- **Hardware Compatibility Tool**—Determine optical interfaces and transceivers supported across all platforms. apps.juniper.net/hct/home

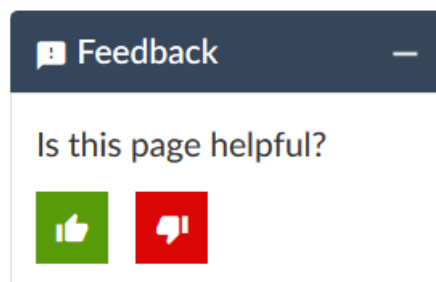
NOTE: To obtain information about the components that are supported on the devices, and the special compatibility guidelines with the release, see the Hardware Guide for the product.

- **Juniper Networks Compliance Advisor**—Review regulatory compliance information about [Common Criteria](#), [FIPS](#), [Homologation](#), [RoHS2](#), and [USGv6](#) for Juniper Networks products. apps.juniper.net/compliance/.

Documentation Feedback

We encourage you to provide feedback, comments, and suggestions so that we can improve the documentation. You can provide feedback by using either of the following methods:

- Online feedback system—Click TechLibrary Feedback, on the lower right of any page on the [Juniper Networks TechLibrary](#) site, and do one of the following:



- Click the thumbs-up icon if the information on the page was helpful to you.
- Click the thumbs-down icon if the information on the page was not helpful to you or if you have suggestions for improvement, and use the pop-up form to provide feedback.

- E-mail—Send your comments to techpubs-comments@juniper.net. Include the document or topic name, URL or page number, and software version (if applicable).

Requesting Technical Support

Technical product support is available through the Juniper Networks Technical Assistance Center (JTAC). If you are a customer with an active Juniper Care or Partner Support Services support contract, or are covered under warranty, and need post-sales technical support, you can access our tools and resources online or open a case with JTAC.

- JTAC policies—For a complete understanding of our JTAC procedures and policies, review the *JTAC User Guide* located at <https://www.juniper.net/us/en/local/pdf/resource-guides/7100059-en.pdf>.
- Product warranties—For product warranty information, visit <https://www.juniper.net/support/warranty/>.
- JTAC hours of operation—The JTAC centers have resources available 24 hours a day, 7 days a week, 365 days a year.

Self-Help Online Tools and Resources

For quick and easy problem resolution, Juniper Networks has designed an online self-service portal called the Customer Support Center (CSC) that provides you with the following features:

- Find CSC offerings: <https://www.juniper.net/customers/support/>
- Search for known bugs: <https://prsearch.juniper.net/>
- Find product documentation: <https://www.juniper.net/documentation/>
- Find solutions and answer questions using our Knowledge Base: <https://kb.juniper.net/>
- Download the latest versions of software and review release notes: <https://www.juniper.net/customers/csc/software/>
- Search technical bulletins for relevant hardware and software notifications: <https://kb.juniper.net/InfoCenter/>
- Join and participate in the Juniper Networks Community Forum: <https://www.juniper.net/company/communities/>
- Create a service request online: <https://myjuniper.juniper.net>

To verify service entitlement by product serial number, use our Serial Number Entitlement (SNE) Tool: <https://entitlementsearch.juniper.net/entitlementsearch/>

Creating a Service Request with JTAC

You can create a service request with JTAC on the Web or by telephone.

- Visit <https://myjuniper.juniper.net>.
- Call 1-888-314-JTAC (1-888-314-5822 toll-free in the USA, Canada, and Mexico).

For international or direct-dial options in countries without toll-free numbers, see <https://support.juniper.net/support/requesting-support/>.

Revision History

10 November 2023—Revision 19, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

26 October 2023—Revision 18, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

18 October 2023—Revision 17, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

6 July 2023—Revision 16, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

9 March 2023—Revision 15, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

10 February 2023—Revision 14, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

25 November 2022—Revision 13, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

29 July 2022—Revision 12, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

31 March 2022—Revision 11, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

24 March 2022—Revision 10, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

17 March 2022—Revision 9, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

24 February 2022—Revision 8, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

17 February 2022—Revision 7, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

28 January 2022—Revision 6, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

25 November 2021—Revision 5, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

11 November 2021—Revision 4, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

14 October 2021—Revision 3, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

7 October 2021—Revision 2, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

16 September 2021—Revision 1, Junos OS Release 20.4R3— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

22 July 2021—Revision 4, Junos OS Release 20.4R2— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

15 July 2021—Revision 3, Junos OS Release 20.4R2— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

1 July 2021—Revision 2, Junos OS Release 20.4R2— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

7 May 2021—Revision 1, Junos OS Release 20.4R2— ACX Series, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

4 March 2021—Revision 6, Junos OS Release 20.4R1— ACX Series, cRPD, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

28 January 2021—Revision 5, Junos OS Release 20.4R1— ACX Series, cRPD, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

13 January 2021—Revision 4, Junos OS Release 20.4R1— ACX Series, cRPD, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

7 January 2021—Revision 3, Junos OS Release 20.4R1— ACX Series, cRPD, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

29 December 2020—Revision 2, Junos OS Release 20.4R1— ACX Series, cRPD, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.

28 December 2020—Revision 1, Junos OS Release 20.4R1— ACX Series, cRPD, cSRX, EX Series, JRR Series, Juniper Secure Connect, Junos Fusion, MX Series, NFX Series, PTX Series, QFX Series, SRX Series, vMX, vRR, and vSRX.